

Axiom

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Chapter 1

Preface

"If you start with the right axioms, the rest will follow."— David Hilbert

This book "Axiom" is a continuation of my previous project Lógos. Which was mostly a collection of essays on several topics; physics, philosophy, religion, civilization, psychology, personal plans, etc.

It was previously meant to work through my ideas, using the fact I was writing it out. It allowed me to keep track of the dozens of ideas and how they all interacted with each other. Developing on the complexity without forgetting and creating a full framework with defined axioms.

Creating this new project is for multiple reasons.

1. To create a more organized book
 - What I mean is have an order, like create a fully formalized epistemology and basic metaphysics and then using that to create a fully organized Ontology. Rather than a chaotic coll
2. To mark the end of my high school career and beginning

of University

3. Lógos had become too big and had too many errors
 - Too big to run on Overleaf, too many errors to run directly in the terminal.
 - This will start smaller, and once it is bigger will be made without errors so it can be run on the terminal once it gets bigger. Also more modular
4. Be able to rewrite with my new knowledge to update it.
5. Write it with a better understanding of the packages, LaTeX, and even create my own custom packages

In the end, it serves the same purpose; to help me better understand things through writing, develop thing iteratively over time using writing, and write out notes to remember things for the future.

As of writing this section, the text will be broken into three parts. Part one being "The Foundation." It would be a axiomatic approach to creating the foundations needed for future systems. Two, being "Notes" on things that I have learned. Three, being writings on my "Projects"; mostly physics research projects but also projects in further developing philosophy or other similar things. Finally, "Life in Practice" being more practical things; plans for the future, self-analysis, ideals on society, etc.

Part I

The Foundation

Chapter 2

Introduction

As mentioned in the preface, this section is made to axiomatically develop the foundations needed.

To illustrate, in order to develop physical theories, mathematics must be developed, a philosophy of science must be made, metaphysics, epistemology, and formal logic. Thus, a complete theory of all of these things and more should be created. One that has fundamental, expressible axioms.

Also, most of these theories rely on one another, so a logical chronology must be defined. At the moment of writing this it is defined to be logic → metaphysics → Epistemology → philosophy of science → philosophy of physics → mathematics → physics → theology → ethics → social philosophy → personal philosophy.

One additional thing to add, you may notice the separation of metaphysics and the philosophy of physics. These are both meant to signify a difference where metaphysics represents extremely fundamental questions like does existence exist. Things that must be answered before epistemology can be fully introduced. The next section, philosophy of physics, are slightly less fundamental, like the ontology of entropy. Ques-

tions that require a theory of epistemology to be answered.

Chapter 3

Logic

“One must be able to say at all times—instead of points, lines, and planes—tables, chairs, and beer mugs.” — David Hilbert

3.1 Introduction

Logic is an important tool for the creation of all formal systems. It is the foundation of mathematics, which is for the most part the foundation of everything else. It is also simply important to anything else by clarify concepts, giving them formal definitions, and insures consistency. Allowing for a bedrock behind all analytical thinking.

Though, before I can get into it too much, I should define what the study of logic is. It is the branch of philosophy that critically examines the fundamental nature, scope, and principles of logic itself.

It has a couple fo subfields as well.

- Philosophy of Logic
 - The study of logic at a high level, analyzing its nature

and scope.

- Formal Logic
 - A creation and usage of symbolic logic symbols to create systems; mainly mathematics.
- Metalogic
 - The technical, mathematical study of the properties of formal logical systems. The philosophy of logic often uses the results of metalogic (like Gödel's incompleteness theorems) to address its conceptual questions, such as the limits of a formal axiomatic system

3.2 Defining an Axiom

This entire text is named Axiom, for it is the foundation. To define systems axiomatically and then build them out into their true complexity is not only one of my favorite things to do, but as an extension of that it is what this book is trying to do. (Obviously, I wouldn't do something like this if I didn't enjoy it.)

So let's define it. As I mention extensively in my last book, *Lógos*, we must define things extremely rigorously. By testing against edge cases and keep it with consistent.

A good way to start is with the dictionary definition. "a statement accepted as true as the basis for argument or inference." For instance, our understanding of scientific realism is the understanding of the fact that our observations of reality are accurate.

A more defined definition can then be defined to say that an axiom is a statement that either cannot be proved or cannot be proved currently that is used within a greater statement

that can be proved upon its edifice. These axioms, while don't require 'proof' generally desire some type of reasoning even if it is not objective or purely analytical in nature. For instance, the above arguments for the objective nature of reality aren't true objective proof, they still exist as semi-arguments. While we cannot prove the objectivity of reality with true and objective reasoning, some reasoning can be applied to get some kind of 'proof' of the statement so that further studies can be applied based upon that said axiom.

An astute observer can find that these mean the same thing, my is more of an explanation so, the dictionary definition does work:

Definition 3.1. *Axiom: a statement accepted as true as the basis for argument or inference.*

3.3 *****

I am currently a little stuck on how to write this chapter, so I will return at a later point.

3.3.0.1 Propositional logic

The most basic form of classical logic, but instead of telling you now, I will develop it.

3.3.0.2 Axiom System for the Propositional Calculus

For this simple system a number of axioms must be assumed:

- $A \implies (B \implies A)$
- $(A \implies (B \implies C)) \implies ((A \implies B) \implies (A \implies C))$

- $(\neg B \implies \neg A) \implies (A \implies B)$

Then one rule of inference is needed. Modus Ponens(MP)

From A and $A \implies B$ infer B

This defines all tautologies and thus the larger system as a whole.

3.3.0.3 Defining Tautologies

A tautology is simply a statement form that is always true, regardless of the truth values of the statement letters.

An example would be the law of the excluded middle. $(A \vee (\neg A))$. This is a statement that is true by definition and there is no case where it can be false. Axioms in logic are generally tautologies(such is the way are the once mentioned in the previous subsection).

3.3.0.4 Truth Tables

Negation is the first and simplest form. The symbol is $\neg$, so $\neg A$ is the negation of A .

so the table can be constructed

$A \quad \neg A$

$T \quad F$

$F \quad T$

Where T, represents true, and F represents false.

A further complication is the conjunction operator $\wedge$ or "and."

It combines to functions into a greater set.(a union of function)

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

$A \wedge B$ is only true when both "A" and "B" are true.

$\vee$ is the inverse, only one of the functions must be true. It is called a disjunction.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

$A \implies B$ means "if A, then B" meaning that A needs to happen for B or that A causes B.

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

There are a couple of other operators used in prepositional logic, but for not I will move on and if needed bring those back up at a later point.

Chapter 4

Philosophy of Mathematics

Chapter 5

Fundamental Metaphysics

5.1 Scope

This chapter will cover fundamental metaphysics, what I mean by this is for my progression of metaphysics I need a theory of epistemology, philosophy of science, inductive and deductive research on physics, etc; however, for all of these things I need some concepts of metaphysics. Things like the existence, reality, consciousness, questions of God, and so much more. Thus, this will serve that purpose.

5.2 Reality

5.2.1 Existence Exists

To truly come to a thorough understanding of a system, one must create it from first principle slowly and carefully. To question every implicit assumption, this for the most part is the quest of philosophy; especially metaphysics.

Thus let us begin our creation of reality.

Existence Exists, this is by definition an irrefutable point. To negate an argument an antithesis must be supplanted, non-existing, but non-existence by definition implies a existence. This is because you cannot have a non-existence without an existence. $\neg E(x) \implies E(x) \iff ((\forall P(Prop(P) \implies E(p))) \wedge (\exists Prop(E(\neg E(x))))).$ By the simple definition of existence to be everything. Another more concrete argument is that to deny the existence is a claim, a claim is an act, an act requires and actor, and thus an actor must exist, an actor exists within the subset of existence, so if it exists so must existence.

Though, this doesn't say anything about the existence, in fact there is a special case in which existence is the absent of all things. Using the second argument, we create a conscious entity. *Cognito, ergo sum*. I think therefore I am. Now upon inspection of this entity it is quite simple, the only capacity it necessarily has is the ability to make a claim, such a claim needs no basis in reality, sense, logic, or anything of such. It could be the dream of a butterfly, random interactions of particles in the dead of space, or anything of the such.

One constraint does exist, that entity is me. While it could still(at least for now) be some random assortment of particles or what ever machinations you may dream of, there is some knowledge we do know. It has the ability to observe, reason, and has innate concepts like space, time, similarities, and differences. One thing to be clear, at this point there is no justification that any of these things have any connection to reality, but they are apart of the entities observations.

5.2.2 The Entity and Observable reality

Now, it is technically true that the entity has no way to truly and faithfully validate his observations and reasoning beyond self-referential, due the fact that these are the entities' only capacity to make claims.

Thus, taking these as true is a fair axiom, unprovable but logical in the necessity of the argument. I do not mean taking all observations as true (in fact always questioning them is of great importance as I will address later). I mean regarding that our observations are a consequence of reality. Taking the reality that existed around us as the "truth." While observations can be complete, wrong, they are not false by the nature of being observations by the entity.

Now this creates a dichotomy between the observed reality and "true reality" (whatever that might be). There is a spectrum of how much these two overlap.

Ranging from:

A: Scientific Materialism; that all of observed reality and true reality overlap because these observations are directly true, and nothing exists beyond what is within the capacity of theoretical observation.

B: Idealism; there is no overlap at all. Such as the Boltzmann brain.

Now, this observable reality, is still apart of existence, and is thus no metaphysically distinct if some greater "true reality" and is thus given the same importance and study. While other sub-existences beyond our observable are lesser for reasons, these are more ethical, aesthetic, and theological rather than truly metaphysical.

5.2.3 Beyond Observable

Given this spectrum, it is entirely possible for there to exist unobservable reality that is true. This reality has unknown capacity and can theoretically interact and cause observables.

5.2.4 Independence of Reality

Given that this entity is me, I observe the continuation of reality beyond my direct observation. Once again this is not direct proof that it is, but makes it indistinguishable from if it was.

5.3 The Entity and Observations

5.3.1 Introduction

Now this entity observes a great many things; matter, other entities, change, relations, space, time, identity, theory, mathematics, difference, categories, individuation, emotions, structure, aesthetics, fundamentalism, layers, evidence, argumentation, possibility, necessity, abstracta, causation, and much more. Categorizing these and understanding them is the foundation of metaphysics.

5.3.2 Properties

All observables are made up of properties, and many of the observables themselves are properties.

5.3.3 Abstraction: Math, sets, and Relations

Given this nature, these properties can act as sets of 1 and can make up object which are thus observables. They thus follow the same mathematical rules of sets.

- Relations to other sets(and themselves)
- Shared Properties with other sets
- can be apart of larger sets(categories)

- and so much more

Now I won't examine all of this because it is mentioned in the earlier chapter so we will move on.

These sets can be whatever, such as the other properties or something new all together.

5.3.4 Matter

One of the most important sets is matter, within this set is a near infinite amount of subsets. These subsets can "theoretically" have about any property. The main one that define it is that it must have the capacity to affect the properties of other matter (and thus be effected too). It must also have the properties of existing within space and time. Direct space or time is not needed, but having the property is required. Finally, it must be independent of a thinker.

5.3.5 SpaceTime

Once again, given the simplicity of this model this examination will be quite simple. Essentially all that can be taken at this moment is that space and time are properties of many things. Extremely important ones. Dimensions to be considered, later an address the geometry and importance of these fundamental dimensions.

5.3.6 Nous

A thinker is an entity like object of matter. These entities are capable of creating, holding, using, being effected by, etc by non-material objects. These thinkers do not need to be human, or even alive for that matter. For future references I will refer to it as nous; based upon the greek word.

5.3.7 Non-material

Already defined to be objects that exist(observable have effects) dependent of a thinker.

5.4 Assumptions(Axioms)

Get it, I said the name of the book, Axiom.

Anyways, here I define and summerize the assumptions I have made.

- Principle of Causal Order
- Time and Spacial symmetry of the laws of nature
- Reality Obeys Fundemental Logic and Tautologies

Many of the other commonly noted axioms like that the uniev-erse follows mathematic or that nature is orderly follow natu-rally from these assumptions.

5.5 Platoism and Realism

With this view of reality, to somewhat conflicting views occur, Platoism and Realism.

Given the view of the surronding reality as 'real,' all other principles of Realsim become natural and logical.

From the fact that this view of reality is simply our visiable abstraction of reality based upon logic and mathematics, Pla-toism becomes natural. Further analysis of these argumen-tations have already been presented and will be presented in Essays and this section may latter be updated.

For the purposes of this book, the the equavleny of these expressions will generally be used. With, given the mathematical and abstract nature of this exersice, Platoism will be used for the majority of the book with equivlent materialistic expressions being used as a translation.

Chapter 6

Epistemology

“Epistemology without contact with science becomes an empty scheme. Science without epistemology is... primitive and muddled.” — Albert Einstein

6.1 Introduction

Here we will derive the basic principles of Epistemology that will later be expanded upon by the next chapter on the philosophy of science.

6.2 Ontology

6.2.1 Statements

There are so many possible statements one can make *“an electron is 511 ± 0.5 keV”* or *“I am a person”* or *brutalism is the best form of artwork*. Given that these statements will be at the corners stone of the structure and development of knowledge; clear definitions, rules, and such must be derived.

One assumption one can take is that these statements exist within the Boolean domain thus that $S \in \mathbb{B} = FALSE, TRUE$. Making it so that not only can truth values be assigned to a statement, a statement can only be true or false. This is clear from any logical analysis that for their to be a logically consistent truth system, that their only be true or false. Nuance, while useful outside of strict epistemology, doesn't actually change the fact that the statement is either true or false.

From here, the rules developed in Formal logic ring true. Addition, negation, or, etc. They are the rules that govern these statements and allow for knowledge to be constructed in the first place.

Another rule to add is equivalence, but there are different types of equivalence. Semantic equivalence(saying the same thing in a different form), Logical equivalence(One proves another, or both prove each other), and finally material equivalence(they both have the same truth value). These tools of equivalence will be useful in the future.

An addition to traditional logic are adding relations like narrower, broader, compatible, or incompatible.

With these in mind we satisfy De Morgan Laws and Boolean algebra infinite distribution with all of the rules we need for the construction of statements. Other similar methods can be used, but this minimal set traditionally holds.

6.2.2 Events

As we have already defined an agent observing objects with observable properties. A new axiom would be as the property of time tends to move forwards, properties tend to change. The position in time state space in which said changes happens is defined to be the *event*

6.2.3 Relations

As we have already defined causal order, a simple extrapolation of simple logic

$$P \rightarrow Q$$

$$\therefore \forall P \exists Q$$

If P causes Q, then if P happens then Q happens. A simple relation between the two.

There are other simple derivations to get other relations; biconditional, common causal variable, and coincidental.

6.3 Verifying Statements

Now we must talk about how to verify these statements for the development of knowledge.

6.3.1 Deduction

Using the principles of formal logic, one may derive statements from one another. These statements will be sound by definition but not technically true by definition. For they rely on the statements, Axioms, taken to be true.

These rules must be fulfilled by how we defined a statement to be. Being Boolean and thus susceptible to the relations of formal logic.

this can be trivially proved from our earlier axiom that reality obeys such logical rules.

One can use these for several examples. Say an understanding of the laws of gravitation, one can derive the trajectories of objects flying on the moon with only information of the mass, radius, and air resistance on the moon. Or derive

General Relativity from special relativity and the principle of equivalence.

6.3.2 Induction

This brings us to our next form of developing logic, induction. From our earlier Axiom of UoN(uniformity of Nature), induction arises. Allowing us to use past observations to inform natural laws and models. Finding that predictive power is evidence for the validity of a model.

Essentially, past actions are valid evidence for future ones.

There is also abduction which is closely combined with induction, this will also be addressed in the next sub-section.

6.3.3 Abduction

While these both give us information. It doesn't give the why, only the fact it happened(or could). The development of theories, causal order is beyond their scope unless it is completely obvious from earlier theories(which is technically still abduction in most cases, unless solely deductive with no inductive reasoning).

Abduction is the use of models, models being human made ideas that are meant to represent the world. Whether they are meant to be ontologically true or just predictive, at the end they are simply predictive. One creates models and evaluates them based upon a set of criteria to influence where they are on a hierarchy of models.

6.3.4 Statement Functions

Statement functions can be identified as variation within a statement, such as the mass of the ball is x , where $5kg < x <$

6kg in this, there are infinitely many individual statements. When attempting to validate it through induction, one cannot truly validate or invalidate all possible statements as all experiments(acting on an object in a way to measure a property.)

6.3.5 Probability Space

We take a manifold of a metric space. This manifold satisfies

$$P(\Omega) = 1$$

$$(\Omega, F, P)$$

Where Ω is defined as sample space, F is events, and P is Probability measure.

From here the capacity of probability space is quite clear

6.3.6 Probability Rules

From here many things are proved

$$P(\emptyset) = 0$$

$$P(A^c) = 1 - P(A)$$

and

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Further, variables become a metric function

$$X : \Omega \rightarrow \mathbb{R}$$

Once these variables are defined:

$$P(X \in B) = P(X^{-1}(B))$$

This makes it so the pushforward measure is the probability distribution. Elementary probability distributions are all simply particular pushforward methods.

After that, the expectation can be clearly defined through the Lebesgue integral

$$E(X) = \int_{\Omega} X dP$$

Where variance follows as

$$Var(X) = E((X - E(X))^2)$$

6.3.7 Bayes

Simple derivation brings

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

This simple theorem holds much of modern science. As $P(A)$, where how strongly evidence should be treated is based upon the likelihood of said thing happening with or without A being true.

6.4 Information

Information is a defined non-material existence, being as it exists in conclusion of a person. Properties of either a material object or non-material object can both be described as information

6.4.1 Information of Distinction

Information is simply a way to distinguish between states, possible and existent.

Formally,

Suppose

$$S = s_1, s_2, \dots$$

is the set of possible states.

Information requires

$$\exists s_i \neq s_j.$$

Without distinguishable alternatives, no information exists. If every possible observation produces the same result, the system conveys nothing.

Suppose a nous initially considers

$$\Omega = s_1, s_2, \dots, s_n.$$

An observation transforms

$$\Omega \longrightarrow \Omega'$$

where

$$\Omega' \subseteq \Omega.$$

Information is therefore the reduction of admissible possibilities. Knowledge will later be defined as appropriately justified reductions that correspond to reality.

6.4.2 Formulisms of Information

6.4.2.1 Shannon Information

Claude Shannon deliberately removed meaning from information.

Suppose an event occurs with probability

$$p.$$

Its self-information is

$$I = -\log p.$$

Rare events contain more information because they eliminate more uncertainty.

A probability of 1 contains no information.

Nothing new has been learned.

Entropy becomes

$$H(X) = -\sum_i p_i \log p_i.$$

Entropy measures expected uncertainty before observation.

Information is the decrease in uncertainty after observation.

6.4.2.2 Algorithmic Information

Algorithmic information defines information by the length of the most compressible form.

6.4.2.3 Semantic Information

The traditional view includes the syntax and semantics of a statement, what properties are transported through activity; ie talking, painting, measurement, etc.

6.4.2.4 Programme

While all of these formulisms work in their respective fields and are all incredibly important; for the purposes of this programme, using Shannon is most efficient. As it works best for physics, scientific studies, and mathematical representation.

6.5 Justification**6.6 Belief**

Chapter 7

Philosophy of Science

Chapter 8

Philosophy of Physics

Chapter 9

Foundations of Physics

Chapter 10

Foundations of Mathematics

Chapter 11

Foundation of Computations

Chapter 12

Foundations of Mathematics

Chapter 13

Theology

Chapter 14

Ethics

14.1 The Primacy of God

Chapter 15

Social Philosophy

Chapter 16

Personal Philosophy

Part II

Notes

Chapter 17

Introduction

This part is for the purpose of writing my notes on other people's work, rather than my own. Whether it be notes from a class, a textbook, a regular book, or anything of that nature. Pretty simple.

Chapter 18

Physics

18.1 Introduction

Here I will write my notes related to physics

18.2 Theoretical Physics by Georg Joos

The vector analysis portion is interesting due to the fact that it takes something as simple as vector analysis and brings rigorous ideas to it like that $\oint ds = 0$ for close surfaces.

I also never really thought about using vector analysis instead of tensors.

The rest of curl, gauss, and such is very simple. Though, now it is getting into tensors though vectors.

$$dv_x = ds \nabla v_x$$

$$dv_y = ds \nabla v_y$$

$$dv_z = ds \nabla v_z$$

So therefore $dv = ds \nabla v$

To calculate ∇v , three vectors or nine scalar must be known.

Another interesting addition is the fact that in physics, symmetric tensors can be represented as a surface to the second degree.

One thing I have noticed is I need a more intuitive grasp into the relationships between curl, div, laplace, and grad. They use it a lot to simplify the calculations.

$$\nabla^2 f = \nabla \cdot (\nabla f)$$

The Laplace operator is equal to the divergence of the gradient.

$$\nabla \times (\nabla f) = 0$$

For smooth scalar fields. The curl of the gradient is zero.

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0$$

The same is true for the divergence of the curl. $\nabla^2 \mathbf{F} = \nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$ For vector fields, the Laplace operator has different relations.

Next, onto calculus of variations. I will derive Euler-Lagrange differential equation.

Let: $\tilde{y}$ be a neighboring function to y . Where ϵ be a small quantity and $\eta(x)$ be a arbitrary function of x . so if $\tilde{y} = y + \epsilon\eta$ then $\tilde{y}' = y' + \epsilon\eta'$. Here we stipulate that the two functions $\tilde{y}$ and y converge at the beginning and end. Thus, η must vanish at the ends. So if we substitute an integral I , we find that it becomes a function of ϵ . Then we require that $I(\epsilon)$ must have an extreme value of $\epsilon = 0$. Here it is in mathematical terms:

$$I(\epsilon) = \int_{x_0}^{x_1} F(x, y + \epsilon\eta, y' + \epsilon\eta') dx = \text{extremum for } \epsilon = 0$$

This gives us a simple way of determining the extreme value for a given integral. The condition is:

$$\left(\frac{dI}{D\epsilon}\right)_{\epsilon=0} = 0$$

We can then expand the integrand function F in Taylor's series, according to the powers of ϵ .

The differentiate with respect to ϵ .

This expression then vanishes for $\epsilon = 0$. Thus then simply remains the condition for the extremum.

Integrate this to get the Euler-Lagrange differential equation:

$$\frac{d}{dx} \frac{\partial F(x, y, y')}{\partial y'} - \frac{\partial F(x, y, y')}{\partial y} = 0$$

For writing constraining forces, we find it to be

$$Z = \lambda \text{ grad } G$$

Where $G(x, y, z)$ is the equation of the surface

18.3 On Sympathetic Reduction in Classical Mechanics

18.4 Magnetic Fields and Magnetic Diagnostics for Tokamak Plasmas by Alan Wooton

18.5 Advanced MHD with Applications to Laboratory and Astrophysical Plasmas by Cambridge

For wide variety of MHD instabilities operating in tokamaks, represented by normal modes of the form (assuming that cylindrical approximation and the toroidal representation may be ignored):

$$f(\psi, \vartheta, \varphi, t) = \sum_m \tilde{f}(\psi) e^{i(m\vartheta + \eta\varphi - \omega t)}$$

Is only unstable for perpendicular wave vectors.

The reason is the enormous field line bending energy of the Alfvén waves

18.6 Hamiltonian description of the ideal Fluid by P.J.Morrison

18.7 Classical Dynamics a Modern Perspective

18.7.1 Chapter 6

A lot more importation stuff I forgot to write down, well I will start in chapter 6: If multiple canonical transformations, directly transforming would be canonical. $w \rightarrow w' \rightarrow w''$ then $w \rightarrow w''$. This is the group Composition Law.

Next is the Associativity of Composition Law: if $w' = \phi(w)$, $w'' = \Phi(w')$, and $w''' = \xi(w'')$ so no matter what order of transformation they will be the same, whether $(T_3(T_2T_1))$ or $(T_3T_2)T_1$ or anything of the such.

Identity: The identity transformation $w' = w$ obviously obeys the conditions for being canonical.

Inverses: If $w \rightarrow w'$ is canonical, then so is the transformation $w' \rightarrow w$.

"Thus all the properties needed to define a group are trivially obeyed. (In Chapter 12 we discuss group structure in a little more detail). "

The canonical group is characterized by the dimension, $2k$, of the phase space.

The subgroups of such a phasespace is very large, though three can be easily defined.

Another subgroup is the contact transformations.(when a poisson bracket can be formed)

Note: changing coordinate systems can satisfy phasespace

rules, but isn't generally considered to be a transformation

"Intuitively speaking, a finite canonical transformation that can be connected continuously to the identity should be built up as a succession of infinitesimal transformations."

To express the relation between Poisson Brackets and canonical transformations through differential equations:

$$\frac{d\omega}{d\theta} = \{\omega^\mu, \phi(\omega)\}_\omega = \epsilon^{\mu\nu} \frac{\partial \phi(\omega)}{\partial \omega^\nu}$$

Let us first clarify the meaning of this equation. We look upon the w^μ as functions of θ ; ϕ is a function with a fixed functional form, and w^μ are the unknowns in the differential equation. Thus on the right-hand side of this equation, the arguments of ϕ are just the quantities we are trying to solve for. From the theory of differential equations, we know that there is a unique solution for w^μ if we are given the values of w^μ at $\theta = 0$. Calling these boundary values of w^μ as w_0^μ , the solution of the equation can be written as:

$$\omega^\mu = \varphi^\mu(\omega_0; \theta), \varphi^\mu(\omega_0; 0) = \omega_0^\mu$$

18.7.2 Chapter 7

...

18.8 Differential Geometry - Robert Davie

18.8.1 1-forms

These are linear functionals that map tangent vectors at a point on a manifold to real numbers. In local coordinates on a manifold M , a 1-form can be written as $\omega = \omega_i dx^i$, where ω_i

are smooth functions and dx^i are basis 1-forms (dual to coordinate basis vectors). For example, the differential of a function $df = \frac{\partial f}{\partial x^i} dx^i$ is a 1-form. They are used to measure vectors, like gradients or line integrals.

18.8.2 2-forms

These are antisymmetric bilinear maps that take two tangent vectors and produce a real number. In coordinates, a 2-form looks like $\omega = \omega_{ij} dx^i \wedge dx^j$, where $\wedge$ denotes the wedge product (ensuring antisymmetry: $dx^i \wedge dx^j = -dx^j \wedge dx^i$). 2-forms are used to measure oriented areas, such as in surface integrals or the electromagnetic field tensor in physics.

18.8.3 3-forms

These are antisymmetric trilinear maps, taking three tangent vectors to a real number. In coordinates, a 3-form is $\omega = \omega_{ijk} dx^i \wedge dx^j \wedge dx^k$, with antisymmetry in all indices. They measure oriented volumes and are used in integrals over 3-dimensional submanifolds, like in fluid dynamics or general relativity.

18.8.4 k-form

This can then be made general as

$$\omega = \sum_{i_1 \dots i_k} \omega_{i_1 \dots i_k} dx^{i_1} \dots \wedge dx^{i_k}$$

18.8.5 Introduction to Exterior calculus

An exterior derivative maps k-forms into k+1-forms.

$$d : \wedge^k(M) \rightarrow \wedge^{k+1}(M)$$

and must follow these rules Linearity: $d(\alpha + \beta) = d\alpha + d\beta$ Leibniz(product rule): $\wedge(\alpha + \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ Nilpotency: $d(d\alpha) = 0$

$$\therefore d\omega = \sum_{i < j} \left(\frac{\partial \omega_i}{\partial x_j} - \frac{\partial \omega_j}{\partial x_i} \right) dx^i \wedge dx^j$$

In 3D, the exterior derivative of the 2-form correspondence to the divergence in vector field.

Allows for integration of manifolds. Shows that the integral of the exterior derivative of a differential form over a manifold of the integral of the form itself over the boundary of the manifold:

$$\int_M d\omega = \int_{\partial M} \omega$$

This unifies vector calculus in areas like divergence or green's theorem.

It also show vector flow

Lie derivatives measure the change of forms

18.8.6 Exterior calculus-2

The exterior for a k-form is

$$d\omega = \sum_{n=1}^k \left(\sum \frac{\partial x_{i_1, \dots, i_n}}{\partial x^{i_n}} dx^{i_n} \right) \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}$$

4-form is the highest form

Back to to integration of manifolds with stokes theorem. While $\int_M d\omega = \int_{\partial M} \omega$ is true $\int dx \wedge dy$ and $\int dx dy$ represent slightly different mathematical concepts. In diff forms, it is the integral of an oriental area. In Multi-variable calculus it is the integral of over a region in the xy-plane

18.8.7 Wedge Product-2

Properties of the wedge product: Anti-symmetry: $\omega \wedge \eta = (-1)^{pq} \eta \wedge \omega$. Where p & q are the dimensionality of ω, η

Symmetric group: S_{p+q} is the group of all possible permutations.

Permutation and sign of permutations: σ is essentially a specific way of ordering a set of elements. The sign of permutation $sign(\sigma)$ tells us if the permutation can be achieved in an even or odd number of swaps. Where odd is negative and even is positive

Role of these two things: The sign tells us the sign of a wedge product given a permutation. $dx^1 \wedge dx^2 \rightarrow -dx^2 \wedge dx^1$. The symmetric groups tells us all possible permutations.

$$\therefore (\omega \wedge \eta)(X_1, \dots, X_{p+q}) = \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} sign(\sigma) \omega(X_{\sigma(1)}, \dots, X_{\sigma(p)}), \\ \eta(X_{\sigma(p+1)}, \dots, X_{\sigma(p+q)})$$

This $\frac{1}{p!q!}$ ensures correct normalization.

18.8.8 Introduction to the Hodge Star Operation

This operator allows us to relate manifolds. Crucial in formulating coordinate free theories. It transforms k -forms into $(n-k)$ -forms. Specifically it is an isomorphism between the spaces of k -forms and $(n-k)$ -forms on an n -dimensional Riemannian (M, g) . Where g is the metric tensor. It is denoted as:

$$\star : \wedge^k(T^*M) \rightarrow \wedge^{n-k}(T^*M)$$

The way it works is hard to write out, so just remember it.

To construct a hodge star

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle \text{vol}_M$$

Where $\langle \cdot, \cdot \rangle$ is the inner product and Vol is the volume form of M. Defined as $\text{vol}_M = \sqrt{|g|} dx^1 \wedge \dots \wedge dx^n$ Where $|g|$ is the determinate of the Metric Tensor.

The equation is long, so just look it up, it is simple but long.

18.8.9 Hodge star 2

Hodge star establishes duality between geometric theories. Involution: Doing it twice returns it to the original Linear transformations: You know what this means Interacts like multiplication with exterior derivative: $\star(d\omega) = d(\star\omega)$ The co-differential is defined using the Hodge Star.

18.8.10 The push forward of vectors on manifolds

A pushforward is a way to map a vector on a tangent space onto another manifold's tangent space.

Properties:

Linearity: The pushforward $\phi_* : T_p M \rightarrow T_{\phi(p)} N$ is a linear map. For vectors $v, w \in T_p M$ and scalars a, b ,

$$\phi_*(av + bw) = a\phi_*v + b\phi_*w$$

Chain rule: If you have another smooth map $\psi : N \rightarrow P$, the pushforward satisfies the composition rule:

$$(\psi \circ \phi)_* = \psi_* \circ \phi_*$$

This follows from the chain rule for derivatives.

Action on curves: If you think of a vector $v \in T_p M$ as the tangent to a curve $\gamma : (-\epsilon, \epsilon) \rightarrow M$ with $\gamma(0) = p$, then $\phi_* v$ is the tangent vector to the curve $\phi \circ \gamma$ at $\phi(p)$.

The pushforward $\phi_* v \in T_{\phi(p)} N$ is given by:

$$\phi_* v = v^i \frac{\partial \phi^j}{\partial x^i} \frac{\partial}{\partial y^j}$$

Here, v^i are the components of v , and $\frac{\partial \phi^j}{\partial x^i}$ are the Jacobian entries. The result is a vector in $T_{\phi(p)} N$, expressed in the basis $\left\{ \frac{\partial}{\partial y^1}, \dots, \frac{\partial}{\partial y^n} \right\}$.

18.8.11 Pull-Back in k-forms

Given a smooth map $\phi : M \rightarrow N$ between manifolds M (dimension m) and N (dimension n), and a k -form $\omega \in \Omega^k(N)$ on N , the pullback $\phi^* \omega \in \Omega^k(M)$ is a k -form on M . The pullback essentially "transfers" ω from N to M by composing it with the map ϕ . The pullback is defined pointwise. For a point $p \in M$, and k tangent vectors $v_1, \dots, v_k \in T_p M$, the pullback $\phi^* \omega$ at p is:

$$(\phi^* \omega)_p(v_1, \dots, v_k) = \omega_{\phi(p)}(\phi_* v_1, \dots, \phi_* v_k)$$

Here:

$\phi_* : T_p M \rightarrow T_{\phi(p)} N$ is the pushforward (differential) of ϕ , which maps each tangent vector v_i to $\phi_* v_i$. $\omega_{\phi(p)} \in \wedge^k T_{\phi(p)}^* N$ is the k -form ω evaluated at $\phi(p)$, acting on the pushed-forward vectors.

This definition ensures that $\phi^* \omega$ is a k -form on M , as it takes k vectors in $T_p M$ and produces a number.

Key Properties of the Pullback

Linearity: The pullback $\phi^* : \Omega^k(N) \rightarrow \Omega^k(M)$ is a linear map.

Preserves degree: The pullback of a k -form is a k -form.

Functoriality: For maps $\phi : M \rightarrow N$ and $\psi : N \rightarrow P$,

$$(\psi \circ \phi)^* = \phi^* \circ \psi^*$$

Commutates with exterior derivative: For any k -form ω ,

$$d(\phi^*\omega) = \phi^*(d\omega)$$

This is a powerful property, making pullbacks compatible with the exterior calculus.

Wedge product: The pullback respects the wedge product:

$$\phi^*(\omega \wedge \eta) = (\phi^*\omega) \wedge (\phi^*\eta)$$

Computing the Pullback in Coordinates To compute $\phi^*\omega$ in practice, we typically use local coordinates. Here's the step-by-step process:

Choose coordinates:

On M , use coordinates $(x^1, \dots, x^m)$ around p .

On N , use coordinates $(y^1, \dots, y^n)$ around $\phi(p)$.

The map ϕ is expressed as $y^i = \phi^i(x^1, \dots, x^m)$.

Express the k -form:

Let $\omega \in \Omega^k(N)$ be written in coordinates as:

$$\omega = \sum_I a_I(y) dy^{i_1} \wedge \dots \wedge dy^{i_k}$$

where $I = (i_1, \dots, i_k)$ is a multi-index with $1 \leq i_1 < \dots < i_k \leq n$, and $a_I(y)$ are smooth functions on N .

Pull back the form:

The pullback is:

$$\phi^* \omega = \sum_I (a_I \circ \phi) \phi^*(dy^{i_1} \wedge \cdots \wedge dy^{i_k})$$

Compute $a_I \circ \phi$: Replace y with $\phi(x)$, so $a_I(y) = a_I(\phi(x))$.

Compute the pullback of the basis k -form:

$$\phi^*(dy^{i_1} \wedge \cdots \wedge dy^{i_k}) = \phi^*(dy^{i_1}) \wedge \cdots \wedge \phi^*(dy^{i_k})$$

For a 1-form dy^j , the pullback is:

$$\phi^*(dy^j) = d(y^j \circ \phi) = d(\phi^j(x)) = \frac{\partial \phi^j}{\partial x^\ell} dx^\ell$$

(using the chain rule, summing over ℓ). Thus:

$$\phi^*(dy^{i_1} \wedge \cdots \wedge dy^{i_k}) = \left(\frac{\partial \phi^{i_1}}{\partial x^{\ell_1}} dx^{\ell_1} \right) \wedge \cdots \wedge \left(\frac{\partial \phi^{i_k}}{\partial x^{\ell_k}} dx^{\ell_k} \right)$$

This is a k -form on M , expressed in the dx^ℓ basis.

Simplify:

Expand the wedge products and collect terms. The result is a sum of terms involving $dx^{\ell_1} \wedge \cdots \wedge dx^{\ell_k}$, with coefficients that depend on the Jacobian $\frac{\partial \phi^j}{\partial x^\ell}$ and $a_I(\phi(x))$.

18.8.12 Pull-Back of volume forms

The pullback of a volume form $\omega = f(y) dy^1 \wedge \cdots \wedge dy^n$ on N under a map $\phi : M \rightarrow N$ is an n -form on M , computed as:

$$\phi^* \omega = (f \circ \phi) \cdot \det \left(\frac{\partial \phi^i}{\partial x^{j_k}} \right) dx^{j_1} \wedge \cdots \wedge dx^{j_n}$$

If $\dim M = \dim N$, and ϕ is a local diffeomorphism, $\phi^*\omega$ is a volume form on M . If $\dim M < \dim N$, the pullback is zero. The Jacobian determinant captures the volume scaling, making this operation central to integration and geometric transformations.

18.8.13 Integration on Manifolds Using the Pullback of Volume Forms

The pullback of volume forms allows for integration of manifolds. Which is useful in the integration of curved coordinate systems.

volumes generalize "dx" or "dxdy."

The steps in integration of manifolds using volume forms is:

Step 1: M : The domain of integration (often $\mathbb{R}^n$ or a simpler manifold). N : The target manifold with a volume form ω . $\phi : M \rightarrow N$: A smooth map, typically a parametrization or diffeomorphism. $f : N \rightarrow \mathbb{R}$: The function to integrate (if $f = 1$, we're computing the volume).

Express the volume form:

Write $\omega = f_\omega(y) dy^1 \wedge \cdots \wedge dy^n$ in coordinates on N .

Compute the pullback:

Compute $\phi^*\omega$, which involves:

Substituting $y = \phi(x)$ into $f_\omega(y)$. Computing the Jacobian determinant of ϕ . Forming $\phi^*\omega = (f_\omega \circ \phi) \cdot \det \left(\frac{\partial \phi^i}{\partial x^j} \right) dx^1 \wedge \cdots \wedge dx^n$ (if $m = n$).

Set up the integral:

The integral becomes:

$$\int_{\phi(M)} f \omega = \int_M (f \circ \phi) \cdot \phi^*\omega$$

In coordinates, if $\phi^*\omega = g(x) dx^1 \wedge \cdots \wedge dx^n$, and M corresponds to a region $U \subset \mathbb{R}^n$:

$$\int_U (f \circ \phi)(x) \cdot g(x) dx^1 \dots dx^n$$

Evaluate the integral:

Perform the integral over U using standard multivariable calculus techniques, accounting for the Jacobian determinant and any orientation changes.

18.8.14 Line Integrals on Manifolds

Parameterize the curve: Define the curve $\gamma(t) = (x(t), y(t), \dots)$ for $t \in [a, b]$.

Define the one-form: Let the one-form be $\omega = f(x, y, \dots)dx + g(x, y, \dots)dy + \dots$

Calculate the pullback: Substitute the coordinate functions $x(t), y(t), \dots$ and their differentials into the one-form. This gives the pullback one-form $\gamma^*\omega$ on the interval $[a, b]$. For example, if $\omega = f(x, y)dx + g(x, y)dy$ and $\gamma(t) = (x(t), y(t))$, the pullback is $\gamma^*\omega = (f(x(t), y(t))\frac{dx}{dt} + g(x(t), y(t))\frac{dy}{dt})dt$.

Integrate the pullback: Integrate the resulting expression with respect to t from a to b .

This gives the value of the line integral:

$$\int_{\gamma} \omega = \int_a^b (f(x(t), y(t))\frac{dx}{dt} + g(x(t), y(t))\frac{dy}{dt})dt$$

18.8.15 Classical Stokes Theorem in Manifolds

18.9 Symplectic geometry & classical mechanics

18.9.1 Lecture 1

Symplectic doesn't work well in dissipative structures.

It also goes over the definition of local euclidian space, which is pretty basic.

They also go over how you can make an object unique by maxionable defining coordinate systems.

This is given by an atlas of coordinate charts

To get a differntiable structure, an atlas is described through coordinate charts (U_α, ϕ_α) such that:

Each $\psi_\alpha : U_\alpha \rightarrow R^n$ is a homeimorphism Also, on overlaps between $U_\alpha \cap U_\beta$ it is continously differentiable.

18.9.2 Lecture 2

They go over catogories in a brief overview, and that this lecture will be going over a map of differential geometry.

So, let us start with a manifold: $f(p) \in N$

Now there isn't much we can say about this basic manifold, though let's add a bit more context. Say that manifolds seem like flat-space(euclidian) locally.

So we can take the charts in the manifold to project euclid-ian space upon it to differentiate.

When k is $\inf f$ is smooth.

Chapter 19

Calculus of Variations by Gelfand & Fomin

19.0.1 Elements of the Theory

A functional is defined as a function where the variable is in it of itself a function. It can also be seen as a class of functions.

While the calculus of functionals most developed and used branch is finding the extrema of variations, this is hardly its extent.

A great example is using the path length formula $L = \int_a^b \sqrt{1 + \dot{y}^2} dx$ and finding which y achieves the minimum value for L , this is by definition the straightest line, or geodesic. It can also attempt to maximize. It is most often applied to any form of optimization theory or minimizing the action for Lagrangian dynamics.

In this book they will primarily use geometric language defining a space, the functional space, to be a space whose elements are functions. The nature and dimensions of such a space are defined by the functional defined. It should be as specific and defined with the problem. Continuity is the main constraint.

To formalize the "closeness" needed for continuity the space will use normed linear space. Which is defined by the following axioms:

- $x + y = y + x$
- $(x + y) + z = x + (y + z)$
- There exists a 0 element
- $\forall x \exists (-x) : -x + x = 0$
- Multiplication of 1
- Multiplative communication

Normed is defined to be the magnitude of the length.

As I am flipping through this book I am seeing that the subject matter is quite trivial but presented in a overtly complex way. I may return to it but that is unlikely.

19.1 MIT 8.323 Relativistic Quantum Field Theory 1

19.1.1 Lecture 1: Classical Field Theories and Principle of Locality

The first part goes over trivial processes in field theory and general mathematics. The main note on this section would be the principle of Locality:

$$L = \int d^3x \mathcal{L}(\phi_a, \partial_t \phi_a, \partial_i \phi_a)$$

Where $\phi_a(\vec{x}, t)$: general fields. $\mathcal{L}$ is a function of ϕ and its derivatives. $\mathcal{L}$ is called the Lagrangian density.

For this form it constrains all dynamics so that all actions must be local. It is used by a single integral to enforce this. Another constraint is one will only use the first derivative in time. This is because including a second order time derivative will make an equation of motion with an initial condition with an acceleration which includes arbitrary complications without need. This rule is less important and fundamental but is used to make things easier. Lastly, in this class, being that it is relativistic one won't need additional spacial derivatives, but in classical models this is not the case.

Canonical momentum density is exactly what one would expect.

$$\Pi_a = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a}$$

Hamiltonian is defined the same way.

One brief note on notation, in this x is simply a label for a degree of freedom, there can be infinite such degrees of freedom. Also, the teacher is using Einstein's notation that repeated indices will be summed.

19.1.1.1 Equations of motion

He gives an example through:

$$\delta S = 0$$

$$S = \int dt L$$

$$0 = \delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta (\partial_\mu \phi_a) \right]$$

One can simplify this through integration by parts

As for boundary conditions. One will choose $\delta\phi_a$ so that boundary conditions will vanish. Thus one will not need to worry about that.

This thus implies that $\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0$.

From now on all things will be translationally invariant and Lorentz invariant.

He briefly goes over the simplest form of a quantum field, a scalar field.(Pion or higgs)

It follows a very similar model:

$$S = \int d^4x [-1/2, \partial_\mu \phi, \partial^\mu \phi - V(\phi)]$$

Transnational symmetry means that all parameters of $V(\phi)$ must be constant. V is the same as the Lagrangian density

Using this equation one can once again find the Hamiltonian Density and thus Hamiltonian. To give a simple equation of motion.

He goes through a famous Klein-Gordan:

$$\partial^2 \phi - m^2 \phi = 0$$

19.1.2 Lecture 2: Symmetries and Conservation Laws

I forgot to save my notes last time, so those are gone. It was generally pretty simple going over Noether theorem and defining types of symmetries at the beginning.

Some how it deleted again... oh well, moving on.

19.1.3 Lecture 3: Why Quantum Field Theory

Non-Relativistic Schrodinger equation

$$\partial^2\psi - m^2\psi = 0$$

Simplest scalar field theory

$$S = -\frac{1}{2} \int dx (\partial_\mu \phi \partial^\mu \phi + m^2 \phi^2)$$

Implies

$$\partial^2\phi - m^2\phi = 0$$

This changes $\phi(t, x)$ where x becomes an eigenvalue rather than a coordinate.

In relativistic quantum mechanics, there are several challenges. For one, no way to define a positive probability density. Also, there exists negative energy states. You must describe a fixed number of particles without changing. As equations change as wavefunctions change. This troubles with relativity as energy can create particles, adding additional difficulty. Finally, there is a fundamental asymmetry between the parameter t and the eigenvalues of x .

Therefore, relativistic quantum mechanics one cannot give a fundamental description.(at best an approximate, no particle creation or annihilation) The solution of this is quantum field theory, simply the quantization of classical field theory.(using t and x as parameters not eigenvalues) Field theory can also arise as a limit of discrete systems.

In classical field theory, one takes the originally discrete Lagrangian and defines $\phi(x_n, t)$ to be the discrete parts. From here a simple Taylor expansion can be used to change all of the interaction terms to in the form of the new continuum. Finally one may derive the field-theoretic Euler-Lagrange equation.

Quantization of harmonic oscillator in Heisenberg picture.

$$L = \frac{1}{2}\dot{x} - \frac{\omega^2}{2}x^2$$

$$p = \dot{x}$$

$$H = \frac{p^2}{2} + \frac{1}{2}\omega^2 x^2$$

$$\ddot{x} + \omega^2 x = 0$$

The quantum solution is exactly the same, except x becomes an operator:

$$\hat{x}(t) = \hat{A} \cos \omega t + \hat{B} \sin \omega t$$

From here one can define the Hilbert space to define through $[\hat{a}, \hat{a}^+]i$

$$|h\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^+)^n|0\rangle$$

One can then use this to quantize a field theory. First, find the most general solution of the EOM. Second, promote the integration constants to constant operators to give full time evolution. Impose canonical quantization conditions $\{q, p\} = 1$ thus, $[\hat{q}, \hat{p}] = i\hbar$. Finally, use this to develop Hilbert Space.

19.1.4 Lecture 4: Canonical Quantization of a Free Scalar Field Theory

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Chapter 20

Numerical Hamiltonian Problems

20.0.1 Chapter 1: Hamiltonian Systems

MacKay and Meiss have compiled great papers on the subject in 1987, Marsen (1992) offers a more geometric introduction.

Usually in mechanics, q is generalized coordinates while p is conjugated generalized momenta. H being total mechanical energy.

I like how they show the nature of the Hamiltonian system to be a logical necessity based upon how the phase space is constructed, rather than purely physical construction.

It is sometimes useful to combine dependent variables into a $2d$ $y = (p, q)$.

$$\frac{dy}{dt} = J^{-1} \nabla H$$

Where, ∇ is the gradient operator and J is skew-symmetric Matrix

20.0.2 Chapter 2: Symplecticness

As mentioned, Hamiltonian problems are exceptional. Most systems do not allow for area preservation (energy conservation, time symmetry, etc), which is a requirement of Hamiltonian systems. That is why exploration into other systems like Nambu, Metriplectic, etc is needed.

This area preservation is not simply required as a general rule but is the basis of the structure. All other properties of Hamiltonian dynamics is derived from the area property.

One can check for the preservation of area by checking if the Jacobian is exactly 1 or a simpler way is to find it is if $dp * \wedge dq = dp \wedge dq$ or that the pull back of the 2-form is equal to the original 2-form.

In multiple-dimensions. Look for multiple area preservations rather than multi-dimensional preservation.

20.1 Chapter 2: Numerical Methods

Stiff problems are defined to be where numerical stability, not accuracy, forces you to take extremely small steps.

Explicit methods requires s number of evaluations per steps while implicit provides coupled system of $s \times D$ algebraic systems and $s \times D$ components of stage vectors.

After reading this I wrote a program in Julia to compare the time it takes for both an explicit RK and Implicit to solve a ODE. It was more challenging than it should have been.

20.1.1 Chapter 4: Order Condition

More so Computer science stuff, not so important at the moment but something to look at later.

20.1.2 Chapter 5: Implementation

Really just the stuff I did earlier when writing my own implicit RK.

20.1.3 Chapter 6: Symplectic Integration

If $b_i a_{ij} + b_j a_{ji} - b_i b_j = 0$ for $i, j = 1, \dots, s$. Then the method is symplectic. They have a nice proof that I can use as a reference.

Most all symplectic methods are implicit.

They go through solutions of various families of symplectic RK integration.

20.1.4 Chapter 7: Symplectic Order Conditions

I still don't get order conditions. I will end it at one point.

20.1.5 Chapter 8: Numerical Experimentations

Page 115 lists multiple symplectic numerical methods, the rest of the chapter examines them. Too much for me to write but useful and interesting information nonetheless.

One thing I will say is that they claimed that Calvo is clearly the best and very little improvement is likely possible.

20.1.6 Chapter 10: Properties of Symplectic Integrators

20.1.6.1 Backward error analysis

How much must the numerical problem be perturbed so that the numerical problem is equal to the analytical. This is espe-

cially useful in uncertain models, in which perturbation is the best we can do.

This only holds if the numerical solution of time t_n is computed by iterating n times 1 in the same symplectic maps. Thus, variable-step integrations fail.

20.1.6.2 Alternative Approach

There is another method through repeating periodically; though, this system is now non-autonomous and is discontinuous.

20.1.6.3 Conserved Energy and Quantities

While there are exceptions for unique cases, generally one must choose whether to have conserved energy or symplectic structure. Obviously, choosing must be done by the researcher, but generally symplectic is superior for the fact that it does not confine the nature of the dynamics.

There is a great deal of research into methods that exactly conserve H . This is not covered in the book.

There are several ways to come with close approximations of H though.

20.1.6.4 KAM

KAM theory proves that symplectic methods can show that nonlinear effects are not destabilizing.

20.1.7 Chapter 11: Generating Functions

Mentioned earlier, S can be used to express symplectic mappings and transformations. S is a single real value function. This function is called the *generating function*.

20.1.7.1 First kind

Let $(p^*, q^*) = \phi(p, q)$ be a symplectic transformation defined in a simply connected Ω domain. For each closed path γ :

$$\int_{\gamma} pdq - \int_{\gamma} p^* dq^* = 0$$

Where pdq is the differential form.

Making it so that

$$dS = pdq - p^* dq^*$$

Further assume that q and q^* are functions independent of Ω . If this is true we can express $S(p, q)$ as function S^1 of q and q^* .. Thus,

$$p = \frac{\partial S^1}{\partial q}, \quad (20.1)$$

$$p^* = \frac{\partial S^1}{\partial q^*} \quad (20.2)$$

20.1.7.2 Third kind

What happened to *second*, I don't know.

If q and q^* are not independent. Then, if p and q^* are independent you can express $S^3(p, q^*)$. The rest is self-explanatory.

20.1.7.3 All kinds

One can use the Poincare generating function.

20.1.7.4 Hamiltonian-Jacobi

This is another way to generate functions by turning dynamics into single nonlinear partial differential equations through the equation:

$$\frac{\partial S^3}{\partial t} + H(\mathbf{p}, \mathbf{q}, t) = 0$$

While in writing this I use the third kind, it can be used for any kind.

20.1.7.5 Symplectic Integrators on Generating Functions

They talk about a lot of different research, not going into detail about any of it

20.1.8 Chapter 12: Lie Formulism

This being something that I have research well in the past is relatively simple. The Poisson Bracket:

$$\{F, G\} = \sum \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right)$$

is defined by a couple of rules

- Bilinearity
- Skew-Symmetry
- Jacobi Condition
- Leibiz' rule

By introducing 2d-dimensional vectors the equation becomes

$$\{F, G\} = \nabla F^T J^{-1} \nabla G$$

Where J is the skew-symmetry matrix.

From here we can see that this works well the Hamiltonian equations of motion as

$$\dot{y} = \{y, H\}$$

Also, the hamiltonian is canonical iff the Poisson Bracket of any two functions remains invariant under changes in variables.

20.1.8.1 Lie Operators & series

Powers of lie operators can be expanded to compositions of lie operators, or vice versa. This can be used as an equivalent to a derivative as a function of time for Hamiltonian systems.

Using the exponential operator on the lie operator, one can find the taylor series expansion of a flow(for integration) by choosing G to be one of the coordinate functions.

20.1.8.2 Multiplication of exponentials

Composing two exponential lie operators represents the composition of flows; specifically $\phi_{t-F_1}(\phi_{t-F_2})$ in that order.

20.1.8.3 Baker-Campbell-Hausdorff formula

If one wishes to find $e^X e^Y = e^Z$ and X and Y are non-commutators, it can be found through the infinite series:

$$Z = X + Y + \frac{1}{2}\{X, Y\} + \frac{1}{12}\{X, \{X, Y\}\} - \frac{1}{12}\{Y, \{X, Y\}\} + \dots$$

20.1.8.4 Fractional-Step Method

I don't get this...

20.1.9 Chapter 13: Higher Order Methods

After reading my book on Lie Algebra I should come back to this

Chapter 21

General Relativity by Robert M. Wald

21.0.1 Manifolds and Tensor Fields

21.0.1.1 Manifolds

Spacetime is a four-dimensional continuum and by that nature a manifold is simply a multi-dimensional continuum.

When going over this concept they first present an open ball, where all points distance from a chosen center is always strictly less than some radius. Essentially the inside of some interval, circle, sphere, etc. Then an open set in $\mathbb{R}^n$ is any set that can be expressed as a union of open balls. This make $\mathbb{R}^n$ a topological space.

Essentially a manifold is a set made up of peices that "look like" open subsets of $\mathbb{R}^n$. These peices can be "sewn together" smoothly. This can be more precisely defined a n-dimensional, C^∞ , real manifold M is a set together with a collection of subsets $\{O_\alpha\}$ that satisfies the following properties: O_α covers the entire manifold, all points can be mapped onto another subset of $\mathbb{R}^n$, such subsets must be differentiable.

These maps ϕ_α are generally called charts by mathematicians and coordinate systems by physicist. They further define the maps to be homeomorphic, or that all mappings preserve topological properties.

In Euclidean space they map a sphere into a two dimensional manifold; because, pieces can be smoothly sewn together. This then satisfies the above conditions.

21.0.1.2 Vectors

Because traditional vectors do not work in geometric curvature, vectors will be instead used as directional derivatives, rather than an intrinsic structure. These directional derivatives are characterized by their linearity and "Leibnitz" rule. Letting F denote collections of smoothly differentiable functions from M to $\mathbb{R}$, $v : F \rightarrow \mathbb{R}$. Also, they give a proof that $\dim V = n = \dim M$. They also define the vector transformation law which is defined as:

$$v'^\nu = \sum_{\mu=1}^n v^\mu \frac{\partial x'^\nu}{\partial x^\mu}$$

21.0.2 Tensors; the Metric Tensor

To define a tensor one must first define a dual vector space. A dual vector space is a space of the functional transformations from vectors to linear measurements. A double vector space takes those linear measurements and turns them into scalars. This double dual space is isomorphic to the original vector space. Reconstructing all measurements. This double dual space can be used to reconstruct the vector space using the dual vector space.

Now to define a tensor, T, of type (k,l):

$$T : V^* \times \dots \times V^* \times V \times \dots \times V \rightarrow \mathbb{R}$$

In other words, given k dual vectors and l ordinary vectors, T produces a number. A number that if one were to fix the values of all of the other vectors and dual vectors, it is a linear map on the remaining variable. Thus, according to this definition, a tensor of type (0,1) is a dual vector. Of type (1, 0) a double dual vector. Because double dual vectors identify with traditional vectors, they are simply vectors. From this identification one can construct higher types of tensors. (1,1) is a multilinear map from V to V or for the other side.

There are two important operations on tensors. The first being called the contraction. Defined as $C : \lambda(k, l) \rightarrow \lambda(k-1, l-1)$. Thus if T is a tensor of type (k,l), then:

$$CT = \sum_{\sigma=1}^n T(\dots, v^{\sigma*}, \dots; \dots, v_{\sigma}, \dots)$$

Where v_{σ} is the basis of V and $v^{\sigma*}$ is its dual basis. These vectors are inserted into the ith and jth slot of T.(note this is specifically the contraction of tensor of type (1,1)). The next operation being the outer product if T and T'. Denoted as $T \otimes T'$, by the following rules. Given (k+k') dual vectors $v^{k+1*}, \dots, v^{k+k'*}$ and the equivalent for l vectors. $T \otimes T'$ is defined as the product of $T(v^{1*}, \dots, v^{k*}; w_1, \dots, w_l)$ and $T'(v^{k+1*}, \dots, v^{k+k'*}; w_{l+1}, \dots, w_{l+l'})$. Thus, one way of constructing tensors is to take the outer product of vectors and dual vectors. A tensor of this would be called simple. From this any tensor of type (k,l) can be expressed as a sum of simple tensors.

$$T = \sum_{\mu_1, \dots, v_l=1}^n T^{\mu_1 \dots \mu_k}_{v_1, \dots, v_l} v_{\mu_1} \otimes \dots \otimes v^{v_l*}$$

The basis expansion of coefficients are called components of the tensor.

This discussion above refers to arbitrary finite-dimensional vector spaces. Now let us move onto where V is the tangent space V_p at point p of a manifold M . In this V_p^* is referred to as the cotangent space. Where at this point the vectors are cotangent vectors. Another term would be in V they are contravariant vectors and V^* as covariant vectors. (remember dx^μ is a symbol that merely is defined as $dx^\mu(\frac{\partial}{\partial x^\nu}) = \delta^\mu_\nu$).

Finally, we have the defining property of all tensors, tensor transformation law. (though a multilinear map on vectors and dual vectors also works)

$$T'^{\mu_1 \dots \mu_k}{}_{\nu_1 \dots \nu_l} = \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial x'^{\mu_k}}{\partial x^{\alpha_k}} \frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}} \dots \frac{\partial x^{\beta_l}}{\partial x'^{\nu_l}} T^{\alpha_1 \dots \alpha_k}{}_{\beta_1 \dots \beta_l}$$

(for future notice this in Einstein summation)

Now we have a tensor field. I will also add that covariant vector field is as smooth as contravariant vector field. The tensor field's smoothness is defined the same way.

Now one introduces a metric. Intuitively the metric gives the "infinitesimal square distance" associated with "infinitesimal displacement." As mentioned earlier this displacement is defined by the tangent vector. Thus a metric g , should be a linear map from $V_p \times V_p$ into numbers, ie a tensor of type $(0,2)$. In addition, the metric must be symmetric and nondegenerate. Meaning $g(v_1, v_2) = g(v_2, v_1)$ and $g(v, v_1) = 0$. Where $v_1 = 0$.

In a coordinate basis, we may expand a metric in terms of its components as:

$$g = \sum_{\mu, \nu} g_{\mu\nu} dx^\mu \otimes dx^\nu$$

Sometimes ds^2 is used in place of g to represent the metric tensor. Given a metric g , we can always find an orthonormal

basis. The number of + and - signs occurring is called the signature of the metric. The Lorentzian is defined as - + + +.

21.0.2.1 Abstract Index Notation

In this indices do not represent components or numbers but rather the type of tensor. Repeated indices imply contraction (as labels not coordinate slots.)

Also, (ab) implies symmetry and [ab] implies antisymmetry.

Lastly, repeated index implies summation automatically.

21.0.3 Curvature

Generally curvature is defined as extrinsic curvature of a "surface" embedded in a higher dimensional space. There is also intrinsic curvature.

Essentially, this intrinsic curvature can be defined through parallel transport. From here a geodesic is a curve whose tangent is parallel-transported along itself, or the "straightest possible" curve. A space is curved if a parallel geodesic is no longer parallel. One can use the definition of a derivative to define parallel transport as when the derivative along a given curve is 0. Thus the definition of curvature can be defined in terms of failure of successive differentiation on tensor fields to commute. Another thing to add is that given a metric manifold there exists a unique notion of parallel transport in which preserves the inner product of all pairs of vectors.

21.0.3.1 Derivative Operators and Parallel Transport

A derivative operator Δ (sometimes called a covariant derivative) takes every differentiable tensor field of type (k, l) to a smooth tensor field of type $(k, l+1)$ and satisfies linearity, Leib-

nitz rule, commutativity with all contraction, consistency, and torsion free (this fifth one is sometimes dropped).

Given this one can derive an expression for the commutator of two vector fields.

$$[\nu, w]^b = \nu^a \Delta_a w^b - w^a \Delta_a \nu^b$$

Next, the Christoffel symbol can be found as

$$\Delta_a t^b = \partial_a t^b + \Gamma_{ac}^b t^c$$

Now as mentioned earlier, parallel transport can be defined as

$$t^a \Delta_a T_{c_1 \dots}^{b_1 \dots} = 0$$

If one chooses a coordinate system it can then be defined as:

$$t^a \partial_a v^b + t^a \Gamma_{ac}^b v^c = 0$$

Or in terms of components in the coordinate system:

$$\frac{dv^\nu}{dt} + \sum_{\mu, \nu} t^\mu \Gamma_{\mu\lambda}^\nu v^\lambda = 0$$

While many forms of derivative operators can be defined from the general structure of a manifold, but metric manifolds privilege a specific kind.

In this case the Christoffel symbol can be defined by:

$$\Gamma_{ab}^c = \frac{1}{2} g^{cd} \left(\hat{\Delta}_a g_{bd} + \hat{\Delta}_b g_{ad} - \hat{\Delta}_d g_{ab} \right)$$

Where $\hat{\Delta}$ refers to an arbitrary derivative operator.

21.0.3.2 Curvature

The Riemann Curvature Tensor is defined by:

$$\Delta_a \Delta_b \omega_c - \Delta_b \Delta_a \omega_c = R_{abc}^d \omega_d$$

From here the total change in the vector v can be found by:

$$\delta v^a = \Delta t \Delta s v^d T^c S^b R_{cbd}^a$$

Where s and t represent infinitesimal changes within the metric during the transport.

Now the key properties of the Riemann tensor are

$$R_{abc}^d = -R_{bac}^d$$

$$R_{[abc]}^d = 0$$

$$R_{abcd} = -R_{abdc}$$

And the Bianchi identity holds

Decomposing the Riemann tensor into "trace part" and a "trace free part" can be useful. As a trace over the first two or last two can vanish

$$R_{abc}^b = R_{ac} = R_{ca}$$

The scalar curvature is further defined as

$$R = R_a^a$$

The "trace free part" is called the Weyl Tensor C_{abcd} and is defined for manifolds $n \geq 3$ by the equation

$$R_{abcd} = C_{abcd} + \frac{2}{n-2}$$

The Riemann Curvature Tensor is defined by:

$$\Delta_a \Delta_b \omega_c - \Delta_b \Delta_a \omega_c = R_{abc}^d \omega_d$$

From here the total change in the vector v can be found by:

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Decomposing the Riemann tensor into "trace part" and a "trace free part" can be useful. As a trace over the first two or last two can vanish

$$R_{abc}^b = R_{ac} = R_{ca}$$

This defines the Ricci Tensor R_{ac} . From here the scalar Curvature, R , is defined as the trace of the Ricci tensor.

$$R = R_a^a$$

The "trace free part" is called the Weyl Tensor, C_{abcd} , and when dimensions are greater than 3.

$$C_{abcd} = R_{abcd} - \frac{2}{n-2} (g_{a[c} R_{d]b} - g_{b[c} R_{d]a}) + \frac{2}{(n-1)(n-2)} R g_{a[c} g_{d]b}.$$

Because it behaves easily in conformal transformations it is

sometimes referred to as the conformal tensor.

Next, Einstein tensor is found from the twice contracted Bianchi identity:

$$G_{ab} = R_{ab} - \frac{1}{2}Rg_{ab}$$

This tensor is used to ensure the consistency of Einstein's equations.

21.0.3.3 Geodesics

One can define a geodesic as a curve whose tangent, T^a satisfies this equation

$$T^a \Delta_a T^b = 0$$

Actually, one does not need to force the same length so this weaker condition may be true

$$T^a \Delta_a T^b = \alpha T^b$$

Where α is an arbitrary function on the curve.

Thus, this means that a geodesic must be affine parameterized. Thus in a coordinate basis this becomes.

$$\frac{d^2 x^\mu}{dt^2} + \sum_{\sigma, \nu} \Gamma_{\sigma\nu}^\mu \frac{dx^\sigma}{dt} \frac{dx^\nu}{dt} = 0$$

Next, in order for simpler calculation, a Riemannian Normal coordinate can be used in which all geodesics are straight lines. It follows that the Christoffel symbol will vanish in this case.

In another form in which the derivative operator arises from the metric. One will use the Gaussian normal coordinate system. It is most useful when given a hypersurface. They satisfy the important property that geodesics remain orthogonal to all the hypersurfaces.

For a curve on a manifold the length can be derived to be:

$$l = \int (g_{ab} T^a T^b)^{1/2} dt$$

Where T is tangent to the curve and t is the curve parameter.

Obviously this is very similar to the traditional length formula.

For Lorenzian space, a curve is said to be timelike if the norm of its tangent is negative everywhere. Null if equal to zero. Spacelike if positive. (for signature(-+++))

When it is null, the length is obviously zero. If it is timelike, we change the sign in the square root and use the term proper times. τ

$$\tau = \int (-g_{ab} T^a T^b)^{1/2} dt$$

Note, since a geodesic tangent is parallel transported and thus the norm is constant. Meaning they will not change signatures.

Next, to find the curve whose length does not change to first order under arbitrary smooth deformations. More concisely put, the extremized length. For this we will take the example of a spacelike curve in coordinate basis.

$$l = \int_a^b \left(\sum_{\mu, \nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \right)^{1/2} dt$$

Where $C(a) = p$ and $C(b) = q$ are the endpoints of the curve. The extremization problem is obviously mathematically iden-

tical to extremization problem in action or in most all cases.

$$\delta l = \int_a^b \left(\sum_{\mu, \nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \right)^{1/2} \sum_{\alpha, \beta} \left(g_{\alpha\beta} \frac{dx^\alpha}{dt} \frac{d(\delta x^\beta)}{dt} + \frac{1}{2} \sum_{\sigma} \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \delta x^\sigma \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} \right) dt$$

Because the length is parameterized independently, we may assume the original curve was parameterized so that

$$g_{ab} T^a T^b = 1 = \sum_{\mu, \nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$$

Thus the equation can be simplified to

$$0 = \int_a^b \sum_{\alpha\beta} \left(g_{\alpha\beta} \frac{dx^\alpha}{dt} \frac{d(\delta x^\beta)}{dt} + \frac{1}{2} \sum_{\sigma} \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \delta x^\sigma \frac{dx^\alpha}{dt} \frac{dx^\beta}{dt} \right) dt$$

It can be further simplified, but essentially the main point is that a curve extremizes the length between its endpoints if and only if it is a geodesic. Similar proof can be used to find that the geodesic equation (with affine parameterization) can be obtained from variation of the Lagrangian

$$L = \sum_{\mu, \nu} g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}$$

The final task of this section is to derive the geodesic deviation equation. Let $\gamma_s(t)$ denote a smooth one-parameter family of geodesics. Σ denote the two-dimensional submanifold spanned by the curve. Let s and t be coordinates of Σ . Vector field $T^a = (\partial/\partial t)^a$ is tangent to the family of geodesics and, thus, satisfies $T^a \nabla_a T^b = 0$. Vector field $X^a = (\partial/\partial s)^a$ represents the deviation vector. There is “gauge freedom” in X^a . In the sense that X^a changes if one changes the affine parameterization.

One may ensure that X and T are orthogonal using the metric. Further ensuring that S is made in such a way that $X^a T_a = 0$.

Next $\nu = T^b \nabla_b X^a$ gives the rate of change along a geodesic of the displacement. Thus the relative velocity. Thus,

$$a^a = T^c \nabla_c v^a = T^c \nabla_c (T^b \nabla_b X^a)$$

Is the relative acceleration.

$$a^a = -R^a_{cbd} X^b T^c T^d$$

This, using the Riemann tensor, is known as the geodesic deviation equation.

Thus, one can find curve if $R^d_{abc} = 0$.

21.0.3.4 Methods of Computing Curvature

To begin, one must choose the coordinate system. Express the derivative operator ∇_a in terms of the ordinary derivative ∂_a of this coordinate system and the Christoffel symbol Γ^c_{ab} . For a dual vector field ω_a , we have:

$$\nabla_b \omega_c = \partial_b \omega_c - \Gamma^d_{bc} \omega_d$$

thus,

$$\nabla_a \nabla_b \omega_c = \partial_a (\partial_b \omega_c - \Gamma^d_{bc} \omega_d), \quad (21.1)$$

$$- \gamma^e_{ab} (\partial_e \omega_c - \Gamma^d_{ec} \omega_d) \quad (21.2)$$

$$- \Gamma^e_{ac} (\partial_e \omega_c - \Gamma^d_{bd} \omega_d) \quad (21.3)$$

Thus, the Riemann curvature tensor can be expressed:

$$R^d_{abc} \omega_d = [-2\partial_a \Gamma^d_{bc} + 2\Gamma^e_{ca} \Gamma^d_{be}] \omega_d$$

This can be simplified to for the coordinate system:

$$R_{\mu\nu\rho}^{\sigma} = \frac{\partial}{\partial x^{\nu}} \Gamma_{\mu\rho}^{\sigma} - \frac{\partial}{\partial x^{\mu}} \Gamma_{\nu\rho}^{\sigma} + \sum_{\alpha} (\Gamma_{\nu\rho}^{\alpha} \Gamma_{\alpha\nu}^{\sigma} - \Gamma_{\nu\rho}^{\alpha} \Gamma_{\alpha\mu}^{\sigma})$$

The Ricci tensor can then be obtained by contracting the above equation.

There are some other important points. One can write the coordinate basis for components, $g_{\mu\nu}$, of the metric as a matrix. Then the components $g^{\mu\nu}$ will then obviously be the inverse. We can define g to be the determinate of $g_{\mu\nu}$.

Then, the natural volume element of the manifold is $\sqrt{|g|} dx^1 \dots dx^n$.

A simple formula may be derived for the contracted Christoffel symbol.

$$\Gamma_{a\mu}^a = \sum_{\nu} \Gamma_{\nu\mu}^{\nu} = \frac{1}{2} \sum_{\nu, \alpha} g^{\nu\alpha} \frac{\partial g_{\nu\alpha}}{\partial x^{\mu}}$$

Thus,

$$\Gamma_{a\mu}^a = \frac{1}{2} \frac{1}{g} \frac{\partial g}{\partial x^{\mu}}$$

This is useful as Γ_{ab}^a is used in the Ricci Tensor as well as formulating the divergence of an arbitrary vector field.

21.0.3.5 Orthonormal Basis (Tetrad) Methods

In tensor calculations an orthonormal basis is in some cases advantageous, thus one develops a nonholonomic or noncoordinate, orthonormal basis of smooth vector field $(e_{\mu})^a$ that satisfies:

$$(e_{\mu})^a (e_{\nu})_a = \eta_{\mu\nu}$$

where $\eta_{\mu\nu} = \text{diag}(-1, \dots, -1, 1, \dots, 1)$. In four dimensions, $(e_{\mu})^a$

is called a tetrad. It is useful in the equation

$$\sum_{\mu, \nu} \eta^{\mu\nu} (e_\mu)^a (e_\nu)_b = \delta_b^a$$

Where δ is the identity map of the vectors and $\eta^{\mu\nu}$

Calculation by tetrad has sufficient differences than the coordinate basis method. In both methods three key ingredients are needed. 1: derivative operator that is compatible with the metric, $\Delta_a g_{bc} = 0$. 2: the derivative operator is torsion free. 3: the Riemann tensor is related to the derivative operator.

21.0.3.6 Einstein's Equations

21.0.3.6.1 Special

Energy is the time component of the 4-vector momentum.

A perfect fluid's Stress-energy tensor can be defined as

$$T_{ab} = \rho u_a u_b + P(\eta_{ab} + u_a u_b)$$

where u^a is a unit timelike vector field representing the 4-velocity of the fluid. ρ and P are the mass energy density and pressure of the fluid measured in its rest frame.

a cool trick is that one can define the Euler's equations using non-relativistic limits.

The stress energy of the Lorentz force:

$$T_{ab} = \frac{1}{4\pi} \left(F_{ac} F_b^c - \frac{1}{4} \eta_{ab} F_{de} F^{de} \right)$$

21.0.3.7 General

In general theory of Relativity, while it is possible to define background observers in such a way to define gravity for Earth. When time symmetry does not hold, such in 3-body problems, this is not the case and tidal forces are the only “force” with a meaningful definition.

From the simple exploration of physical energy conservation in gravitational fields where the dimensions is small compared to the curvature of space, it can be found that all matter obeys the equation of motion:

$$\nabla^a T_{ab} = 0$$

Through logical examinations the equation comes to be

$$G_{ab} = R_{ab} - \frac{1}{2}Rg_{ab} = 8\pi T_{ab}$$

Using a trace we find that

$$R = -8\pi T$$

and thus:

$$R_{ab} = 8\pi \left(T_{ab} - \frac{1}{2}Tg_{ab} \right)$$

Thus, the entirety of GR can be summed up as followed. *Spacetime is a manifold M on which there is defined a Lorentz Metric g_{ab} . The curvature of the metric is related to matter distribution via the Einstein's equation.*

21.0.3.8 Linearized Gravity

I am not sure what to write here...

21.1 Homogeneous, Isotropic Cosmology

...

21.2 The Schwarzschild Solution

21.2.1 Derivation

For an object to be static, the time-like killing vector field must satisfy the following equation:

$$\xi_a \Delta + b \xi_C = 0$$

From here, the fact that this shows that spacetime is camve-riant in the time dimension leads to the metric can be defined as

$$ds^2 = -V^2(x^1, x^2, x^3)dt^2 + \sum_{\mu, \nu=1}^3 h_{\mu\nu}(x^1, x^2, x^3)dx^\mu dx^\nu$$

Where $V^2 = -\xi_a \xi^a$, and the absence of $dt dx^\mu$ cross terms expresses the orthogonality of ξ^a and σ .

From here time translation and reflection are here. In most cases something that is time invarient will also be time reflected but not when rotation is involved. If one takes it to be spherically symmetric:

$$ds^2 = r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

In curved space, r does not have to be in relation to the radius, but for the purposes of this, we will take it as such.

the rest...

Now that we have the metric, we see that the solution is asymptotically flat, ie, as r approaches infinity spacetime approaches flat. Allowing the interpretation of an isolated body. Where C is equivalent to $-2M$. (Remember, natural units)

A striking feature is that the metric components become singular in both $r = 2m$ and $r = 0$. This is caused by either a breakdown of coordinate, or a singularity of the spacetime structure. We will see later that the singularity of $2M$ is simply a breakdown of coordinates while at 0 it is seen as a physical singularity.

It can also be found that for all spherically symmetric spacetimes with $R_{ab} = 0$ are static.

21.2.2 Interior Solutions

Now take a static, spherical, stress energy tensor

$$T_{ab} = \rho u_a u_b + P(g_{ab} + u_a u_b)$$

Where u^a must point in the same direction as the static killing vector.

We can now take three independent forms of T_{00} , T_{11} , and T_{22} using the metric derived earlier. To get h to be

$$h(r) = \left(1 - \frac{2m(r)}{r}\right)^{-1}$$

Here to have staticity, where $h \geq 0$, $r \geq 2m(r)$

Where $m(r)$ isn't exactly the proper mass but rather difference between the gravitational binding mass and the proper mass.

From here they derive the Tolman-Oppenheimer-Volkoff equa-

tion or TOV. Which is the hydrostatic equilibrium with relativistic corrections. This equation comes with a limit where pressure required goes towards infinity as is the case of the singularity.

21.2.3 Geodesics

Not too much other than deriving gravitational lensing

21.2.4 Kruskal Extention

N/A

21.3 Methods for Solving Einstein's Equation

21.3.1 Stationary, Axisymmetric Solutions

Earlier we have derived that a spacetime is said to be stationary if there exists a one-parameter group of isometries whose orbits are timelike curves. Thus every stationary spacetime possesses a timelike killing vector field.

A spacetime is called axisymmetric if there exists a one-parameter group of isometries whose orbits are closed space-like curves. To have both

$$[\xi, \psi] = 0$$

these solutions are useful as they describe

Chapter 22

The Geometry of Kerr Black Holes by Barrett O'Neil

22.1 Beginning Kerr Spacetime*

One can define the difference between rotation types to be given by the horizon function $\Delta = r^2 - 2Mr + a^2$ (this is in geometrized units)

The kerr metric does not fail at $r = 0$ like the Schwarzschild solution does. Though it adds the two horizons and the ring singularity.

Really complex math for relatively trivial solutions. I plan to go back at some point, just not today.

22.2 Maximal Extention

Traditional kerr uses ingoing principle null geodesics, but to get maximal extention, one must “glue” both the traditional ingoing formulation and outgoing.

Chapter 23

Assumptions of Physics

23.0.1 Overview

The overall goal is to explore the minimal set of physical assumptions are necessary and sufficient to derive different theories within a unified framework. This is almost identical to my Axiom project with the exception that while not as broad their research is much deeper than mine.

It is broken into two parts

Reverse Physics: Looks at the main theories and finding the common and the different aspects of those theories.

Physical Mathematics: Connecting physical objects and theories to be mapped onto the correct mathematical objects

23.0.2 Reverse Physics

It was a great read with a lot of interesting ideas, but I don't want to write it all, especially since I am planning on writing something very similar in part 1.

So, I will write a summary of the summary

The first assumption is Infinitesimal Reducibility(IR). This tells us the domain of classical physics is of those that can be thought of as made of arbitrarily small parts. Mathematically this tells us that the state space of the particles is a differentiable manifold, together with a volume measure that defines the count of states for each region. We can then add the assumption IND that tells us that the system is decomposable into independent degrees of freedom. For this to work the manifold must have even dimensions. That any state variable q^i has a conjugate state p_i . This give a symplectic manifold. If we assume DR, that evolution is deterministic and reversable, we obtain Hamiltonian Mechanics. This enforces preservation of area. Thus, the displacement vector field is H , which corresponds with the Hamiltonian. This can be generalized to the time dependent case, which recovers relativistic features even without the noting of a metric tensor.

Assuming KE, that dynamics are recoverable from kinematics, we get Newtonian. If we assume both KE and DR we get Lagrangian.

23.0.3 Physical Mathematics

This one is a bit more abstract and newer stuff, so I will have more notes.

Chapter 24

A Concise Introduction to Geometric Numerical Integration

For this I will skip around as I already have knowledge on the subject.

24.1 Variational Integrator

Essentially, finding the discrete Lagrangian

$$L_h(q_n, q_{n+1}, h) \approx \int_{t_n}^{t_{n+1}} L(q(t), \dot{q}(t))$$

Then summing them over a discrete curve with N points

$$S_h(\{q_n\}_{n=0}^N) = \sum_{n=0}^{N-1} L_h(q_n, q_{n+1}, h)$$

Now assume continuous and find the variation of this sum

held fixed.

24.2 Volume-Preserving Methods

Volume preservation is used in many places; particle tracking in in-compressible flows, perturbations in Hamiltonian systems.

Take a divergence free function of velocity. Then a volume perserving flow can be described by this satisfied by the derivative $X(t) = \frac{\partial \varphi_t(x)}{\partial x}(x_0)$

$$\dot{X} = A(t)X$$

$$X(0) = I$$

Known as the variational equation. Here $A(t) = f'(x(t))$ repersenting the Jacobian matrix of f evaluated at $x(t) = \varphi_t(x_0)$.

Thus, if X is divergence free, then the determinate is equal to 1. All Hamiltonians are divergence free.

If one takes a numerical look. X becomes using the implicit midpoint rule.

$$X_{n+1} = (I - \frac{h}{2}A)^{-1}(I + \frac{h}{2}A)X_n$$

So the numerical flow is volume-perserving if.

$$\det \left(\frac{\partial X_{n+1}}{\partial X_n} \right) = 1$$

24.3 Lie Group Method

Assume another differential equation of the same type as before, but $A(t)$ and I are $d \times d$ matrixies

With this equation, despite its simplicity, closed form solutions only exist in particular situations. In scalars($d=1$), it is trivial. For $d > 1$, it is true when $A(t)$ and $\int_0^t A(s)ds$ commute.

In non-commutable cases expressed

$$X(t) = \exp(\Omega(t))$$

One can derive the differential equation

$$\dot{\Omega} = d \exp_{\Omega}^{-1}(A(t))$$

$$\Omega(0)$$

where

$$d \exp_{\Omega}^{-1} = \sum_{k=0}^{\infty} \frac{B_k}{k!} ad_{\Omega}^k(A)$$

Here B_k are Bernoulli Numbers and ad^k is shorthand for iterative commutators.

Solving, one gets a very long infinite series that I will just look up if needed(Magnus Expansion)

I think I am going to more on now.

24.4 Time-Splitting PDE

This is simply the same as traditional splitting but using time with evolution equations.

Chapter 25

The Relativistic Boltzmann Equation

25.1 Special Relativity

It is convenient to define the four-velocity of a particle as

$$U^\alpha = \frac{dx^\alpha}{d\tau}$$

Where τ is proper time, which is scalar invariant. The contra and co variant four velocity as

$$(U^\alpha) = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{V}{\sqrt{1 - v^2/c^2}} \right)$$

$$(U_\alpha) = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{-V}{\sqrt{1 - v^2/c^2}} \right)$$

25.1.1 Minkowski Force

Four velocity is defined in the same manner

Then the four momentum contravariant component is given as

$$(p^\alpha) = \left(\frac{E}{c}, \mathbf{p}\right)$$

Here we can derive the famous $E = mc^2$

Taking the acceleration we get the Minkowski force's contravariant components

$$(K^\alpha) = \left(\frac{\mathbf{F} \cdot \mathbf{v}}{c(1 - v^2/c^2)^{\frac{1}{2}}}, \frac{\mathbf{F}}{(1 - v^2/c^2)^{\frac{1}{2}}}\right)$$

25.1.2 Electrodynamics in free space

25.1.2.1 Maxwell Equations

The electromagnetic field tensor is defined through:

$$F^{\alpha\beta} = (-\mathbf{E}, c\mathbf{B})$$

This tensor is antisymmetric.

With its transformation laws being

$$F'^{\alpha\beta} = \lambda^\alpha_\gamma \lambda^\beta_\delta F^{\gamma\delta}$$

$$F^{\alpha\beta} = \bar{\lambda}^\alpha_\gamma \bar{\lambda}^\beta_\delta F'^{\gamma\delta}$$

$$\begin{aligned}
 p^\mu p_\mu &= -m^2 c^2 && \text{(mass shell),} \\
 F^{\mu\nu} &= \partial^\mu A^\nu - \partial^\nu A^\mu && \text{(EM field),} \\
 \partial_\mu F^{\mu\nu} &= \mu_0 J^\nu, \quad \partial_\mu {}^* F^{\mu\nu} = 0 && \text{(Maxwell),} \\
 \frac{dp^\mu}{d\tau} &= q F^{\mu\nu} u_\nu && \text{(Lorentz force),} \\
 dP &= \frac{d^3 p}{p^0} && \text{(invariant measure),} \\
 J^\mu &= \sum_s q_s \int p^\mu f_s dP && \text{(kinetic current),} \\
 p^\mu \partial_\mu f + q F^{\mu\nu} p_\nu \frac{\partial f}{\partial p^\mu} &= C[f] && \text{(relativistic Boltzmann),} \\
 T^{\mu\nu} &= \int p^\mu p^\nu f dP && \text{(kinetic stress-energy).}
 \end{aligned}$$

25.2 Relativistic Boltzmann Equation

25.2.1 Single Non-degenerate Gas

Note that momentum and number of particles in a volume element are scalar invariant.

We make the volume element

$$d\mu = d^3 x d^3 p$$

We can approximate the Jacobian to be

$$J = 1 + \frac{\partial F^i}{\partial p^2} \Delta t + O[(\Delta t)^2]$$

We assume the following

- Only binary collisions are considered (reasonable if the gas is dilute)
- Molecular Chaos assumption
- One particle distribution function does not vary much over the time interval

Remember the momentum four vector is

$$p^\alpha = (p^0, \mathbf{p})$$

where p^0 is

$$p^0 = \sqrt{|\mathbf{p}|^2 + m^2 c^2}$$

where p^0 is the energy component and $\mathbf{p}$ is the 3-vector spatial momentum vector.

This is all pretty trivial and can be looked up later. I am going to skip ahead.

25.3 Vlasov

25.3.1 The Vlasov-Maxwell system

The Vlasov can be written in form

$$p^\mu \frac{\partial f}{\partial x^\mu} - \Gamma_{\mu\nu}^\sigma p^\mu p^\nu \frac{\partial f}{\partial p^\sigma} = 0$$

Here the phase space is not a product of the two spaces but rather a sub-manifold of the tangent bundle.

Thus the derivative is taken in the direction tangent to the manifold $g_{\mu\nu} p^\mu p^\nu = m^2 c^2$.

One can couple with Einstein's field equation by applying the

stress energy tensor to be:

$$T_{\mu\nu} = c \int p_\mu p_\nu f \sqrt{g} \frac{d^3 p}{p_0}$$

Creating the Vlasov-Einstein system.

25.3.2 The Threshold of Black hole formation

Chapter 26

Introduction to 3+1 Numerical Relativity

26.1 3+1 formulism

26.1.1 3+1 split

The 3+1 split of the metric can be defined as

$$g_{\mu\nu} = \begin{pmatrix} -\alpha^2 + \beta_k \beta^k & \beta_i \\ \beta_j & \gamma_{ij} \end{pmatrix}$$
$$g^{\mu\nu} = \begin{pmatrix} -1/\alpha^2 & \beta_i/\alpha^2 \\ \beta_j/\alpha^2 & \gamma_{ij} - b^i b^j/\alpha^2 \end{pmatrix}$$

The volume can then be expressed as

$$\sqrt{-g} = \alpha \sqrt{\gamma}$$

with g and γ being the determinate of the tensors.

The spacial metric γ_{ij} is simply defined as the metric induced

on each hypersurface Σ by the full spacetime

$$\gamma_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$$

This spacial metric is a full four-tensor but when written in adapted coordinates its time component becomes trivial. This spacial metric is simply the projection operator on the spacial hypersurface.

The lapse function is

$$\alpha = (-\nabla t \cdot \nabla t)^{1/2}$$

The unit normal vector space can be written as

$$n^\mu = -\alpha \nabla^\mu t$$

Where the minus sign enforces that the unit vector is future pointing.

26.1.2 Extrinsic Curvature

The extrinsic curvature tensor $K_{\alpha\beta}$ is a measure of the change of the normal vector under such parallel transport.

Before we do that the projection operator onto the spacial hypersurface:

$$P_\beta^\alpha := \delta_\beta^\alpha + n^\alpha n_\beta$$

which is simply the induced spacial metric.

Using this, the extrinsic curvature tensor is

$$K_{\mu\nu} := -P_\mu^\alpha \nabla_\alpha n_\nu$$

The tensor is symmetric

In the coordinate system created the Hamiltonian and Mo-

momentum constraints take the form:

$$R + K^2 - K_{ij}K^{ij} = 16\pi\rho$$

$$D_j(K^{ij} - \gamma^{ij}K) = 8\pi j^i$$

with

$$\rho := n^\mu n^\nu T_{\mu\nu}, \quad j^i := -P^{i\mu} n^\nu T_{\mu\nu}$$

26.1.3 The ADM evolution Equation

With a lot of math I am not going to write down the Hamiltonian density can be defined as

$$\mathcal{H} = -\sqrt{\gamma} [\alpha(R + K^2 - K_{ij}K^{ij}) + 2\beta_i D_j(K^{ij} - \gamma^{ij}K)]$$

Then the hamiltonian is obviously

$$H := \int \mathcal{H} d^3x$$

Variations of this hamiltonian with respect to α and β^i yields the Hamiltonian and Momentum constraints.

The evolution equations for the gravitational field now follow to be

$$\dot{\gamma}_{ij} = \frac{\delta \mathbf{H}}{\delta \pi^{ij}} \quad \dot{\pi}^{ij} = -\frac{\delta \mathbf{H}}{\delta \gamma_{ij}}$$

where gamma is the spacial metric and pi is the cononical momenta.

26.1.4 BSSNOK formulation

ADM are only weakly hyperbolic and those are highly unstable. BSSN is also called conformal ADM

Consider the conformal rescaling of the spacial metric of the form (where, Conformal rescaling means multiplying a metric by a positive function)

$$\tilde{\gamma}_{ij} := \psi^{-4} \gamma_{ij}$$

This can be any arbitrary chosen scalar field, but a convenient choice would be a factor that the conformal metric has a unit determinate, that is

$$\psi^4 = \gamma^{1/3} \implies \psi = \gamma^{1/12}$$

With γ being the determinate of γ_{ij} .

There are some other values to be taken into account when using differing spacetimes.

The BSSNOK also separates the extrinsic curvature into its trace K and its tracefree part

$$A_{ij} = K_{ij} - \frac{1}{3} \gamma_{ij} K$$

Which under conformal rescaling

$$\tilde{A}_{ij} = \psi^{-4} A_{ij}$$

The crucial point is that BSSNOK introduces three auxiliary variables known as the conformal connection functions.

$$\tilde{\Gamma}^i = \tilde{\gamma}^{jk} \tilde{\Gamma}_{jk}^i = -\partial_j \tilde{\gamma}^{ij}$$

After some more math the equation simplifies to the evolution equations

$$\frac{d}{dt} \gamma_{ij} = -2\alpha \tilde{A}_{ij}$$

$$\frac{d}{dt}\phi = -1/6\alpha K$$

$$\frac{d}{dt}\tilde{A}_{ij} = e^{-4\phi} \left\{ -D_i D_j \alpha + \alpha R_{ij} + 4\pi\alpha [\gamma_{ij}(S - \rho) - 2S_{ij}] \right\}^{\text{TF}} + \alpha \left(K\tilde{A}_{ij} - 2\tilde{A}_{ik}\tilde{A}^k_j \right)$$

$$\frac{d}{dt}K = -D_i D^i \alpha + \alpha \left(\tilde{A}_{ij}\tilde{A}^{ij} + \frac{1}{3}K^2 \right) + 4\pi\alpha(\rho + S)$$

$$\frac{d}{dt}\tilde{\Gamma}^i = \tilde{\gamma}^{jk} \partial_j \partial_k \beta^i + \frac{1}{3}\tilde{\gamma}^{ij} \partial_j \partial_k \beta^k - 2\tilde{A}^{ij} \partial_j \alpha + 2\alpha \left(\tilde{\Gamma}_{jk}^i \tilde{A}^{jk} + 6\tilde{A}^{ij} \partial_j \phi - \frac{2}{3}\tilde{\gamma}^{ij} \partial_j K - 8\pi\tilde{j}^i \right)$$

26.2 Gravitational Waves

26.2.1 Gauge invariant Perturbations of Schwarzschild

When referring to multipoles, if the transformation is $(-1)^l$ then it is even or polar. It is odd or axial if it is instead $(-1)^{l+1}$.

The authors define the Zerilli-Mathews tensor harmonics, useful tensors for looking at quadropoles. It works spesifically for even tensors

$$Z_{ab}^{l,m} := D_a D_b T^{l,m} + \frac{1}{2}l(l+1)\Omega_{ab}Y^{l,m}$$

where the scalar sphere harmonics are eigenfunctions of the laplace operator on the sphere. The above tensor is in fact traceless, we can also find its divergence to be given by

$$D^b Z_{ab}^{l,m} = -\frac{1}{2}(l-1)(l+2)Y_a^{l,m}$$

Odd tensors can only be constructed in one way

$$X_{ab}^{l,m} = \frac{1}{2}(D_a X_b^{l,m} + D_b S_a^{l,m})$$

With the divergence being

$$D^b X_{ab}^{l,m} = -\frac{1}{2}(l-1)(l+2)X_a^{l,m}$$

This can be expanded into multipoles that find that the vector harmonics vanish at monopoles while the tensor ones vanish at both mono and di - poles.

Next the Regge-Wheeler gauge is a choice that $G^{l,m} = H_A^{l,m} = 0$ and K and H_{AB} stay the same.

Here we have the even and odd parity master equation. I won't write it down but I am sure I can look it up quite easily.

26.2.2 Gravitational Radiation in the TT Gauge

I don't want to write all of this, it is relatively simple math(not really), but a lot of it.

26.2.3 Energy and Momentum of GRW

Take the second order perturbation to derive the Ricci Tensor as

$$R_{\mu\nu} = \epsilon R_{\mu\nu}^{(1)} + \epsilon^2 R_{\mu\nu}^{(2)}$$

Where $R_{\mu\nu}^{(n)}$ can be derived through F rather simply.

This can be used to construct a psuedo-tensor that can then be used to find the real tensors.

Chapter 27

Applications of Lie Groups to Difference Equations

27.1 Finite Differences and Transformation Groups in Space of Discrete Variables

One can take a differential operator and expand using a Taylor series.

Chapter 28

Topological Methods in Hydrodynamics

28.1 Group and Hamiltonian Structures

Chapter 29

Chaos and Structure in Nonlinear Plasma

29.1 Nonlinear Oscillations

29.1.1 Harmonic Generations in Small Amplitude Oscillations

N/A

29.2 Hamiltonian Dynamics

In plasma physics a special case for 1D would be when the Hamiltonian is equivalent to the stream function ψ .

Having a physical problem reduced to a 3-dimensional vector space is tantamount to having the problem solved do to the clear research in the area.

29.2.1 Poincare Surface of Section

29.2.2 Fixed Points and Invariant Curves

29.2.3 KAM Theory

Take the two dimensional map

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = T(x, y)$$

produced from the flows of systems such as given above are common in plasma physics. The theorems used are called KAM for convinences based on the three founders of many of the theorems. To apply maps for twist conditions that produce monotonically increasing rotation of the point in the surface of section.

$$\frac{\partial y'}{\partial x} > 0 \quad \forall x$$

The twist conditions guarantees that around a fixed point there is a monotonic increase of the rotational angle.

29.2.4 KAM Torii

Essentially, KAM is the study of how nearly integratable Hamiltonians remain similar to their integratable counterparts. Resonance refers to ones that remain.

29.2.4.1 Ergodic Theorem

The Ergodic theorem states that the long time average of a function over phase space is equal to the ensemble average.(essentially the average of one function is the same as the average over the collection of possible states of a system)

This theorem is valid for dynamics $\theta' = \theta_\alpha(r)$, provided α is irrational.

We may compute the difference between f_0 from the theorem would be

$$\delta f(\theta) = \sum_{k=1}^{\infty} [a_k e^{ik\theta} + a_k^* e^{-ik\theta}]$$

assuming that $|a_k| < \frac{a_1}{k^2}$ to eliminate discontinuous functions. Summing this we can get that the sum diverges when $k\alpha = 2\pi m$. The sum will converge provided that

$$\left| \frac{\alpha}{2\pi} - \frac{m}{n} \right| > \frac{\epsilon}{n^3}$$

not here that m and n are integers, particular resonance between the frequencies $\alpha = (m/n)2\pi$

29.2.5 KAM theory in Laboratory Plasmas

The $1\frac{1}{2}$ D abstraction can be mapped out for electron in toroidal confinement. Also, incompressible drift motion by $E \times B$ plasma flow.

29.2.6 Visualization of the Magnetic Flux surfaces by Electron beam mapping

The winding number in the phase space becomes directly to the rotational transformation $t(r)$, this “safety factor” $q(r) \equiv 1/t(r)$ of tokamak theory.

They used this methodology to shrink instability islands in the tokamak.

29.2.7 Magnetic Islands and the Unstable-Chaotic Trajectories of Field Lines in Tokamaks

Most dangerous magnetic islands are developed by helical plasma currents from unstable resistive MHD motions.

Chapter 30

Papers

I can't believe how few papers I have written notes about, I should do better.

30.1 Operator Splitting Methods: Numerical Solutions of Ordinary Differential Equations via Separation of Variables

30.1.1 General Idea

It is deriving the mathematical underpinning of operator splitting from Koopman-Lie semigroup rather than physical systems. Generalizing it beyond Hamiltonian theory

30.1.2 Notes On Introduction

N/A

30.1.3 Strongly Continuous and bi-continuous semigroups

You know what, at the moment this paper may be beyond my graps.

30.2 Splitting methods for differential equations

30.2.1 Notes on introduction and Abstract

Some argue there are only ten big ideas in numerical analysis. They make a cool little connection to Descartes rules of logic and how splitting lays the foundation of science.

They then use a simple argument to derive strang splitting

Adjoint method refers to stepping forward and then stepping backwards with the inverse of its adjoint. Adjoint simply refers to that if $\psi_h = \varphi_1(\varphi_2)$ then $\psi_h^* = \varphi_2(\varphi_1)$

This is called time symmetric or self-adjoint method.

Some disadvantages of higher order splitting is that they go backwards in time which is troublesome in dissipative systems. extensively used in fields as distant as molecular dynamics, simulation of storage rings in particle accelerators, celestial mechanics, astronomy, quantum (statistical) mechanics, plasma physics, hydrodynamics, and Markov Chain Monte Carlo methods.

30.2.2 Higher Order Splitting and Composition

30.3 Core Collapse Supernova Modeling: The Next Ten Years

30.4 Physical, numerical, and computational challenges of modeling neutrino transport in core-collapse supernovae

30.4.1 Structure Preserving

30.5 Discontinuous Galerkin Methods

30.6 Energy conserving and well-balanced discontinuous Galerkin methods for the Euler–Poisson equations in spherical symmetry

30.6.1 Numerical Methods

Poisson equations is derived to be

$$\Phi_h(r) = \Phi_h(r_{j+1/2}) - \left(\sum_{i=1}^{k+1} \frac{g_{j,i} \tau^{i+1}}{i+1} - \frac{g_{j,-2}}{\tau} \right) \Big|_{\tau=r}^{r_{j+1/2}}$$

The euler-Poisson equation in DG

Chapter 31

Colloquia: UTK

31.1 August 24: Controlling Topological Quantum Phases through Epitaxy

31.1.1 Main notes

How can one material have topological phase? Essentially the electronic band structure. Things to modify are strain, dimensionality, crystal, structure, and composition.

He uses a molecular beam epitaxy, the growth of crystalline overlayer on a crystalline substrate.

- Phase and strain control
- Dimensionality control

31.1.1.1 Phase and strain control

First one cools to avoid bubbling.

Odd points that were later understood via computational experiments via energy ...

31.1.1.2 Strain-controlled phase stabilization

IDK

31.1.1.3 Quantum Oscillation in strained α -Sn

Heavy-hole state and surface state have negative correlation.

31.1.1.4 Dimensionality Control

IDK

31.1.1.5 Geometry and Interfaces

Changing the arranging of mask, InGaAs, InAs and InP(001). In order to protect, enhance electron mobility, and phase coherence length.

These create kinds of pseudo-shapes like trenches and bridges that can grow continuously retaining their general shape.

You can suppress induction effects post fabrication through something(not sure)

31.1.1.6 Emerging

Decouple synthesis from integration: Find what materials best lead to growth

Reconfigure strain: Find new ways to reduce growth strain

Expand Heterogeneous integration: IDK

Main ways of doing this was through remote epitaxy and van der Waals epitaxy on flexible mica

31.1.2 Connections

IDK

31.1.3 Summary

I think the general idea is an overview of creating superior method of developing superconducting topological material through epaxity. Studying the best arrangements, conditions(strain, dimensions, temp, etc). I really didn't understand beyond this vague overview.

31.1.4 Questions

What are electronic band structure?

What are buffer layers?

What are α -Sn and β -

How do these things grow?

31.2 August 31 - Electron Pairing and Fractionalization in age of AI and Quantum Computing

31.2.1 Main notes

31.2.1.1 Introduction

Takes the traditional billiard-ball electron scattering to either pairing up or quantum hall, making it so that Resistance can come to zero. Also, conduction is an exact fraction of e^2/h .

These Cooper pairs are caused by quantum pairs via photon glue. These permutate lattices that make it “easier” for other particles to do the same and move without resistance.

He introduces, the fact that the last ten years, people aren’t trying to increase temperature but rather greater mathematical machinery. Looking at Chiral 2D SC. Allowing for accuracy with both counter and clockwise translations.

On the side of quantum hall and fractionalization.

Found that conductors can effectively channel $1/3$ of electrons through “anyons,” a type of quasiparticle.

31.2.2 Concepts

Using a Fourier transformation can define N atoms using K space.

Solving the encoding problem with the quantum states is near impossible in classical computers so he hopes that quantum computers can do it much easier.

Current theory can only work for around ten atoms. It also becomes a struggle when the Hilbert space goes up exponentially. This is particularly troubling when real matter has 10^{23} atoms.

They work on this through using smooth Moire potential space that is more smooth than real solids.

31.2.3 Learning Hamiltonians

Learning Hamiltonian AI systems are meant to bring such material simulating to $\mathcal{O}(N)$, which is quite out there.

Encoding Hamiltonian through one section of the idea of locality as on the scale of matter, electron become local.

They have a “white box”

They have a scale of $10^6 - 10^9$.

31.2.4 Kinetic Magnetism

To lower the energy in matter, you can polarize so that all spins are facing the same direction.

Singlets entangle between spin polarization.

Connecting them all together is what develops the cooper pair.

Neural networks can also be used to compute wavefunctions. This is done by pairing-native electron antzan.

Geminal works to machine precision in very few states. Much lower then previous models.

31.2.5 Quantum Computing

Four main different methods of quantum computing.

31.2.6 Connections

Open frontier being:

Finite-q photons

One thing he mentioned is the idea that programming for quantum computers will end up something primarily done by mathematicians and physicist for a while as AI or traditional CS won't have the skills needed at the moment.

31.2.7 Summary

AI for quantum matter:

Controlled strong-coupling theory:

Quantum Proccessors:

Next comes:

Further AI-Hamiltonian

AI Wavefunction

Simulating anyon braiding and topological quantum computation

31.2.8 Questions

31.3 Curiosity and Perseverance: Martian Surface

31.3.1 Main notes

Using fluid dynamics one is able to more accurately figure out the past of mars surface. This is highly challenging due to limited data and people's tendency to apply Earth principles to the Earth.

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31.3.2 Connections

31.3.3 Summary

31.3.4 Questions

Chapter 32

Coding

32.1 *Linux Terminal

32.2 *Github

32.3 *FORTRAN

32.4 *Python

32.5 *Emacs

32.6 *LaTeX

Chapter 33

Bible Reading

33.1 2026-16-02

33.1.1 Genesis 41:17-42:17

God protects Joseph through establishing gifts within him. He does the same for us.

Why does Joseph test his brothers?

33.1.2 Matthew 13:24-46

The parables of Jesus, establish stories and myth as the methodology of developing the abstract concepts of God's will.

It refers to the weeds as everything that causes sin and all who do evil, what exactly does this mean? Everyone does evil and everyone has within that which tempts them. Taking out the "weeds" prematurely will destroy the wheat.

The final stories is about how the kingdom of God gives everything for the grace of those who are lost.

33.1.3 Psalm 18:1-15

Love of God

33.1.4 Proverbs 4:1-6

Community, family, and God fosters wisdom. Wisdom that protects you.

33.2 2026-17-02

33.2.1 Old testament

Why is Joseph testing with them even after hearing what they had to say while also giving them money

33.2.2 New Testament

The kingdom of God takes in all, but not all remain.

Interesting how such a radical take on ones word is used.

33.2.3 Psalm

God rescues you and protects you from the enemies that are greater than you

33.2.4 Proverbs

Pursue wisdom over all other virtues

33.3 2026-24-03

33.3.1 Genesis 46:1-47:31

Joseph built his family and tradition up in accordance with God's will. He also did his duty with extreme diligence.

33.3.2 Matthew 15:1-28

While tradition matters, God's commandments are far above.

Your words matter and define you. Disregarding this defiles your nature. Acting in a virtuous manner in accordance with the idea matters much more than rules. Beyond that, a critical examination of such things is important as people lead one another astray.

Actions can corrupt you, they are not in isolation but rather, not just a consequence of what is beneath but create what is beneath.

Jesus was here for the Jewish people, yet his words go far beyond.

33.3.3 Psalm 19:1-14

To be truly free is to be free of sin, while this is impossible on Earth, God's commands lead to this by our creation of our own self though the actions he has put forth.

33.3.4 Proverbs 4:14-19

Don't think of evil for it will corrupt you without thought. For, while not always, that is what evil comes from, lack of thought. Impulse and action without conscious morality driving ones

will. One must then separate oneself from this evil that surrounds them.

33.4 03/25/26

33.4.1 Genesis 48:1-49:33

I continually attempt to find greater understanding of God's will for us through the traditions and things he set forth. How should I go about doing it? I mean I would make the claim being an endogamous clan based family structure does not hold deeper wisdom, but how do I make sure my cultural preferences are hiding things from me in other areas.

In this story as in others, God overturns natural human hierarchy as he chooses the secondborn above the first.

In the prophecies it mentions the children's clans fate that mirror their temperament, flaws, and virtues. Meaning or fate is either our own or God has put it in front of us purposefully.

Juda's past is messy but he brings out the Davidic monarchy and eventually Jesus.

Joseph's blessing begins by his attack but also being saved through God, not himself.

33.4.2 Matthew 15:29-16:12

Interpret the signs of our time to understand God's will.

Those that have turned for Jesus will need signs to see, but with those with faith that is not needed.

The Pharisees only offer short 'bread' coming from gratification of tradition and lineage, to go beyond one must follow Jesus.

The disciple worry about food even after seeing the miracle Jesus had just done. In the same way we so simply forget and ignore the acts Jesus has already done for us.

33.4.3 Psalm 20:1-9

To quickly we think our victories are our own, to devote ourself to Christ is the antidote to this pride.

33.4.4 Proverbs 4:20-22

Pay attention to the wisdom of others, especially what God says. Attempt to fully understand it, beyond even what they have defined for the wisdom.

Guard your heart tightly, even listening to the world around you too closely can corrupt you. This heart, the virtue inside of you guides your entire being, your life in ways that aren't clear at first glance. Sacrificing this is far more dangerous than it seems.

Even something as trivial as degrading language can have a profound impact on our heart, for this reason one must make a plan and follow it.

33.4.5 Take-away

I have already started on my essays on self-authorship, these passages even more so highlight how important this is for my development with my relationship with Christ.

33.5 4/24/26

33.5.1 Exodus

"Am I God, that I can punish you?" Only God has the right and capacity to punish in a just manner.

One theory I had was that they misunderstood years and that is why they lived so "long," but according to the bible they legitimately lived to see multiple generations worth of people.

33.5.2 Matthew

The disciples were told that Jesus would raise himself from the grave and Jesus even talked of the cross, and yet they doubted when the time came. We can too miss what is directly in front of us.

Peter tried to control Jesus even knowing he is the son of man, just as we do without thought.

What does Jesus mean by Peter is the Rock, why does he have this authority. What does it mean for God to give the authority over heaven to a mortal man and a singular one at that?

33.5.3 Psalm

We do not have enemies for victories, we only have reflections of God's. His victories is our honor, his enemies are ours. We are extensions of his will upon the Earth, empty without his will and might.

33.5.4 Proverbs

You will express the wisdom you have gained. While one can fake wisdom through prideful intellect, it becomes obvious over time as it falls as poison to the mind of those who hear.

33.6 4/25/26

33.6.1 Exodus

God has Moses take his shoes off to walk upon the holy ground. What does this mean?

God understands and condemns the slavery and abuse of his people. Takes upon an unworthy person and gifts him with the abstract skill of public speaking. Just as he does with all of us. Our natural talents are a product of him. It does make one question what is not a direct product and what is an indirect product of him and us.

I AM and Yahweh are God's names, his eternal names. Yahweh is sometimes directly translated to I AM, so what does this mean? The common interpretation is that he is self-referential and the beginning of all things, that he is unchanging. One thing I am thinking of is that I am is a relational function, so this blankness is complex with multiple ideas; he is everything, he is by definition relational, he is beyond categorification, he has no attributes, all attributes are defined by him, etc. The main one I am thinking of is that he is beyond relations, categories, attributes, and all things as he defines them, he even defines tautologies.

33.6.2 Matthew

It is said that John the Baptist came in the spirit and power of Elijah, what exactly does this mean; did he reincarnate, guided by Elijah, take on the power as a lineage to Elijah?

"How long must I put up with you?" Jesus says this to the disciples as he criticizes their lack of faith. Our sin is insignificant compared to the fatigue of our disbelief. I, more so than most, switch this around, but the truth is in Jesus not morality dressed up.

33.6.3 Psalm

The world will tear you down completely, beyond what is thought possible, but God remains.

33.6.4 Proverbs

As people fall into lust it clouds their judgment more than most vices. Tearing themselves apart and bringing them to ruin as they stagger forward like an animal. This is what makes lust a sin, it is the very death of a man's soul, mind, and discipline if it ensnares itself. Making one a slave to others and their own flesh.

33.7 4/29/26

33.7.1 Exodus

I remember one theory that the reason he was not good with words was because he was raised speaking the Egyptian language rather than Hebrew.

When referring to the who God is by his creation of all things. He uses an infinite tense rather than on of the past. Does this suggest classical theism or occasionalism?

Why does he harden the Pharaohs heart. One interpretation is the above occasionalism as everything and every action is a result of his power. Another is simply that he permits it, and had foreknowledge of it. Another could be that hardening his heart could refer to an act of judgment. Another could be it is a narrative purpose to lead out and show his power to the Jews. Also, classical theism of God being the primary cause of all things. Lastly, a literary device to simply show God is in control.

"They are lazy," referring to slaves doing backbreaking work that kills them. We can see this in the same way throughout society and history.

33.7.2 Matthew

Jesus refers to turning from sin to be like children, but children tend to be the most hedonistic and foresake all others in their immediateness, but this is not on purpose. They are blameless not because they share no blame but because their heart and intentions have not been sorrowed by evil.

"Sorrow awaits the world." How can the world feel sorrow?

Do not let your body control you but cut it off. Do not let others tempt you but fear them.

Show forgiveness, but do not barter with those that abandon the Church.

Jesus refers to their Angles, making a case for guardian angles, at least for children.

33.7.3 Psalms

They already had inklings of God's gloabal salvation.

33.7.4 Proverbs

The Lord sees clearly everything we do, our intentions, and where it leads us. This path is the reason for all of the moral rules, we degrade ourselves which opens us up to more and corrupts our power.

He refers to your wife as yourself and another as a stranger, highlighting the extreme close bound between two spouses.

Having relations with others gives them a part of you, a power over you that you cannot completely take back. In this case, of that of a stranger, that part of you is destroyed in another.

33.8 5/24/26

33.8.1 Exodus

In this story God directly influences the people through physical force. What does this tell us about him when he doesn't use it in the same way now? Should one take this story literal or figertively. Showcasing who God is and his nature?

He also influences the Pharaoh's mind in order to show his capacity. Why? This is also related to communion. Maybe before Jesus's resurection, gentiles didn't have souls. This was a communion of his people and to bring them against an enemy to unite them. Afterwards, Jesus brings them into the community.

Later they use the term predicted rather than cause the

hardening of his heart. Maybe it means more his actions will bring about the desired goal, so hust as before the connection of a national identity that forms into the Church but without the idea that the gentiles don't have souls, that was rather silly.

33.8.2 Matthew

Forgive everyone from the heart as our grevines towards the world are far more. Every tiny, of what we call imperfection, is a violent blast to the perfection of God. God desires ous to be social.

He also desires a holy marriage, but this life is not for everyone. Many are not meant for marriage. A great example of this would be those that are gay, but this then begs the question of whether it is the marrige, physical relations, or impulse that is wrong. Further examination is needed.

33.8.3 Psalm

...

33.8.4 Proverbs

Self-control is the heart of all things. Without it one becomes a slave of their impulses that destroy their minds and body.

33.9 5/26/26

33.9.1 Matthew

Become like children. Children are often wrong, selfish, and mildly cruel, but they do so without malice. They do actions

out of nature than conscious deliberation. So does God mean be in that way, open like children, do little sin(for while they do it relativley often their actions are small), or dependence on God and shape ourselves in a new way unlike we have in the past. I think the last one.

33.10 6/8/26

33.10.1 Exodus

“So you can tell our children and their children”

Here he is creating Jewish identity through the pass-over. Not to waste, to share, cook fully, frugal and healthy choices, urgency in what you do, and be prepared/

33.10.2 Matthew

Serving others comes first before all. The further down the higher up.

33.10.3 Proverbs

Take initiative and work for your life, rather than be absent of it all.

33.11 6/16/26

33.11.1 Exodus

All things belong to the Lord. In this he creates traditions for us to remind ourselves of this. Habits to live by and learn from.

It is filled with symbols, such as yeast symbolizing moral decay.

33.11.2 Matthew

Temples should have dignity just as people do. Protected from things. The act of selling and buying is important but it is not of the Church.

Why did he make the fig tree wither? Metaphor for us doing our part in stewardship and work?

33.11.3 Psalms

Our integrity protects us as it is in the God almighty.

33.11.4 Proverbs

You will understand people as you see them. Wicked makes themselves clear without thought.

33.12 6/23/26

33.12.1 Exodus

This reminds me of an idea. That the reason God does not reveal himself is to protect us from ourselves. When we look at the Israelites disobeying God when he is there with them in physical proof, we are outraged. It seems insane, but in many ways we do the same. For those that aren't Christian for reasons other than lack of proof, how would they live their lives with God manifest before them. Even for Christians, this would be a sudden and drastic difference.

33.12.2 Matthew

Jesus tells us for our actions and lives rather than what we boast outwardly.

Now, he will take new people rather than the Jews who had desecrated his kingdom. We should be careful in our dogma but remain in our loyalty of the Lord, this is the primary paradox of Christian teachings.

33.12.3 Psalms

Our integrity is what we have for the Lord. To protect it, one takes a further extreme than thought. To separate oneself and mind from their environment and become more than a slave to their impulses and history.

33.12.4 Proverbs

Why did they write “six... no seven”? What could possibly been their purpose?

Anyways, the gifts that God has given us can just as easily be used for evil. This being the curse of our free will, our purpose and function exist but the capacity to function in its inverse is there just as well. As someone cannot be good without the capacity towards evil, nor can they truly do good.

It gives us seven things: haughty eyes, lying tongue, hands that kill innocence, heart that plots evil, feet that race to do wrong, a false witness who pours out lies, and a person who sows discord in a family.

On the other side: Eyes to observe and learn, a tongue to teach and express, hands to steward the world, a mind to meditate on what is right, feet to bridge the gap with others, witness to protect one another, and a person to build up and sharpen

a family.

33.13 Meeting

33.13.1 Order of events

- General opening
- Open the table
- Things I noticed and thought of
- questions

33.13.2 Things I have noticed

- Jesus parable of taking out the weeds will kill the crops
- Wisdom does not only come from God alone but community and family
- God has designed our nature in his image, a nature of mind and virtue, but such capacity can just as easily be in vice that defiles your nature in such the way that it is inline with it.
- Actions influence who you are and aren't just in isolation, guard your heart
- To be truly free is to be free of sin
- Be fully aware at all times as when we are happy is when our impulses sneak up on us.
- Our fate is in line with our nature, both as our actions create our lives and God creates our lives in accordance to who we are

- One must interpret the signs of our times
- Pay attention to the wisdom of others, don't just assume you know what they mean. Look into it, even if you know what they mean God may have more in it for you.
- We often try to control Jesus subconsciously, just as Peter does in Matthew trying to tell him what to do even after knowing that he is God manifest.
- We are an extension of God's will on Earth
- People and their nature are obvious
- God's name is I AM
- Our sin is insignificant compared to the fatigue of our disbelief, Jesus is not morality dressed up
- Why does God harden Pharoah's heart
- Jesus and us acting like Children
- The passover and communion
- the idea that God is not here to protect us from ourselves

33.13.3 Discussion Questions

- What do you think are the signs of our time
- What does God referring to himself as I AM mean
- Why does Jesus ask us to be like children, and how does this relate to Genesis
- The relation of passover to communion
- What does it mean to be made in the image of God

- Does civilization progress towards the glory of heaven or do we delude ourselves
- What is our nature and how does that relate to the ideas presented
- Exodus and freedom, difference between freedom from and freedom for, why do the Israelites want to go back to Egypt
- Jesus's teachings and virtue ethics
- What does it mean for the fruit to be the knowledge of good and evil
- Do communities need rituals, and what rituals do we do or should do
- What do God's and Jesus's miracles and actions reveal about them
- what responsibilities do we have towards society
- If one were to collapse the morality God puts out for us, what would it be and why?
- in just the same way that the stories of the bible reveal things about truth, God's nature, and morality; what does your life as a story tell?

33.14 6/24/26

33.14.1 Exodus

This story is a great symbol of why God does not make himself manifest. People complain, challenge, question, and become

overtly used to him instead of traditional reverence. Not taking it upon themselves to become virtuous.

33.14.2 Matthew

First, God calls out to all, not all come, but some who come do not there truly, and will be forced to leave.

Next, we will not be married in Heaven, what does this mean for our relationships?

33.14.3 Psalms

It is right to give our thanks and praise. Some question the fact that what he protects us from is a result of him, we should still praise for our existence and that which challenges us if for a greater purpose.

33.14.4 Proverbs

Take upon our parents wisdom and let it create who we are as a person. This is what will truly protect us, our integrity and character.

33.15 7/27/26

33.15.1 Exodus

God is coming in his form, further reading is needed for this.

33.15.2 Matthew

Call no one rabbi or teacher as there is only one teacher. I have trouble with this as I often search for knowledge outside of religion.

33.15.2.1 Psalm**33.15.2.2 Proverbs**

Lust is inherently dangerous as it makes you do objectively foolish things.

Chapter 34

Philosophy

34.1 Scope

Here I will write general notes from sources on the subject of philosophy.

34.2 Nicomachean Ethics

34.2.1 3/26/26

While I have constantly defended virtue ethics, and even Aristotle's ethics, I am a bit critical of his methods. Rather than defining a platonic ideal, he observes those around him and defines ethics based upon that. It works quite well and is something I also do but this as the primary method tends to lead to some differences.

I am also surprised how similar his views of ethics are so similar to 21st century ethics in many ways; considering how different the views of people just a couple decades ago were.

I definitely find coherence with his idea of magnanimity with

my philotimic virtue. While I had originally conceived of the idea from listening to a summary of the idea, reading it directly from him shows that it was incredibly similar. Making me almost wonder if I should forget having my own word and just use that one. The only real difference is that I also blend philotimic with my specific views of morality and the idea of self-authorship; obviously an Aristotelian magnamonious man would value self-authorship, it isn't directly imbued with the word. Also, his spesific definition has a more prideful connotation.

Kinda funny how even in that time older folks found young comedians as vulgar and inappropriate while the younger found the otherside as offensive and cruel.

Obviously his views of justice, while different are quite well.

Part III

Projects

Chapter 35

Introduction

Chapter 36

Previous Research

These are brief/lazy copies/explanations of things I have written previously made projects. Copied from my last book Logos

36.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method

The construction of a theoretical model, magnetohydrodynamic lattice boltzmann method(MHD-LBM) model for 3D compressed plasma, using a finite volume scheme is constructed. The hyperbolic Maxwell equations, which satisfy the elliptic constraints of Maxwell's equations and the constraint of charge conservation, are used to simulate the electromagnetic field. The flow field and electromagnetic field are coupled to simulate a compressible plasma through the electromagnetic force and magnetic induction equations. This model can further be applied to create a quantitative simulation to model complex nonequilibrium effects of compressed plasma to provide mesoscopic physical insights into the flow mechanism of a shock

wave in a supersonic plasma.

Theoretical model, Finite volume scheme, hyperbolic maxwell equations, and nonequilibrium effects

36.1.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method Introduction

The study of nonequilibrium dynamics in compressible plasma is a critical area of research in plasma physics, with significant implications for both theoretical understanding and practical applications. Compressible plasmas are found in a variety of contexts, such as astrophysical phenomena(1), industrial processes(2,3), and fusion. Understanding the complex interactions and behaviors of plasma under nonequilibrium conditions is essential for advancing these fields.

Previous research has primarily focused on two-dimensional models(4) While these models have provided valuable insights, they often fall short in capturing the full complexity of three-dimensional plasma dynamics. The limitations of these models highlight the need for more advanced simulation techniques that can accurately represent the intricate behaviors of compressible plasma in three dimensions.

The Lattice Boltzmann Method (LBM) offers a powerful tool for modeling fluid dynamics at the microscopic level.(4,5) Unlike traditional computational fluid dynamics methods, such as the Navier-Stokes equation, the LBM is particularly well-suited for simulating nonequilibrium effects and complex boundary conditions. Its ability to handle multiphase flows and incorporate microscopic interactions makes it an ideal choice for studying compressible plasma dynamics.(5)

This paper aims to develop a comprehensive 3D model using the LBM to simulate nonequilibrium effects in compress-

36.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method 173

ible plasma. By leveraging the strengths of the LBM, we seek to overcome the limitations of previous models and provide a more accurate and detailed representation of plasma behavior. Our research focuses on the coupling of flow and electromagnetic fields, exploring the interactions and dynamics that arise under nonequilibrium conditions.

36.1.2 Physical Model

Kinetic Equation

Li and Zhong(10) introduced the potential energy distribution function as well as a compressed DDF Lattice Boltzmann equation. A potential energy distribution function can be added so the Boltzmann BGK can obtain an adjustable specific heat ratio or Prandtl number(4)

This new Boltzmann Kinetic equation can be written as followed:

$$\frac{\partial f_k}{\partial t} + (\mathbf{e}_k \cdot \nabla) f_k + \mathbf{a} \cdot \nabla_e f_k = -\frac{1}{\tau_f} (f_k - f_k^{eq})$$

$$\frac{\partial h_k}{\partial t} + (\mathbf{e}_k \cdot \nabla) h_k + \mathbf{a} \cdot \nabla_e h_k = -\frac{1}{\tau_h} (h_k - h_k^{eq}) + \frac{z_k}{\tau_{hf}} (f_k - f_k^{eq})$$

Where k is the direction of discrete velocity. f_k is the density distribution function. h_k is the potential energy distribution function, while f_k^{eq} is the equilibrium distribution function. $\mathbf{e}_k$ is the discrete velocity component, and τ_f is the relaxation time of the density distribution function. τ_h is the relaxation time of the potential energy distribution function.

This can also be defined as:

$$\tau_{fh} = \frac{\tau_f}{\tau_h}$$

The force term can be approximated as:

$$\mathbf{a} \cdot \nabla_e f \approx \mathbf{a} \cdot \nabla_e f^{eq} = -\frac{a \cdot (\mathbf{e}_k - \mathbf{u})^2}{RT} f^{eq}$$

36.1.3 Density Distribution Function

$$\begin{aligned} f_k^{n+1}(x_i, y_j, z_l) = f_k^n(x_i, y_j, z_l) & - \Delta t \left(\frac{F_{k,i+1/2,j,l} - F_{k,i-1/2,j,l}}{\Delta x} + \frac{F_{k,i,j+1/2,l} - F_{k,i,j-1/2,l}}{\Delta y} \right. \\ & \left. - \Delta t \cdot \frac{1}{\tau} (f_k - f_k^{eq}) \right) \end{aligned}$$

36.1.4 Potential Energy Distribution

$$\begin{aligned} h_k^{n+1}(x_i, y_j, z_l) = h_k^n(x_i, y_j, z_l) & - \Delta t \left(\frac{G_{k,i+1/2,j,l} - G_{k,i-1/2,j,l}}{\Delta x} + \frac{G_{k,i,j+1/2,l} - G_{k,i,j-1/2,l}}{\Delta y} \right. \\ & \left. - \Delta t \cdot \frac{1}{\tau_h} (h_k - h_k^{eq}) + \frac{z_k}{\tau_{hf}} (f_k - f_k^{eq}) \right) \end{aligned}$$

36.1.5 Hyperbolic Maxwell Equations

The previously used Maxwell equations have been enhanced with Lagrange multipliers, Ψ and Φ , to combine with the evolution equation. These are the new Maxwell equations in derivative form:

$$\begin{aligned} \frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} + \gamma \nabla \psi &= 0 \\ \frac{\partial \mathbf{E}}{\partial t} - c^2 \nabla \times \mathbf{B} + \chi c^2 \nabla \phi &= -\frac{\mathbf{J}}{\epsilon_0} \end{aligned}$$

36.1 3D Modeling of Non-Equilibrium Dynamics in Compressed Plasma Using Lattice Boltzmann Method 175

$$\frac{\partial \Psi}{\partial t} + \gamma c^2 \nabla \cdot \mathbf{B} = 0$$
$$\frac{\partial \Phi}{\partial t} + \chi \nabla \cdot \mathbf{E} = 0$$

Where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, Ψ and Φ are introduced divergence variables, γ and χ are divergence error propagation speeds, c is the speed of light, and ϵ_0 is the vacuum permittivity.

36.1.6 Kinetic Non-Equilibrium Method

The nonequilibrium effects on compressed plasma are observed through differences in molecular speed. Each non-equilibrium kinetic moment can be represented as the difference between the corresponding kinetic moment and the local equilibrium kinetic moment. The equations of kinetic moments may be used to find the nonequilibrium quantities.

$$M_{f,m}^{neq} = M_{f,m} - M_{f,m}^{eq}$$
$$M_{h,m}^{neq} = M_{h,m} - M_{h,m}^{eq}$$

36.1.7 Discussion

In this paper, a 3D MHD-LBM analytical model was constructed for compressed kinetic plasma. Future research must be done in developing a computer model and validating said computer model. A third-order finite volume MUSCL could be applied to this analytical model combined with a D3Qx density equilibrium distribution function to create an easily solvable numerical solution. While electromagnetic fluxes were evaluated through Steger-Warming flux vector splitting. MHD-LBM models have shown to have greater accuracy when it comes to plasma shockwaves, therefore MHD-LBM models have good

reason for continual development.

36.2 A Hamiltonian Framework on ICF Implosions Rocket Equation Based on Rayleigh–Taylor Instabilities

36.2.1 Background

Inertial Confinement Fusion (ICF) Implosions: One of the primary methods of heating fusion environments is through lasers. In this approach, high-energy lasers heat spherical fuel capsules, causing them to implode. This implosion leads to an increase in pressure and heat on the fuel, creating the necessary environments for fusion.

Hamiltonian: A Hamiltonian is a function in classical mechanics that describes the total amount of energy in the system. From here, equations of motion are derived.

Rayleigh–Taylor Instabilities: Rayleigh–Taylor instabilities are specific types of instabilities caused by forceful interactions between higher and lower density fluids. In the context of ICF Implosions, when the higher density outer shell interacts with the lower-density inside, it creates perturbations through finger-like structures and bubbles that interfere with the efficiency of the implosion.

Mass Ablation (Rocket Effect): During the process of ICF Implosion, the interaction between the outer shell and the inner fuel causes mass to be sprayed off, giving a rocket-like effect.

36.2.2 Problem Statement

Achieving commercial fusion energy through inertial-confinement-fusion (ICF) remains one of the most significant scientific and engineering challenges of our day. One of the key obstacles is optimizing the interactions between high-energy lasers and fuel capsules, which are prone to Rayleigh–Taylor instabilities during implosions. These instabilities can lead to inefficient energy confinement and hinder the overall success of the fusion process. Traditional numerical simulations are computationally expensive and often lack the accuracy needed to address this issue. Therefore, there is a need for a robust analytical framework to model these instabilities and provide new insights for optimizing the interactions. This research aims to develop a Hamiltonian framework for ICF implosions, specifically focusing on Rayleigh–Taylor instability and mass ablation. By leveraging this theoretical approach, we seek to provide deeper insights and practical solutions for optimizing laser-capsule interactions in fusion experiments.

36.2.3 Assumptions

Thin-Shell Approximation: The thickness of the shell is assumed to be insignificant compared to the size of the imploding object. This approximation simplifies the creation of the model and the subsequent numerical analysis. It is valid given that the perturbations caused by the instabilities are much larger than the actual thickness of the shell.

Deceleration: This model is specifically for an imploding spherical shell that decelerates as it converges onto the compressed fluid within its interior.

Acceleration: This model does not consider the acceleration phase of an ICF capsule implosion at the beginning of the capsule-laser interaction.

36.2.4 Framework

In order to analyze the dynamics of the imploding shell, it must first be parameterized through two Lagrangian coordinates ϑ and ϕ . Here, ϑ corresponds to the polar angle θ , and ϕ corresponds to the azimuthal angle. The position vector is $\mathbf{X} = \mathbf{X}(t, \phi, \vartheta)$. Thus, the parameterization can be expressed as:

$$\mathbf{X} = \mathbf{X}(t, \phi, \vartheta)$$

The derivative of the surface is given by:

$$\frac{d\mathbf{X}}{dt}$$

Reference: D. E. Ruiz; Degradation of performance in ICF implosions due to Rayleigh–Taylor instabilities: A Hamiltonian perspective. Phys. Plasmas 1 December 2024; 31 (12): 122701.

36.2.5 Shell Kinematics

The force differential can be described by:

$$dF = p(t) \cdot d\mathbf{A} \times \mathbf{n}$$

where $p(t)$ is the function of pressure with respect to time, and the cross product involves the vectors following the surface of the shell. This leads to the derivation of several important quantities, such as the velocity vector field, position vector, and centrifugal forces, which are relevant to the shape and dynamics of the shell.

An example of the centrifugal force is:

$$F_{\text{centrifugal}} = m \cdot \omega^2 \cdot r$$

36.2.6 Shell Areal Density

The change in density of the shell is critical to its dynamics. Given that total mass is not conserved in this equation, deriving it becomes more complex.

$$\rho_{\text{areal}} = \frac{M(t)}{A}$$

where $M(t)$ is the mass at time t , and A is the area of the shell's surface.

36.2.7 Compressed-Fuel Pressure

The pressure that decelerates the shell can be described in several ways depending on the known information and boundary conditions. One example is:

$$P = P_0 \left(\frac{V_0}{V} \right)^\gamma$$

where P_0 is the initial pressure, V_0 is the initial volume, γ is the adiabatic index, and V is the current volume.

36.2.8 Variational Principles

To mathematically solve the Hamiltonian framework, given that the equation is asymptotic, variation principles are applied.

36.2.8.1 Phase-Space Lagrangian

This formulation describes the dynamics of a system by combining its configuration space (position variables) and momentum space into a single framework, called phase space.

Euler-Lagrange Equations

Through the principle of least action, a set of differential equations are derived to provide an analytical solution to complex nonlinear functions. For example, for radial velocity:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0$$

where L is the Lagrangian of the system.

36.2.8.2 Conservation Laws

Generally, conservation laws are employed to show that certain parts of the Euler-Lagrange or phase-space Lagrangian are conserved. However, due to mass ablation, parts of the fuel and capsule are lost, which complicates these conservation laws.

36.2.9 Conclusion

Summary: A Hamiltonian framework was developed to include both Rayleigh–Taylor Instabilities and mass ablation in an inertial confinement fusion implosion, leading to a set of variational principles for future work on optimizing ICF implosions in practical applications.

Significance: Analytical analysis can provide future insights into optimizing ICF implosions in fusion reactors. Additionally, nonlinear numerical models can be developed based on these principles, leading to less computationally expensive models.

Limitations:

1. This model does not account for more complex laser-plasma interaction phenomena (e.g., bremsstrahlung x-ray losses, alpha heating).

2. This model does not consider the initial acceleration phase of the interaction.
3. This model uses the thin-shell approximation, which may not be accurate when perturbations caused by instabilities are smaller than the shell thickness.

Future Works:

1. Quasilinear Models To study additional insights from this model, a quasilinear model must be developed for analytical study.
2. Acceleration Phase Incorporating the initial acceleration phase into the model will increase its accuracy.
3. Nonlinear Growth Calculations Developing nonlinear numerical simulations will be useful in generating further insights and advancing current models.

36.3 Basic MHD

36.3.1 MHD Module

The first section of it is initializing the values of things like magnetic field, pressure, and velocity. Future work will be done to have it so such values can be edited in a config file.

Next is computing the current density. The analytical equation can be written as

$$J = \frac{1}{\mu} \nabla \times B$$

Which in finite difference is:

$$J_z(i, j) = \frac{1}{\mu_0} \left[\frac{B_y(i+1, j) - B_y(i-1, j)}{2\Delta x} - \frac{B_x(i+1, j) - B_x(i-1, j)}{2\Delta y} \right]$$

Next we have

Update: I have advanced the MHD simulations time steps, magnetic field calculations, current, pressure calculations, and density calculations. Beyond that I have created animations based on magnetic field and made graphical outputs for heat, velocity, temperature, and magnetic field.

I also created another one specifically made for Tokamak geometry.

36.3.2 Hamiltonian MHD

My next project was creating a Hamiltonian based MHD simulation also in FORTRAN.

I explicitly created simulation of plasma thrusters with it.

36.4 A Metriplectic Formulation of Reduced Magnetohydrodynamics with Resistivity

36.4.1 Introduction

Here I will use the Strauss formulation of RMHD. Then I will incorporate resistivity. This resistivity typically breaks Hamiltonian structure, but I will use metriplectic to preserve the geometric identities through adding a separate antisymmetric bracket.

36.4.2 Defining the Hamiltonian

For Strauss MHD, a reduced MHD model that works for strongly magnetized Plasmas, we will formulate the Hamiltonian system.

$$\mathcal{H}[\omega, \psi] = \frac{1}{2} \int (\phi\omega + \psi j) dx dy$$

Here the $\phi\omega$ represents the kinetic energy with ψj represents the magnetic energy.

Now onto the Noncannonical Poisson Structure for this ideal Hamiltonian.

$$\{F, G\} = \int \omega \left[\frac{\delta F}{\delta \omega}, \frac{\delta G}{\delta \omega} \right] dx dy + \int \psi \left(\left[\frac{\delta F}{\delta \omega}, \frac{\delta G}{\delta \psi} \right] - \left[\frac{\delta G}{\delta \omega}, \frac{\delta F}{\delta \psi} \right] \right) dx dy$$

36.4.3 Entropy and Dissipation

Now to add dissipation. While generally adding this will break Hamiltonian structure, we will use a couple of tools.

Though first we must define our function we are using, or more accurately functional. This is our functional for entropy:

$$S[\psi] = \frac{1}{2} \int \psi^2 dx dy$$

36.4.4 Metric Bracket and Metriplectic Structure

To incorporate we define our symmetric bracket. We make sure to construct it so that it only affects the magnetic flux variable ψ , since resistivity only affects the magnetic field lines and not

vorticity directly.

$$(F, G) = \int \frac{\delta F}{\delta \psi} \eta \nabla^2 \frac{\delta G}{\delta \psi} dx dy$$

This has

- Symmetry $((F, G)) = ((G, F))$
- Positive semi-definiteness: $((S, S)) \leq 0$
- Energy Conservation: $((\mathcal{H}, F)) = 0$ for any F

36.4.5 Combined Dynamics

36.5 Death Star

Not to long ago(from when I wrote this section) I tried to simulate the death star using FLASH-X(which is really a testament to who I am as a person, getting my hands on government codes and the fist thing I do is play around with pop-culture based planetary destruction) anyways. It failed. The laser dynamics were based upon old code that was outdated and I couldn't figure it out. Though I did learn a lot about Flash-X as a system, which is good.

So for it... what am I doing, this doesn't make sense if I am trying to teach myself through practical action of writing but I already did the most practical action of all, doing it. This is kinda useless.

I later went back and ended up actually making the simulation.

36.6 Interstellar travel

36.6.1 Speeds and forces

Before we can get to all of the more complex stuff about design and the effects of extreme environments of specific interstellar space travel we must first look at how forces effect speeds at relativistic speeds. Here is the math stuff in simplified form: We all know that force is the derivative of the momentum over time:

$$F = \frac{dp}{dt}$$

Where momentum can be expand for relativistic momentum:

$$p = \gamma mv$$

You can then expand force to be (this is a constant mass, we will look at that in a second)

$$F = m \frac{d(\gamma v)}{dt}$$

One thing to keep in mind for future maths is that in simplifying $\dot{\gamma}$ is equal to $\frac{\gamma v}{c^2} \frac{dv}{dt}$ With the force you can calculate acceleration with

$$a = \frac{F}{m\gamma} \frac{1}{1 + \frac{\gamma^2 v^2}{c^2}}$$

Due to the fact most rockets use propulsion which lowers the mass, we must include mass variations:

$$F = \frac{\gamma mv}{dt}$$

Where you must use the product rule to expand

$$F = \dot{\gamma}mv + \gamma\dot{m}v + \gamma m\dot{v}$$

Which can be further expanded to

$$F = mv \frac{\gamma v}{c^2} \frac{dv}{dt} + \gamma v \frac{dm}{dt} + \gamma m \frac{dv}{dt}$$

With m being the infinitesimal change in mass over time.

To calculate acceleration you must find $\frac{dv}{dt}$. First subtract the middle equation without acceleration within it.

$$F - \gamma v \frac{dm}{dt} = mv \frac{\gamma v}{c^2} \frac{dv}{dt} + \gamma m \frac{dv}{dt}$$

Then factor for acceleration

$$F - \gamma v \frac{dm}{dt} = \frac{dv}{dt} (mv \frac{\gamma v}{c^2} + \gamma m)$$

Then divide the parentheses:

$$(F - \gamma v \frac{dm}{dt}) \frac{1}{mv \frac{\gamma v}{c^2} + \gamma m} = a$$

Now, this is all well and good, now let us find the position. To take position from acceleration you must take a double integral, but this integral can only be solved numerically(maybe it could be approximated)

Another possible way to solve this is through Hamiltonian-s/Lagrangians (I will choose Hamiltonians)

Let us first start with the Hamiltonian expressed for relativistic speeds in covariant form

First the Lagrangian must be calculated: The four-momentum is

$$P^\mu = m\gamma(c, v)$$

Energy:

$$E = \gamma mc^2$$

Therefore the Lagrandian is:

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}$$

Now to find generalized momentum:

$$p = \frac{\partial L}{\partial v}$$

$$p = \gamma m v$$

Now the Hamiltonian

$$H = p \cdot v - L$$

Therefore a relativistic Hamiltonain can be represented as

$$H = \sqrt{(pc)^2 + (mc^2)^2}$$

Now to include acceleration

$$H(r, p) = \sqrt{(pc)^2 + (mc^2)^2} + V(r)$$

Now to make mass variable for the rockets:

$$H(r, p, t) = \sqrt{(p(t)c)^2 + (m(t)c^2)^2} + H_{thrust}$$

Now to incorporate with equations of motion:

$$\frac{dr}{dt} = \frac{pc^2}{\sqrt{(p(t)c)^2 + (m(t)c^2)^2}}$$

Now, this works for the time for most people, now let us look at proper time

$$H(r, p, \tau) = \sqrt{(p(\tau)c)^2 + (m(\tau)c^2)^2} + H_{thrust}$$

For specific cases, these equations will be evolved to include them, but this is the general theoretical framework.

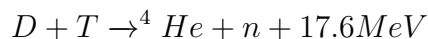
One thing I will add, is the force will likely be variable, a couple of days of 2-3 g, most of the time in g then reversing after the half-way point.

36.6.2 Magnetic Fusion Plasma Drive

36.6.2.1 Theoretical Basis

Magnetic Fusion Plasma Drive is a type of propulsion system specifically for interstellar travel.

The basis of this concept is that there is a fusion reactor undergoing Thermal Nuclear fusion:



The spent fuel then being shot out the nozzle as propulsion.

36.6.3 *Optimization

36.6.4 *Special Circumstances

36.6.4.1 *Strong Magnetic Fields

36.6.4.2 *Gravitational Fields

36.7 Kerr Black hole stuff

36.7.1 Energy (finish)

Kerr Black holes, or spinning black holes are a great source of energy. They have extreme amounts of radial kinetic energy that we are actually capable of 'bleeding' off through several methods.

First, we must ask why we are able to do this. Well, because a rotating object with a strong gravitational field 'drags' spacetime, that spacetime can affect the direction, speed, and even waves in a way that can amplify them.

First, Penrose(really like that dude) found that if you break apart an object rotating a Kerr black hole, the object that escapes gains energy. This process is challenging given that the object that escapes must have a velocity of $v > \frac{c}{2}$. There is a whole lot of math involved in proving this, but I will skip ahead.

The next that is found is that electromagnetic waves can be amplified. That if they are reflected around the black hole, they will gain energy and be amplified(this process is different from the Doppler shift and does not change the frequency.

After a lot, and I mean a lot of very complected mathematics you can find the amplification factor as

$$Z = 8r_+^2 T_{H(2l+1)}(r_+ - r_-)^{2l} \left[\frac{\Gamma(1+l-s)\Gamma(1+l+s)}{2l+1)!!\Gamma(l+1)\Gamma(2l+1)} \right]^2 \times,$$

$$\sinh\left(\frac{m\Omega_H}{r_+T_H}\right)\Gamma\left(l - \frac{im\Omega_H}{\pi r_+T_H} + 1\right)\Gamma\left(l + \frac{im\Omega_H}{\pi r_+T_H} + 1\right)$$

For more information visit [here](#)

Now the point of this is to take strange theoretical and apply it, so I will apply it for actual application. For this I will assume the average mass of $10 M_\odot$ or ten solar masses. A spin of 0.98 (if 1 refers to an extremal Kerr black hole)

First we must find out what a spin 0.98 really means for a $10M_\odot$ black hole. We can find this with a simple equation with a being the spin parameter, r being the radius, and M being the mass

$$\Omega = \frac{ac^3}{2GM r_h}$$

Though, for a Kerr black hole the is slightly different to the Schwarzschild radius.

$$r_H = \frac{GM}{c^2} \left(1 + \sqrt{1 - \frac{ac}{GM}}\right)$$

You know, I wanted to optimize the black hole engine, but I don't want to do that anymore, so I choice to move on.

36.7.2 *Multiversal Travel**36.7.3 *Other effects****36.8 *Extreme Theoretical Work****36.9 Astroid Game****36.9.1 Introduction**

I have recently created a Python game based on the Atari game Asteroid.

36.9.2 Classical

Just like the regular game, nothing special.

36.9.3 Newtonian Gravity

This adds that the asteroids and UFO's mass, so they attract each other and add more challenges.

36.9.4 Dark Matter

Add the challenge of invisible mass attractors, though when you are close by they light up.

36.9.5 Relativistic

This adds time dilatation(time skips or slow downs), length contractions(changing shapes of objects), black holes, and even doppler effect.

Chapter 37

Omega-X

37.1 Pre-planing(part 1)

37.1.1 Ideas & Overarching Plan

Start with replicating 1D thermal fluid model, then start with more complex models, then experiment with other ways to do it, then dive deeper into the mathematics and hopefully come up with something(maybe metriplectic integrator), then create a code base, compare these systems to other traditional models, and create tokamak simulation(maybe, not sure if possible yet) that is it so far. Hopefully these ideas will evolve.

Basically, the whole idea is to create a Flash-X like project with Metriplectic formulism underlying the the dynamics rather than lagrangian or Newtonian. Obviously much simpler though.

37.1.2 Metriplectic 4-bracket algorithm for constructing thermodynamically consistent dynamical systems

37.1.2.1 Math

1: Identify dynamical variables 2: Propose energy and entropy functionals 3: Find Poisson bracket F, G for which entropy S is a Casimir invariant, $F, S = 0 \forall F$ 4: Construct metriplectic 4-bracket $(F, K; G, N)$ via Kulkarni-Nomizu product by a new method that separates local thermodynamics from phenomenological quantities, giving the EoMs as Poisson bracket + 4-bracket

To find the Σ and M a new method is derived.

$$\partial_t \xi^\alpha = \{\xi^\alpha, H\} + \nabla \cdot J^\alpha$$

$$\alpha = 1, 2 \dots N - 1$$

$$\partial_t \xi^N = \{\xi^N, H\} + \nabla \cdot J^N + Z_\alpha \cdot \tilde{L}^{\alpha\beta} \cdot Z_\beta$$

where $\xi_N = \sigma$, the entropy density. Above splits Hamiltonian and conservative. Which leads to:

$$M(dF, dG) = F_{\xi_N} G_{\xi_N}$$

$$\Sigma(dF, dG) = \nabla F_{\xi_\alpha} \frac{L^{\alpha\beta}}{H_{\xi_N}} \nabla (G_{\xi_\beta})$$

37.1.3 A thermodynamically consistent discretizations of 1D thermal-fluid models using their metriplectic 4-bracket structure

37.1.3.1 Math

Edit: so I don't keep forgetting, ρ is mass density, m is momentum density, and σ is entropy density. Then for the thermodynamic quantities: η is specific entropy, u is velocity, and T is temperature. Let $V_h = v_h \subset H^1(\Omega)$ be the degree- p continuous Galerkin finite element space defined over a uniform grid, τ_h , on Ω : i.e.

$$V_h = v_h \in H^1(\Omega) : v_h|_K \in P_p(K), \forall K \in T_h$$

The discretizations is accomplished using the method of lines by positing that all dynamical fields have spatial dependence modeled in this Galerkin subspace. However, rather than discretizing the equations of motion themselves, we discretize the weak forms implied by the metriplectic formulation.

Let $(\rho_h, m_h, \sigma_h) \in V_h \times V_h \times V_h$. So that the discretized Hamiltonian and entropy can be given as:

$$H^h[\rho_h, m_h, \sigma_h] = \int_{\Omega} \left[\frac{1}{2} \frac{m_h^2}{\rho_h} + \rho_h U(\rho_h, \frac{\sigma_h}{\rho_h}) \right] dx$$

$$S^h[\sigma_h] = \int_{\Omega} \sigma_h dx$$

Then the Metriplectic 4 Bracket can be expressed through:

$$(F^h, K^h, G^h, N^h)_h = -\frac{1}{\text{Re}} \int_{\Omega} \frac{1}{T_h} \left[(K^h \partial_x F_{m_h}^h - F^h \partial_x K_{m_h}^h) \right],$$

$$(N^h \partial_x G_{m_h}^h - G^h \partial_x N_{m_h}^h) +,$$

$$\frac{1}{Pr\gamma - 1} \frac{1}{T_h} (K^h \partial_x F^h - F^h \partial_x K^h) [(N^h \partial_x G^h - G^h \partial_x N^h)] dx$$

Then the poission bracket is thus:

$$\{F^h, H^h\}_h(u_h) = -(m_h \partial_x u_h, \phi_m)_{L^2} + (m_h u_h, \partial_x, \phi_m)_{L^2} - (\rho_h \partial_x \eta_h, \phi_m)_{L^2} + (\rho_h u_h, \partial_x, \phi_\rho)_{L^2} - (\sigma_h \partial_x T_h, \phi_m)_{L^2} + (\sigma_h u_h, \partial_x, \phi_\sigma)_{L^2}$$

Now, one thing I will say is that this is not technically a Poisson bracket, because it fails to satisfy the Jacobi identity. This is because it is impossible(or at least nobody has figured out a way to discretize fluid poission brackets.

To then find equations of motion, adding them together $F^h = \{F^h, H^h\}_h + (F^h, H^h; S^h, H^h)_h$ where F^h is the observable.

In the case of momentum we must let $\phi_\rho = \phi_\sigma = 0$ so that only ϕ_ρ the momentum test function in the finite element(FE) test space V_h .

$$(\phi_m, \partial_t m_h) + (m_h \partial_x u_h, \phi_m)_{L^2} - (m_h u_h, \partial_x, \phi_m)_{L^2} + (\rho_h \partial_x \eta_h, \phi_m)_{L^2} + (\sigma_h \partial_x T_h \phi_m)_{L^2}$$

Next, I will have $\phi_m = \phi_\sigma = 0$ to get continuity equations:

$$(\phi_\rho, \partial_g \rho_h)_{L^2} - (\rho_h u_h, \partial_x \phi_\rho)_{L^2}$$

Finally the entropy equation is derived in a similar fashion:

$$(\phi_\sigma, \partial_g \sigma_h)_{L^2} - (\sigma_h u_h, \partial_x \phi_\sigma)_{L^2} + \frac{1}{Re} [(\frac{(\partial_x u_h)^2}{T_h}, \phi_\sigma)_{L^2} - \frac{1}{Pr} \frac{\gamma}{\gamma - 1} [(\frac{\partial_x T_h}{T_h}, \partial_x \phi_\sigma)_{L^2} - ($$

To prove energy conservation

$$\left(\frac{u_h^{n+1} - u_h^n}{\Delta t}, \delta h_h\right) = \frac{H(u_h^{n+1}) - H(u_h^n)}{\Delta t} = 0$$

Positive entropy production can be found through:

$$\frac{s_h^{n+1} - s_h^n}{\Delta t} = \frac{1}{Re} \left[\left(\frac{(\partial_x u_h^n)^2}{T_h^n}, 1 \right)_{L^2} + \frac{1}{Pr} \frac{\gamma}{\gamma - 1} \left(\frac{(\partial_x T_h^n)^2}{T_h^n}, 1 \right)_{L^2} \right] \geq 0$$

Let's figure out all we need to find for each equation, starting with momentum. Start with $M_{ii} = \phi_m$, this should be equal to dx , but I will confirm. Also need the time derivative of the momentum variational, u_h variational, ρ_h variational, T_h , η_h , and $\frac{1}{Re}$, reynolds number, decide later

$$M_{ii} = dx$$

$$u_h = (\rho_h, m_h, \sigma_h)$$

ρ_h , T_h , and η_h are the respective fields, initialized simply and evolved through the equation. Obviously ρ_h, m_h , and σ_h evolve through their metriplectic equations of "motion." T_h , η_h , are evolved through these equations:

$$\left(\eta_h + \frac{m_h^2}{2\rho_h^2} - U\left(\rho_h, \frac{\sigma_h}{\rho_h}\right) - \rho_h \partial_1 U\left(\rho_h, \frac{\sigma_h}{\rho_h}\right) + \frac{\rho_h}{\sigma_h} \partial_2 U\left(\rho_h, \frac{\sigma_h}{\rho_h}\right), \phi_\eta \right)_{L^2} = 0$$

$$\left(u_h - \frac{m_h}{\rho_h}, \phi_u \right) = 0$$

$$\left(T_h - \partial_2 U\left(\rho_h, \frac{\sigma_h}{\rho_h}\right), \phi_T \right)_{L^2} = 0$$

Nothing new is added in continuity

Entropy adds: Pr and γ

37.1.3.2 Coding

The first thing is to use a Galerkin Method for projecting PDEs into a finite-dimensional function space. (can base off of Firedrake, FEniCS, Galerkin, or another) M_{ii} in mesh.f90

The second thing to do is to initialize the parameters and states(use mesh to calculate field) in states.f90 (Pr , Re , γ for constants; initialize $(\rho_h, \sigma_h, m_h, \eta_h, T_h, u_h)$)

Then functionals: calculates pointwise functional derivatives of the Total Hamiltonian and Entropy. in functionals.f90 (Update $\rho_h, \sigma_h, m_h, u_h$)

EOS.f90 is used to update the thermodynamic points(η_h, T_h)

Time step the program, by using Gauss-Legendre implicit Runge-Kutta methods in time_integration.f90

Run the Program(using the driver and makefile)

Input the outputs in io.f90

Some notes: $(f, g)_{L^2} = \int_{\Omega} f(x) \cdot g(x)$

An example would be, for a point in the momentum equation $(\phi_m, \partial_t m_h)_{L^2}$, or the mass matrix times the derivative of momentum functional as a function of time. Thus it makes it so $\phi_m = M_{ii} = dx$ thus, $(\phi_m, \partial_t m_h)_{L^2} = dx \cdot \frac{m_h}{\partial t}$

37.1.3.3 Future Work

The paper presents several ways to improve, or at least look into improving through future work.

Mainly, they mention that many different structure preserving methods exist and may be more suitable.

I also need to figure out how to make this codebase work for more examples

37.1.4 Metriplectic Framework for Dissipative Magneto-Hydrodynamics

37.1.4.1 My Introduction

In my time trying to figure out how to numerically solve metriplectic equations, someone beat me to dissipative magneto-hydrodynamics. Oh well, I can build off of.

Beyond that, the abstract introduces the paper better than I could, "The metriplectic framework, which permits to formulate an algebraic structure for dissipative systems, is applied to visco-resistive Magneto-Hydrodynamics (MHD), adapting what had already been done for non-ideal Hydrodynamics (HD). The result is obtained by extending the HD sym metric bracket and free energy to include magnetic field dynamics and resistive dissipation. The correct equations of motion are obtained once one of the Casimirs of the Poisson bracket for ideal MHD is identified with the total thermodynamical entropy of the plasma. The metriplectic framework of MHD is shown to be invariant under the Galileo Group. The metriplectic structure also permits to obtain the asymptotic equilibria toward which the dynamics of the system evolves. This scheme is finally adapted to the two-dimensional incompressible resistive MHD, that is of major use in many applications."

37.1.4.2 Notes on introduction

They have an interesting word choice, "with its noble descendants of path integral representations," or Algebrization of dynamical systems appears to be the final destination of that virtuous route.

MSTDof means "microscopic, statistically treated, degrees of freedom"

In this introduction they derive the metriplectic existence

quite simply. Starting with the poisson bracket

$$\dot{f} = \{f, g\}$$

In this case

$$0 = \{S, H\}$$

Thus it is the Casimir functionals of the Poisson bracket

$$\{C, f\} = 0 \forall f$$

From here we can define the Hamiltonian free energy.

$$F = H + \lambda C$$

Where λ is an arbitrary constant left. In MSTDOF, it is the minus of the temperature.

To then expand the system, $\langle\langle f, g \rangle\rangle$ via F . So that $\langle\langle f, g \rangle\rangle = \{f, h\} + (f, g)$ and (f, g) is symmetric, bilinear and semi-definite.

Evolution is then given by

$$\dot{f} = \langle\langle f, F \rangle\rangle$$

(the symmetric bracket (f, g) will be defined so to cancel out the presence of the coefficient λ , removing it from the equations of motion).

(f, g) is defined so that H is conserved.

$$(H, f) = 0 \forall f$$

Thus metriplectic evolution reads:

$$\dot{f} = \{f, H\} + \lambda(f, C)$$

The Casimir to mimic entropy undergoes an evolution described

by:

$$\dot{C} = \lambda(C, C)$$

Now, I like this introduction. Clearly showing the basic understanding on Metriplectic dynamics.

37.1.5 METRIPLECTIC FORMULATION OF VISCO-RESISTIVE MHD

The system observed is fully ionized plasma. Where dissipation takes place due to the finite viscosity and resistivity of the fluid. Also, heat conductivity is finite. Here is the SO(3)-covariant form.(expressing physical equations in a way that is invariant under rotations in three-dimensional space)

$$\begin{aligned}\partial_t \rho + \partial_i(\rho u^i) &= 0 \\ \partial_t B^i + \varepsilon^{ijk} \partial_j E_k &= 0, \quad \partial_i B^i = 0 \\ \partial_t(\rho u^i) + \partial_j(\rho u^i u^j + p \delta^{ij}) &= \varepsilon^{ijk} J_j B_k + \partial_j(\eta \partial_j u^i) \\ \partial_t s + \partial_i(s u^i) &= \frac{\eta}{T} \partial_i u^j \partial_j u^i + \frac{\kappa}{T^2} \partial_i T \partial_i T + \frac{\mu}{T} J^i J^i\end{aligned}$$

There is quite a bit to this equation, but I will move on.

One thing I will add is a new notation. $\dot{H} \stackrel{\partial}{=} 0$, it simply means that means that $\dot{H}$ and 0 only differ by a boundary term.

They then describe the poisson bracket for ideal MHD(when dissipation does not happen, thus traditional poisson bracket).

This equation has multiple Casimir observables. The paper mentions:

$$C[\rho, s] = \int_{\mathbb{D}} \rho \varphi(s) d^3x$$

Which within is both the total mass and entropy.

$$M[\rho] = \int_{\mathbb{D}} \rho d^3x$$

$$S[\rho, s] = \int_{\mathbb{D}} \rho s d^3x$$

Thus, mass and entropy are conserved. There are some other, but obviously for this purpose, use the entropy.

Taking the dissipative equations from the to combine $D = (\vec{D}^\rho, \vec{D}^v, \vec{D}^B, \vec{D}^s)$ something called an 8-uple. So we must want to define our metriplectic bracket as $\psi = (\psi, D)$ where ψ is the dynamic variables.

Uhm.. I just realized this wasn't the paper I thought it was. Good read, but I don't think I will continue these notes.

37.1.6 A least Action principle for visco-resistive Hall Magnetohydrodynamic with metriplectic reformulation

37.1.6.1 Abstract

The abstract persented says,

"We present a new variational formulation for Viscous and resistive Hall Magnetohydrodynamic. We first find a variational principle for ideal HMHD by applying the physical assumptions leading to HMHD at the lagrangian level, and then we add the viscous and resistive terms by the means of constrained variations. We also provide a metriplectic reformulation of our formulation, based on two canonical Lie-Poisson brackets for the ideal part and metric 4-brackets for the dissipative part."

They then go into all of these, I will begin with the Metriplectic for notes alone.

37.1.6.2 Metriplectic bracket, Hamiltonian and Entropy for Viscoresistive Hall MHD

They use the Inclusive curvaturelike tensor found in Morrison's recent work. Where the dissipation is included using a 4-bracket that have the same symmetries as a curvature tensor.

They take the assumption that the viscous tensor can be found to be

$$\sigma = \Lambda \nabla u_i$$

Where Λ is symmetrize and positive.

37.1.7 Variational discretizations of viscous and resistive magnetohydrodynamics using structure-preserving finite elements

37.1.7.1 Abstract

We propose a novel structure preserving discretization for viscous and resistive magnetohydrodynamics. We follow the recent line of work on discrete least action principle for fluid and plasma equation, incorporating the recent advances to model dissipative phenomena through a generalized Lagrange-d'Alembert constrained variational principle. We prove that our semi discrete scheme is equivalent to a metriplectic system and use this property to propose a Poisson splitting time integration. The resulting approximation preserves mass, energy and the divergence constraint of the magnetic field. We then show some numerical results obtained with our approach. We first test our scheme on simple academic test to compare the results with established methodologies, and then focus specifically on the simulation of plasma instabilities, with some tests on non Cartesian geometries to validate our discretization in

the scope of tokamak instabilities.

This paper

37.2 part 2

37.2.1 Coding(1D fluid)

37.2.1.0.1 Introduction My first step to creating my code-base is to create it for the simplest form. This is specifically made for the purpose of figuring things out, documenting it, and such things. So, I will be writing down each file.

37.2.1.0.2 mesh_1d Obviously, this mesh program creates the mesh for the program. The size N , the nodes x_nodes and the mass matrix M_{ii} . The rest are self explanatory but I will go into what the mass matrix is. It represents the discrete inner product of basis functions.

$$M_{ii} = (\phi_i, \phi_j)_{L^2} = \int_{\Omega} \phi_i(x) \phi_j(x) dx$$

So, M_{ii} becomes a metric tensor for program. It turns continuous inner products into discrete sums over coefficients.

It can be approximated through the "lumped mass matrix" where $M_{ii} = dx$ Though, I use the trapezoidal mass matrix:

$$M_{ii}(1) = 0.5d0 * dx$$

$$M_{ii}(N) = 0.5d0 * dx$$

$$M_{ii}(2 : N - 1) = dx$$

For more information visit, <https://arxiv.org/pdf/2008.03883>

Listing 37.1: mesh_1d.f90

```

1 module mesh_1d
2   implicit none
3   private
4   public :: initialize_mesh, x_nodes, dx, N, Mii
5
6   ! Parameters
7   integer, parameter :: N = 100           ! Number of
      spatial grid points (nodes)
8   real(8), parameter :: L = 1.0d0        ! Length of
      the domain
9
10  ! Mesh data
11  real(8), dimension(N) :: x_nodes        !
      Coordinates of each mesh node
12  real(8) :: dx                          ! Uniform
      cell spacing
13
14  ! Mass matrix (diagonal approximation here, for
      full FEM use full M)
15  real(8), dimension(N) :: Mii           ! Lumped
      mass matrix diagonal
16
17 contains
18
19  subroutine initialize_mesh()
20    integer :: i
21
22    dx = L / (N - 1)
23
24    ! Initialize node positions
25    do i = 1, N
26      x_nodes(i) = (i - 1) * dx
27    end do
28
29    ! Lumped mass matrix: each diagonal entry is dx
30    Mii = dx

```

```

31
32      ! Uncomment below to use trapezoidal mass matrix
33      Mii(1) = 0.5d0 * dx
34      Mii(N) = 0.5d0 * dx
35      Mii(2:N-1) = dx
36
37      end subroutine initialize_mesh
38
39
40 end module mesh_1d

```

37.2.1.0.3 states_1d States then initializes the constants and fields. First the constants(parameters) used throughout the code. Re, Reynolds numbers. γ , compression of the fluid. Pr , Prandtl number, ratio of viscous to thermal diffusion.

Then, it initializes the discretion fields of entropy, momentum, mass, temperature, and specific entropy.

This should be modularized for the future to make this codebase iterative.

Now, what exactly to do. I think for this, have constants in a separate one. Then for the fields, I am not sure at this moment.

Listing 37.2: states_1d.f90

```

1 module states_1d
2   use mesh_1d
3   implicit none
4   private
5   public :: m_h, rho_h, sigma_h, eta_h, T_h
6   public :: initialize_states
7
8   ! Field arrays
9   real(8), allocatable :: m_h(:)           ! Momentum
      density

```

```

10  real(8), allocatable :: rho_h(:)      ! Mass
    density
11  real(8), allocatable :: sigma_h(:)    ! Entropy
    density
12  real(8), allocatable :: eta_h(:)      ! Specific
    entropy (diagnostic)
13  real(8), allocatable :: T_h(:)        ! Temperature
    (diagnostic)
14  real(8), parameter, public :: gamma = 1.4
15  real(8), parameter, public :: Pr = 0.71
16  real(8), parameter, public :: Re = 1000.0
17
18  contains
19
20  subroutine initialize_states()
21      implicit none
22      integer :: i
23
24      ! Allocate fields
25      allocate(m_h(N))
26      allocate(rho_h(N))
27      allocate(sigma_h(N))
28      allocate(eta_h(N))
29      allocate(T_h(N))
30
31      ! Set initial conditions (example: Gaussian
    density bump)
32      do i = 1, N
33          rho_h(i) = 1.0d0 + 0.2d0 * exp(-100.0d0 * (
    x_nodes(i) - 0.5d0)**2)
34          eta_h(i) = 1.0d0
35          sigma_h(i) = rho_h(i) * eta_h(i)
36          m_h(i) = 0.5d0
37          T_h(i) = (0.4d0) * eta_h(i) ! Ideal gas !
    Uses EOS
38      end do
39

```

```

40   end subroutine initialize_states
41
42 end module states_1d

```

37.2.1.0.4 projection_matrix_1d Here the creation of two subroutines to simplify future calculations. First, `initialize_projection_matrix` that simply creates matrix. Next, `apply_weak_derivative(f, d_proj)`. This does:

$$proj = \sum_{i=1}^n \left(\sum_{j=1}^n D_{i,j} \right) * f$$

Where f is the observable. For future work, I think all I need to do for this one is create multiple for different dimensions.

Listing 37.3: `projection_matrix_1d.f90`

```

1 module projection_matrix_1d
2   use mesh_1d
3   implicit none
4   real(8), allocatable :: D(:, :)
5   public :: initialize_projection_matrix,
6             apply_weak_derivative
7
8 contains
9
10  subroutine initialize_projection_matrix()
11    integer :: i
12    real(8) :: dphi_dx
13    if (allocated(D)) then
14      deallocate(D)
15    end if
16
17    allocate(D(N, N))
18    D = 0.0d0
19
20    dphi_dx = 1.0d0 / dx

```



```

21      ! Loop over elements and assemble D
22      do i = 2, N-1
23          ! Local contributions from basis functions
            over each element
24          D(i, i-1) = -dphi_dx / 2.0d0
25          D(i, i ) =  0.0d0
26          D(i, i+1) =  dphi_dx / 2.0d0
27      end do
28
29      ! Optional: Neumann (zero derivative) at
            boundaries
30      D(1,:) = 0.0d0
31      D(N,:) = 0.0d0
32  end subroutine initialize_projection_matrix
33
34      ! Applies D to f to compute (phi_i, partial_x f)
35  subroutine apply_weak_derivative(f, d_proj)
36      implicit none
37      real(8), intent(in)  :: f(N)
38      real(8), intent(out) :: d_proj(N)
39      integer :: i
40
41      d_proj = 0.0d0
42      do i = 1, N
43          d_proj(i) = sum(D(i,:) * f(:))
44      end do
45  end subroutine apply_weak_derivative
46
47  end module projection_matrix_1d

```

Listing 37.4: mass_1d.f90

37.2.1.0.5 mass_1d

```

1  module mass_1d
2      use mesh_1d
3      use projection_matrix_1d
4      implicit none

```

```

5 contains
6
7   subroutine compute_mass_flux(N, rho_h, u_h,
8     rho_rhs)
9     integer, intent(in) :: N
10    real(8), intent(in) :: rho_h(N), u_h(N)
11    real(8), intent(out) :: rho_rhs(N)
12    real(8) :: flux(N)
13    integer :: i
14
15    ! Compute flux
16    do i = 1, N
17        flux(i) = rho_h(i) * u_h(i)
18    end do
19
20    ! Apply weak derivative
21    call initialize_projection_matrix()
22    call apply_weak_derivative(flux, rho_rhs)
23
24    ! Divide by dx
25    do i = 1, N
26        rho_rhs(i) = -rho_rhs(i) / dx
27    end do
28
29    ! Boundary conditions
30    rho_rhs(1) = 0.0d0
31    rho_rhs(N) = 0.0d0
32
33    end subroutine compute_mass_flux
34 end module mass_1d

```

Listing 37.5: momentum_1d.f90

37.2.1.0.6 momentum_1d

```

1 module momentum_1d
2   use states_1d, only: Re

```

```
3   use mesh_1d
4   use projection_matrix_1d
5   implicit none
6   contains
7
8   subroutine compute_momentum_rhs(N, rho_h, m_h,
9       sigma_h, eta_h, T_h, rhs_m)
10      integer, intent(in) :: N
11      real(8), intent(in) :: rho_h(N), m_h(N), sigma_h
12          (N), eta_h(N), T_h(N)
13      real(8), intent(out) :: rhs_m(N)
14      real(8) :: u_h(N), du_proj(N), mu(N), dmu_proj(N)
15          , deta_proj(N), dT_proj(N)
16      integer :: i
17
18      ! Compute velocity
19      do i = 1, N
20          if (rho_h(i) > 1.0d-12) then
21              u_h(i) = m_h(i) / rho_h(i)
22          else
23              u_h(i) = 0.0d0
24          end if
25      end do
26
27      ! Apply weak derivatives
28      call initialize_projection_matrix()
29      call apply_weak_derivative(u_h, du_proj)
30      do i = 2, N-1
31          mu(i) = m_h(i) * u_h(i)
32      end do
33      call apply_weak_derivative(mu, dmu_proj)
34      call apply_weak_derivative(eta_h, deta_proj)
35      call apply_weak_derivative(T_h, dT_proj)
36
37      ! Compute RHS
38      do i = 2, N-1
39          rhs_m(i) = 0.0d0
```

```

37     rhs_m(i) = rhs_m(i) - m_h(i) * du_proj(i) * dx
38     rhs_m(i) = rhs_m(i) + dmu_proj(i) * dx
39     rhs_m(i) = rhs_m(i) - rho_h(i) * deta_proj(i)
      * dx
40     rhs_m(i) = rhs_m(i) - sigma_h(i) * dT_proj(i)
      * dx
41     rhs_m(i) = rhs_m(i) - (1.0d0 / Re) * du_proj(i)
      * dx
42     end do
43
44     ! Boundary conditions
45     rhs_m(1) = 0.0d0
46     rhs_m(N) = 0.0d0
47
48     end subroutine compute_momentum_rhs
49
50 end module momentum_1d

```

Listing 37.6: entropy_1d.f90

```

1 module entropy_1d
2     use states_1d
3     use mesh_1d
4     use projection_matrix_1d
5     implicit none
6 contains
7
8     subroutine compute_entropy_rhs(N, sigma_h, u_h,
      T_h, rhs_sigma)
9         implicit none
10        ! Dummy arguments
11        integer, intent(in) :: N
12        real(8), intent(in) :: sigma_h(N), u_h(N), T_h(N)
      )
13        real(8), intent(out) :: rhs_sigma(N)
14

```

```

15      ! Local variables
16      real(8) :: flux(N), dx_u(N), dx_T(N)
17      real(8) :: visc_prod(N), heat_prod(N), heat_grad
18              (N), heat_term(N)
19      integer :: i
20
21      ! Compute flux
22      do i = 1, N
23          flux(i) = sigma_h(i) * u_h(i)
24      end do
25
26      ! Apply weak derivatives
27      call initialize_projection_matrix()
28      call apply_weak_derivative(flux, rhs_sigma)
29      call apply_weak_derivative(u_h, dx_u)
30      call apply_weak_derivative(T_h, dx_T)
31
32      ! Viscous and thermal terms
33      do i = 1, N
34          visc_prod(i) = (dx_u(i)**2) / T_h(i)
35          heat_grad(i) = dx_T(i) / T_h(i)
36          heat_prod(i) = (dx_T(i)**2) / (T_h(i)**2)
37          heat_term(i) = (gamma / (gamma - 1.0d0)) * (
38              heat_prod(i) - heat_grad(i) / dx)
39          rhs_sigma(i) = rhs_sigma(i) - (1.0d0 / Re) *
40              visc_prod(i) * dx
41          rhs_sigma(i) = rhs_sigma(i) - (1.0d0 / (Re *
42              Pr)) * heat_term(i) * dx
43      end do
44
45      ! Divide by dx
46      do i = 1, N
47          rhs_sigma(i) = -rhs_sigma(i) / dx
48      end do
49
50      ! Boundary conditions
51      rhs_sigma(1) = 0.0d0

```

```

48     rhs_sigma(N) = 0.0d0
49
50     end subroutine compute_entropy_rhs
51
52 end module entropy_1d

```

Listing 37.7: velocity_1d.f90

37.2.1.0.8 velocity_1d

```

1 module velocity_1d
2     use states_1d
3     use projection_matrix_1d
4     implicit none
5 contains
6
7     subroutine compute_velocity(N, rho_h, m_h, u_h)
8         implicit none
9
10        ! Dummy arguments
11        integer, intent(in) :: N
12        real(8), intent(in) :: rho_h(N), m_h(N)
13        real(8), intent(out) :: u_h(N)
14
15        ! Local variables
16        integer :: i
17        real(8), parameter :: eps = 1.0d-12 ! Small
18        ! number to avoid division by zero
19
20        ! Compute velocity safely
21        do i = 1, N
22            if (rho_h(i) > eps) then
23                u_h(i) = m_h(i) / rho_h(i)
24            else
25                u_h(i) = 0.0d0
26            end if
27        end do

```

```

28     end subroutine compute_velocity
29
30 end module velocity_1d

```

37.2.1.0.9 Listing 37.8: EOS_1d_thermal.f90

```

1  module eos_1d
2      use states_1d
3      implicit none
4      private
5      public :: compute_temperature, compute_eta
6
7
8
9  contains
10
11      !
12      ! -----
13
14      ! Compute temperature T from specific entropy eta
15      ! T = (gamma - 1) * eta
16      !
17      ! -----
18
19      function compute_temperature(eta_h) result(T_h)
20          real(8), intent(in) :: eta_h
21          real(8) :: T_h
22
23          T_h = (gamma - 1.0d0) * eta_h
24      end function compute_temperature
25
26      !
27      ! -----
28
29      ! Compute specific entropy eta from temperature
30      ! eta = T / (gamma - 1)

```

```

25      !
26      -----
27
28      function compute_eta(T_h) result(eta_h)
29          real(8), intent(in) :: T_h
30          real(8) :: eta_h
31
32          eta_h = T_h / (gamma - 1.0d0)
33      end function compute_eta
34
35 end module eos_1d

```

37.2.1.0.10 Listing 37.9: time_integrator_1d.f90

```

1  module time_integrator_1d
2      use mesh_1d
3      use states_1d
4      use velocity_1d
5      use eos_1d
6      use mass_1d
7      use momentum_1d
8      use entropy_1d
9      implicit none
10     private
11     public :: advance_one_step
12
13 contains
14
15     subroutine advance_one_step(dt, u_h, rhs_m,
16                                rho_rhs, rhs_sigma)
17         real(8), intent(in) :: dt
18         real(8), intent(INOUT) :: u_h(N), rhs_m(N),
19             rho_rhs(N), rhs_sigma(N)
20         integer :: i
21
22         ! Step 1: Compute velocity

```



```

21  call compute_velocity(N, rho_h, m_h, u_h)
22
23  ! Step 2: calculate temperture
24  do i = 1, N-1
25      T_h(i) = compute_temperature(eta_h(i))
26  end do
27  ! step three: calculate Eta
28  do i = 1, N-1
29      eta_h(i) =compute_eta(T_h(i))
30  end do
31  ! Step 4: Compute Galerkin RHS terms
32  call compute_momentum_rhs(N, rho_h, m_h, sigma_h
33      , eta_h, T_h, rhs_m)
34  call compute_mass_flux(N, rho_h, u_h, rho_rhs)
35  call compute_entropy_rhs(N, sigma_h, u_h, T_h,
36      rhs_sigma)
37
38  ! Step 5: Advance in time (Euler / RK1)
39  do i = 1, N
40      m_h(i)      = m_h(i)      + dt * rhs_m(i)
41      rho_h(i)    = rho_h(i)    + dt * rho_rhs(i)
42      sigma_h(i)  = sigma_h(i)  + dt * rhs_sigma(i)
43  end do
44
45  end subroutine advance_one_step
46
47 end module time_integrator_1d

```

37.2.1.0.11 io_1d

Listing 37.10: io_1d.f90

```

1  module io_1d
2      use mesh_1d
3      use states_1d
4      use velocity_1d
5      use time_integrator_1d
6      implicit none

```

```

7   private
8   public :: write_state_to_csv
9
10  contains
11
12  subroutine write_state_to_csv(filename)
13      implicit none
14      character(len=*), intent(in) :: filename
15      integer :: i
16      logical :: wait
17      real(8) :: u_h(N)
18      integer :: ierr
19
20      ! Ensure the output directory exists
21      wait = .true. ! Command should run
22               synchronously
23      call execute_command_line("mkdir -p output",
24                               wait, ierr)
25      if (ierr /= 0) then
26          print *, "Error: Could not create output
27                  directory!"
28          stop
29      end if
30
31      ! Compute velocity  $u = m / \rho$ 
32      call compute_velocity(N, rho_h, m_h, u_h)
33
34      ! Open file for writing
35      open(unit=10, file=filename, status="replace",
36           action="write", form="formatted")
37
38      ! Header
39      write(10, '(A)') 'x,rho,m,sigma,u,T,eta'
40
41      ! Data
42      do i = 1, N

```

```

39      write(10, '(F12.6,1x,F12.6,1x,F12.6,1x,F12.6,1
      x,F12.6,1x,F12.6,1x,F12.6)') &
40      x_nodes(i), rho_h(i), m_h(i), sigma_h(i),
      u_h(i), T_h(i), eta_h(i)
41  end do
42
43      close(10)
44  end subroutine write_state_to_csv
45
46 end module io_1d

```

37.2.1.0.12 Listing 37.11: momentum_1d.f90 1D Thermal driver

```

1  program omega_x_driver_1d
2      use mesh_1d
3      use states_1d
4      use time_integrator_1d
5      use io_1d
6      implicit none
7      real(8) :: dt, t, t_end
8      integer :: step = 1
9      real(8), dimension(N) :: u_h, rhs_m, rho_rhs,
      rhs_sigma
10     character(len=100) :: fname
11
12     dt = 1.0d-3
13     t_end = 1.0d0
14     t = 0.0d0
15
16     call initialize_mesh()
17     call initialize_states()
18     u_h = 0.0d0
19     rhs_m = 0.0d0
20     rho_rhs = 0.0d0
21     rhs_sigma = 0.0d0  ! Explicitly initialize
      rhs_sigma here

```

```

22
23   do while (t < t_end)
24       call advance_one_step(dt, u_h, rhs_m, rho_rhs,
25                               rhs_sigma)
26       t = t + dt
27       print *, 't = ', t
28
29       if (mod(step, 100) == 0) then
30           write(fname, '(A,I4.4,A)') "output/state_",
31               step, ".csv"
32           call write_state_to_csv(fname)
33       end if
34
35       step = step + 1
36   end do
37 end program omega_x_driver_1d

```

37.2.1.1 Plans

I have now create it in its most basic form, here are my next steps.

- Break states up into constants and fields(multi-dimensional fields)

- Make operators folder for projection, weak derivative, boundary conditions

- Add checks(mainly energy conservation)

- Upgrade time integration,(figure out)

- Make different spacial discretization

- More boundary conditions

- Add more physics

Though, the main thing I need to work on is module registry + driver pattern.

37.3 part 3

37.3.1 Notes

37.3.1.1 Notes on module FORTRAN

I should have a main driver like:

Listing 37.12: Omegax.f90

```
1 program Omegax
2   use Driver_module
3   implicit none
4
5   call Driver_init()
6   call Driver_evolve()
7   call Driver_finalize()
8
9 end program Omegax
```

There should also be a config file, saying which modules should be used, and a python script capable of parsing this registry file. Config would also be where parameters would be stored.

37.3.1.2 Diminsionaleess programing

In the config file I can have a section to input the diminisions.

Then have all variable allocatable based upon the diminisions

then have an allocation based upon such diminisions(either preprocessor or generic shape handling)

Instead of hardcoding loops, create cases for different diminisions.

Generaliz derivatives and fluxes.

Also, while physics sections should be generalized, initializing between 3D, would be beneficial.

37.3.1.3 Generalize

Due to Metriplectics dyanmics complex mathematics, a fully generalized system, like FLASH-X, is impossible. This is because it is a dynamical theory based upon Hamiltonian Mechanics, rather than forces in Newtonian.

I am thinking the way to fix this is to create a symbolic python code to be able to generate the function I need.

Also, since these would be in py

37.3.2 Plan

I think the steps I am going to take are:

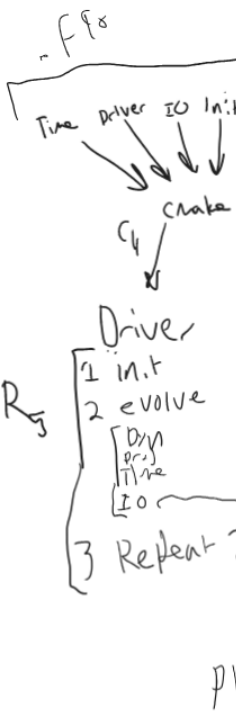
1: Develop a python code that can generate the math I need, I would also like for it to be able to read a LaTeX code for it.

2: I need to look into how to create a Config file for my problem at hand

3: Fix integration for this new system

4:

37.3.3 New Structure



Here is the basic idea of what it should look like and do:

37.4 Part 4

It has been quite some time since I had last worked on this project so I am in a way starting anew.

37.4.1 Summary On the Past

37.4.2 AI Attempt

Recently, I tried to use Claude AI to attempt to do something of a similar vein of this project. To see how capable it was, it quite surprised me. Here I will analysis its attempt; my initial view is that it misinterpreted what I wanted and relied on well developed past developments rather than the novel project I was attempting, but its ability to optimize and clearly write things out outpaces my ability, also using things like kernel development was and parallel computing was suprising.

First, to help with my analysis I used the AI tool to develop its own analysis, then I will do a separate one on it.

37.4.2.1 AI Analysis

37.4.2.2 Code

37.4.2.3 My Analysis

Chapter 38

Axiom

38.1 *Structure

38.2 LaTeX

38.2.1 Division of Structure

In this program I have used this structure to divide the book up:

```
\part{}  
\chapter{}  
\section{}  
\subsection{}  
\subsubsection{}  
\subsubsubsection{} % custom made  
\paragraph  
\subparagraph
```

I may have to learn new ways to divide in the future, but I will try to avoid that.

38.2.2 Commands

Here are some commands that I will have here for reference:

`\begin{verbatim}` writes things exactly, used mostly
for writing LaTeX commands in this method

`\label{}` → Set a reference point

`\ref{}` → Refer to numbered items

`\pageref{}` → Refer to page number

`\nameref{}` (requires hyperref) → Refer to name/title
of chapter/section

`\autoref{}` (requires hyperref) → Smart referencing
(automatically adds `\Chapter`", `\Section`")

`\begin{equation} ... \end{equation}` → Numbered
equations

`\[ ... \]` → Displayed, unnumbered equations

`$ ... $` → Inline math

`\frac{a}{b}`, `\sqrt{x}`, `\sum`, `\int` → Standard
math symbols

`\begin{align} ... \end{align}` (from amsmath)
→ Multi-line aligned equations

`\begin{cases} ... \end{cases}` → Piecewise functions

`\bra{}`, `\ket{}`, `\braket{}` →
Use physics package for quantum notation

`\tensor{}` → Use tensor package

for tensor indices

`\dv{}`, `\pdv{}` → Derivatives (from physics)

`\begin{figure}[h!] ... \end{figure}` →

Insert figures

`\includegraphics[width=0.8\textwidth]{file}`

→ Image inclusion (`graphicx`)

`\caption{}` / `\label{}` → Caption and reference

`\begin{table}[h!] ... \end{table}` → Tables

I also have `\begin{lstlisting}` to write both FORTRAN and make. I will likely add python and Julia

Chapter 39

Linux & Customizing my Computer

Now that I am dual booting my computer I am going to make plans for customizing it and making sure I have all of the things I want installed.i

39.1 Introductory Plans

I want the libraries, package management tools, set up emacs, Terminal Multiplexing, compilers, toolchain, git, setting up Windows/Linux workflow, and automation

39.2 Package Manegment

ripgreg, fd, bat, fzf

39.3 Terminal Emulator

I am thinking of choicing alacritty

39.4 Editor

While I plan to put all of my Emacs stuff in the GitHub to transer, what I am thinking is for this time to use NeoVim and then from there decide which to use.

39.5 AI summary

Rather than write the rest of it I had ChatGPT write a summary:

39.5.1 Core System Setup

39.5.1.1 Installation

```
sudo apt update && sudo apt upgrade -y

sudo apt install -y \
build-essential cmake make ninja-build \
git curl wget \
htop tmux openssh-client openssh-server
```

39.5.1.2 Purpose

- build-essential: Provides compilers and standard libraries
- cmake: Cross-platform build system

- tmux: Persistent sessions for long simulations
- ssh: Remote access for distributed workflows

39.5.2 Compilers

39.5.2.1 Installation

```
sudo apt install -y gcc g++ gfortran clang
```

39.5.2.2 Why Multiple Compilers?

Different compilers provide:

- Different optimization strategies
- Better compatibility with legacy codes
- Access to vectorization and architecture-specific tuning

39.5.2.3 Usage Example

```
gfortran -O3 -march=native simulation.f90 -o sim
```

39.5.3 Scientific Libraries

39.5.3.1 Installation

```
sudo apt install -y \  
libopenblas-dev liblapack-dev \  
libfftw3-dev \  
libhdf5-dev \  
libopenmpi-dev openmpi-bin
```

39.5.3.2 Mathematical Role

- **BLAS/LAPACK**: Linear algebra (matrix solves, eigenvalues)
- **FFTW**: Fast Fourier transforms
- **HDF5**: Structured scientific data storage
- **MPI**: Parallel distributed computation

39.5.3.3 MPI Example

```
mpirun -np 4 ./simulation
```

39.5.4 Python Scientific Environment

39.5.4.1 Installation (Conda)

```
conda create -n physics python=3.11
conda activate physics

conda install numpy scipy matplotlib pandas numba
               sympy h5py
```

39.5.4.2 Role in Workflow

Python acts as:

- Data analysis layer
- Visualization interface
- Rapid prototyping environment

39.5.4.3 Example

```
import h5py
import numpy as np

f = h5py.File("output.h5", "r")
data = np.array(f["density"])
```

39.5.5 Fortran Development

39.5.5.1 Tools

```
pip install fpm
```

39.5.5.2 Why Fortran?

- Efficient array operations
- Strong numerical performance
- Widely used in legacy HPC codes

39.5.6 Visualization

39.5.6.1 Installation

```
sudo apt install gnuplot paraview
```

39.5.6.2 Purpose

- **gnuplot**: Quick plotting
- **ParaView**: Large-scale simulation visualization

39.5.7 LaTeX Environment

39.5.7.1 Installation

```
sudo apt install texlive-full
```

39.5.7.2 Purpose

LaTeX is used for:

- Research papers
- Technical documentation
- Mathematical derivations

39.5.8 Editors

39.5.8.1 Installation

```
sudo apt install emacs
```

39.5.8.2 Usage Philosophy

- Emacs: Integrated research environment

39.5.9 Containers and Reproducibility

39.5.9.1 Installation

```
sudo apt install docker.io
```

39.5.9.2 Purpose

- Reproducible environments
- Dependency isolation
- Portable experiments

39.5.10 Workflow Architecture

39.5.10.1 Full Pipeline

1. Write simulation code (Fortran/C++)
2. Compile with optimized libraries
3. Run locally or via MPI
4. Output data in HDF5 format
5. Analyze using Python
6. Visualize results
7. Document in LaTeX

39.5.11 Directory Structure

```
~/projects/  
~/simulations/  
~/data/  
~/papers/  
~/notes/
```

39.5.12 Advanced Topics

39.5.13 Performance Optimization

- Use `-O3 -march=native`
- Profile with `perf` or `gprof`
- Optimize memory access patterns

39.5.13.1 Parallel Scaling

- Domain decomposition
- MPI communication patterns
- Load balancing

39.5.13.2 Reproducibility Strategy

- Version control (git)
- Environment locking (conda)
- Containerization (Docker)

39.6 3/20/26

I had some trouble with having different scaling for different displays. I ended up figuring out to change the GNOME from X11 to Wayland.

I was not able to make sure that the cursor switching works as I wanted, as far as I can tell there is nothing I can do to fix that.

Chapter 40

Emacs

This will be a summary of some of the main packages and customizations that I have either installed or created. As well as whatever I think about

Chapter 41

Category-Theoretic Metriplectic

41.1 Plans

41.2 Papers

Here I will begin with a list of papers to use for my research in this study

- Baez, Weisbart & Yassine — Open systems in classical mechanics
- John C. Baez & Christopher L. Rogers — Categorical Symplectic Geometry and the String Lie 2-Algebra (2008/2009)
- Alan Weinstein — Symplectic categories (survey/paper, 2009)
- Brendan Fong — Decorated Cospans (2015) & related work (Baez–Fong, Baez–Fong–Pollard)

- John C. Baez — expository / historical pieces on n-categorical physics
- P. Guha — Metriplectic structure, Leibniz dynamics and dissipative systems
- Work on double categories / structured cospans for open dynamical systems (Courser thesis; Kenny Courser; “Open Systems: A Double Categorical Perspective”)

41.3 Review of Metriplectic systems

41.4 Understanding of Category-Theoretic Systems

41.5 My work

Chapter 42

Gravitational Memory study

42.1 First meeting

We talked in general about a couple of things. My next “mini-project” was to create a python code to read the strain off of the gravitational wave data and use that to create some audio. I have developed the code to work for hdf5 files and proved it worked with a black hole merger, I do not have the gravitational wave data for CCSN yet.

Also, the bigger project is to work on figuring out how to use 3Gen gravitational wave detectors to find CCSN. They even estimate these new projects may be able to see throughout the Virgo Cluster. We are presumably basing our work off of the paper **Detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova**

42.1.1 First go: Detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova

General notes from my first look at the paper detecting Gravitational Wave Memory in the Next Galactic Core-Collapse Supernova. Using data from GWOSC and several CCSN simulations they use some kind of linear prediction filtering system to develop what the waves should look like and how far away we could see them.

42.1.1.1 Action Notes

- Learn Linear Prediction Filtration
- Learn librosa python module
- What is a whitened signal
- What is amplitude spectral density

I found a MITOpenCourseWare on signal theory that I will likely try to do

42.1.1.2 Code

Got some audio stuff working

42.1.1.3 General Readings/Videos

Convolution is defined via

$$\sum_{k=-\infty}^{\infty} x[k]h[x - k] = (x \star h)[n]$$

There is also a lot of other information listed but I am not sure what is important, the equations are generally pretty simple so I think I will just look them up.

42.2 September 10

add whiten to gw, zero out wavelets

Looking into trying without interpolation.

Look into generating gravitational waves from a 2d checkpoint file, calculate quadropole moments. Density variable multiply by coordinates. Stencile for time derivative or look into Pedro.

42.2.1 Code

I got the code for the audio of the CCSN. I did successfully whiten the gravitational wave data but was unable to get the black hole merger right. I verified what I did using the graph though

42.2.2 Quadrupoles

Calculating the quadrupole mass moment is fairly simple

$$I_{2m} = \frac{16\sqrt{3}\pi}{15} \int \rho Y_{2m}^* r^2 dV$$

The star superscript means we are using the complex conjugate.

Rather than write everything out, calculate mass moment very simply, find its second order time derivative, then plug that into the gravitational wave equation.

Chapter 43

Website

43.1 Brainstorm

I had thought of a couple of ideas, but had forgot to write them down. I guess I will just explain my final one.

My plan is to have it so it can parse one of my daily org files for a schedule, then can paste that schedule into the website to view. I would like to allow for checking things off, in progress, notes, and some other things I will get into at a later point.

I would also like to access my book easily through the website

43.2 2/16/2026

I meant to document what all I did and why through my journey, but I hadn't. Thus, at the moment I am finished.

Chapter 44

YouTube

44.1 Scope and Goals

As a kid, I had always wanted a YouTube Channel. Now I will be working towards that, on a bit more practical level. For the purposes of learning more personally, leveling up my skill as a

44.2 Getting Started

44.2.1 Ideas

At some point I want to dive into Kerr Black holes but I am not ready for that.

- Numerical Hamiltonian(done)
- A Physical Look Into Hamiltonian Mechanics
- Hidden Assumptions in Modern Physics
- Journey through General Relativity

- Positive Mass Warp Drive
- Axioms of Hamiltonian Mechanics
- Understanding of Metriplectic Dynamics
- Gravitational Waves
- MagnetoHydrodynamics Propulsion
- MagnetoHydrodynamics in a nebula
- Deeper Numerical Methods
- Scientific Programme
- Destroying a Universe with a punch(done)
- Best way to simulate a solar system(done)
- Metriplectic simulation
- In-depth operator splitting(done*)
- Deriving Plasma kinetic equations
- Simple GR solver
- ADM
- Symplectic geometry: A physicist's perspective
- DG
- Deriving Gravitational Wave Memory

44.3 Introduction to Symplectic Integrators

44.3.1 General overview of direction

- Introduction to me
- Motivation
- Hamiltonian Mechanics introduction
- Geometric perspective
- Symplectic Euler compared to traditional Euler
- Leapfrog/Verlet
- Practicule guidance

44.3.2 Opening

Before I begin this video; I am not an expert. Most of my understanding comes from Problems In Hamiltonian Dynamics by (.....). This is more for my own education than others, for a better understanding of the concepts and development of my teaching abilities. I highly recommended continue to learn far beyond this simple introduction I will lay out. (picture of the book)

44.3.3 Motivation

Symplectic Numerical methods are focus inheritily on the structure and methods of Hamiltonian mechanics. Thus, to understand them, one must understand why. Why it is used at all.

At first glance, it seems quite contrived. Hamiltonian Mechanics is quite mathematically abstract and complicated. Beyond that, expanding it in differing ways such as dissipative or entropic spaces requires creative solutions. Thus, begs the question, why is it such a developed field?

(put really complex hamiltonian)

The short answer is that Hamiltonian Mechanics geometry enforces energy conservation in a way others formulisms do not. The long answer... is well this video.

(put graph showing energy conservation)

44.3.4 Hamiltonian Mechanics Review

First, a quick review on Hamiltonian Mechanics. If you are comfortable feel free to skip ahead.

A hamiltonian system is defined by the function $H(q,p,t)$ - the hamiltonian - which repersents something analogous to total energy. Not exactly though, hopefully in a future video we will get to that, but for not thinking of it in energy works great for this purpose.

$$H(q, p, t) = T + V$$

Here q repersents generalized position, which is the minimum amount of set parameters that fully spesify where a system is in configuration space. Where configuration space is a space whose points repersent all possible configurations. Meaning that generalized position is not confined to distance, but also things like angles, diminsionless quantaties, or what ever other convenient quantities that can be used.

The q , repersents generalized momenta. Which holds the same rules as position does, but repersenting momenta instead of position.

The evolution can thus be derived to be

$$\frac{dq}{dt} = \frac{\partial H}{\partial p}$$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q}$$

Let us look at a concrete example: a simple pendulum
(picture of pendulum)

$$H(q, p) = \frac{p^2}{2m} + \frac{1}{2}kq^2$$

Such, one can take the partial derivatives to find:

$$\frac{dq}{dt} = \frac{p}{m}$$

$$\frac{dp}{dt} = -kq$$

In phase-space, the space of all generalized pairs, the system traces out a trajectory. For harmonic oscillators these are ellipses. Each ellipse corresponds to a different energy level. (show phase-space)

Here are some key points.

- energy remains constant along the trajectories. As trajectories are defined by conserved energy

- the phase-space volume is conserved. This is essentially (shade in one region)

- finally, the most important for this, the manifold retains a symplectic structure. Which is defined at the 2-form $\omega = dq \wedge dp$. This will be discussed further shortly.

44.3.5 Why standard models fail

To understand the symplectic methods, one must find their motivation. Why traditional methods fail in certain scenarios, no matter the type. To understand this, we will look at the simplest integrator. Forward Euler

$$q_{n+1} = q_n + h * \frac{\partial H}{\partial p_n}$$

$$p_{n+1} = p_n + h * \frac{\partial H}{\partial q_n}$$

Where h is the time step.

Let us apply this to our harmonic oscillator and see what happens... Trajectory spirals outwards! This is simply because traditional methods aren't made to conserve energy, they are meant to model physics. The main reason is that tiny errors add up without the ability to contain them. Also, traditional methods add unphysical correction terms that over time lead to unphysical effects. Being dissipative in certain areas and adding energy in others. Adding artificial energy to the system in order to have temporary stability. Now, obviously higher order corrections help with these problems and making the models more physical helps, but these problems are fundamental. Adding higher order corrections simply increases the computational workload without ever fully resolving the problem.

Even sophisticated methods like RK4 have this problem. It is accurate to the 4th order, but that means that the error per step is $O(h^5)$. This is tiny, but adds up. For something like a celestial simulation over millions or even billions of years, this creates distorted simulations.

(Log-log plot showing error growth)

44.3.6 Symplectic Integrators

(illustration of symplectic maps)

Symplectic integrators fix this problem by preserving the symplectic structure I mentioned earlier.

$$\omega = dq \wedge dp$$

The symplectic form is a mathematical object that measures oriented areas in phase-space. For a simple system with one degree of freedom in the position and momentum each one gains this equation.

This is a 2-form, when integrated over any region of phase space, you get an oriented area. (physical deformation)

As the hamiltonian flow evolves, the region over phase-space gets stretched, rotated, and deformed. But, the symplectic form remains preserved.

From this basic axiom, one can derive something called the Shadowing Theorem, that says:

A trajectory computed by symplectic integrator, while not exactly true, stays close to the trajectory of a slightly modified Hamiltonian.

(visual: compared integration to true)

Meaning, standard integrators essentially create a new equation that is close to the original.

Symplectic integrators create another hamiltonian that is exactly equal to the original hamiltonian plus the order of the error.

(energy conservation plot)

Such, the energy does not increase monotonically, it oscillates within a bounded region. Over exponentially long time, the error grows logarithmically rather than linearly.

44.3.7 Symplectic Euler's method

To compare, we will once again take the simplest possible case.

Take the traditional Forward Euler

$$q_{n+1} = q_n + h * \frac{\partial H}{\partial q}_n$$

$$p_{n+1} = p_n + h * \frac{\partial H}{\partial p}_n$$

Now turn it into what is called a backward Euler

$$q_{n+1} = q_n + h * \frac{\partial H}{\partial q}_{n+1}$$

$$p_{n+1} = p_n + h * \frac{\partial H}{\partial p}_{n+1}$$

Notice what has changed, we updated p using the old position but the momentum by using the new momentum. One thing to note, backwards or implicit integration is commonly used outside of symplectic integration, it is simply a general requirement for symplectic integration. There are some special cases where explicit numerical methods satisfy the requirements of a symplectic integrator. Another requirement being that one using a Hamiltonian or Hamiltonian like formalism such as Nambu mechanics.

Now let us see why it is symplectic:

$$H(q, p) = T(p) + V(q)$$

where T is kinetic energy and V potential.

The symplectic Euler step can be decomposed into pure position-step, which is the flow of the potential. The a pure momentum step, which is exactly the flow of kinetic energy. The composi-

tion of these is the exact flow of the Hamiltonian, which is by definition symplectic.

The exact requirements are quite involved and mathematically difficult to derive, so an independent video on the topic is required. For the purposes of this video the general rules described and the types of integrators are sufficient.

Let us implement this for the harmonic oscillator.

In this, the orbit remains stable. Energy oscillates but doesn't drift.

One can also make the equation implicit by updating q before p .

44.3.8 The LeapFrog/Verlet Method

The most commonly used symplectic integrator is the leapfrog method. It is second order accurate.

The core idea: Evaluate positions and momenta at staggered time points.

(equation)

(visual)

Think of it as half-step momentum backward. Full step position forward, using the half step. Then half-step momentum backward again.

This is symmetric: if you run it backward, you get the exact same algorithm. This time-reversibility is what gives second order accuracy.

There is an equivalent position formulation that is often more convenient. (equation)

Where v is velocity and a is acceleration. Note, that while it uses variables commonly used in Newtonian dynamics, it is still fundamentally hamiltonian. (visual) One can compare

performance after 1000 orbits. For a timestep of 0.01 Forward euler has energy error of 15.3%, symplectic euler 0.8%, and leapfrog 0.002%.

As you can see much smaller, as the error scales by the square of the time step. Allowing for larger time steps for the similar accuracy.

A derivation of this equation is outside of the scope of this video but is an interesting idea that may be developed into the future.

44.3.9 Splitting Method

To build higher order symplectic integrators, one must learn operator splitting.

The key idea: if you can write the Hamiltonian as a sum

$$H = H_A + H_B$$

and you know how to integrate each piece exactly, you can build approximate integrators by composing these exact functions of these two.

where $\phi_A(h)$ is the exact flow generated by H_A over time h . and ϕ_B is the exact flow over H_B .

Then the true hamiltonian flow is:

where $\{\cdot, \cdot\}$ denotes the Poisson Bracket.

The simplest approximation is obtained by composing the two exact flows, this is known as the Lie-Totter splitting, this method is first order accurate. Higher order methods can be built using splitting, such as the 4th-order Forest-Ruth:

$$\psi_4(h) = \psi_2(\theta h) \circ \psi_2((1 - 2\theta)h) \circ \psi_2(\theta h)$$

A derivation of this is outside the scope of this video but is

something we will get into.

44.3.10 Beyond the Basics

Symplectic integration goes far beyond the introduction given in this video.

- Dynamics on a manifold
- variational integration
- Backward error analysis
- Adaptive symplectic methods
- Methods for Hamiltonian adjacent frameworks(ie. Nambu, metriplectic, etc)

Some of these I would like to explore in future videos but many of them are currently beyond my expertise.

44.4 Physical Interpretations of Hamiltonian Dynamics and Beyond

44.4.1 Planning

1. Opening
2. What is the Hamiltonian
3. What physically is phase space
4. Dissipation and the Non-Hamiltonian World
5. Beyond Symplectic: Metriplectic and Generic

6. Beyond Symplectic: Contact Geometry and Geometrization of Irreversibility
7. Beyond Symplectic: Poisson Manifolds, Casimirs, and Nambu
8. Structural Realism
9. Open Problems

44.4.2 Opening

Often in higher level mathematical physics, you will here many phrases thrown around with generally little to no explanation. "the hamiltonian is analougus to total energy but isn't literally", "Hamiltonian mechanics forbids dissipation", and many more.

These complex ideas are pushed aside for latter, well if you are like me, you think that later is now, so that is the purpose of this video.

This video will require a good intuitive and understanding of Hamiltonian mechanics and the basic structural underpinnings. Also, be warned, this is a fun project, mistakes are entirely possible.

44.4.3 What is the Hamiltonian?

In most courses, you will be introduced to the Hamiltonian functions through this equation

$$H = V + T$$

where the Hamiltonian is the addition of kinetic and potential energy, and is thus the total energy.

A few years later you may hear terms like generator of the flow, with only the explanation that H is a function that generates the correct observables, whatever those are.

So, how do these definitions fit together, and what do they even mean.

Essentially, the idea is that H uses Noether's theorem's time translation to make it so that the Hamiltonian is always equal to:

$$\frac{\partial A}{\partial t} = \{A, H\}$$

where H is the observable. An observable is a pretty self explanatory concept. Something that can be physically observed; momentum, energy, velocity, entropy, or even non-physics measurements like personality. Though classical Hamiltonians have trouble with entropy as we will get into.

As you can figure out with some careful playing; the Hamiltonian is only equal in a special case; in classical systems where the observables are position and momenta. In other cases, it no longer is that. It could be something else, or simply a number with no traditional physical analogy.

44.4.4 What is Phase Space?

Now, to actually use the Hamiltonian, now we need phase space. So, what physically is that.

Well, a good analogy is imagine the universe is frozen in time, what information do you need to predict what happens next?

Now that is just an analogy, if you are watching this video I suspect you want something more than that.

Well, we must first start with a state, which is really just a property, the properties needed for physics. Newton defined

these properties through forces, mass, position, time, and their respective derivatives to arbitrary orders.

Hamilton, defined it through configuration momenta and position. It really can be anything but as we will motivate later in most physical systems these are the generally the most useful. Then, he found you are able to for every possible physical state, there could be one point in this phase space of position and momenta that corresponds to it. Such as a 2D body can be plotted in 4D space through

$$(x, y, p_x, p_y)$$

There, objects no longer viewed as moving through space but rather trajectories in phase space. Generally this view is considered to be a mathematically significant but metaphysically trivial change, but some people disagree, which is something that will be covered later in the video.

44.4.5 Is Momentum more Fundamental than Velocity

The question of which is more fundamental; momentum or velocity seems quite arbitrary, and in many ways it is, but we will look into it regardless.

In traditional Newtonian Mechanics; momentum is traditionally derived by velocity, but the inverse is just as true.

In hamiltonian mechanics, this isn't the case. Momentum is more than just mass and velocity, it also includes think like electric fields or even other fundamental forces like gravity in some cases. In quantum mechanics this connection becomes even more muddled for obvious reasons.

Now to really answer the question. I am sad to say that it depends on what you are looking for. Velocity is much more

intuitive, but momentum is a generator of translations and naturally connected with symplectic structures. Likely not a very satisfying answer, but the only one I have.

44.4.6 Hamiltonian Mechanics and Teleology

44.4.7 Noether's Theorem

44.4.8 Canonical Transformations

44.4.9 The Question of Dualism

44.4.10 Hamiltonian Branch offs

44.4.11 Hamiltonian and Dissipation

44.4.11.1 Metriplectic and Others

44.4.12 Structural Realism

44.5 Variational Integrator for RMHD

44.5.1 Planning

1. Opening
2. Introduction
3. RMHD
4. Formal Lagrangian
5. Conservation Laws
6. Electron Inertia

7. Symmmetrisation
8. Discrete Action Principle
9. Discrete Conservation Laws
10. Numerical Methods
11. Validation
12. Random Testing

I am thinking for random testing either a video of one of the earlier simulations or do something random.

44.5.2 Opening

Before I begin, as always, I am not an expert. Most of my information comes from the paper “Variational Integrators for Reduced Magnetohydrodynamics”, by Michael Kraus, Emanuele Tassi, and Danueka Grasso. For more information their paper will be in the description. Anyways, on to the video.

44.5.3 Introduction

For all purposes, the subject of this video is the creation of a numerical method that takes a simplified model of magnetized plasma and preserves the conservation laws with greater accuracy than most other methods. The tool used to do so is called a variational integrator. That descretizes the action, as in the action in Lagrandian Mechanics, rather than the equations of motion.

44.5.4 Variational Integrators

Variational integrators work in the inverse of traditional methods. In traditional methods you start with the model

$$F = kx$$

then define the equations of motion

$$\ddot{x} + \frac{k}{m}x = 0$$

From here one discretizes the equation of motion.

For variational derivatives one takes the action:

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q}) dt$$

The nature selects where

$$\delta S = 0$$

From here derivation of things such as

- Euler-Lagrange
- Hamiltonian
- Conservation laws

From here instead of discretizing the differential equation, one discretizes the lagrangian

$$S_d = \sum_{k=1}^{N-1} L_d(q_k, \dot{q}_{k+1})$$

Where

$$L_d \approx \int_{t_k}^{t_{k+1}} L dt$$

From here, varying across the sequence rather than a smooth curve gives you the Discrete Euler-Lagrange Equation. Here the equation is already an algorithm, no additional methods needed(though we will use some in this video)

44.5.5 RMHD

To do this, we must define what RMHD is, to do that we must first define MHD. MHD treats plasma as a single conductive fluid coupled with a magnetic field. While it is a massive simplification of “Billions of charged particles” it is still quite accurate and obviously much less computationally expensive.

RMHD, further simplifies, for a specific, useful physical situation. In plasma with a strong background magnetic field, like in a tokamak, where the field mostly points one way (call it “toroidal”) and everything interesting happens in the plane perpendicular to it. Under those conditions, the fast physics along the strong field direction basically stops mattering, and you’re left with an effectively two-dimensional problem.

With only four fields remaining as useful:

- ϕ , the stream function, describes how the fluid flows
- ω , the vorticity, how the flow swirls
- ψ , the magnetic potential
- $\mathbf{j}$, the current density, how the magnetic field swirls

The equation then simplifies to:

$$\omega_t \{\phi, \omega\} + \{j, \psi\} = 0, \quad -\Delta \phi = \omega$$

$$\psi_t + \{\phi, \psi\} = 0, \quad -\Delta\psi = j$$

With the poisson bracket being simply two dimensional so that

$$\{\phi, \omega\} = \phi_x \omega_y - \phi_y \omega_x$$

Here, it can be seen that RMHD has Casimir Invariants, quantities that are conserved, not from traditional physical symmetries but from the structure of the Lie algebras. There are families with infinite many subsets, for the purposes of this we will focus on Magnetic helicity (how tangled the magnetic fields are) and Cross helicity (how aligned the flow is with the magnetic field)

Now, for this paper to work, rather than use Hamiltonian structure, it uses Lie-Poisson, Hamilton's weird cousin. It still conserves energy but standard symplectic methods simply do not work.

This is because fluid mechanics observables do not naturally describe motion along a phase space, but live on a Lie Algebra with symmetries within it. So the Poisson operator changes with the state rather than is a constant symplectic structure.

44.5.6 Formal Lagrangian

From here comes a snag, variational integrators need a Lagrangian, an action to discretize. But, for RMHD, there is no obvious Eulerian-Lagrangian. So what does one do?

They make one up. There is a trick called a formal Lagrangian. Essentially, take every equation in your system and multiply it by a made up variable called an auxiliary variable that has no physical meaning. Then add them

up

$$L = \xi (\omega_t + \{\phi, \omega\} + \{j, \psi\}) + \mu (\omega + \Delta\phi) + \chi (\psi_t + \{\phi, \psi\}) + \zeta (j + \Delta\psi)$$

Here four variables ξ, μ, χ, ζ , are all ghost variables that only exist to make the arithmetic work.

This helps by giving stationary variables to allow for one to vary along the physical variables using Hamilton's Principle leading to a brand new "adjoint equation."

Leading to the Euler-Lagrangian:

Where φ denotes all of the variables. Many of the auxillary variables can be found to be to be functions of the original variables. So that φ can be defined as

$$\varphi = (\omega, \psi, \phi, j, \psi\omega, 0, 0)$$

44.5.7 Conservation Laws

Now, the whole purpose of the creation of the Lagrangian was to apply Noether's theorem. We restrict our attention to conservation laws which are generated by vertical transformations, that is transformations which only affect the field variables but leave the base space invariant. From here with some complex math I won't get into, it is found that the symmetries are

- Magnetic Helicity
- L^2 norm of ψ
- Cross Helicity
- Energy

44.5.8 Electron Inertia

This current RMHD, has a defect. It does not include magnetic reconnection, field line snapping and reconnecting. Which tend to be one of the most important studies in plasma physics. So, we will add a small correction, electron inertia. Real electrons have mass, just a small one, ones you treat them as such a new term develops in Ohm's law, one that allows reconnection. Thus the new model changes ψ to $\bar{\psi} = \psi + d_e^2 j$ to give the electron canonical momentum where d_e denotes the electron skin depth.

That's it, that is the whole mortification. From here, small bit of derivation shows that the Hamiltonian stays the same, so does the Casimir invariants and the conservation laws, only expressed in for of $\bar{\psi}$ rather than ψ .

44.5.9 Symmetrisation

Before we can put anything into our computer, there is some tedious house keeping needed. Our formal Lagrangian has a second derivative, which is highly challenging to discretize directly and even more costly to compute. To do this we use symmetrisation, essentially integration by parts but for functional rather than function's. We do this on the terms that produce the Laplacians, and get an equivalent Euler-Lagrange equation that only depends on the first order derivative of the fields.

Further care can be taken when discretising the Poisson Bracket. It can be seen through integration by parts when assuming appropriate boundary conditions that the cyclic permutations of the functions in the integrand are all identical

Insted of selecting on of those forms, a convex combination

can be considered.

$$\int \xi \{ \phi, \omega \} dx dy = \int [\alpha \xi \{ \phi, \omega \} + \beta \phi \{ \omega, \xi \} + \gamma \omega \{ \xi, \phi \}]$$

Where $\alpha = \beta = \gamma = \frac{1}{3}$ to conserve symmetry. From here we find the motified Lagrangian

$$L'''(\varphi, \varphi_t, \varphi_x, \varphi_y) = \frac{1}{2} \varphi_t^T \Lambda \varphi \quad (44.1)$$

$$+ \frac{1}{3} (\xi \{ \phi, \omega \} + \phi \{ \omega, \xi \} + \omega \{ \xi, \phi \}) \quad (44.2)$$

$$+ \frac{1}{3} (\xi \{ j, \psi \} + j \{ \psi, \xi \} + \psi \{ \xi, j \}) \quad (44.3)$$

$$+ \frac{1}{3} (\chi \{ \phi, \psi \} + \phi \{ \psi, \chi \} + \psi \{ \chi, \phi \}) \quad (44.4)$$

$$+ \mu \omega - \nabla \mu \cdot \nabla \phi + \zeta j - \nabla \zeta \cdot \nabla \psi. \quad (44.5)$$

44.5.10 Discrete Action Principle

As mentioned earlier we will discretize the action rather than the equation of motion, for simplicity we will use Veselove-type finite difference to split into two space and one time. But the extention to more elaborate approaches is fairly straight forwards from here.

Here we can quite easily find that

A large deal of algebra allows us to get to the point of employing the Hamilton-Pontryagin Principle. Which states when transferring from Lagrangian to Hamiltonian we should not assume velocity and momentum's relation but derive it after the fact of treating them as independent.

The using what is called the Arakawa's discretization for the

Poisson bracket, which is given by

$$A_{i,j}(\phi, \omega) = \frac{1}{3} \left(A_{i,j}^{++}(\phi, \omega) + A_{i,j}^{+\times}(\phi, \omega) + A_{i,j}^{\times+}(\phi, \omega) \right). \quad (44.6)$$

Where the bracket is the averaging of the A^{++} , the centered difference of both X and y . $A^{+\times}$, the diagonal interaction. And $A^{\times+}$, another diagonal rotated relative to the other.

Using these principles and a lot of mathematics we gain these massive equations.

```

0 &=
\frac{\omega^{n+1}_{i,j} - \omega^n_{i,j}}{h_t}
+ \frac{1}{4} \left[
    A_{i,j}(\phi^{n+1}, \omega^{n+1})
    + A_{i,j}(\phi^n, \omega^{n+1})
    + A_{i,j}(\phi^{n+1}, \omega^n)
    + A_{i,j}(\phi^n, \omega^n)
\right] \nonumber \backslash

\quad + \frac{1}{4} \left[
    A_{i,j}(j^{n+1}, \psi^{n+1})
    + A_{i,j}(j^n, \psi^{n+1})
    + A_{i,j}(j^{n+1}, \psi^n)
    + A_{i,j}(j^n, \psi^n)
\right], \tag{86a}
\backslash

\quad + \frac{1}{4} \left[
    A_{i,j}(\phi^{n+1}, \psi^{n+1})
    + A_{i,j}(\phi^n, \psi^{n+1})

```

```

+ A_{i,j}(\phi^{n+1}, \psi^n)
+ A_{i,j}(\phi^n, \psi^n)
\right], \tag{86b}
\

\ [1em]
\omega^{n+1}_{i,j} &=
- \Delta_x \phi^{n+1}_{i,j}
- \Delta_y \phi^{n+1}_{i,j}, \tag{86c}
\

\ [1em]
j^{n+1}_{i,j} &=
- \Delta_x \psi^{n+1}_{i,j}
- \Delta_y \psi^{n+1}_{i,j}. \tag{86d}

```

If you wish to see the derivation out, I would recommend reading the paper.

44.5.11 Discrete Conservation Laws

Using the same methodology we can find our earlier conservation laws

Total energy

$$E = \frac{hx\,hy}{2} \sum_{i,j} (\phi_{i,j} \omega_{i,j} + \psi_{i,j} j_{i,j}) = \text{const.}$$

Magnetic Helicity

$$C_{\text{MH}} = hx\,hy \sum_{i,j} \psi_{i,j} = \text{const.}$$

L^2 norm of ψ

$$C_{L^2} = hx hy \sum_{i,j} \psi_{i,j}^2 = \text{const.}$$

And cross Helicity

$$C_{\text{CH}} = hx hy \sum_{i,j} \omega_{i,j} \psi_{i,j} = \text{const.}$$

44.5.12 Numerical Methods

So now we have out implicit forms tangled in non-linearity.

To solve this we use Newton's Method linearize the residual, solve the resulting linear system, update your guess, repeat until the residual is essentially zero. That linear system at each Newton step gets handed off to GMRES, an iterative Krylov-subspace solver, and to make GMRES converge fast, they build a custom, physics-informed preconditioner — following earlier work by Chacón and collaborators — that exploits the specific structure of the vorticity and Ohm's law equations.

The elliptic bits — inverting the Laplacian to get the variables — get handled by whatever general-purpose elliptic solver you have on hand (they use PETSc); if your domain happens to be doubly periodic, there's an even cleaner trick available, since periodic boundary conditions mean the discrete Laplacian is exactly diagonalized by a Fast Fourier Transform, so you can invert it directly and exactly in Fourier space, no iteration required.

I mimic this in python by setting up a file to create the initial condition, the operators of the spectral solver and Arakawa's method, and integrator file to run it through the actual equations derived earlier.

44.5.13 Validation

Here I make sure my model matches the paper through Orszag-Tang Vortex

(image)

Current sheet

(image)

Current sheet with Electron Inertia

(image)

Here the models match that of the paper, and matches within machine precision the analytical solution.

44.5.14 Random Test

Now that we have a working model, let's use it for something cool. Here is Kinetic-Alfven Turbulence

(movie here)

44.6 Why: Developing Physics

44.6.1 Planing

Essentially, one person teaching another parabolic trajectories but the student keeps asking why in a generally eurdite way. Analogous to Socratic Dialog.

1. Introduction to Parabolic trajectories

44.6.2 Introduction

Here we set the stage, a stage. An introductory physics teacher is having a one-on-one lesson with a young student. A highly

intelligent student that wants to get to the base of everything. The most fundamental of the fundamental.

44.7 Can you Destroy a Universe with a Force?

44.7.1 Introduction

I am sure everyone knows of the extreme forces created by fictional characters. Blowing up planets, stars, galaxies, and even Universes. Now these obviously aren't physically realistic, but it begs the question, how much? What would such extreme forces actually look like?

44.7.2 Planck Force

To start with this, we will start with the theoretical highest force that still works with modern physics. The Planck Force. To explain this, the Planck Units are essentially normalizing and using units and physics starting with only the speed of light, gravitational constant, and Planck's Constant. Beyond the simple exercise of mathematical reasoning and the possibility of simpler mathematics. As is the case with other Natural Units, these units have been found to also coincide with when both quantum effects and gravity become significant. Making traditional theories fall apart, as no one knows how these two things will interact.

Using this method, the force unit becomes

$$\frac{c^4}{G}$$

Obviously an extremely large number, in newtons it is roughly

two times ten to the power of 44.

Many have conjectured that this is the largest possible force that can occur. Others have disputed. Others have found that according to traditional General Relativity it is simply a fourth of the Planck Force.

Until a theory of Quantum Gravity has been resolved, no one will know.

44.7.3 Striking the Earth with a Planck Force

While obviously, without a theory of quantum gravity, no one really know what will happen with a Planck Force, but using General Relativity to guess should be fun.

I used Einstein's toolkit, a very helpful open source numerical general relativistic code base, and added a thorn to strike an object at a point with a force. Then I edited their Neutron Star simulation to be Earth adjacent, obviously not exact but with such extreme cases I doubt it would make much of a difference.

Doing this, was surprisingly difficult. I thought it would be the easiest part of this video, but I was wrong.

after running it I go this

(Insert movie)

While you likely can't read the numbers, and they are very strange units at that. So I will explain, very quickly the of contact very quickly explodes into a vacuum at relativistic speeds. Then, as the relativistic mass adds up, gravity slowly overcomes the pressure wave. Collapsing it into a small point. A point roughly the size of a single mesh in the simulation. Thus, including things such as the likely fusing of particles, neutron degenerate matter, or if it would collapse into a black hole would require a more advanced simulation.

There is another problem. The force being so great would lead to particles reaching super-luminal speeds before the next time step. Leading to the simulation coming to errors as the mass become imaginary. The creators of the toolkit obviously came to a solution, but I am not exactly sure what their solution is. Thus, how accurate this simulation is, is knowledge beyond mine.

I tried to decrease the computational complexity and decrease the time-step substantially. I made the simulation symmetric so that I only had to simulate one eighth of the Earth, 2 dimensional rather than 3d, and increase the size of the mesh.

I eventually got to a single time step being one ten trillionth of a second. Still getting the same warning. So, I eventually had to accept the fact that I don't know. I would think that a collapse would happen but not into a black hole. Likely a mini-star that would eventually explode out, but that is something that I can't confirm.

To do so would require much more than my computer and messing with Einstien toolkit.

44.7.4 Assumptions about a Universe

While that was rather disappointing, there is still the question of the universe. Which is surprisingly much simpler.

Anyways, the we need to make assumptions about said universe.

One, our current observable universe, 93 billion light years in diameter, is the size we will use. Assuming that matter outside of such volume would have a negligible effect on the simulation.

Two, we assume that spacetime is mostly flat. Most observations put that the universe has an extremely small overall curvature, but the small change likely wouldn't change the re-

sults of the simulation much.

Three, take dark energy to be constant, such as a vacuum energy model. While recent studies dispute this, they have trouble coming up with exact values so for now I will leave it as is to simplify.

Four, with dark matter, we take the estimated values of both Hubble's estimations of dark matter and Shao's as well.

Five, we take a relatively isotropic density in the universe. Meaning a relatively constant one, rather than variable.

Six, when we later talk about the energy of the force, assume that it is some kind of magic or from another universe. As it is "creating" or adding energy to the universe that was not there previously.

Now that we have these.

44.7.5 Modeling a Universe

Now, how to actually destroy a universe. Being filled with so much empty space it is pretty obvious you can't just punch it and blow it up. The small quarks in the plasma might hit a couple of things, but would be pretty useless and definitely wouldn't make it to other galaxies. So, that leads us to our solution. In the previous simulation, the Earth collapsed from gravity, so it is conceivable that a more intense force would collapse into a black hole.

So, first we work backwards. To get force we can take energy, and to get energy we can find the analogous mass needed to overcome dark energy.

Starting with the FRW Friedmann equation:

$$H(a)^2 = H_0^2 (\Omega_{r0} a^{-4} + \Omega_{m0} a^{-3} + \Omega_{\Lambda 0})$$

Here we will compress the parentheses into $E(a)$ for simplifi-

cation.

Next, we take the Schwarzschild-de Sitter metric function for a relatively flat expanding space time.

$$f(r) = 1 - \frac{2GM}{c^2 r} - \frac{\Lambda r^2}{3}$$

Here, one can take the derivative and find when that derivative is equal to 0 to find the point in which the universe becomes static to be

$$r_c = \left(\frac{3GM}{\Lambda c^2} \right)^{\frac{1}{3}}$$

Then solving for mass when the radius is the current size of the universe we get

$$M_{crit} = \frac{\Lambda c^2}{3G} R_{obs}^3 = \frac{\Omega_{\Lambda 0} H_0^2}{G} R_{obs}^3$$

Now for the exact solution we take

$$ds^2 = -c^2 dt^2 + \frac{R'(r, t)^2}{1 + 2E(r)/c^2} dr^2 + R(r, t)^2 d\Omega^2$$

Here we take the Lemaitre-Tolman-Bondia metric, an exact solution for Einstein's equation for a lump of matter with no assumed symmetry.

(show algebraic manipulation)

Here we see that the structure of the shell makes no difference, only the total mass of it.

From here simple algebra shows allows us to find the energy needed to lead to an eventual collapse and the function that can be solved to find when it crosses.

To actually solve, we need values. For the hubble constant and the mass and energy of the universe. Using the two values

from Hubble telescope and Shoe, we use Bayes uncertainty to create a Monte Carlo simulation to find the list of values.

44.7.6 Results

From here, we run the simulation. Getting the below results (results) with the total energy needed to overcome

44.7.7 Interpreting the Results

Now, the purpose of this, wasn't to find the energy, but rather the force. I know most people watching this know the general work formula, but in the case of relativity, this doesn't always work. Though, with a couple of simplifications, we will end up with a just as simple formula.

We start with a practically flat spacetime to make the calculations simple.

From here, the fact of the conservation of energy and momentum, makes it so that the stress energy tensor, which has both energy and momentum is clear. Derived to be

$$\nabla_\mu T^{\mu\nu} = f^\nu$$

Where we take the contravariant derivative of the stress energy tensor to get the force.

where we can expand to

$$f^\nu = \frac{\partial T^{0\nu}}{\partial x^0} + \frac{\partial T^{1\nu}}{\partial x^1} + \frac{\partial T^{2\nu}}{\partial x^2} + \frac{\partial T^{3\nu}}{\partial x^3}$$

where we can take the first term to find that this is simply the energy added by the force by volume. Then the other three

terms simplify to be

$$f^i = \frac{1}{c} \frac{\partial T^{0i}}{\partial t} + \partial_j T^{ij}$$

which is essentially the change in momentum density plus the momentum flowing through boundary.

To simplify the calculations, we will focus just on the energy density portion. The result would be the same just differing math. Say the change in energy is 3.6 times ten to the power of 71. This force is spread through a volume of 1 meter, energy density and thus force would be 3.6 times 10 to the power of 71 Newtons.

As I have mentioned, this is an extreme simplification, but when working with energies such as these, there is really no physical object that would lead to the initial energy or the pressure gradient to have any non-negligible effect.

So now we have the force. 36 followed by 70 zeros Newtons.

About 67 orders of magnitude beyond the average punch of 1,000 Newtons. About 66 orders of magnitude above the highest ever recorded punch of 69,000 Newtons.

For comparison, this is 25 orders of magnitude above the estimated energy release of the big bang.

On the order of comics, the best feat that I could come up with a real number too, not estimations based on the wacky comic physics would be superman pulling a chain of planets at very quick speeds. With other people's estimates coming at 10^{38} to 10^{40} newtons. Still on the upper end being 31 orders of magnitude away. Humans are closer to the superman's feat than superman is to the calculated feat.

44.7.8 General Questions

There are a couple of questions that I can anticipate. If gravity moves at the speed of light, how would that changes things. The answer is not an insane amount. Red shifting and losing energy is only for waves, the constant gravity would not lose energy. Our model make sure the gravity overcomes dark energy by definition. The main change would be the change in the radius of the universe and the possibility of parts of the currently observable universe going past the cosmic horizen where gravity would never be able to reach it. The two aspects conetract one another, and since we are working with such large numbers I assume they wouldn't change much. Though, as always, I could always be wrong.

Thank you for watching

44.8 Over Engineering Celestial Mechanics: WHCKL model

44.8.1 Planning

1. Introduction
2. Scope
3. Deriving WH
4. WH Maping
5. Deriving Symplectic correctors
6. using variations
7. Splitting method

8. Lazy Kernel
9. General Improvements
10. Comparing to other methods
11. video of solar system

44.8.2 Introduction

(general animate introduction)

Hi everyone

44.8.3 Scope

Essentially the scope and goals of this project are to develop a celestial mechanics numerical model that is mathematically complex enough to be a challenge. To slightly constrain the scope, the goal will be to allow for highly efficient long term keplarian celestial mechanics. On the order of millions of even billions of years with relative accuracy.

After some research, the clear, for very long term celestial mechanics the clear modern winner is Wisdom-Holman

Then, when researching this program I found that REBOUND, an open source code for celestial mechanics found ways to improve its accuracy that introduced new fun challenges. Such as symplectic correctors, variational methods, splitting methods, and even something they call a lazy kernel. So, let us get into it.

44.8.4 Deriving WH

To derive Wisdom-Holman, we start with a traditional N-body hamiltonian in inertial cartesian coordinates

$$H(q, p) = \sum_{i=0}^{N-1} \frac{P_i^2}{2m_i} - G \sum_{i < j} \frac{m_i m_j}{\|q_i - q_j\|}$$

With $i = 0$ to be the central star, and the rest being the planets.

From here, the hamiltonian is not sperable, so we make a canonical transformation into heilocentric canonical coordinates.

Where we define $Q_i = q_i - q_0$ where essentially Q is the position of the planets in refrence to the sun. The corresponding canonical momenta doesn't change. $P_i = p_i$.

With the exeption of the sun which changes by the transformation. $P_0 = p_0 + \sum_{i=0}^{N-1} P_i$.

Then we define capital M to be the cumulative mass

From here it is simply derived that the Hamiltonian becomes

$$H = \frac{P_0^2}{2M} + \sum_{i=1}^{N-1} \left[\frac{P_i^2}{2m_i} - \frac{Gm_0 M_i}{Q_i} \right] + \sum_{i=1}^{N-1} \frac{P_i^2}{2m_0} - G \sum_{i < j, i, j > 0} \frac{m_i m_j}{\|Q_i - Q_j\|}$$

Here we can divide the hamiltonian into $H = H_{sun} + H_{kepler} + H_{int}$.

This is the traditional method; though, for later reasons we will get into, rather than Heilocentric coordinates we will use Jacobi, meaning in refrence to the center of mass rather than the central body.

Where the center of mass of all particles interior to the i -th particle is R_{i-1} , thus their coordinates become $r'_i = r_i - R_{i-1}$.

Because jacobi coordinates are linear functions of cartesian, the rest of the quantities, such as velocity and accelera-

tion change in the same way. Momenta is different however.

$$p'_i = m'_i \dot{r}'_i$$

where the reduced mass is equivalent to.

$$m'_i = m_i \frac{M_i - 1}{M_i}$$

Using these coordinates the N-body hamiltonian instead becomes

$$H = \sum_{i=0}^{N-1} \frac{p_i'^2}{2m'_i} - \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{Gm_i m_j}{\|r_i - r_j\|}$$

Keeping the potential term in cartesian coordinates and only writing the kinetic in jacobi.

Next, for the sake of the algebra we add and subtract the term

$$H_{\pm} = \sum_{i=1}^{N-1} \frac{Gm'_i M_i}{\|r'_i\|}$$

This grouping leads to the Hamiltonian

$$H = \underbrace{\frac{(p'_0)^2}{2m'_0}}_{H_0} + \underbrace{\sum_{i=1}^{N-1} \left(\frac{(p'_i)^2}{2m'_i} - \frac{Gm'_i M_i}{|r'_i|} \right)}_{H_{\text{Kepler}}} + \underbrace{\sum_{i=1}^{N-1} \frac{Gm'_i M_i}{|r'_i|} - \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{Gm_i m_j}{|r_i - r_j|}}_{H_{\text{Interaction}}}.$$

Here one can further split the kepler term into

$$(H_{\text{kepler}})_i = \frac{p_i'^2}{2m'_i} - \frac{Gm'_i M_i}{\|r'_i\|}$$

each Hamiltonian describes the keplerian motion of the i-th particle around the center of mass of all interior particles.

After some more algebra we can split the jacobian into two parts, one that can be easily computed in cartesian and the other in Jacobi coordinates

$$H_{int} = \sum_{i=2}^{N-1} \frac{Gm'_i M_i}{\|r'_i\|} - \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{Gm_i m_j}{\|r_i - r_j\|}$$

We now have three speratable equations that a all much easier to integrate than the highly non-linear one we stated with. Where H_0 is motion along a straight line, H_{Kepler} is the hamiltonian for kepler orbits, and H_{int} is the interaction of particles outside of the i-th particle. To note, this methodology only works for near keplarian orbits, in say a three body binary this method would be quite unphysical quite quickly.

44.8.5 WH Mapping

We do this splitting as the original equation's equation of motion have no analytical solutions.

$$H(q, p) = \sum_{i=0}^{N-1} \frac{P_i^2}{2m_i} - G \sum_{i < j} \frac{m_i m_j}{\|q_i - q_j\|}$$

Thus, approximate solutions are needed. Many methods exist but the method we have been developing is obviously splitting the Hamiltonian into smaller parts that can be integrated easily.

Analytical solutions can be found for the evolution of the system under each individual Hamiltonian.

Thus, to create the symplectic integrator using operator split method.

To do this we start with the familiar Liouville operator

$$L_H f = \{f, H\}$$

If you are not familiar, this is simply the same as saying that

$$\dot{f} = \{f, H\}$$

This obviously looks like an ordinary differential equation that has the formal solution of

$$f(t+h) = e^{hL_H} f(t)$$

Because this equation is for a differential operator not a variable, this exponential operator is instead defined through:

$$e^{hL_H} = I + hL_H + \frac{H^2}{2!} L_H^2 + \dots$$

where the key difference is that differential operators do not commute as traditional variables do. This here is the backbone of the symplectic integrator, to approximate this non-commutable equation.

Using this we can take our original Hamiltonian in Poisson form to be:

$$L_H = L_0 + L_{Kep} + L_{int}$$

let us turn those into A, B, and C:

$$L_H = A + B + C$$

Then add A and B to make a and turn C into B

$$L_H = A + B$$

then the exact propagator is obviously

$$e^{h(A+B)}$$

Now that we have that we can use the generic strang method.

$$e^{\Delta t(A+B)} \approx e^{\frac{\Delta t}{2}A} e^{\Delta t B} e^{\frac{\Delta t}{2}A}$$

Or the drift-kick-drift. Drifting along the kepler motion and motion of the center of mass and kicking using the inteaction between planets.

44.8.6 Symplectic Correctors

Another method we will add are symplectic correctors. A symplectic corrector is simply a small adjustment to the coordinates that one does before and after running a symplectic integrator so that the numerical trajectory more closely matches the real physical trajectory.

Because they are only applied at the very beginning and end, one can add arbitrarily higher order correctors with negligible effects on the speed.

We choose an eleventh order corrector that effectively brings the effective order of the integrator from second order to fifth, while barely changing the time it takes to run the program.

The exact benefits of the correctors change. As in Earth mass planets, beyond a fifth order corrector there is little change, but in Jupiter mass ones an eleventh order method is quite helpful.

44.8.7 Building the Code

When building this code, taking optimization to heart I used FORTRAN. A computational science coding language that is very fast. Optimized it is comparable and arguably faster than optimized C++, it also has the added benefit of being much easier to read.

I found a paper that already derived the best ways to optimize the subcalculations, specifically coordinate transformations between Jacobi and Cartesian, Newtons Method, c-functions, and a couple of other things.

I won't bore you with the details, but I eventually added the fifteen of the most massive bodies that orbit the sun. The eight planets, Ceres, Pluto, Haumea, Makemake, Gonggong, and Erirs. Yes, those are real names of dwarf planets.

I didn't add moons as this simulation, by using the kepler hamiltonian, assumes that the body orbits the center of mass of interior object, but moons obviously don't do that. Such, the simulation is not as physical for say the Earth whose moon is a significant compared to its mass. But it still works as a fun project with beyond that being very accurate.

After that I set the time step to roughly 18 days, the max speed while still being accurate. I did underestimate how many years the simulation was able to complete so Mercury did become unbounded near the end

44.8.8 Results

Here we now come to the fruits of our labor. I really should have worked on another program to have something to compare to, as these numbers are pretty meaningless on their own. Just that they are about as efficient as it gets...

Well, the purpose was the process not the result

44.8.9 Watching it

Though, to get a little extra out of it, let us watch the animation of a couple of the planets.

44.9 Lie Operator Splitting: One of the ten Pillars

44.9.1 Introduction

Hi, welcome to a semi-advanced Introduction into Lie Operator splitting, a mathematical Approach.

44.9.2 Scope

Essentially, the scope of this video being an introduction into Lie Operator splitting, a method that I have used several times in past videos.

A good introduction to its importance is a quote by Dr. Strang, “If, as has sometimes been argued, there are only ten big ideas in numerical analysis and all the rest are merely variations on those themes, splitting is undoubtedly one of them.” as it is at the foundation of so much of numerical analysis. I primarily have used it for symplectic integrators, but its uses are far beyond that. Other structure preserving methods, statistical mechanics, nearly integratable systems, optimization, quantum mechanics, and really anything that can be described by differential equations without easily found analytical solutions.

In fact, one could say that, at least since Descartes stated his four rules of logic in the Discours de la méthode, the notion of sub-dividing a complicated problem into its simpler constituent parts, solving each one of them separately and com-

binning those separated solutions in a controlled way to get a solution to the original overall problem constitutes a guiding principle in all areas of science and philosophy.

44.9.3 Motivating Operator Splitting

As in most cases in applied mathematics, the easiest start to defining new mathematics is to motivate their existence. To confine this agumentation, let us look at the simplest case of a non-integratable Hamiltonian. An Anharmonic oscillator defined through this equation:

$$H(q, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 + \lambda q^4$$

Here, this equation's derivatives have no elementary closed-form solution

But we can split it into its potential and kinetic energy

$$H_T = \frac{p^2}{2m}$$

$$H_V = \frac{1}{2}m\omega^2 q^2 + \lambda q^4$$

Here these problems have trivial solutions.

Thus, if one wishes to solve them together one can trivially see that the Hamiltonian vector field is

$$X_H = X_T + X_V$$

(show vector fields adding together)

The corresponding Liouville operators are

$$L_H = L_T + L_V$$

Then if we introduce a phase-space vector

$$z = \begin{pmatrix} q \\ p \end{pmatrix}$$

Solving the differential equation we get

$$z = e^{h(L_T + L_V)}$$

Solving this may seem easy. Your first instinct would probably be solve the inside the parenthesis, but this is obviously the same thing as integrating the whole, making the earlier splitting useless.

Another methodology would be through commuting

$$e^{hL_T} e^{hL_V}$$

But, as this doesn't work as they do not commute.

$$[L_T, L_V] \neq 0$$

We can see this directly for solving for q

$$[L_T, L_V]q = -\frac{V'(q)}{m} - 0$$

Thus this take naive one-step composition to be problematic.

I will note, this traditional splitting can work, it is only first order accurate and has a local error of $O(h^2)$

44.9.4 Deriving Strang Splitting

Taking our differential equation earlier, replacing T and V with A and B :

$$\dot{z} = (A + B)z$$

Using Taylor expansion we can expand to:

$$z(t + h) = \left[I + h(A + B) + \frac{h^2}{2}(A^2 + AB + BA + B^2) \right] z + O(h^3)$$

Thus, any second order method must reproduce the middle section.

To do that, let us assume a three state composition that has factors a and b

$$S(h) = e^{ahA} e^{bhB} e^{ahA}$$

We can thus expand the exponentials to

$$e^{ahA} = I + ahA + \frac{a^2 h^2}{2} A^2 + O(h^3)$$

$$e^{bhB} = I + bhB + \frac{b^2 h^2}{2} B^2 + O(h^3)$$

Multiplying together we get

$$S(h) = I + h(2aA + bB) + O(h^3)$$

Where as we previously noted that the exact solution should be

$$I + h(A + B) + O(h^3)$$

Therefore, a should equal one half while b should equal 1.

Creating a second order method with a local error of $O(h^3)$.

An important thing to note, while if you assume that the lower case a's could have differing values, we gain infinitely many solutions, but these solutions lose symmetry that would then lead to another first order equation.

44.9.5 Adjoint formulism

self adjoint formulisms are very simple, they are when a numerical method satisfies

$$\psi_h = \psi_h^* = (\psi_{-h})^{-1}$$

Where ψ^* is the inverse composition of ψ . ie if $\psi = x^1 \circ x^2$ then $\psi^* = x^2 \circ x^1$.

These self-adjoint is sometimes called time-symmetric as ...

44.9.6 Higher Order Splitting and Composition

"it is a meta-theorem of numerical analysis that second order methods often achieve the right balance between accuracy and complexity." - Strang

While most often 2nd order is used, it isn't the only one. There are methods for developing higher order methods, particularly used many places. The most obvious being simulations of the solar system.

So, if we start with the strang splitting

$$S_h^{[2]} = \varphi_{h/2} \circ \varphi_h \circ \varphi_{h/2}$$

The composition is then

$$\psi_h = S_{\gamma_s h} \circ S_{\gamma_{s-1} h} \circ \dots \circ S_{\gamma_1 h}$$

The order is atleast three if

$$\sum_{j=1}^s \gamma_j = 1$$

$$\sum_{j=1}^s \gamma_j^3 = 0$$

The smallest number of j that satisfies this equation is three. By imposing a symmetry where $\gamma_3 = \gamma_1$, one can relatively easily derive a solution for γ_1 and γ_2 .

Futher, in general one can take the equation

To find arbitrarily high orders of order $2k$ where $k \geq 2$.

There are many other ways to find higher order splitting, a more detailed observation of these will be in the description.

44.10 Discontinuous Galerkin Method: Self Gravitating Euler Equations

44.10.1 Planning

- Introduction
- Scope
- Motivating DG
- General Introduction to DG
- Physical Model Model
- Deriving Projection
- Structure preserving LDG

- Time Discretization
- Oscillation Elimination
- Structure Preserving Properties
- The Code itself(Flowchart)
- Results

44.10.2 Introduction

Hi, welcome to Discontinuous Galerkin Method: a focus on Self-Gravitating Euler Equation

44.10.3 Scope

Here we address the scope of this video. It is meant as an introduction into DG, using an example to help facilitate understanding. For both the purposes of challenging myself, I have chosen a particularly challenging one I have recently read about.

Taking a self-gravitating fluid and using the structure preserving capacity of DG to preserve positivity, ie that pressure, mass, energy, etc are all positive at all points, and well balanced, ie compressible fluid that admits hydrostatic equilibrium. For this video I won't go too far into the math of these more advanced subjects.

Also, for the rest of the video I will refer to the DG as "DG"

44.10.4 Introduction to DG

Before we get into the math, we will take a more intuitive look before we get into the math. A good way to start is to look

at the whole family tree. I assume anyone who is watching this has heard of and understands finite difference models; approximate spacial derivatives with stencils on point values. Further, we get to finit volume which trackes cell averages and evolve them via fluxes computed at cell faces from a Riemann solver. Then we come to continuous finite element or Galerkin who represents the solution as one global piecewise-polynomial function, continuous across element boundaries. Higher order and structural preserving methods are baked into the equations. The separation between DG and CG is that these polynomials are discontinuous, exactly as you would expect. The reason for this is that in many cases, the physical situations are partially discontinuous. Think of an explosion in a fluid, it takes time for the explosion to effect outer areas of the fluid, more time than would be found through a continuous polynomial.

essentially, we divide the space into elements

$$\Omega = \cup_e \Omega_e$$

We approximate the solution to the differential equation using polynomials

$$u_h(x, t) \sum_{j=0}^N u_j(t) \phi_j(x)$$

Next we write the PDE in its weak form

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{F} = S(u)$$

Multiply the test function ϕ then integrate by parts

$$\int_{\Omega} \phi \frac{\partial u}{\partial t} dV - \int_{\Omega} \nabla \phi \cdot \mathbf{F} dV + \int_{\partial \Omega} \phi \mathbf{F} \cdot \mathbf{n} ds = S(u)$$

They communicate between elements via fluxes, that can be defined via your physical system

Then we can generalize the fluxes into the residuals from the weak form equation used previously used to rearrange into

$$\mathbf{R}(\mathbf{U}) = \mathbf{R}_{\text{vol}} + \mathbf{R}_{\text{surf}} + \mathbf{R}_{\text{src}} \quad (44.7)$$

$$\mathbf{R}_{\text{vol}} = \int_K \nabla \phi \cdot \mathbf{F}(\mathbf{u}_h) dV \quad (44.8)$$

$$\mathbf{R}_{\text{surf}} = - \int_{\partial K} \phi \mathbf{F}^* dS \quad (44.9)$$

$$\mathbf{R}_{\text{src}} = \int_K \phi \mathbf{S}(\mathbf{u}_h) dV \quad (44.10)$$

$$\mathbf{R}(\mathbf{U}) = \int_K \nabla \phi \cdot \mathbf{F}(\mathbf{u}_h) dV - \int_{\partial K} \phi \mathbf{F}^* dS + \int_K \phi \mathbf{S}(\mathbf{u}_h) dV \quad (44.11)$$

$$M \frac{d\mathbf{U}}{dt} = \mathbf{R}(\mathbf{U}) \quad (44.12)$$

from here, using the basis functions from the mesh we get the mass matrix which is the matrix representation of the inner product on your finite-dimensional approximation space.

From here, the derivative of the functions as a function of time can be derived via

$$\frac{\partial u}{\partial t} = M^{-1} rhs(u_h)$$

Now, listening to all of this may seem fairly confusing, this is doubled by my admitted brevity, but I allow this brevity as I am going to take a concrete example as mentioned previously.

44.10.5 Physical Model

As mentioned earlier, we take the Euler-Poisson for a self-gravitating fluid, such as a toy model of a star. While keeping the simplicity of the model, I add a personal challenge by adding constraints of energy conservation and positivity. Using the general Euler-Poisson equation

insert long equation here

Then adding Polytropic steady-states and Lane-Emden equations in order to enforce EOS and positivity.

equation

I know I once again was very brief, but that is simply because the actual equations don't matter too much, just the general shape and view of them so that we can create the code.

44.10.6 Mesh and Projection

We start by assuming a uniformly space square/cube meshing
(image of square meshing)

We can then define n^+, n^- as the normal of the flux.

Next to define the projection, but to do that we define our test problem. Simply a weighting function within the polynomial space that allows us to integrate more easily.

(show examples in traditional methods)

Then we can define the projection matrix P via the L^2 projection:

$$\int_K uv dx = \int_K P(u)v dx$$

Now we can start moving backwards by rearranging the equa-

tion to get

$$P(u)(x) = \sum c_j v_j(x)$$

By defining the mass matrix as

$$M_{ij} = \int_K \phi_i v_j$$

where ϕ_i is the basis function, we use the nodal basis function, coefficients fairly easily looked up.

we can also define the right side vector as

$$b_j = \int_K u v_j dx$$

We can rearrange our original projection function

$$P(u)(x) = \sum (M^{-1}b)_j v_j(x)$$

44.10.7 Flux

We start with a basic piecewise function for flux of a fluid.

$$F^{hllc}(U_L, U_R; \mathbf{n}) = \begin{cases} \mathbf{F}(U_L), & \text{if } 0 \leq S_L, \\ \mathbf{F}_{*L}, & \text{if } S_L \leq 0 \leq S_*, \\ \mathbf{F}_{*R}, & \text{if } S_* \leq 0 \leq S_R, \\ \mathbf{F}(U_R), & \text{if } 0 \geq S_R, \end{cases}$$

This is the standard HLLC or Harten–Lax–van Leer–Contact flux solver. As this is more an introduction into DG than fluid dynamics I won't go through and expand all of these out, but essentially, it approximates the Riemann problem to approximate the flux of conserved quantities like energy, momentum, mass, etc.

I also slightly modify it in order to enforce Well-balancedness by adding the earlier Lane-Emden equation to the flux equation.

44.10.8 Scheme

Now building the actual numerical method. Before we get to the well-balanced, energy-conserving version, I want to build the plain scheme first, so that when I show you exactly how it fails near equilibrium later, you can point to the specific term responsible, rather than just taking my word for it.

Let's write out our conserved variables, flux, and source term the way we did generically before, but now specifically for the self-gravitating Euler system in d spatial dimensions:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho \mathbf{u} \\ E \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}_d \\ (E + p) \mathbf{u} \end{bmatrix}, \quad \mathbf{S}(\mathbf{U}, \nabla \phi) = \begin{bmatrix} 0 \\ -\rho \nabla \phi \\ -\rho \mathbf{u} \cdot \nabla \phi \end{bmatrix}.$$

Here $E = \rho e + \frac{1}{2} \rho |\mathbf{u}|^2$ is the usual total (non-gravitational) energy, e is the specific internal energy fixed by an equation of state, and for an ideal gas $p = (\gamma - 1) \rho e$. With this notation, our full model compresses down to

$$\mathbf{U}_t + \nabla \cdot \mathbf{F}(\mathbf{U}) = \mathbf{S}(\mathbf{U}, \nabla \phi), \quad \Delta \phi = 4\pi G \rho.$$

ϕ is the gravitational potential and G the gravitational constant. That second equation, Poisson's equation, is the entire reason this problem is harder than an ordinary Euler solve: it isn't fixed but rather generated by the fluid, it changes shape every time the density field changes, and every single timestep we need to re-derive a self-consistent gravitational field from whatever the density happens to look like at that instant.

44.10.8.1 Why local DG, specifically

There are two broad strategies exist for handling the gravity term numerically. One is to treat the Poisson equation as a pseudo-hyperbolic problem, giving it an artificial wave speed and a relaxation time so the same Riemann-solver machinery built for the Euler equations can be reused directly on the gravity sector too. The other, which is the one this scheme takes, is to solve the elliptic Poisson equation directly. Direct solves used to be avoided in this setting because they historically meant either an FFT-based Poisson solver (fast, but restricted to very particular, usually periodic, domain shapes) or a multipole expansion (flexible, but expensive and awkward to make high order). Local DG sidesteps both complaints: it was originally invented specifically to let a DG method handle equations with second derivatives without ever forming a second derivative explicitly, and it inherits all of the DG machinery's flexibility with boundary conditions and geometry for free. That's the actual technical justification for reaching for LDG here rather than, say, a finite-difference Poisson solver bolted onto a DG hydro code.

44.10.9 Standard LDG

So concretely: how do we get ϕ out of a DG method when Poisson's equation has a second derivative in it, and DG's whole weak-form machinery is built assuming one derivative per equation? This is exactly the "Local" in Local DG. Following Cockburn and Shu's original construction, we introduce an auxiliary variable standing in for the gravitational force itself,

$$\mathbf{g} = \nabla \phi,$$

and rewrite the whole system as a genuinely first-order system, no second derivatives anywhere:

$$\begin{cases} \mathbf{U}_t + \nabla \cdot \mathbf{F}(\mathbf{U}) = \mathbf{S}(\mathbf{U}, \mathbf{g}), \\ \mathbf{g} = \nabla \phi, \\ \nabla \cdot \mathbf{g} = 4\pi G \rho. \end{cases}$$

Now every equation is first order, so we throw the exact same weak-form recipe from earlier at each piece independently. We work on a mesh $\mathcal{T}_h$ of elements K , with $\mathcal{E}_I$ the interior faces and $\mathcal{E}_B$ the domain boundary. On a shared face $\mathcal{E}$ between two elements K^+ and K^- , with outward normals $\mathbf{n}^+, \mathbf{n}^-$, define the usual average and jump of a scalar function u and a vector function $\mathbf{q}$:

$$\{\{u\}\} = \frac{1}{2}(u^+ + u^-), \quad [[u]] = u^+ \mathbf{n}^+ + u^- \mathbf{n}^-,$$

$$\{\{\mathbf{q}\}\} = \frac{1}{2}(\mathbf{q}^+ + \mathbf{q}^-), \quad [[\mathbf{q}]] = \mathbf{q}^+ \cdot \mathbf{n}^+ + \mathbf{q}^- \cdot \mathbf{n}^-.$$

On a single element K , testing the three equations against $v \in V_h^k$, $\mathbf{w} \in \Sigma_h^k$, and $\psi \in V_h^k$ respectively (piecewise polynomial spaces of degree up to k , scalar and vector valued), and integrating by parts exactly the way we did for the single scalar conservation law earlier, we get:

$$\int_K (\mathbf{U}_h)_t v \, d\mathbf{x} - \int_K \mathbf{F}(\mathbf{U}_h) \cdot \nabla v \, d\mathbf{x} + \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \widehat{\mathbf{F}} v^{\text{int}(K)} \, ds = \int_K \mathbf{S}(\mathbf{U}_h, \mathbf{g}_h) v \, d\mathbf{x},$$

$$\int_K \mathbf{g}_h \cdot \mathbf{w} \, d\mathbf{x} = \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \widehat{\phi}_h \mathbf{w}^{\text{int}(K)} \cdot \mathbf{n}_{\mathcal{E},K} \, ds - \int_K \phi_h \nabla \cdot \mathbf{w} \, d\mathbf{x},$$

$$\int_K \mathbf{g}_h \cdot \nabla \psi \, d\mathbf{x} = \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} \psi^{\text{int}(K)} \widehat{\mathbf{g}}_h \cdot \mathbf{n}_{\mathcal{E},K} \, ds - \int_K 4\pi G \rho_h \psi \, d\mathbf{x}.$$

The superscript “int(K)” just means: take the value from

inside the element K we're currently integrating over, as opposed to the value carried over from the neighbor on the other side of the face ("ext(K)"). This distinction matters constantly in DG because our solution is discontinuous across faces by construction, so "the value at the face" is genuinely ambiguous until we pick a numerical flux to resolve it.

The first equation is exactly the fluid solve we already built, with $\widehat{\mathbf{F}}$ the HLLC flux from the previous section, evaluated using $\mathbf{U}_h^{\text{int}(K)}$ and $\mathbf{U}_h^{\text{ext}(K)}$ as the left and right Riemann states. The other two equations are the "elliptic half" of the scheme: a mixed formulation of Poisson's equation.

44.10.9.1 Fluxes for the elliptic part

Here's a subtlety that trips people up the first time they see LDG for an elliptic problem: gravity, unlike a fluid wave, doesn't propagate in a preferred direction. There's no upwind side to a Poisson equation the way there's an upwind side to a shock. So HLLC, or any Riemann-solver-based flux, simply doesn't apply here. Instead, following the standard LDG-for-elliptic-problems recipe, we use plain averages of ϕ_h and $\mathbf{g}_h$ across the face, corrected by a jump-penalty term that keeps the resulting linear system well posed:

$$\widehat{\mathbf{g}}_h = \{\{\mathbf{g}_h\}\} - C_{11}[[\phi_h]] - \mathbf{C}_{12}[[\mathbf{g}_h]], \quad \widehat{\phi}_h = \{\{\phi_h\}\} + \mathbf{C}_{12} \cdot [[\phi_h]],$$

with C_{11} a scalar penalty and $\mathbf{C}_{12}$ a vector penalty, both free parameters we get to choose. On a tensor-product (rectangular) mesh, a standard, well-behaved choice is $C_{11} = 1$ with $\mathbf{C}_{12} = (0.5, 0.5)^T$ in 2D or $(0.5, 0.5, 0.5)^T$ in 3D.

If you assemble the full linear system for $(\mathbf{g}_h, \phi_h)$ implied by

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these two weak forms, it has a block, saddle-point structure,

$$\begin{bmatrix} K & B \\ -B^T & S \end{bmatrix} \begin{bmatrix} \mathbf{g}_h \\ \phi_h \end{bmatrix} = \begin{bmatrix} g \\ f \end{bmatrix},$$

which is annoying to solve directly because it's not symmetric. The standard move is a Schur complement: eliminate $\mathbf{g}_h$ algebraically to get a smaller system purely in ϕ_h ,

$$A\phi_h = (S + B^T K^{-1} B) \phi_h = f + B^T K^{-1} g_h,$$

and now A is symmetric positive definite, which means a Cholesky factorization or a (preconditioned) conjugate-gradient solve both work reliably. Since our mesh never moves or changes over the course of the simulation, that Cholesky factorization can be computed exactly once, at the very start of the run, and reused at every single timestep afterward — the sparsity pattern and the matrix entries here depend only on geometry and the penalty parameters, never on the current fluid state. (For a periodic domain the block structure changes slightly, the reduced system loses its symmetry, and an LU factorization is used instead — same idea, slightly different linear algebra.) This is a genuinely nice practical property of building gravity this way: the expensive linear-algebra setup work happens once, not once per timestep.

Initial conditions for the fluid variables are set by the L^2 projection $\mathbf{P}$ we derived earlier: $\rho_h^0 = \mathbf{P}\rho^0$, $(\rho\mathbf{u})_h^0 = \mathbf{P}(\rho\mathbf{u})^0$, $E_h^0 = \mathbf{P}E^0$. Wrapping the whole thing in a forward-Euler step in time gives a complete, working numerical scheme:

$$\int_K \left(\frac{\mathbf{U}_h^{n+1} - \mathbf{U}_h^n}{\Delta t} \right) v \, d\mathbf{x} - \int_K \mathbf{F}(\mathbf{U}_h^n) \cdot \nabla v \, d\mathbf{x} + \sum_{\varepsilon \in \partial K} \int_{\varepsilon} \widehat{\mathbf{F}}^n v^{\text{int}(K)} \, ds = \int_K \mathbf{S}(\mathbf{U}_h^n, \mathbf{g}_h^n) v \, d\mathbf{x}.$$

Density, momentum, and energy evolve; gravity is re-derived

from the freshly updated density at the top of the next step, and the loop repeats. This is what I'll call the *standard scheme* for the rest of the video. Let me be fair to it: this is a real, working, high order accurate method. Run it on a smooth manufactured solution and you'll see the convergence rates you'd expect from the polynomial degree you chose. Run it on a genuine shock and, with an appropriate limiter, it captures the shock. It is not a broken scheme.

It does, however, have one specific, well-defined blind spot, and that blind spot is the entire reason the rest of this video exists.

44.10.10 Structure preserving DG

A huge fraction of astrophysically interesting configurations spend almost all of their time sitting extremely close to hydrostatic equilibrium: gravity pulling material inward, pressure pushing back outward, velocity essentially zero, the two forces balanced to enormous precision. That exact balance is

$$\mathbf{u} = 0, \quad \nabla p = -\rho \nabla \phi, \quad \Delta \phi = 4\pi G \rho.$$

A common and analytically tractable family of such equilibria is the *polytropic* equilibrium, where pressure and density are locked together by $p = \kappa \rho^\gamma$ for some positive constant κ . Substitute this and the hydrostatic balance into Poisson's equation and, after the classic Lane-Emden change of variables $\rho = \lambda \theta^n$ with $\gamma = (n+1)/n$, you land on the Lane-Emden equation. In spherical symmetry, with r the (rescaled) radius, it reads

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = -\theta^n.$$

This has closed-form solutions only for a few special integer values of n : $\theta_0(r) = 1 - r^2/6$ for $n = 0$ (an incompressible, $\gamma = \infty$

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fluid); $\theta_1(r) = \sin(r)/r$ for $n = 1$ (a $\gamma = 2$ fluid, and the one this video's numerical tests will actually use, since it's smooth, non-trivial, and exact); and $\theta_5(r) = 1/\sqrt{1+r^2/3}$ for $n = 5$ ($\gamma = 6/5$). For any of these, the full equilibrium profile is recovered from

$$\rho^e = \lambda (\theta(\mathbf{x}))^n, \quad \mathbf{u}^e = \mathbf{0}, \quad p^e = \kappa(\rho^e)^\gamma, \quad \phi^e = -\frac{\kappa\gamma}{\gamma-1}(\rho^e)^{\gamma-1}.$$

This is the analytic “star in a box” that our scheme will be asked to hold perfectly still.

Here's the actual failure mode, stated precisely. The standard scheme discretizes the pressure-gradient contribution to the momentum flux and the gravity source term $-\rho\nabla\phi$ using two completely independent numerical procedures: one is a DG surface/volume flux integral, the other is a DG volume source integral. There is no algebraic reason these two, separately truncated approximations should cancel exactly, even when the exact continuous quantities they're approximating do cancel exactly. What's left behind is $O(h^{k+1})$ discretization error sitting right on top of the equilibrium, at every single timestep. If the physical phenomenon you actually care about — a stellar oscillation mode, a slowly growing instability — is smaller in amplitude than that leftover truncation error, it's simply drowned out, and the only fix available to the standard scheme is to keep refining the mesh until the noise floor drops below your signal, which gets expensive fast, especially in 3D. This is the well-balanced (WB) problem, and a scheme that avoids it is called well-balanced.

There is a second, related failure worth separating out clearly, because it has a genuinely different cause: total energy conservation (TEC). Because gravity here is self-generated rather than externally imposed, the gravitational potential energy is honestly part of the system's total energy budget, and near equilibrium it's typically a large negative number nearly can-

celling a large positive kinetic-plus-internal energy. Any small, systematic energy leak in the numerical scheme is no longer a rounding-error-level nuisance; it can be comparable to, or larger than, the actual dynamical energy of the perturbation you're trying to resolve, especially over long integrations or during genuine gravitational collapse, where nonlinear growth will happily amplify a small numerical energy leak into a qualitatively wrong answer.

We want both properties simultaneously, and it turns out they require two largely independent fixes.

44.10.10.1 Fix one: splitting the potential for well-balancedness

The core idea is to split both density and gravitational potential into a known, static equilibrium piece and an evolving perturbation piece:

$$\rho = \rho^e + \rho^\delta, \quad \phi = \phi^e + \phi^\delta, \quad \mathbf{g}^e = \nabla \phi^e, \quad \mathbf{g}^\delta = \nabla \phi^\delta,$$

with ρ^e, ϕ^e computed once at initialization from the analytic equilibrium above (via the L^2 projection: $\rho_h^e = \mathbf{P}\rho^e$, and ϕ_h^e from actually solving the discrete Poisson problem for $4\pi G\rho_h^e$) and then frozen, unchanged, for the entire simulation. Only the perturbation fields $\rho_h^\delta, \phi_h^\delta$ are recomputed every timestep. On its own this split buys us nothing; the trick is entirely in how the numerical flux and source term are built around it.

44.10.10.1.1 The modified HLLC flux. Recall the contact property of HLLC: feed it two states at rest with equal pressure, $\mathbf{U}_L = (\rho_L, 0, p/(\gamma - 1))^T$ and $\mathbf{U}_R = (\rho_R, 0, p/(\gamma - 1))^T$, and it collapses exactly to

$$\mathbf{F}^{hllc}(\mathbf{U}_L, \mathbf{U}_R; \mathbf{n}) = (0, p \mathbf{n}^T, 0)^T.$$

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You can check this directly from the definitions: with zero velocity on both sides, the estimated wave speeds satisfy $S_L = -c_L \leq 0 \leq c_R = S_R$ and the middle speed $S_\star$ works out to exactly 0, which collapses the star-region states to something that leaves only the pressure term surviving in the momentum flux, with mass and energy fluxes vanishing identically. This “nothing is happening, only pressure pushes on the face” behavior is precisely the discrete analogue of hydrostatic balance, and it’s the property the well-balanced modification is built to exploit.

The fix: instead of feeding HLLC the raw conserved states, we rescale each side by the ratio of a face-averaged equilibrium pressure to that side’s own cell-interior equilibrium pressure:

$$\widehat{\mathbf{F}} = \mathbf{F}^{hllc} \left(\frac{p_h^{e,\star}}{p_h^{e,int(K)}} \mathbf{U}_h^{int(K)}, \frac{p_h^{e,\star}}{p_h^{e,ext(K)}} \mathbf{U}_h^{ext(K)} ; \mathbf{n}_{\mathcal{E},K} \right), \quad p_h^{e,\star} = \frac{1}{2} \left(p_h^{e,int(K)} + p_h^{e,ext(K)} \right)$$

Why this particular rescaling? Because at exact equilibrium, $\mathbf{U}_h^{int(K)}$ and $\mathbf{U}_h^{ext(K)}$ are both zero-velocity states with pressures $p_h^{e,int(K)}$ and $p_h^{e,ext(K)}$ respectively, and generally $p_h^{e,int(K)} \neq p_h^{e,ext(K)}$ — the equilibrium pressure genuinely varies across a face, that’s just what a stratified star looks like. The contact property only fires cleanly when both sides share the *same* pressure. So we rescale each side up (or down) to the common reference value $p_h^{e,\star}$ before calling HLLC, apply the contact property there (where it gives exactly $(0, p_h^{e,\star} \mathbf{n}^T, 0)^T$), and this becomes the discrete flux we actually use. The rescaling is undone implicitly, because it’s baked directly into the momentum source term derivation next, so the two exactly cancel at equilibrium.

44.10.10.1.2 The well-balanced source term. On the momentum equation, the full source is $\mathbf{S}^{[2]} = -\rho \nabla \phi = -\rho \nabla \phi^e - \rho \nabla \phi^\delta$. The perturbation piece is discretized in the ordinary, direct way. The equilibrium piece is where the actual balance

ing act happens, and it's worth walking through the algebra in full, because every line of it exists for a specific reason:

$$\int_K -\rho \nabla \phi^e v \, d\mathbf{x} = \int_K \rho \frac{\nabla p^e}{\rho^e} v \, d\mathbf{x}$$

(using the exact equilibrium relation $\nabla p^e = -\rho^e \nabla \phi^e$ to trade $\nabla \phi^e$ for $\nabla p^e / \rho^e$, a quantity we can evaluate directly from the known equilibrium profile). Now add and subtract the cell-average ratio $\bar{\rho}_K / \bar{\rho}_K^e$ inside the integral:

$$= \int_K \nabla p^e \left(\frac{\rho}{\rho^e} - \frac{\bar{\rho}_K}{\bar{\rho}_K^e} \right) v \, d\mathbf{x} + \frac{\bar{\rho}_K}{\bar{\rho}_K^e} \int_K \nabla p^e v \, d\mathbf{x}.$$

The first term is the genuinely new part of the perturbation source, and notice it vanishes identically the instant ρ / ρ^e is constant across the cell — exactly what happens at equilibrium, where $\rho_h = \rho_h^e$ everywhere. The second term is integrated by parts, using precisely the same weak-form move that built DG in the first place, turning a volume integral of a gradient into a boundary flux minus a volume integral against ∇v :

$$\frac{\bar{\rho}_K}{\bar{\rho}_K^e} \int_K \nabla p^e v \, d\mathbf{x} = \frac{\bar{\rho}_K}{\bar{\rho}_K^e} \left(\sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} p^e v \, \mathbf{n}_{\mathcal{E},K} \, ds - \int_K p^e \nabla v \, d\mathbf{x} \right).$$

Putting it together, and writing $\langle \mathbf{S}^{[2]}, v \rangle_K$ for the fully quadrature-discretized version (quadrature points $\mathbf{x}_K^{(q)}$ with weights $w_K^{(q)}$ inside K , and $\mathbf{x}_{\mathcal{E}}^{(\mu)}$ with weights $w_{\mathcal{E}}^{(\mu)}$ on each face):

$$\begin{aligned} \langle \mathbf{S}^{[2]}, v \rangle_K &= \sum_q w_K^{(q)} \left(\frac{\rho_h(\mathbf{x}_K^{(q)})}{\rho_h^e(\mathbf{x}_K^{(q)})} - \frac{\bar{\rho}_K}{\bar{\rho}_K^e} \right) \nabla p_h^e(\mathbf{x}_K^{(q)}) v(\mathbf{x}_K^{(q)}) \\ &+ \frac{\bar{\rho}_K}{\bar{\rho}_K^e} \left(\sum_{\mathcal{E} \in \partial K} \sum_{\mu} w_{\mathcal{E}}^{(\mu)} p_h^e(\mathbf{x}_{\mathcal{E}}^{(\mu)}) v^{\text{int}(K)} \mathbf{n}_{\mathcal{E},K} - \sum_q w_K^{(q)} p_h^e(\mathbf{x}_K^{(q)}) \nabla v(\mathbf{x}_K^{(q)}) \right) - \sum_q w_K^{(q)} \rho_h \end{aligned}$$

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That last line is just the ordinary perturbation-gravity contribution, $-\rho \nabla \phi^\delta$, tacked back on with no special treatment. The whole point of everything above it is that, at exact equilibrium, this discrete source term is built term-for-term to exactly cancel the modified HLLC flux from the previous paragraph. That's not a coincidence or a lucky numerical accident; it's a designed algebraic identity, and it's the reason we can actually *prove* the well-balanced property later instead of just observing it empirically on a few test cases.

44.10.10.2 Fix two: a conservative form for total energy

For energy conservation, the move is to stop evolving the ordinary non-gravitational energy E directly, and instead evolve a total energy that folds in gravitational potential energy from the start:

$$E_{tot} = E + \frac{1}{2} \rho \phi.$$

Why does this help? Because with some care you can rewrite the evolution equation for E_{tot} in genuinely conservative form — divergence of a flux, with no leftover volumetric source term at all:

$$\frac{\partial E_{tot}}{\partial t} + \nabla \cdot \left((E + p) \mathbf{u} + \mathbf{F}_g \right) = 0,$$

where the new piece, the *gravity energy flux*, is

$$\mathbf{F}_g = \frac{1}{8\pi G} \left(\phi \nabla \dot{\phi} - \dot{\phi} \nabla \phi \right) + \rho \mathbf{u} \phi.$$

It's worth actually seeing why this identity is true, rather than taking it on faith, because the derivation is short and it explains where every piece of $\mathbf{F}_g$ comes from. Start from the time derivative of the gravitational energy piece, $\partial_t(\frac{1}{2} \rho \phi)$, and add and subtract the divergence of the proposed flux and the ordi-

nary source term $-\rho \mathbf{u} \cdot \nabla \phi$ from the plain energy equation:

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \rho \phi \right) + \nabla \cdot \mathbf{F}_g - \rho \mathbf{u} \cdot \nabla \phi = \frac{1}{2} \dot{\rho} \phi + \frac{1}{2} \phi \dot{\rho} + \frac{1}{8\pi G} \left(\phi \Delta \dot{\phi} - \dot{\phi} \Delta \phi \right) + \nabla \cdot (\rho \mathbf{u}) \phi.$$

Now use mass conservation, $\dot{\rho} = -\nabla \cdot (\rho \mathbf{u})$, to cancel the $\phi \dot{\rho}$ term against the last term above, and use $\Delta \phi = 4\pi G \rho$ together with its time-derivative $\Delta \dot{\phi} = 4\pi G \dot{\rho} = -4\pi G \nabla \cdot (\rho \mathbf{u})$ to rewrite the remaining two terms. Every surviving term cancels in pairs, and the whole expression collapses to exactly zero. Add this identity to the ordinary energy equation and you land on the conservative form above. It's a genuinely satisfying piece of algebra: the messy, seemingly ad hoc combination of terms in $\mathbf{F}_g$ isn't arbitrary at all, it's precisely what's needed to make two independent physical statements (mass conservation and Poisson's equation) conspire to produce a clean conservation law for energy.

There's an obvious wrinkle: $\mathbf{F}_g$ contains $\dot{\phi} = \partial \phi / \partial t$, a time derivative of a field, which isn't something we normally track explicitly in a DG code. The way around it is to note that $\dot{\phi}$ itself satisfies another Poisson equation, obtained by differentiating $\Delta \phi = 4\pi G \rho$ in time and using mass conservation again:

$$\Delta \dot{\phi} = 4\pi G \dot{\rho} = -4\pi G \nabla \cdot (\rho \mathbf{u}).$$

So we solve one additional elliptic problem, every stage, using the exact same LDG mixed formulation as before, now with $-4\pi G \nabla \cdot (\rho \mathbf{u})_h$ as the source instead of $4\pi G \rho_h$, to obtain both $\dot{\phi}_h$ and its gradient $\dot{\mathbf{g}}_h$.

There's one more, purely numerical, subtlety worth being upfront about, because it's the kind of thing that will cost you a full order of accuracy if you don't know to watch for it. If you discretize the right-hand side $-4\pi G \nabla \cdot (\rho \mathbf{u})_h$ naively — literally differentiating the DG polynomial $(\rho \mathbf{u})_h$ pointwise inside each cell — you lose an order of accuracy, because differen-

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tiating a degree- k polynomial approximation generically only gives you a degree- $(k - 1)$ -accurate approximation of the true derivative. The fix, again, is to reach for the DG weak form itself rather than a pointwise derivative: apply summation-by-parts to move that derivative off of $(\rho \mathbf{u})_h$ and onto the test function ψ instead, at the cost of picking up a boundary term built from the already-computed mass flux $\widehat{f}^{[1]}$:

$$\int_K \dot{\mathbf{g}}_h \cdot \nabla \psi \, d\mathbf{x} = \int_{\mathcal{E}} \psi^{\text{int}(K)} \widehat{\mathbf{g}}_h \cdot \mathbf{n}_{\mathcal{E},K} \, ds + \int_K 4\pi G (\rho \mathbf{u})_h \cdot \nabla \psi \, d\mathbf{x} - \sum_{\mathcal{E} \in \partial K} \int_{\mathcal{E}} 4\pi G \widehat{f}_{\mathbf{n}_{\mathcal{E},K}}^{[1]} \psi^{\text{int}(K)} \, ds$$

This one small technical patch restores the full design order of accuracy for $\dot{\phi}_h$, and by extension for the whole energy flux; skip it and you'll still get a scheme that conserves energy (that property doesn't depend on this fix), but one whose overall convergence rate quietly degrades, especially on coarser meshes.

With E_{tot} as the evolved variable, we can repackage the whole system uniformly:

$$\mathbf{W} = \begin{bmatrix} \rho \\ \rho \mathbf{u} \\ E_{tot} \end{bmatrix}, \quad \mathbf{F}(\mathbf{W}) = \begin{bmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I} \\ (E + p) \mathbf{u} \end{bmatrix}, \quad \mathbf{G}(\mathbf{W}) = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{F}_g \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ -\rho \nabla \phi \\ 0 \end{bmatrix},$$

$$\frac{\partial \mathbf{W}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{W}) + \nabla \cdot \mathbf{G}(\mathbf{W}) = \mathbf{S}.$$

One bookkeeping detail that turns out to matter enormously in practice: at every stage we still need the ordinary energy E (to get pressure out of the equation of state), but we're now evolving E_{tot} . The conversion is $E_h = (E_{tot})_h - \mathbf{P}(\frac{1}{2} \rho_h \phi_h)$ — and that $\mathbf{P}$, the L^2 projection, is not optional decoration. If you instead compute $\frac{1}{2} \rho_h \phi_h$ pointwise at quadrature points and skip projecting the product back into the DG polynomial space V_h^k , the formula *looks* identical but the well-balanced cancellation from the previous section silently breaks, because the pres-

sure recovered this way is no longer guaranteed to equal p_h^e exactly at equilibrium. This is exactly the kind of easy-to-miss implementation detail that separates “it looks right and mostly works” from “it’s provably well-balanced,” and it’s the sort of thing I’d genuinely only have caught by trying to prove the property myself and watching the proof break.

44.10.11 Time Discretization

Everything above was written semi-discretely: space is discretized into the DG polynomial spaces, but time is still continuous, giving us an ODE system of the schematic form

$$\mathbf{W}_h^{n+1} = \mathbf{W}_h^n + \Delta t \mathcal{H}(\mathbf{W}_h^n, t^n),$$

where $\mathcal{H}$ is shorthand for the entire right-hand side we just spent two sections deriving: the modified HLLC flux, the well-balanced source term, and the gravity energy flux. (The standard, non-structure-preserving scheme has an analogous operator, which I’ll call $\mathcal{L}(\mathbf{U}_h^n, t^n)$.) Forward Euler, as written, is only first order accurate in time, which would badly undercut all the care we just took getting high spatial order — there’s no point running a P^3 DG method in space if a first-order time integrator is going to wash out the accuracy anyway. The standard fix, and the one used throughout this scheme, is strong-stability-preserving Runge-Kutta (SSP-RK), specifically chosen because it’s built as a convex combination of forward-Euler steps, which means any stability or positivity property you can prove for forward Euler carries over automatically to the full multistage method, with no extra work.

Second order SSP-RK:

$$\mathbf{W}_h^{(1)} = \mathbf{W}_h^n + \Delta t \mathcal{H}(\mathbf{W}_h^n, t^n), \quad \mathbf{W}_h^{n+1} = \frac{1}{2} \mathbf{W}_h^n + \frac{1}{2} \left(\mathbf{W}_h^{(1)} + \Delta t \mathcal{H}(\mathbf{W}_h^{(1)}, t^n + \Delta t) \right).$$

Third order SSP-RK:

$$\begin{aligned}\mathbf{W}_h^{(1)} &= \mathbf{W}_h^n + \Delta t \mathcal{H}(\mathbf{W}_h^n, t^n), \\ \mathbf{W}_h^{(2)} &= \frac{3}{4}\mathbf{W}_h^n + \frac{1}{4}\left(\mathbf{W}_h^{(1)} + \Delta t \mathcal{H}(\mathbf{W}_h^{(1)}, t^n + \Delta t)\right), \\ \mathbf{W}_h^{n+1} &= \frac{1}{3}\mathbf{W}_h^n + \frac{2}{3}\left(\mathbf{W}_h^{(2)} + \Delta t \mathcal{H}(\mathbf{W}_h^{(2)}, t^n + \frac{1}{2}\Delta t)\right).\end{aligned}$$

Each stage is literally just another forward-Euler-style evaluation of $\mathcal{H}$, meaning the entire elliptic solve, modified HLLC flux, and gravity-energy-flux pipeline from the previous section gets called again, unchanged, at every stage. This reusability is one of the more pleasant features of how the scheme was built: none of the well-balanced or energy-conserving modifications care what order of Runge-Kutta method is wrapped around them, so raising the temporal order later is purely a matter of adding stages, with zero changes to the spatial operator itself.

Practically, the timestep is chosen from a standard CFL condition based on the fastest wave speed present in any cell,

$$\Delta t = \text{CFL} \cdot \frac{h}{\max_K (|\bar{\mathbf{u}}_K|_\infty + \bar{c}_K)}, \quad \bar{c}_K = \sqrt{\gamma \bar{p}_K / \bar{\rho}_K},$$

with h the smallest element diameter in the mesh and $\overline{(\cdot)}_K$ denoting the cell average. Notice gravity doesn't directly enter this CFL restriction — the elliptic solve is implicit (we're solving a linear system exactly, not marching it forward explicitly), so the only explicit wave-speed restriction left is the ordinary hydrodynamic one.

44.10.12 Ossilation Elimination

High order polynomials are wonderful at representing smooth flow and genuinely terrible at representing a shock. Near a real discontinuity — the blast wave from our earlier explosion example, say — a high-degree DG polynomial will ring: classic Gibbs-phenomenon over- and under-shoots that, in the worst case, can drive density or pressure negative at a quadrature point and crash the whole simulation outright.

The traditional fix here is a slope limiter, but limiters can be blunt instruments: applied indiscriminately, they can easily destroy the very fine cancellations we spent the last section engineering into the well-balanced source term and modified flux. Instead, this scheme reaches for a more recent idea, the oscillation-eliminating (OE) technique, which works by letting the high-degree moments of the polynomial solution decay through an artificial damping ODE, applied as a separate fractional step tacked on after each Runge-Kutta stage, rather than by clipping or reconstructing the polynomial directly

$$\frac{d\mathbf{U}}{dt} + \mathbf{L}_f(\mathbf{U}) + \Sigma(\mathbf{U})\mathbf{U} = 0 \quad \implies \quad \frac{d\mathbf{U}}{dt} = -\mathbf{L}_f(\mathbf{U}), \quad \frac{d\mathbf{U}_\sigma}{dt} = -\Sigma(\mathbf{U})\mathbf{U}_\sigma.$$

The first piece is just the ordinary DG spatial operator we already have. The second, $\Sigma(\mathbf{U})\mathbf{U}_\sigma$, is a purely local, per-cell damping term, and its coefficient ϑ_K^m is built from a smoothness indicator, summed over each cell's faces:

$$\vartheta_K^m(\mathbf{U}_h) = \sum_{e \in \partial K} \beta_e \frac{\sigma_{e,K}^m(\mathbf{U}_h)}{h_{e,K}},$$

where β_e is a local estimate of the maximum wave speed across face e (the spectral radius of the flux Jacobian in the normal direction), $h_{e,K}$ is roughly the distance from face e to the far side of the cell, and the indicator itself, measures how big the jump

in the m -th derivative across that face is, relative to how much the solution varies overall. In a smooth region, that jump is tiny compared to the solution's global variation, so ϑ_K^m stays small and the damping barely touches anything. Right next to a shock, the jump is large, ϑ_K^m spikes, and the damping aggressively suppresses the offending high-order moments — but, and this is the important structural feature, it leaves the lowest moment, the cell average, completely untouched by construction (ϑ_K^0 effectively doesn't act on the mean). The damping equation actually has a closed form: writing the solution in a hierarchical orthogonal basis $\phi_K^{(\alpha)}$, each higher moment simply decays exponentially at a rate set by the local indicator,

$$\mathcal{F}_{\Delta t} \mathbf{U}_h = \mathbf{U}_K^{(0)} \phi_K^{(0)}(\mathbf{x}) + \sum_{j=1}^k e^{-\Delta t \sum_{m=0}^j \vartheta_K^m(\mathbf{U}_h)} \sum_{|\alpha|=j} \mathbf{U}_K^{(\alpha)} \phi_K^{(\alpha)}(\mathbf{x}).$$

Because the cell average $\mathbf{U}_K^{(0)}$ is left alone, mass, momentum, and total energy on each cell are completely unaffected by the OE step; only the shape of the polynomial *within* the cell gets smoothed. That's exactly the property we need to keep conservation intact.

The remaining piece of care, specific to this well-balanced scheme, is where the damping is allowed to act. It's applied only to the perturbation part $\mathbf{W}^\delta$ of the solution, never to the frozen equilibrium background $\mathbf{W}^e$:

$$\mathbf{W}_h = \mathbf{W}_h^e + \mathcal{F}_{\Delta t}(\mathbf{W}_h^\delta).$$

If the damping were instead applied to the full solution, including the equilibrium piece, it would gradually smooth away the very steady-state structure the well-balanced property exists to protect — a hydrostatic star has real, physically meaningful higher moments in its density profile, and those aren't the "oscillations" we're trying to eliminate. Restricting the damping to

the perturbation alone means: on a problem that's exactly at equilibrium, $\mathbf{W}^\delta \equiv 0$, the damping has literally nothing to act on, and the well-balanced property survives the OE step trivially. On a genuinely perturbed or shocked problem, one can show the damping only introduces an error of the same order as the underlying DG truncation error itself, so smooth problems don't quietly lose accuracy — you only pay a price near an actual discontinuity, which is exactly where you want to be paying it.

For the very strongest shocks — the blast wave test later, where the central pressure jumps by two orders of magnitude — OE alone isn't quite enough to guarantee density and pressure stay positive everywhere, so it's paired with a separate, more surgical positivity-preserving (PP) limiter, which only activates at a quadrature point if density or pressure there is actually about to go negative, and otherwise leaves the solution untouched.

44.10.13 Structure Preserving Properties

It's worth stating plainly, and proving carefully, exactly what all of this extra machinery buys us, because the algebra above is dense enough that the punchline can get lost.

44.10.13.1 Theorem: well-balancedness

Claim. If the scheme is initialized exactly at a known hydrostatic equilibrium — $\mathbf{u}^0 = \mathbf{0}$, $p^0 = -\rho^0 \nabla \phi^0$, $\Delta \phi^0 = 4\pi G \rho^0$ — then for every subsequent timestep n ,

$$\rho_h^n = \rho_h^e, \quad (\rho \mathbf{u})_h^n = \mathbf{0}, \quad p_h^n = p_h^e, \quad \phi_h^n = \phi_h^e,$$

exactly, up to floating-point round-off, regardless of mesh resolution.

Proof sketch, by induction on n . Assume the claim holds at step n . Walk through the flowchart from the next section term by term. Since $\rho_h^n = \rho_h^e$, the perturbation density is exactly zero, so the perturbation Poisson solve returns $\phi_h^\delta = 0$ and hence $\phi_h^n = \phi_h^e$ exactly. From this, the recovered non-gravitational energy is $E_h^n = p_h^e/(\gamma - 1)$ and the recovered pressure is exactly p_h^e — this is precisely where the consistent L^2 -projection book-keeping flagged earlier becomes essential; without it this line fails. Feeding $\mathbf{U}_h^{n,\text{int}(K)} = (\rho_h^{e,\text{int}(K)}, \mathbf{0}, p_h^{e,\text{int}(K)}/(\gamma - 1))^T$ (and the analogous exterior state) into the modified HLLC flux, the contact property fires and gives $\widehat{\mathbf{F}}^n = (0, p_h^{e,*} \mathbf{n}^T, 0)^T$ — in particular the mass-flux component $\widehat{f}^{n,[1]} = 0$. Since the mass flux is zero everywhere, the second Poisson solve for $\dot{\phi}_h, \dot{\mathbf{g}}_h$ has a zero right-hand side, so $\dot{\phi}_h^n = 0, \dot{\mathbf{g}}_h^n = \mathbf{0}$, and the entire gravity energy flux vanishes identically, $(\widehat{\mathbf{F}}_g)^n = 0$. Finally, substitute all of this into the momentum update: the four resulting pieces — the pressure-gradient volume term, the “add-and-subtract” perturbation term from the source, the rescaled-pressure boundary term, and the perturbation-gravity term — cancel exactly in pairs, term for term, leaving $(\rho \mathbf{u})_h^{n+1} = (\rho \mathbf{u})_h^n = \mathbf{0}$. Since the mass and energy fluxes were already shown to vanish, $\rho_h^{n+1} = \rho_h^e$ and $(E_{\text{tot}})_h^{n+1} = (E_{\text{tot}})_h^n$ follow immediately, closing the induction. ■

The proof is mechanical once you have it laid out, but the punchline is genuinely strong: this isn’t an asymptotic statement about the error shrinking as $h \rightarrow 0$. The equilibrium is preserved exactly, on any mesh, at any polynomial order, because every single term that could disturb it has been shown to vanish identically rather than merely become small.

44.10.13.2 Theorem: total energy conservation

Claim. Under periodic boundary conditions, or compactly-supported boundary conditions (velocity vanishing on the bound-

ary), the scheme conserves discrete total energy exactly at every timestep:

$$\int_{\Omega} (E_{tot})_h^{n+1} d\mathbf{x} = \int_{\Omega} (E_{tot})_h^n d\mathbf{x}.$$

Proof sketch. Take the discrete E_{tot} update equation, set the test function $v = 1$, and sum over every cell K in the mesh. Every interior face appears in this sum exactly twice, once from each of the two cells sharing it, with opposite orientations of the outward normal — so every interior-face contribution from the energy flux and the gravity energy flux cancels exactly against its neighbor: whatever leaves one cell enters the other, unchanged. What’s left after that cancellation is purely a sum over boundary faces $\mathcal{E}_B$. Under periodic boundary conditions, boundary faces are identified in matching pairs on opposite sides of the domain and cancel each other the same way interior faces do. Under compactly-supported boundary conditions, velocity vanishes identically on $\mathcal{E}_B$, which makes both the ordinary energy flux and the gravity energy flux vanish there directly, since every term in each flux carries at least one explicit factor of $\mathbf{u}$. Either way, the total boundary contribution is exactly zero, and we’re left with $\int_{\Omega} (E_{tot})_h^{n+1} = \int_{\Omega} (E_{tot})_h^n$.

■

Notice this proof didn’t need the well-balanced machinery at all — TEC is a genuinely independent property, coming entirely from writing the energy equation in conservative form. In practice this shows up as a total-energy trace that’s flat to machine precision on a well-resolved, stable configuration, and — arguably more usefully — as an energy diagnostic you can actually trust while something dramatic is unfolding, like a gravitational collapse, precisely the regime where the standard scheme’s energy error tends to grow fastest and become least trustworthy.

44.10.14 The Code itself (Flowchart)

Putting all of the above in the order you'd actually implement it, here is one full Runge-Kutta stage, advancing from a known state $(\rho_h^n, (\rho\mathbf{u})_h^n, (E_{tot})_h^n)$ to the corresponding stage value:

- **Step 0 — initialization (once, at $t = 0$ only).** Project the physical initial conditions into the DG space: $\rho_h^0 = \mathbf{P}\rho^0$, $(\rho\mathbf{u})_h^0 = \mathbf{P}(\rho\mathbf{u})^0$, $E_h^0 = \mathbf{P}E^0$. Solve the equilibrium Poisson problem once, $(\mathbf{g}_h^e, \phi_h^e) = \mathcal{D}_1(4\pi G\rho_h^e)$, and freeze it for the rest of the run. Solve the initial perturbation Poisson problem to get ϕ_h^0 , and set $(E_{tot})_h^0 = E_h^0 + \mathbf{P}(\frac{1}{2}\rho_h^0\phi_h^0)$ — using the projection here, not a pointwise product, for exactly the reason flagged above.
- **Step 1 — gravity.** Given ρ_h^n , solve the perturbation Poisson problem $(\mathbf{g}_h^\delta, \phi_h^\delta)^n = \mathcal{D}_1(4\pi G(\rho_h^n - \rho_h^e))$ and recombine with the frozen equilibrium piece to get the total field, $(\mathbf{g}_h^n, \phi_h^n) = (\mathbf{g}_h^e, \phi_h^e) + (\mathbf{g}_h^\delta, \phi_h^\delta)^n$. Separately, solve the summation-by-parts-corrected Poisson problem $\mathcal{D}_2$ for the time-derivative fields, $(\dot{\mathbf{g}}_h^n, \dot{\phi}_h^n) = \mathcal{D}_2(4\pi G \nabla \cdot (\rho\mathbf{u})_h^n)$. Recover the ordinary energy, $E_h^n = (E_{tot})_h^n - \mathbf{P}(\frac{1}{2}\rho_h^n\phi_h^n)$, and from it the pressure via the equation of state.
- **Step 2 — fluid flux.** At every interior face, compute the equilibrium-pressure-rescaled states on each side and evaluate the modified HLLC flux $\widehat{\mathbf{F}}^n$ from those rescaled states.
- **Step 3 — gravity energy flux.** Using $\phi_h^n, \mathbf{g}_h^n$ from Step 1, $\dot{\phi}_h^n, \dot{\mathbf{g}}_h^n$ also from Step 1, and the mass-flux component $\widehat{f}^{n,[1]}$ from Step 2, assemble both the volume contribution to $\mathbf{F}_g^n$ inside each element and its face contribution $(\widehat{\mathbf{F}}_g)^n$.
- **Step 4 — assemble and update.** Combine the fluid flux (Step 2), the well-balanced momentum source term (built the same way as in Step 1's rescaling), and the gravity

energy flux (Step 3) into the full right-hand side $\mathcal{H}$, and take one Runge-Kutta stage update for $(\rho_h, (\rho \mathbf{u})_h, (E_{tot})_h)$.

- **Step 5 — oscillation control.** Apply the OE damping (and, for strongly discontinuous problems, the positivity-preserving limiter) to the perturbation part of the newly updated solution only, leaving the frozen equilibrium piece untouched.

Repeat Steps 1 through 5 once per Runge-Kutta stage (twice for the second-order method, three times for third order), then move on to the next timestep with a freshly computed CFL-limited Δt . The only genuinely expensive part of this whole loop is the pair of elliptic solves in Step 1 — but because the mesh is fixed for the entire simulation, the factorization of the underlying stiffness matrix (Cholesky for the compactly-supported case, LU for the periodic case) is computed exactly once, before the time loop even starts, and reused at every stage of every timestep, which keeps the per-step cost dominated by the same sparse triangular solves you'd need for any implicit elliptic problem, not by repeated factorization.

44.10.15 Results

Here I have the results in the form of fun cool animations

44.11 What is Velocity(joke video)

- Basic
- Average speed
- Derivative form
- Vector form

- Newtonian
- Newtonian + inertial reference frame
- Newtonian Covariance/Galilean Relativity
- Newtonian-Cartan/ general covariance
- Fluid
- Special Relativity
 - Coordinate Velocity
 - Four-Velocity
 - Proper Velocity
 - Rapidity
 - Relative Velocity
- General Relativity
 - Coordinate velocity
 - GR four-velocity
 - Tangent vector
 - Covariant velocity
 - Geodesic
 - Warp Drive
- Dynamics
- Lagrangian
- Hamiltonian
- Phase-Space Flow
- Poisson Structure
- Symplectic flow

44.11.1 Introduction

Let's define velocity, I mean its a pretty simple concept, you could ask a fifth grader what it is and they could define it with ease. I mean it is the simple equation...

$$V \equiv (Q_\xi, X) = \mathfrak{X}_{Q_\xi}[X]$$

$$\delta[Q_\xi] = K(\delta, \mathfrak{X}_{Q_\xi})$$

$$= \Omega_{\text{ext}}(\delta, \mathfrak{X}_{Q_\xi}) - G(\delta, \mathfrak{X}_{Q_\xi})$$

$$= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta)$$

$$+ \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta) - G(\delta, \mathfrak{L}_\xi)$$

$$= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta)$$

$$+ \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta)$$

$$= \Omega_{\text{ext}}(\delta, \mathfrak{L}_\xi) - G(\delta, \mathfrak{L}_\xi)$$

$$= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\delta) \lrcorner \vartheta(\mathfrak{L}_\xi)$$

$$- \Omega(\mathfrak{L}_\xi, \delta) - Q_{[\mathbb{X}(\mathfrak{L}_\xi), \mathbb{X}(\delta)]} - \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta)$$

$$- \left[- \oint_{\partial\Sigma} \mathbb{X}(\delta) \lrcorner \vartheta(\mathfrak{L}_\xi) - \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta) \right]$$

$$\begin{aligned}
 &= \delta \left[\int_{\Sigma} (\vartheta(\mathfrak{L}_{\xi}) - \xi \lrcorner L) \right] - Q_{\delta[\xi]} - \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta) \\
 &\quad + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_{\xi})]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_{\xi}) \lrcorner \vartheta(\delta) \\
 &= \delta \left[\int_{\Sigma} (\vartheta(\mathcal{L}_{\xi} g_{ab}, \mathcal{L}_{\xi} \psi^I) - \xi \lrcorner L) \right] - Q_{\delta[\xi]} \\
 &\quad + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_{\xi})]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_{\xi}) \lrcorner \vartheta(\delta) \\
 &= \delta \left[\int_{\Sigma} (\vartheta(2\nabla_{(a}\xi_{b)}, \mathcal{L}_{\xi} \psi^I) - \xi \lrcorner L) \right] - Q_{\delta[\xi]} \\
 &\quad + Q_{[\mathbb{X}(\delta), \xi]} + \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta)
 \end{aligned}$$

$$\begin{aligned}
&= \delta \left[\int_{\Sigma} \left(\vartheta(\nabla_a \xi_b + \nabla_b \xi_a, \mathcal{L}_{\xi} \psi^I) - \xi \lrcorner L \right) \right] - Q_{\delta[\xi]} \\
&\quad + Q_{[\mathbb{X}(\delta), \xi]} + \oint_{\partial \Sigma} \xi \lrcorner \vartheta(\delta) \\
&= \delta \left[\int_{\Sigma} \left(\vartheta(\partial_a \xi_b + \partial_b \xi_a - 2\Gamma^c_{ab} \xi_c, \mathcal{L}_{\xi} \psi^I) - \xi \lrcorner L \right) \right] \\
&\quad - Q_{\delta[\xi]} + Q_{[\mathbb{X}(\delta), \xi]} + \oint_{\partial \Sigma} \xi \lrcorner \vartheta(\delta) \\
&= \delta \left[\int_{\Sigma} \left(\vartheta(\partial_a \xi_b + \partial_b \xi_a - g^{cd}(\partial_a g_{db} + \partial_b g_{da} - \partial_d g_{ab}) \xi_c, \mathcal{L}_{\xi} \psi^I) - \xi \lrcorner L \right) \right] \\
&\quad - Q_{\delta[\xi]} + Q_{[\mathbb{X}(\delta), \xi]} + \oint_{\partial \Sigma} \xi \lrcorner \vartheta(\delta)
\end{aligned}$$

$$\dot{F} = (Q_{\xi}, F)$$

$$(Q_{\xi}, g_{ab}) = \mathcal{L}_{\xi} g_{ab} = 2\nabla_{(a} \xi_{b)}$$

$$(Q_{\xi}, \psi^I) = \mathcal{L}_{\xi} \psi^I$$

$$(Q_{\xi}, x^{\mu}) = \xi^{\mu}$$

$$\Gamma^c_{ab} = \frac{1}{2} g^{cd} (\partial_a g_{db} + \partial_b g_{da} - \partial_d g_{ab})$$

$$g^{ac} g_{cb} = \delta^a_b, \quad \nabla_c g_{ab} = 0$$

$$\begin{aligned}
(Q_{\xi}, Q_{\xi}) &= -G(\mathfrak{L}_{\xi}, \mathfrak{L}_{\xi}) \\
&= \oint_{\partial \Sigma} \xi \lrcorner \vartheta(\mathfrak{L}_{\xi})
\end{aligned}$$

... uhm, well... maybe I am wrong about that.

44.11.2 Basic

Ok, well then, let's start at the beginning. Velocity can generally be defined by the change in the distance traveled divided by the time it takes to travel.

$$v = \frac{\Delta x}{\Delta t}$$

If we take the infinitesimal transformation of such we get

$$v = \frac{dx}{dt}$$

But, as your middle school science teacher likely drilled in your head, velocity isn't just speed, it also has a direction, making a vector the easiest way to represent it.

$$\dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle$$

44.11.3 Newtonian

Now that we have a basic understanding of the math behind velocity we can start putting into context. Likely everyone here knows Newton's second law

$$\mathbf{F} = m \frac{d^2 \mathbf{r}}{dt^2}$$

solving for velocity gives

$$\dot{\mathbf{r}} = \int \frac{\mathbf{F}}{m} dt$$

Now this is all nice and good, but it forgets relativity, I don't mean special or general, I mean Galilean relativity. Keeping

that in track is very simple

$$\dot{r}' = \int \frac{\mathbf{F}}{m} dt - \dot{r}_{\text{reference frame}}$$

This brings us to our next problem, as having Newtonian mechanics rely on frames of references comes up with several problems, I am sure many people have heard high school science teachers telling you that there is no centrifugal forces, as it is a fictitious force created by frames of reference, to solve this problem we make Newtonian mechanics general covariant by using the Newton-Cartan method. Essentially taking a space-time approach similar to general relativity but making time absolute rather than relative.

Defining velocity as

$$V^\mu = \frac{dx^\mu}{dt} = (1, \frac{dx^i}{dt})$$

While this equation may look pretty simple, it is an extreme simplification. I really don't know the purpose of the Newton-Cartan other than to simply prove Einstein wrong.

44.11.4 Special Relativity

This talk or relativity branches cleanly into special relativity, as velocity truly begins to change. As time becomes another relative dimension to move through, which is a really funny idea to think about. Running through time.

Anyways, the velocity vector obviously becomes 4-d adding time to the mix

$$U^\mu = \frac{dx^\mu}{d\tau}$$

Where τ is proper time, or the subjective time of the travelers rather than those observing them. Take the tiny people on a relativistic baseball as compared to the batter waiting for them.

We can relatively easily then expanded this to our traditional relative speed.

Taking galelian relativity

$$v_{\text{rel}} = v_1 - v_0$$

and adding relativistic corrections

$$v_{\text{rel}} = \frac{v_1 - v_0}{1 - v_1 v_2 / c^2}$$

Because why not

44.11.5 GR

But, this method isn't enough, there is now the problem of absolute acceleration, creating paradoxes and and preferential frames of refrence, something physically nonsensical in a relative world without absolute space or time. Though, using our earlier covariant newtonian framework and combining that with Einstienian proper time we get an equation that looks the same

$$U^\mu = \frac{dx^\mu}{d\tau}$$

but now

$$g_{\mu\nu} U^\mu U^\nu = -c^2$$

Thus adding a curved geometry of space-time that adds further separation between observers. The rest also stays the same as SR but with further corrections, and can be expressed through the language of remanian geometry rather than Lorentzian

or geodesics, I won't get into all of that though.

Though, before we move on, I want to talk about the velocity of warp drives. If you haven't heard, a solution of GR allows for apparent faster than speed of light movement, really messing with our definition of velocity which in all cases must be less than the speed of light.

It doesn't really break anything as most solutions have it where the matter doesn't move at all but spacetime shrinks and grows behind it. Making its actual movement 0 m/s yet to an outside observer moving several times the speed of light as it moves a great "distance" in a very short time. Making definitions a bit confusing.

44.11.6 Dynamics

Ok, all of this spacetime stuff is really confusing and make a great deal of a mess. Lets move onto a completely different approach. Moving away from kinematics and into dynamics. Involving phase-space, where there aren't some defined set of dimensions of space and time, but an infinite amount of dimenisons of anything you could observe; momentum, position, time, heat, entropy, well... not really traditionally entropy. Anyways, rather than taking an infinite amount of dimensions, let's shrink it down to two, the two most generally used, momenta and position. Then we will add mass but as it is constant in our situation it is infinitely then and thus can be expressed in 2d

(show graph)

Here, momenta and position are not like their Newtonian cousins, but rather generalized. They are related pairs and aren't constrained in the way position and momentum generally are. Position can mean angle, position plus direction rolled in one, or all three dimensions rolled into one vector field. The

same with momenta, but to make this example concret, we will take the traditional picture of position and momenta to give that if one represents the flow of the vector field as H , or commonly referred to as the Hamiltonian, we can express the derivative of position as a function of time as

$$v = \dot{q} = \frac{\partial H}{\partial p_i}$$

Why, that is a really big question, just trust me.

Thus saying that velocity is a part of the vector field on phasospace

(show on graph)

but what is phase space, well... that is an even bigger question but I will take a quick jab at it.

First we introduce something called a Poisson Structure

$$\dot{f} = \{f, H\} = \frac{\partial f}{\partial q^i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q^i}$$

Where the entire vector field can thus be written as

$$X_H = \{\cdot, H\}$$

Where the dot represents all possible observables, like position, momenta, or really anything else. The geometry behind the vector field is the phase-space. This geometry is something called symplectic, but I won't get into that.

44.11.7 Other

Now, let's speed run some others. Fluid velocity is a vector function that attaches a newtonian velocity to every point in the velocity.

Quantum mechanics adds fuzzy probability to velocity

$$\mathbf{v} = \frac{\mathbf{j}}{|\psi|^2}$$

Kinetic theory combines probability of QM with the continuum of fluids

Geometric mechanics defines it as a “natural element of a tangent bundle,” which seems hard to explain

Chapter 45

GHMDH: Kerr black hole inside of Star

45.1 Scope

Here I outline my plan to create a simulation of a kerr black hole inside of a star, in order to develop my understanding of BSSN while adding th extra challenge of MHD.

45.2 Stage 1: Vacuum Kerr Black Hole

45.2.1 Initializing Values

We start with taking in the data, then transferring it into ADM and then BSSN.

By follutiong the 4D hypersurface by

$$M = \cup_t \Sigma_t$$

Taking

$$g_{\mu\nu} \rightarrow (\gamma_{ij}, \alpha, \beta^i)$$

The spatial metric, the lapse, and the shift.

Also the introducing the extrinsic curvature.

The decomposing it.

Next we introduce the BSSN formalism, which creates the unit-determinant condition, or the contradiction of the conformal Christoffel symbol.

45.2.2 Hamiltonian Constraints

From here we end up at the Hamiltonian constraints, that force conservations of momentum and energy just like traditional hamiltonians, they just aren't the only enforcement in BSSN that keep things only partially hyperbolic.

45.2.3 Integration

45.2.3.1 RK4

We start with a traditional RK4 model for integration. Inside each section we have the traditional residual, Sommerfeld radiative boundary conditions, and k0 dissipation

45.2.3.2 RHS

Chapter 46

DG CCSN

46.1 Scope and goal

In general, I want to have a psuedo-toy simulation of a CCSN that uses DG. To constrain myself further, for simplification I will make it fully Newtonian(no GR or SR) but to add challenge I enforce positivity and well-balanced.

46.2 Mathematical Model

Rather than writing down everything, in general the math starts with the Compressible Euler, adds Polytropic steady-states and Lane-Emden equation to the hydrostatic equilibrium equation. Finally, add boundary conditions in order to enforce energy conservation.

46.3 General DG

First one creates a mesh of the given space to calculate the projection martrix.

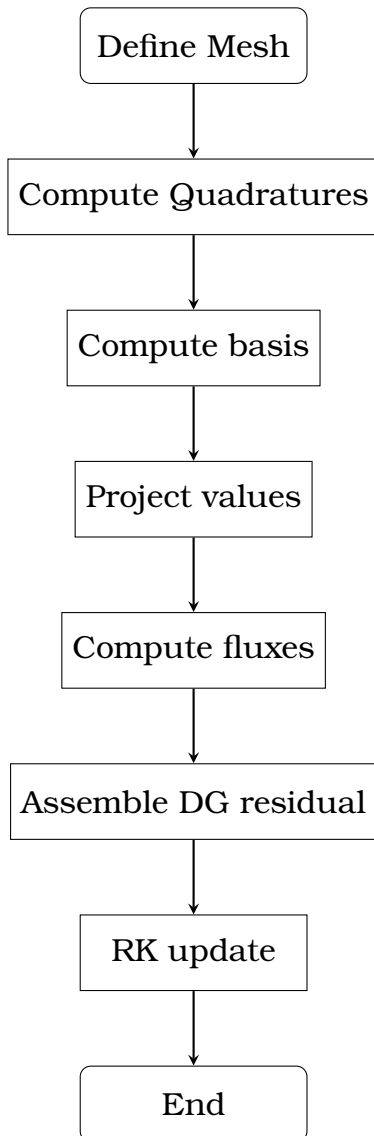
DG projects a PDE onto a functional space of polynomials using the projection matrix. Compute the fluxes of the fluid, then fluxes of energy. Then combine this all together for the residual.

Finally, time step with choosen model.

46.4 My DG

Essentially what I have talked about with the equations I use. For the sake of brevity the link to the paper that has this all writen out is here.

One thing I will add is a flow chart for my code



46.4.1 AI overview

Admittedly, much of this code is written by AI, I had it create documentation which I have here.

46.5 General Notes

Running the algorithm is very slow, far slower than anticipated, taking in total n hours to run the solution for first order polynomial, $16 \times 16 \times 16$, 3rd order Runge-Kutta, and Newtonian iteration.

46.6 Results

Chapter 47

WarpDrive Collision

47.1 Scope

Using Einstienstoolkit, I plan to develop a simulation of a warp drive, then use that to collide with an Earth like object.

Chapter 48

Projects for YouTube

48.1 Planck Force on Earth

48.2 Collapsing Universe

48.3 Post-Newtonian Simulations

48.3.1 General idea

To test the effects of differing modified GR theories. I would like to create N-body post-Newtonian codes. From here I would like to simulate a simple version of the solar system.

I would like to test:

- $f(R)$
- Higher dimensional Brans-Dicke gravity
- Kaluza-Klein Compact dimensions with that are large enough to effect the solar system

48.3.2 Notes on Brans-Dicke

48.4 CCSN via DG

- Newtonian gravity
- hydrodynamics
- SFHo EOS for Nuclear effects
- I need to transfer energy to temperature for the SFHo
- I am going to use Nodal-LGL for my basis

Chapter 49

The Eudaimonia Society

49.1 Initial Exploration

49.1.1 General Thoughts & Scope

Originally, I had begun this exercise by thinking of, if I were to create an institution like the Freemasonry, what would it look like. What doctrine would I lay at the foundation, traditions I would create, values to instill, how to even go about it, and much more. This is an attempt to write it out. One more pragmatic train of thought, I would likely use the general idea in creating a club at school, very similar to my previous SVA, but more evolved. This club would likely be more constrained, focusing on intellectual virtues and their relation to research rather than the broad idea of a society based upon my fundamental values and then using them to grow to be something more.

I will break this up into Moral development, intellectual development, social cohesion, and social impact.

I am thinking of using this idea for the YouTube Channel to an extent. Maybe...

49.1.1.1 Moral Development

The core of the entire project is to take the fundamental approach towards perfection to be the absolute goal. To take all things seriously, never do things half-heartedly, and don't mistakenly take irony to that which is sacred.

That we are all within our own personal design and can take control of our lives and personalities through the continuous cultivation of habits.

On a more direct statement; take accountability for all things (even that which many would consider beyond you), never do anything anti-social (in the psychological sense) even in the most slight way, value truth, be accurate and develop yourself epistemically and intellectually, pursue the development of your community in your own image and values, develop clearly defensible and defined values that are not vague bromides, hold yourself to your standards rigorously, think in the long term for your community, only acception to the anti-social rule would be when it conflicts with written values, and be fully conscious at all times, fully aware of your actions and their relation to your moral identity.

I personally find virtue ethics to be my way of thinking, so this may be a core tenet or may be too constricting.

Beyond that general view of altruism, self-reliance, and logic being virtues to pursue.

49.1.1.2 Intellectual Development

To understand things at their most fundamental level. To understand epistemology, statistical thinking, critical thinking, interdisciplinary thinking, metaphysics, and have a fully simulated world-view rather than implicitly taking on assumptions. This is the core tenet, not to take on others assumptions passed through pop culture.

49.1.1.3 Social Cohesion

Generally thinking a somewhat brotherhood of intellectuals or in close enough fields. To discuss and understand each other and build one another up.

Beyond that many questions arise of what structure would best work for this.

49.1.1.4 Social Impact

Any general social impact will do

49.1.2 First layer of Refinement

The first questions I want to ask consist of:

- Mission Statement
- Structure
- Symbols(if any)
- Meetings

49.1.3 Mission Statement

IDK...

49.1.4 Structure

A both heirercical and collaborative. With strict order on tasks; programmers of the mission, chapter directors, etc. These people have clearly defined stewardships and authority with complete authority over their specific stewardship; though, a

culture of collaboration. Meetings will be scheduled to have democratic-like discussion on actions.

49.1.5 Meetings

In the action of the goal, meetings consists of discussions, debates, and lectures.

Part IV

Essays

Chapter 50

Introduction

This part is used as a collection of essays on whatever topic.
Whether it be philosophy, physics, economic, whatever.

Chapter 51

Post-Surgery

51.1 Overview

51.2 1/9/2026 Essay: Cultivation in Degregation

As mentioned in previous chapters, the development of virtue stands at the cornerstone of the existence of man. Thus, how to develop it is key.

The question of how to develop virtue, what virtues to develop, etc; differ from person to person. I will focus on myself for this essay, on a practical development, rather than in the abstract general way I would traditionally go for.

Specifically, I will focus on my recent 'degregation' in many fields. What I mean is I have been struggling with health problems that caused constant pain, nausea, and brain fog. To combat this I used "quick" and "automatic" gratification. Watched lots and lots, of Instagram videos and YouTube shorts. With quick and simple comedy TV in the background. While it is difficult to divide what is caused by using these simple

causes of gratification and what is caused by my still recovering body, but I am having trouble focusing, get tired easily, less intrinsically motivated, and other similar malignities.

I have a belief that a person can create themselves based upon their habits. Thus, that is what I design to do, create myself in the image of my highest values. I desire that I am someone who enjoys challenge for its own sake (especially intellectual challenge), persistent high energy, optimistic outlook, pro-social tendencies, high focus, and the continual push towards greater virtue in all fields of life.

To reach these goals I have decided to pick up “The Karamazov Brothers,” a book I had abandoned earlier due to my declining condition. I did this not only for enjoyment, but to accomplish my goal. To slowly focus my mind on something that lasts longer than a minute, to remind myself of my enjoyment of intellectual type activities and engage my mind in something that could challenge it, and to reconnect myself with my faith.

I am also reading “Numerical Hamiltonian Problems,” trying to learn to code a website, code personal pet projects, watch educational and long-form youtube videos rather than shorts, exercise each day, and write each day. I do all of these things to re-develop my attention, enjoyment of effort, regain my energy, and to develop my skills.

Developing myself on the social side of things is a bit more challenging. Currently as of writing this essay I am still isolated from my peers, but I have a plan. Teaching Sunday school, volunteering for church outreach groups, returning to my scout troop, participating in JROTC volunteer activities, teaching karate, and so much more. Things to redevelop my enjoyment and understanding of social responsibilities and natural altruism.

With this current plan in motion, I will start making pur-

poseful and willful actions in order to develop in the way that I design.

51.3 3/.. /2026: New Beginnings and Failures

Chapter 52

Metaphysics

The verbal interpretation, on the other hand, i.e. the metaphysics of quantum physics, is on far less solid ground. In fact, in more than forty years physicists have not been able to provide a clear metaphysical model. — Erwin Schrodinger

52.1 Scientific Materialism and God

Despite the fact that there exist no formal term for such philosophies(I should come up with a name for it, Christian Scientific Materialism?)it in a broad stroke is most likely the most common view of meta-physics of all, though I plan to expand upon the concepts more deeply than that which is common. In essence this view can be summarized by the combination of two seemingly contradictory concepts:

First, is scientific materialism, this is the idea that there is nothing further than the material, that reality can be derived empirically through the laws of science and nature.

Second, the idea of the Christian God. That there exist an external omnibenevolent, omnificent, omnipotence, omnipres-

ence, and omniscience as described by the Christian Bible. Later I will explain my faith in more detail, but for now this is sufficient.

You may be wondering how these concepts can be combined due to their seemingly contradictory nature. This is a simple enough question, if the material is all that exist how can a non-material God exist? The answer in all reality is very complex, but it can be simplified to the fact that I view the material world to follow the philosophy of scientific materialism but there to also be a non-material "world" that God exists within. The reality of such a non-material world is sadly beyond my grasps due to my life within the material, I make no assertion that I understand such things. Partly because by definition such things are not possible to understand, like a 4th spacial dimension. While we can understand it through analogies and abstraction based upon prior more concrete idea, but sense we have no clear or objective priors we have no capacity to do such.

You can simplify such thoughts through formal logic

Let:

X=Material

Y=Non-Material

C=Laws of nature and reality

C(X)=Matter/Energy/Both interactions(all material action)

C(X,Y)= Observable phenomena A=Scientifically explainable phenomena

$$X \implies C$$

$$C(X) \iff C(X) \models A(X) \top$$

$$\forall C(X, Y) \exists C$$

$$\nexists Y(C) \therefore \nexists A(Y)$$

52.2 Scientific Realism vs Anti-realism

What is scientific realism and anti-realism? It is the debate between whether our theories of science are ontologically real, or simple predictive models.

My view is that it depends, when it comes to macroscopic and easily observable and prove able; these theories aren't true. You may be thinking the opposite, that it is those that are most definitively true and the rest that aren't. Though I would argue these are the models; a ball don't fall; a collection of subatomic particles follow their geodistic, which is restricted by other particles and forces, and so much more. This can be applied to all things other than the most fundamental of physics. It can especially true for the softer sciences; chemist, biology, psychology, and so many more.

While I think science does study the metaphysical truth through fundamental physics and abstract mathematics, on a broad stroke it does not.

52.3 Causality

Now I am sure you are confused, I mentioned I would dive into deeper questions related to free will, determinism, what is real, divine voluntarism, occasionalism, how quantum mechanics plays within this and other 'deeper' concepts, but we must take things one at a time.

So, let us dive in. What is causality, it is very simple. It is the concept that things happen because other things had happened first. That all things have a 'cause' and that that thus causes the effect(then they later become the cause of a later thing)

I can hear you screeching that this is non-sense, who in

their right mind would question something as basic as cause and effect? Why even look into it. First, that is extremely intellectually lazy, all things have debates against it, you should always look into it. Though, more importantly, there actually are debates against it.

The easiest to dismiss is Hume's skepticism, that causality is simply a human made concept. That things doesn't really happen because of another action but simply that both actions are just sequences of events. Though, I would argue that this pedantic argument of definitions doesn't truly work. On an epistemological basis, causality is easier for humans to understand, and though breaking things into chunks we may better understand the world. That either concept is equally true, but one concept as a greater pragmatic 'truth' and you can argue that that correlates to a better empirical truth. That there is even a speed of causality written in the nature of reality. (The speed of light) A bit more deeply, once you quantize to the plank time, things truly do divide. Something happens, then a next thing happens. The complexity and abnormal reality of quantum mechanics makes this argument a little fuzzy, and thus leads me to my next paragraph.

Now those unfamiliar with quantum mechanics may be confused, how does quantum mechanics may be a little confused. I will attempt to explain it to the best of my ability, but the reality of quantum mechanics is extremely complex and without a clear understanding of differential equations, discrete mathematics, linear algebra, and much more it is impossible to understand. I know many science communicators claim to teach "quantum physics and mechanics" to they layman, but this is a lie. These half backed analogies are not quantum mechanics. It does not come close, but luckily for me I am the only one reading this book I will be able to understand my attempt of analogizing(?) the philosophical question.

In essence, in quantum mechanics. Things are not local, for

many reasons. The most basic is the idea that particles do not exist in the way we can imagine. They exist in multiple places at the same time[not really but useful analogy], simply with different probabilities(don't pretend to understand, nobody does, they just prove the math and accept as is.) You know what, that analogy was not enough. I will explain in more detail. Particles do not exist(not really). Simply quantum fields exist. The electron field for electrons, the photon for photons, and so on. The particles are not particles as we imagine, but excitations within the field. It is a bit more complex than that, they can act as particles when observed but that is a whole other can of worms. Now back to the matter at hand, now waves act in certain ways that is hard to explain. Due to their quantum nature they must be decomposed first through Fourier modes to under stand more clearly. This is an example of a very simple form:

$$\psi(x) = \frac{1}{2\pi} \int \phi(p) e^{ipx} dp$$

Now I won't explain the equation in its entirety but I will say this. That you cannot solve it for an absolute certainty of position(not that you could get a "true" answer even if you could given that is is a Taylor Mode and not an analytical equation). Now you may think this is just a quirk of the math, but it is not. It is real for reason that could take up the entirety of this book if I truly understood them.

Next, quantum mechanics goes against causality through entanglement. Now what do I mean, when people refer to the speed of light, they are actually referring to the speed of causality. To explain this is relatively simple with a good understanding of Special Relativity. It is obvious that the universe is invariant to position and velocity(this has been previously proven thoroughly through experiments in high speeds, specifically through electromagnetism but others can apply), but for this to be true there must be a finite constant cos-

mic speed(because it requires a Lorenz transformation[there is more explained why this si the only answer but I will leave this hear because I have much much more physics to talk about in a philosophy book.]) This is the speed of causality, the speed of information. If something were to go above this speed it would in essence go back in time. This can be proven rather simply through simple algebra:

$$t' = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \left(t - \frac{vx}{c^2} \right)$$

From here you can look at the square root and easily find that if $v > c$ then γ (the value of the first part with the $\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$) would be imaginary. Thus the t value(of time would then become negative, this obviously breaks causality.

Finally, back to quantum mechanics, as you know entanglement transfers information at faster than the speed. This seems to break causality.

Now I could keep on talking about arguments between quantum physics and causality, but I will only talk about one more. Virtual particles. Now first of all, what are virtual particles. Virtual particles are non-existent but existent particles that exist within the math of quantum field theory(and thus above my pay grade) but I will attempt to explain it. The idea in essence is that due to the uncertainty principle there exist small amounts of excitements within their fields. These excitements are not enough for an actual particle to exist in the way we imagine, but occasionally a particle can appear, primary through the use of "borrowing" energy from another particle or through creating a negative particle(not to be confused with anti-particles those are negatively charged, these have negative energy/mass) These virtual particles are especially important to particle interactions, what I mean is the fact that when

two particles are "close" enough together they excite the fields around them to create bosons(force carriers) to cause the actual interaction. As an analogy, imagine two electrons, when close enough together they excite the fields around them to create a virtual photon, the photon then interacts with one electron(and also effects the other electron negatively[it is almost impossible to explain]) and cause the electromagnetic repulsion between the two. Now, back to philosophy, what in the world does this ramble have to do with causality. That is a great question. It is that all of our interactions are based upon something that isn't real. Electrons don't actually create virtual photons, these photons don't create negative photons. Despite having a tangible effect, they don't actually exist. How can something cause another thing if when you break it down to its component parts, particles are behaving 'irrationally' being effected by mathematical objects.

Now after I have listed the critiques, you are probably very aware about how you assumption of the obvious state of causality was intellectually lazy, but you probably also realize I wouldn't have written all of this unless I have an answer to these critiques, or at the very least a possible answer. In all reality I could be wrong about this and causality could be false. Though I am never wrong, so that must not be true.

The uncertainty of particles doesn't necessarily falsify causality, it simply evolves it. We previously see causality as something that is caused by a specific thing, but it can be caused by a probability of things(if that make sense).

Next I will talk about virtual particles, and yes I hear you saying that I should go in the order of which I introduced the ideas, but I do not care. Now, virtual particles. How does causality work with things that are not 'real'? Now the most astute among you might claim, why would causality not apply, and you are exactly right! You all may feel a sense of betrayal, that it was simply a trick of the light. But now, listen

carefully, this is how philosophy works. Though asking questions that are hard to understand, questions that play tricks on our language, on our mind. Though these questions we refine our thoughts. This is the problem with modern philosophy, they confuse critiques with criticism. While criticism has a place within philosophy, it should not be as big as it is now. Now before I continue I will define why I consider a critique and a criticism as different things. A criticism takes a part an idea, concept, action, or really anything for the sake of taking it down. This can be justified if such ideas have no worth and the criticism is an attempt to prove it. A critique is a question used to refine a thought not destroy it.(though if the critique can not be resolved then it may.) This critique of virtual particles is not supposed destroy causality by any means, but like the earlier critiques, it is meant to refine our thoughts. It asks us to question what a cause is, what can be a cause. Can virtual particles be causes, modern science suggests yes, but this requires a refinement of our concept of causality. Of what is 'real.' In a later section I will delve into this in a future section, but now I will leave you to ponder.

Now finally, to entanglement. Does this refine our definition or break it? It breaks it, but due to the evidence surrounding causality, including the fact that the special and general relativity are simply logical derivatives of the idea of a finite special to causality suggests that our idea of entanglement is flawed not our idea of causality. Now what is the answer? In short, nobody knows. There are many theories trying to come up with a solution. These include Einstein-rosen bridge, superdeterminism, and local hidden variables theory. Though since these ideas are both extremely complex and offer little beyond the problem at hand I have elected to move on. Now you may say that my Faustian arrogance has lead me astray to ask questions I can't answer, and to that I say... No dummy.

Now you may think we are done, we are not. We haven't

even mentioned the nature of God in causality and the teleological nature of Hamiltonian mechanics. There is also further refinements of time which must be discussed to even further define causality.

52.4 Divine Law Theory of Causation

You thought I was done with with causality. You thought wrong.

Now, what is the Divine Law Theory of Causation. It is very simple, it is that God predefines the laws of physics and nature. This differs from the ideas of occasionalism. Occasionalism is that all actions are the conscious decision of God, that the uniformity of the laws of nature is simply a coincidence.

Now, why do I choice Divine Law Theory of Causation? While I do not attempt to venture into the idea that I can even begin to comprehend the machinations of the God most high. Though in my mind the idea of predetermined laws make more sense, a more efficient method for God. Though this is not my primary reason for this belief. As I will mention latter in this chapter I will explain several beliefs of my predicated upon divine law theory of causation. Also, it is more in-line with scientific materialism

52.5 Material, Immaterial, and Reality itself

Now we are getting to the meat of things. What makes something real? What makes something itself? How do we define such things?

These are all great questions that I will go through one by one.

I will begin with what makes something real. While it is easy to look at things in the materialistic view, that of which that can be observed and have an observable effect is "real." No this has some implications, what about what about God(this is based on my assumption that he is 'real') but more importantly what about things that have an effect but of the concept of virtual particles.

First, the immaterial. Given the fact that by definition, these things are beyond us I will ignore them because we cannot gain any intuition from it. Just remember that it must be included within the concept of real, simply we are unable to define it accurately.

M= Material

I=Immaterial

$\mathbb{R}$ = *Real*

$$M \wedge I \subset \mathbb{R}$$

Finally, back to virtual particles, I have kept you in suspense enough. How do we satisfy the existence of virtual particles? Just as we have several times before, say it with me, we must refine the definition. Now, how to do this. We must first realize that at the subatomic level, no particles are particles in the way we describe it, as I mentioned earlier they are simply excitations within a field. They are energy in a specific manner. From here we can divide the material into two types; true material and partial material. The true material refers to what we generally consider matter, and the the partial(photons, gluons, and other massless but still interacting objects.) Which leads me to my final definition; material is that with observable effect[observable does not just simply mean seen, it can be any form of observation].

52.6 Identity

While many have abstract ideas on the identity of object, I will simply say that all subatomic particles are identical and thus it is not hard to define, and the others are simple as humans (another word defined by humans) define it. Good day gentlemen

52.7 SpaceTime

Space and time, the two things most clearly and least clearly understood at the same time. Things that we know do to our interactions with it every second (ha-ha) of our lives but still don't truly understand.

Many philosophers have questioned the reality of space and time. Are they simply human made? I would suggest not, the mathematical empiricism and importance of them in higher level physics points to their real nature.

You can see this clearly through special relativity, there is an implied "proper" time at which things like electromagnetism operate correctly and without it our modern view of physics collapses.

Now you may ask about quantum mechanics, where such ideas of concrete space and time become fuzzy. Where particles exist within multiple place, some theories suggest "time travel" (not real time travel) and other. Though, as I have mentioned before, I think this more evolves our ideas matter than our ideas of space and time. Also, there are examples of the objectiveness of space and time through things like quantum field theory (combines special relativity) and the Planck constants. You can't have constants of space and time without an objective reality of space and time. Now, what is the Planck length

and time. They are the units of which distance and time become quantized, and some suggest the smallest units possible. Now, that seems like an odd idea, especially considering how much smaller it is than anything real and the random constants within it c , G , $\hbar$. Speed of light(or causality), gravitational constant, and reduced Plank constant. This is because as you get smaller and smaller there is a new uncertainty, a 'plank' uncertainty. Beyond this the uncertainty becomes absolute. That the realities of a new quantum gravity take hold. The easiest way to look at this is through finding at what point the wave-length of light(our primary method of observation) becomes a black hole.

52.8 Interpretation of Quantum Mechanics

The long awaited further explanation of my views on quantum mechanics. Throughout this chapter, I have hinted at these ideas of interpretations of quantum mechanics. Also, throughout this I have primarily relied on the Copenhagen interpretation of quantum mechanics. Which may lead several into seeing this as my view of quantum mechanics, it is not(I will elaborate further later.) Though let us first go through the different interpretations of quantum mechanics.

The Copenhagen interpretation. In essence the Copenhagen, the most popular view among scientists, is that quantum physics is impossible to understand through our classical views of metaphysics. That our theories of quantum mechanics work, and to ponder further is a waste of time given its impossibility. It takes our ideas of wave functions and entanglement as reality, not because it is real reality, but because the model works. That at the end of the day physics is not the study of reality in the way metaphysics is, but the study of models of the

universe.

Though this explanation isn't really fair, so I will expand upon it. The key concepts are things like wave-particle duality, that particles are both particles and waves and that things like observation can effect how they interact with itself(yes particles interact with themselves) and with other things.[Though observation is never truly defined] Quantum superposition, something we all know fairly well and requires no further explanation. That quantum physics is probabilistic by nature, and truly probabilistic not governed by hidden laws berried beneath. There is a lot more to it but given that I have already explain part of it previously and how popular it is, you can fill in the gaps on your own. While this theory has been extremely important for our modern views, and important to explaining how quantum mechanics works without esoteric 'nonsense' it has a couple flaws like observer dependence and the fact that it quite literately states that it is not the ontological truth but rather a model.

Next, the many worlds theory. One of the popular interpretations by the general public. Basically, it is that when probabilities collapse; whether it is an interaction, wave-particle duality, superposition, or many of the several other things; the universe splits into two(or however many are used to describe the 'whatever'). That one universe has one and the other has the other. This is quite outlandish idea; for its reliance on the observer and the fact that there is no mechanism of 'universe splitting.' One thing I would like to add, is that I have always hated how people use the idea of a multiverse in this context, it is multiple verses it is one with multiple observable components. A better term, generated by Sir Roger Penrose and a Latin Professor that he was friends with(which I think is absolutely amazing, he made up a new word because he didn't like the word being used) he now calls is an omnium(meaing of all)

An even more outlandish theory is many-minds... which is

exactly what it sounds like... yeah

The Pilot wave interpretation is that there are hidden variables; that just as things like coin flips, brownian motion, plasma instabilities, and so much more seem random from the outside, if you really break them down they aren't. Now what are these hidden variables. Now what is this hidden variable, you might ask. Well lets take a step back and explain a bit deeper. First, lets define a quantum wave function, this is different to the excitations of quantum fields mentioned earlier. The wave-function is a complex variable(I mean complex as in in the complex field not challenging to understand) that depends on the particles position and can be used for several things. It can be described:

$$i\hbar \frac{\partial \psi(r, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right) \psi(r, t)$$

where $\hbar$ is obviously the reduced Plank constant. r is the position vector. ∇^2 is the Laplacian operator. m is mass. V is the potential energy. t is time.

Once you have the wave function you can do several things with it, square it for probability distribution, collapse the wave function, etc. But I am getting away from myself, you want philosophy so I will get back on that.

Basically, pilot wave theory suggest that this wave function influences this particle more than we think, and thus can lead to the particle's behaviors being deterministic through equations like this for velocity, v :

$$v = \frac{\dot{j}}{\psi}$$

Now, in all reality the equation is obviously bigger; must expand j and ψ , but I didn't want to write it out(also there are expanded versions for relativistic and whatnot.) though back

to what we had in mind, what did we have in mind, oh yeah super-determinism. Basically, quantum mechanics is deterministically not probabilistic.

One final thing I will say about pilot wave theory, is that some people suggest that the many worlds interpretation and pilot wave theory are one in the same just describing from different points.

Next, we have environmental decoherence theory. Sadly, I know very little about this one at the moment, I will attempt to learn more to better understand it better(for its own sake and given that it could convince me) But it is basically that like how a coin isn't truly random, neither is the quantum world. That there are density matrices made by interactions that effect future actions leading to a theoretically deterministic world. At the moment I don't really understand the mechanisms so I can't prove, disprove, critique, or criticize.

Lastly, Objective collapse theory. The objective collapse theory is similar to Copenhagen, with it's more literal view of quantum mechanics. Though, it has two differences. It has a true literal and ontological belief that goes further into everything within it. Also, it does not rely upon the observer. It relies upon 'random' collapses of wave-functions, such as gravity, scholastic field, or the interactions of other particles.

This is all well and good, and I do truly suggest you try to figure it out yourself(even though you don't exist), but this is a book MY philosophy. So what do I believe you might ask. I find the objective collapse theory to be best with the pilot function being a close second. Now why do I believe this?

First, for reasons I will explain later; the idea of non-determinism is important to other theories of mine and thus I need it for them to function. Though I am sure you all don't think that is enough, so I will go further. Second, in all reality given we cannot empirically prove any of this it doesn't really matter.

Three, given that it doesn't really matter, objective collapse theory makes the most intuitive and philosophical sense to me intuitively. Four, you know what, I give up(for know) these justifications are kind of weak, even in my eyes. I plan to do more research and come up with further justifications. Right now it is basically which sounds right.

52.9 *Other Questions in the MetaPhysics of Quantum Mechanics

...(will update later)

52.10 Quantum Consciousness, Arminianism, Super-Gödel thinking, OR Orch, why I even decided to write this book, and how long can I make the title for a section within a chapter

As much as I love how hilarious the title is; with its length, random buzz word, and joke on the end. I also kind of hate it; quantum consciousness seems like some esoteric nonsense. It feels like I am admitting to believing in a specific theory just to justify my religious beliefs, but this chapter will attempt to prove that that isn't the case. That it is rigorous, while maybe not on a true empirical scale, at least on a formal logic one.

While there are many reasons for the creation of this book, the main ones is to evolve this very idea. It was that I had this idea at one point and it challenged a lot of previous thoughts,

52.10 Quantum Consciousness, Arminianism, Super-Gödel

~~thinking, OR Orch, why I even decided to write this book,
thus, I wanted to evaluate my philosophical thoughts from the
ground up, then apply that. (While it was the original reason, I
plan to continue this book much further and continue it.~~

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Let me first explain what my theory is in broad strokes. It is the belief (more so a belief than a full theory given its unprovable nature, unscientific background, and basis on the belief in an omni-potent God unconstrained by the confines of physics, formal logic, or even abstract-mathematics.) It is that our consciousness and free will is preserved by God through the use of quantum mechanics. This theory in it-of-itself is not groundbreaking, original, or even scientific in nature, but I intend to go about explaining some possible mechanisms of this (though sadly, by definition this theory will be unprovable, which I will explain why in a second.)

First, is there any justification for this idea of quantum consciousness? There is some, it is weak but existent nonetheless. It is related to Gödel's incompleteness theorem. Gödel's incompleteness theorem has two parts. First, "For any consistent formal system F that is capable of expressing elementary arithmetic, there exists a statement G in the language of F such that if F is consistent, then G is true, but G is not provable within F ." The second is, "For any consistent formal system F that is capable of expressing elementary arithmetic, the consistency of F cannot be proven within F ." This claims that formal logic has its own limits; you cannot prove an axiom with itself and you cannot prove a derivative with an axiom alone, you must prove such axioms with other axioms (because you cannot prove an axiom with itself.) Now what does this have to do with free will, well to have free will you must first have some form of consciousness, something outside of our ideas of formal logic. We have some proof of this, the fact that we are able to derive pure math at all seems to prove that we are not algorithmic based, something beyond. Now there are a lot of arguments against this, namely that our ability to do mathematics

comes from our informal thinking rather than beyond-formal logic thinking. In all reality, I like this explanation better, but I place Penrose's argument of consciousness here as a sample of somewhat scientific rigour rather than simply religious thought.

You may now be asking how this super-Gödel thinking proves free will. It doesn't, but it implies a deeper idea of consciousness allows us to theoretically side-step determinism in order to 'allow' free will. Now, before we get into how God can protect free will, and why I think he would, let us first explain some possible mechanics of explaining this seemingly unexplainable phenomena. Though before that, let me say the fact that by definition it goes beyond Gödel's theorems, we by definition can't prove it so take all of this with a grain of salt. It is an idea, not a law, not even a true theory, and idea.

Let us first go through the most popular explanation, the reason I even came up with this idea. Sir Roger Penrose's OR Orch(I seem to be mentioning him a lot) Essentially it is that microtubals, a portion of the brain's neurons, and theoretically a part of consciousness, are highly subjective to quantum 'strangeness' in a way that most other complex structures aren't. That the interactions of quantum mechanics can somehow safeguard consciousness and allow super-Gödel thinks and even possibly free will.

While this idea is specifically for objective wave collapse theory, it also works for things like environmental decoherence and kind of work for many-worlds, but not for Copenhagen and pilot-wave.

For Copenhagen, no extra theories are required, by definition the idea of consciousness in 'beyond science' and thus no extra loop-holes are needed.(they don't specify what an observer is so therefore other explanations are possible, but they hold no further ground than this one.)

Now how does quantum 'brains' lead to free will. It doesn't, not in the most literal meaning, but it does give us extra wiggle room. While most modern views see our view of determinism as falsifying free will, but these theories give us room to say that there are other possible answers; dualism, God's choice, beyond material, and others.

I would say that God, by some manner, protects free-will. Whether it through some hidden ideas of quantum mechanics or it is that our consciousness is beyond the material and effects the material through the quantum world. I don't know, as I mentioned earlier, I am not even confident it is correct. It is an idea.

Now you may or may not be asking, why would God use quantum mechanics to protect consciousness, if by definition he is beyond formal logic and thus wouldn't need such things. While I cannot comprehend the machinations of God, I would suggest that a possible explanation would be to create our logically, self-fulfilling world that we are able to comprehend(for the most part) and live in a true free will.

Now to end, I want to reiterate how theoretical and disconnected from regular science this is, most of my ideas are at least respected by a decent portion of the scientific community, this is not. It is an idea, nothing more. This is an exercise to see if it holds weight, it currently does not seem to.

52.11 Compatibilist Free Will

Even though I wrote a whole lot about quantum libertarian free will. I don't necessarily believe in that. I view the world as either semi or super-deterministically and that that doesn't actually effect free will. Though I still wanted to examine the idea in more detail nonetheless.

52.12 *Super-determinism, semi-determinism Retro-causality, and Chaos theory

52.13 Role of God

I have already gone over the role of God in many ways; creator of the universe, its laws of nature, protecting free will, but now I will go further.

The first question is the relation of miracles, in the Christian faith, we believe that our God is a personal God that creates miracles for us. Now how does he do that if he predetermined the laws of nature?

Well that is fairly simple, God is omnipresent, meaning he not only knows what you are going to pray for before he created the universe, but he also knows how to create such miracles from the creation of the universe. Thus he 'created' the miracles when he created the universe itself; with knowledge of all that will ever transpire.

How does the divine foreshadow effect our idea of free will. Well, God is beyond our free will, he is beyond whatever quantum mechanisms protect our free will, so he can know without interfering with our free will.

Is something good because God says it is, or because it is objectively. It is because it God says it is, beyond God there is not mechanism for the creation of truly objective morals, there are logical models based upon subjective values, but they cannot stand upon themselves. God makes them right because he is beyond formal logic.

52.14 Post-Axiomatic Nature of God

I guess pre, would be the better prefix given what I am about to say, but Post-axiomatic sounds better. Now back to the actual concept.

This idea is simple. That our axioms, as later examined are derived by God rather than God being subservient to the nature of these axiomatic principles, "A line is the shortest distance between two points," "Existence exist," "consciousness in its totality," "A is A," and many more. He, God, derives them. Now this twists our mind given that axioms by definition cannot be imagined without their existence. It is like inventing a new dimension for us to see, it doesn't exist.

This concept enforces many ideas already present; God omni-(something that cannot exist with current axioms), God being beyond comprehension, that God is self defining(see non-infinite definition of words and axiomatic semantics), and being outside of causality, outside of formal logic. He creates existence(something that makes no sense in our current axiomatic understanding of 'existence'), he is three in one(against A is A), and much more.

While they don't 'solve' these questions in the traditional sense, they open the door in a more esoteric sense. While by definition we can never understand these concepts intuitively they do give rise to much of our understanding of God.

On to our esoteric identification(I know must of what I write is intended to be rational and examined this is one of the exceptions.) This idea of pre/post-axiomatic identification leads to our idea of faith. God, isn't just beyond us, he is ineffable and unprovable though our axiomatic thought processes. Beyond that, this further goes into my idea of the "non-material." We can further define this as that of which is based upon our known axioms and that of which is not.

52.15 Metaphysics of cosmology

While I could go on and on about physics of things like the big bang, entropy, arrow of time, anthropic principle and how they could theoretically effect our views of metaphysics. I don't need to, my belief in God nullifies such arguments. That all possible incites have already been satisfied by God and thus have little need for exploration.

52.16 Ontology of Information and Entropy

Once again, we must first define what our concepts are rigorously.

A classical perspective. Within this concept information is the physical quantities that en essence give you traditional information. That allow you to decode past events and even future ones. Entropy is the inverse of this affect. Geometric instabilities and 'randomness' that hide 'information' from observers.

From a quantum perspective; quantum information even includes superpositions, wavefunctions, density matrices, and more. The quantum bits, qubits, can exist and superpositions and exist within the same quantum state(position)

In this view, entropy then becomes a byproduct of entanglement, a process that 'hides' the information from view.

There is much more on the semantics of this. Going into Geometric identities, black hole information paradox, and quantum thermodynamics. Though, for now I will move on to the more metaphysical parts.

One common debate is if information is truly fundamental

or simply a human made connection. That the universe has a 'computational structure' where information is the substrate and entropy the measure of the complexity. With modern understanding of the geometric identities and the fundamentalness of information, the theory of fundamentalism seems to take the lead.

Next is the arrow of time. The second law of thermodynamics/entropy states that in isolation, states drive towards entropy. This necessitates an arrow of time. It breaks the symmetry of time.

Now many question whether or not entropy is a by-product of the arrow of time or the other way around.

The way I see it, the direction of entropy is the consequence of the already time symmetry breaking existence of dark energy, black holes, and other similar objects in this preview. That the direction towards entropy precludes low-entropy existance.

Though this is just one theory. Holographic principle, Gravitational Entropy, Quantum Mechanics and Unitary Evolution, and more. In-fact the quantum mechanics and unitary evolution makes more sense and less problems.

The symmetric theory ignores that some forms of entropy are consequences of statistical physics that doesn't deny time symmetry

52.17 Dualism

I have already mentioned this in interpretations of quantum mechanics, but in physics many possible mathematical(and thus metaphysical) interpretation exist. Different ways to look at the world. They can't both be true, can they?

52.18 The Problem of the one-way speed of light

Quantum Mechanics is not the only physical model with inherent uncertainty. Special relativity has a similar principle. Essentially, the problem is that it is impossible to find the one way speed of light. Basically, while we can use a clock to find the two way speed of light, to find the one way would require two synchronize clocks. To synchronize clocks along a distance, the speed of light is a required number. Thus, the speed of light could theoretically be different in different directions. In one way it could be $0.75c$ and the other could be $1.25c$ or $0.5c$ and ∞ . There is no possible way to actually prove it. Not only does this theory not prove it, but by definition changing the directional speed of light would change the transformations, effectively making everything equal. Just like changing the coordinate system (there are some versions where it does change things, it will get to that)

Now why does it matter if the observables stay the same? There are several reasons I will get into.

One is the question of scientific realism and instrumentalism. A kind of epistemic humility about how one of our singular most verified theories is simply convention. Not ontological truth, simply convention. That there could be hidden mechanisms causing these changes and speed and we could have no way of knowing.

Beyond that, the different theories based upon this fact offer wildly different views upon the ontological nature of space and time. Einsteinian, space and time are relative and this relativity causes the physical views we see. Lorentzian, there is simultaneity and objective time. Relationalism, time is a network of relations and the speed of light is a consequence of that relation not the opposite. Block, everything happens at

the same because simultaneity is frame dependent and so is the speed of light. Presentism, there is an objective present, but the nature of light diseludes human observation by its nature. There are many other theories beyond this, but I think it envelopes the idea.

Lastly, there are some theories that while equivalent in most situations(situations that we have empirical data for) but is different. Different in not just ontological but physical differences. Differences that could lead to a unified field theory.

52.18.1 *Phase-space and the Euclidean Mind

52.19 *Ontology of Numbers

52.20 Reality and the Abstract

When reading this chapter, one can easily get a esoteric/gnostic sense of reality. Of pure abstraction with little regards to reality. Though this is not the case. In fact I find the abstract "non-material/pre-axiomatic" world as inconsequential. The only true consequence being God, but he matter because he influences the material. The focus on this abstraction being the fact that the ideas of reality are largely explored and are easily understood and so large amounts of writing and exploration on the topic is relatively inconsequential.

The focus on abstract, structuralism, post-axiomatic is simply to satisfy my need for cognition(as later explored) rather than being core parts of who I am. These theoretical ideas that I am not even sure if they are right or not(post-axiomatic God, Consciousness, etc) are explored not because they are core to who I am and what I believe(maybe the God one) but because it is fun.

52.21 Dialects' Critique of Relativity

I have recently been watching a YouTuber Dialect, he is quite intelligent and well spoken, but one thing that quite surprised me was the fact that he, and fellow researchers, had an intense disdain of Einstein's Relativity. While I think his extreme disdain and inflammatory writing it untoward, the essential idea is worth exploring. While there is much more that he goes on, but I will focus on two things that he mentions; critiquing physical positivism and Platonism in modern physics.

Let us examine positism first. Positivism, essentially says that science should be relegated to only observable phenomena. That physics does not and should not concern itself with ontology and metaphysics beyond what is strictly observable. Things like the one-way speed of light or the nature of simultaneity are strictly convention and should not be taken seriously. This is seen in a much harsher way in the field of interpretations of quantum mechanics. The idea of positism is obviously closely linked with instrumentalism as mentioned in other sections.

Opponents argue that while we should have epistemic humility and understand that from a view of instrumentalism this is as far as science can realistically go, we should nonetheless continue to find ontological arguments that posit real underlying entities. That science should not renege itself to measurement bookkeeping and remain at its heart natural philosophy that looks into explanatory mechanisms. Looking for something semi-physical like ether that it a medium for a field rather than the nature of the field itself. That viewing gravity as a nature of how we measure things within spacetime is overly simplistic and lacks realism.

Another common argument being as mentioned that relying on the nature of time and simultaneity as simply convention as

a fact given the mathematical nature of time is similarly quick to judge. As mentioned, other theories like modern ether, matrix, etc. explain these things in a much more natural physiccally way rather than give further causal explanation than Relativity does.

For the first argument, I do agree in the abstract. While I recognize that as physics becomes more and more advanced, and further and further from Galilean intuition, connecting abstract physics with ontology is hard if not impossible. Especially given how even these most basic ideas are open problems in the field. Though, modern physics has become too divorced, in many cases ignoring physiccism entirely. In the case of Relativity, I would not say this is the case. While some of the more mechanistic theories like ether can technically be considered to be "more mechanistic," it isn't truly. Not in a truly paradigm shifting way. Relativity does offer a true mechanism, the curvature of SpaceTime. A more intuitive medium that comfines better with the quantum fields would, in my mind, be better fit, but that is far from saying that their are true misgiving in the nature of Relativity that must be rectified. It is just as likely that quantum field theories nature is what is misplaced.

As for the second point, I once again agree in the abstract. I do not like relying on such a fundamental nature to be at best convention and at worst be convention; though, we cannot rely on what we cannot prove. To take one of the other equivalent theories as fundamental would once again be convention. Thus, the problem is not of Einstienian Relativity but of the basic uncertainty principles of nature.

One final note, it that in the instance of many sets of Ether theories, it can be taken to be that Relativity and Ether are equivalent and to simply have relativity be a descriptive theory and ether as an explanatory theory. While I do like the deeper explanatory power of ether theory, given its unprovable nature

it is simply a deeper more intuitive explanation of the same thing but with no backing. One can argue that the Einstienian view has no deeper backing and thus no better. Thus choosing one above the other is simply from an aesthetic choice, mathematical elegance or clear ontology. In this case, there is no answer. From a personal stance, while I generally find clearer ontology to be superior to mathematical elegance, given that the majority of the physics community is on the opposite side, it is more pragmatic to build upon Einstienian Relativity to avoid confusion and to have the tools set forth by those that have come before. For instance, a general Relativity equivalent Ether theory is still an open problem with it being quite underdeveloped, far from the clearly developed theory that has set up itself in dozens of different formalism's, metrics, and tools like numerical solvers.

While in this writing I have also addressed much of the arguments related to Platoism, I will address that here as well.

In many interpretations of Relativity, namely Minkowski, the mathematical four-dimensional geometric object known as SpaceTime is considered to be real. Critics claim that it is not testable and thus not scientific and that this over emphasis on mathematical elegance is foolhardy. That to take a literalist approach to the consequences of mathematical symmetry and reduction is going too far.

This has quite some merit, for instance to take a literalist view of Hamiltonian reduction of fluid dynamics would give quite a different view. The way scientist avoided this was the fact that reduction through inductive observation came first, this is obviously not possible for the nature of SpaceTime. Thus, having trying to look for deeper mechanism, like how particle kinetic theory or molecular theory does, is likely a worthwhile investigation. Two problems arise though. Once, given the nature of SpaceTime, modern science sees no way to study it in such a way that it is possible to go beyond simply

mathematical interpretation. The other being that mathematical exploration is argueably a better way to explore the nature of reality. In other chapters(namely the next one) I explore a more mathematically abstract view of ontology than traditional newtonian view.

52.22 Structure as Fundamental

Chapter 53

Epistemology

"Man is neither infallible nor omniscient; if he were, a discipline such as epistemology—the theory of knowledge—would not be necessary nor possible: his knowledge would be automatic, unquestionable and total." — Ayn Rand

53.1 Limited Objectivism

Before I begin upon explaining this, I must first specify the influence of Ayn Rand is limited to Epistemology. It has not influenced my politics nor ethics. Now lets begin with the philosophy.

In a short simplified sense, objectivistic Epistemology is the inverse of transcendental idealism(though despite what many say, not the opposite). Both philosophies take both reason thought and empirical data as useful tools in the accumulation of knowledge.

Though they differ in importance and direction. Transcen-

dental idealism takes reason as the most fundamental and such abstract thought can then be applied to the outside world through empiricism. Objectivism takes the inverse, that we create our abstract models based upon our interactions with the world. Now these may seem like small differences, but they expand themselves through their logical conclusions. I could explore how they differ in more detail, but I will focus on objectivism.

Now, where did the limited part come from. As I have mentioned before, I am a religious person. I believe there are something beyond our sense and possibly our mortal comprehension, therefore we are limited by such things when acquiring knowledge. I do believe what we are capable of discovering is 'objective' and just as metaphysically true as what we can't, but that does not take away the fact that some abstractions are beyond us.

53.2 Consciousness and Reality

A common continental question, is the question upon whether our consciousness truly models objective reality. While these questions can be useful when directed in a purposeful and specific way; though most people don't direct it in a useful way, but rather in a simply abstract way with little refinement of knowledge than the ability to trick others through strange and useless question.

Now, why do I think this is a useless question? Well, for several reasons I will address categorically. Now I will preface this by saying this is a criticism, not a critique, I may later may later add a more refining critique but for now it is an acknowledgment of the stupidity of this, and to show that all derived concepts have no true analytical framework.

The most basic argument against this is the pragmatic the-

sis. In essence, it doesn't matter that reality is consistent with the observations of the mind, the mere fact that our observations closely associate enough for practical purposes is reality enough. There is no purer form of reality than this functions.

Though, this isn't enough for me, I think it could be enough if it was all there was but I want more. I do believe in truly objective reality and not just our semi-subjective observations of reality.

A refinement of the pragmatic argument to include the existence of an objective world would be a question of statistics. If a further world with separate physical laws existed it wouldn't work with our vast and extreme observations. Though, this wouldn't work with some more extreme theories wouldn't work with more radical and complete ideas, like Recreant's dream theory. Though, this implies that whatever objective reality is it must not be capable of interacting and effecting our reality. This can be attributed to nothing that could interact with our world, or our world is independent of the interactions of reality by its own definition. This may seem like a minor and obvious reduction but nevertheless it such reductions are the essence of philosophical discussion.

Next, such dream theories are useless. Derive no further discussion, only exist as a way to trick people up and thus serve no purpose. For these, the original pragmatic argument is all that can be derived secularly.

Now, you may say this is a good way to question our scientific premises. An argument against scientific realism, but in reality it doesn't change anything. Scientific realism stands or doesn't regardless of the 'validity' of such theories.

My final secular argument is that such philosophical arguments are harmful. That philosophy should not start with doubt but rather with perception. Such perception must be doubted, but doubt existing in its own self-containerized muni-

tion is useless. It defines nothing and leads to strange and possibly dangerous ideas. For instance Desecrate lead to Spinoza who lead to Hegel, who lead to Italian idealism, who lead to Giovanni Gentile the philosophical founder of fascism. Now, you may criticize that this was one complex branch, you could also derive similar lines through Aristotle or just any philosopher far back enough, and this is a good argument. Though, there is a clear line, also to other dangerous ideas like communism, NAZIs, and so much more. These things are predicated on these strange cartesian/continental questions that derive no true thought, and thus should be understood in this way. Simply as pathetic and dirty tricks and shouldn't be seen as true paths to knowledge. While yes, we should still engage sometimes, truly placing it on equal footing (or higher as we have been) is dangerous for our collective understanding reality and derived sense based upon it.

Though, many of you may have noticed I specify secular. There is a religious argument as well. If God created the reality as reality, then it is by definition reality. The only argument otherwise is that God doesn't exist (either entirely or he is not God)

53.3 Nature and Validation of Axioms

First, like always, we must define axioms thoroughly. The dictionary definition is "a statement accepted as true as the basis for argument or inference." For instance, our understanding of scientific realism is the understanding of the fact that our observations of reality are accurate (there are some observations that can be morphed due to other things that are observable.)

A simplified and more observable definition can then be defined to say that an axiom is a statement that either cannot be proved or cannot be proved currently that is used within a

greater statement that can be proved upon its edifice. These axioms, while don't require 'proof' generally desire some type of reasoning even if it is not objective or purely analytical in nature. For instance, the above arguments for the objective nature of reality aren't true objective proof, they still exist as semi-arguments. While we cannot prove the objectivity of reality with true and objective reasoning, some reasoning can be applied to get some kind of 'proof' of the statement so that further studies can be applied based upon that said axiom.

53.4 Non-infinite Definition of words and Axiomatic Semantics

In the field of axioms, there becomes a problem. Part of the reason that 'some' words cannot be further defined. They are defined by themselves, they can only be intuitively understood, and even there is a limit.

A great example is existence, because to define this thesis, you need an antithesis. That doesn't exist(non-existence by definition needs to exist to be something and thus cannot exist because it would need to exist to be in existence.), so therefore existence is defined by itself.

This brings up a great deal of problems, paradoxes, and much more that will be explored.

53.5 Induction & Deduction

In the expansion of human knowledge there are two main ways in which they can be reached; induction and deduction.

Deduction is by taking premises/axioms and derive conclusions based upon this. Induction is the opposite, from observ-

ing 'effects,' define whatever knowledge is desired.

This can be seen in physics, deduction is taking a fundamental theory and applying it in a new way to find novel uses, predict, or whatever may be needed. Induction would be taking data to come up with a way to model reality or come up with an entirely new theory.

Both of these are extremely important. Without induction we could never go beyond basic logic, but without deduction we can never find objective truth. The way I have found to be the most useful is to use induction to attempt to find our 'axioms' and then use these to deduct the reality around us, though this is only really works for my fields of interest. In other fields, like psychology this strategy would fail.

53.6 The Problem of Induction

Nothing can be proven true. As controversial of an idea as this may seem, it isn't. This basic axiom is the foundation of modern scientific epistemology. That theories can only be proven wrong, never correct. That is why they will always remain theories.

Now, you might think something as foundational as something like Newton is true. Though, there are questions through MOND(Modified Newtonian Mechanics) and others. Given, Newtons empirical evidence, there would need to be a lot of evidence for MOND for it to be taken anywhere near the same level

Chapter 54

Ethics

”Finally, brothers and sisters, whatever is true, whatever is noble, whatever is right, whatever is pure, whatever is lovely, whatever is admirable—if anything is excellent or praiseworthy—think about such things.” — Philippians 4:8

54.1 Introduction to ethics

Many that know me, know the question I love to ask, “what is your moral philosophy?” I love this question for many reasons. First, it intrigues people, they love to talk about their own morals and beliefs, but engages their curiosity with these newish words. Second, Morality is something that is both simple and fundamental enough that everyone has thoughts of it but still complex enough that it can have engaging discourse. Three, it get to explain stuff, most people haven’t heard of virtue ethics, deontology, egoism, etc.

Despite this fundamentalness to ethics, many questions are still unanswered, which fundamental theory? Many questions are raised for each theory that must be addressed? Even people question if Morality is even objective or not?

54.2 Virtue Ethics

My fundamental theory is virtue ethics

Virtue ethics is based upon the idea that Morality is derived from living a virtuous life rather than the adherence to rules or a logical stance. That you must pick a set of virtues and live by those principles.

Now why do I choose this over the other types of theories of morality and ethics. To understand this I will first criticize (yes criticize not critique) other theories then I will defend virtue ethics.

First, deontology. Deontology comes from the Greek word *deon* meaning duty. That you have a duty to follow certain 'rules.' That morality is based upon a set of rules and concepts rules and concepts to follow. Say, don't kill, don't steal, give to others.

My problem with this is not that it is too ridged, but that it is impossible. It would be impossible. You can't make rules for all possible interactions so most theories of deontology tend to just be contorted virtue ethics. Also, that both the action and consequences should be included when making an ethical decision but generally rules can lose this complexity.

Next, egoism. Yeah no

OK, fine, I'll do a little more. Most theories of egoism fall into either three camps; nonsense hedonism (which needs no criticism), logical consequentiality based concepts, or altruistic egoism. First, the consequentialism type eventually leads to normal consequentialism but with more steps to find the conclusion so I will focus on that when I get to it.

Next, consequentialism. First, for consequentialism to work a set of values must first be prescribed (so it isn't fundamental) there are others like utilitarianism but they aren't self-

justifying nor truly justified by other means. There are some that exist but I will show that to analyse all in-depth is a waste of time. My reasoning is that both actions and consequences should be analyzed. Why, good question... Well, first there is religion obviously. Though a more secular reasoning is that ethics is used as a binding agent, people emotionally don't bind well to 'immoral' actions, and those that do tend not to bode well with other things.

Now for justification of virtue ethics. The first and most important is that I am a Christian, from my analysis most Christian ethics is highly tied in virtue ethics. The fruits are virtues. Most sins are vices. Most ethical advice is based upon abstract virtues rather than a simple value(consequentialism) or rule(deontology). While there are many rules, most of them are based upon the virtues later mentioned rather than justified by themselves.

Next, on a more secular level, virtue ethics are far more complete and level for people to analyze, create, and apply to your life. They can include anything you want to have in them without creating paradoxes and problems(as long as the two concepts are not paradoxical in themselves)

I will show this through the future sections that go through a list of virtues and why I feel they should represent my ideas.

One final thing I will talk about, is something that may be a strength or weakness depending on how you look at it. Virtue ethics has a much more 'subjective' connotation. A lot of virtues that I value another may not, also these virtues can be analyzed in your own way.

The way I see it, some virtues are fundamental and should be applied to all given they are principles derived by God, others I feel as personal virtues and feel no sense that they should be forcefully applied to others(though I feel things would be better for all that way.)

In essence, the idea of virtue ethics is to pursue perfection of the mind, body, moral thought, etc in order to create a purity of intention. This is the goal, not absence of sin, but the perfection of moral intentions. To create yourself in the image of your own ideal and to pursue it. That morality isn't just a responsibility but ontological alignment. To pursue the ideal of Jesus.

54.3 Stewardship

Throughout the rest of this chapter, many times I reflect on morality as my values. Values that are observed because of some 'internal promise.' This is because it is the way I have naturally thought, though I am trying to evolve past this.

I have always seen the internal promise as more pure than any other form of moral enforcement, though I have found this not to be true in the highest extent.

This obsession with dignity and following through with my own values alone is cornered in the vice of pride. It sits upon it, being thus controlled by it, and by proxy then so am I.

Now I have found, not responsibility, but existence in stewardship. In stewardship to the will of God, his grace, his mercy, his creation, and most importantly each other.

You see, God brought Adam and Eve into the world, as images of himself: builders, molders, and beings capable of reason. A creation to tend the garden, name and care for the animals, and to care for each other. This was the reason for humanities existence.

After the fall of mankind, humanities purpose shifted, but not much. We are still arbiters of his will and creation, by authority he has given us. Though, because of the imperfections made by us, our responsibility is to fix them. To perfect the

world around us through the words that God has authored for us.

For God himself has given us this divine commission; though he hasn't left us to do it alone. He gives us gifts, hope, and healing. He feeds us and nurtures us so we made be sanctified in his love for us.

Now in practice, this becomes our purpose of moral perfection. Both for ourselves and for those around us. To create, not only pure intentions for ourselves but for the society and people around us. I will discuss this more throughout this book, especially this chapter, but here I will leave with this.

Though our actions, we may be able to co-create with God to bring Heaven on Earth. Bring about social holiness caused by our own actions with others. This is the purpose of Christianity, of ourselves. Only secondary to a relationship with God.

54.4 Categorical Good

Based upon my religious beliefs I propose this axiom, "that all of reality is designed in such a way in that "good" can be logically inferred and the actions to find this good is what we call virtue."

$$\forall x(R(x) \rightarrow \exists P(I(P, G(x))) \wedge \forall A(V(A) \iff \exists y(L(A, y) \wedge G(y))))$$

In essence, there exists a good for all categories(ex. communication, get information delivered clearly). Thus, there are good ways to do this, virtue(ex. use precise language when needed but don't use overtly complex language when not). Finally, not only can iterations of this build up an idea for how to "be a good person" based upon a complex model of different circumstances throughout life, but that what virtues exist re-

veal what makes a person "good" by what God has consciously divined, giving the actor "wisdom". This is all centered around the Stewardship given to us by God as mentioned in the previous section.

Obviously, this isn't a purely religious view. Aristotle came through with an idea very similar idea, but even his idea had a pantheistic view, and other secular versions generally make it an unquestionable axiom without further backing other than the fact it creates a non-paradoxical system. Also, this system allows further development, like founding it in stewardship or having all things reveal "wisdom" to further develop virtue.

54.5 Cardinal Virtues

Now, while my cardinal virtues have basis in the traditional cardinal virtues of the catholic faith they are not the same. The name comes from the fact that cardinal comes from the Latin word 'cardo' which means hinge. In essence all other virtues hinge on these concepts.

Here is a short list of the main ones

- Temperance: Excess in most matters proves detrimental; self-control is paramount in one's life.
- Prudence: Deliberation is essential before action in all circumstances.
- Fortitude: Mastery over one's emotions is a paramount virtue, as the inability to do so inflicts harm upon oneself and others.
- Faith: Maintain strong belief not only in God but also in oneself and one's values and beliefs; these should remain immutable unless confronted with supreme evidence.

- Duty: One has obligations to oneself, family, others, and one's values, which should drive one's life. Duty to oneself includes ambition and adherence to personal values; duty to family involves support and politeness; duty to others is similar but to a lesser extent.
- Individualism: Embrace self-reliance, self-respect, and ambition, and adhering to personal values.

As you can see all of these virtues are simple and widely acknowledged and don't require much additional justification. The only true exception being individualism....

54.6 *Salient Virtues

54.7 The Philotimic Virtue

Now Philotimic is a word I have made up, I will go into more detail about what exactly it means later, for now assume it means pride.

Now, I hear your vapid and lost mind screaming, "isn't pride a vice, what are you doing" Though I will examine why I think pride in some forms is a virtue not a vice(at least certain types of pride)

First to examine we must first define pride and what it is. One definition is , "feeling of deep pleasure or satisfaction derived from one's own achievements, the achievements of those with whom one is closely associated, or from qualities or possessions that are widely admired." or "consciousness of one's own dignity." Now what does this mean and how do they fit together.

Let us start with the first one, a deep feeling of pleasure in ones own achievements. This can manifest itself in many ways;

work, morality, intellectual achievement, and so many others. Beyond that it can effect people's actions in both positive and negative ways. Let's start with negative, it can lead to obsession with one's own capacity and the product of such. On the flip side, a healthy obsession will lead to personal growth on whatever they feel pride in. Also, it can give them confidence in their own capacity. An example would be with morality, if they feel pride in their own sense of morality it can lead to self-assurance in their own senses leading to them being able to apply and do them, but it can lead to close-mindedness and looking at just your own thoughts and nothing more.

Now let's look at the second definition, consciousness of one's own dignity. Meaning that they are aware and influenced by concepts of their own dignity. The belief that some things are within their ability and others are beneath them.

Now what I mean by philotimic, as essentially the positive aspects of this. That to have philotimic virtue is to take your own life seriously. To be principled, to be absolute. Not just in behaving moral in a moral and dignified manner, though this is definitely a part of it. In fact, even within this manner this virtue is partly about fanatical pursuing such moral perfection. Though, outside of this there are many things. Most notably is having such fanatic belief in things beyond morals, other values like being early, being prepared, working hard, being mature, and being other similar concepts. That while there are values that God doesn't arbitrate, but you do, and thus you should keep them to almost the same esteem.

This concept, as mentioned earlier is about taking your own dignity, self-esteem, self-respect, virtue, and above all life seriously. Because that is what you should do, take it seriously in a dignified and ridged manner. To never compromise upon even the most minimalist value, because to compromise on such things is far worse than any other conceivable interaction.

Another concept within it is control. Control over your inner-mind. Your thoughts, emotions, temperament, and even personality being bent to the will of your own 'ego.' Not ego in the regular sense, but as in the inner consciousness of your own self. The self of morality, values, reason, and more (more closely related to the idea of 'superego')

In essence, recursively creating your own self based your higher values rather than letting your environment impact your supposed values.

Philotimia in its highest sense, is not merely pride, but the love of what is worthy of love. To find what is worthy of such esteem and pursue it with fanaticism and create your soul around such values. To take responsibility for your own existence, morality, self-esteem, and so much more.

An even further simplification is it is taking upon yourself the responsibility to be human. The responsibility to think, to fear God, to live accordance with your values, to have values, and to live in the image of your highest self. Not only that but bend the world into the image of your highest self.

One thing I will add, is in the conflict of universal morality (derived by God) and personal values. This conflict doesn't really exist. Both of these values exist in my mind. While universal morality is obviously stronger, this doesn't dissipate the importance of personal values. The largest differentiator is others; for universal morality, others disobeying. is an infringement of morality, while for personal values only you 'break them,' no one else.

54.8 *The Internal Promise and Dignity

54.9 Good Life and Eternal Struggle

There are two parts to this. Obviously the good life and the eternal struggle as defined in the title. This are both extremely connected, as I will show momentarily.

First, the good life. To live a virtuous life is to live a good life, but there is challenge. One, you should train yourself to desire this good and enjoy it. as defined in the philotimic virtue, take this training upon yourself rather than simply passively let culture and external figures train for you. While yes, culture can do a good job, training yourself creates a better feedback loop and consequence; beyond that, training yourself is your own moral responsibility as a human being.

Find this good to be your highest desire, so that you not only can but will naturally follow it and ignore sin and vice around you. To desire it fanatically. While many consider fanaticism to be a bad thing, in the pursuit of virtue, it isn't just a good thing but a lack of it is evil.

Beyond that, this struggle will be enternal. Virtue by its very nature is never perfected. Therefore, you should always struggle. To take this further, you should love the struggle against it.

54.10 *Personal Virtues, Values, and God's Morality

54.11 Intentions

In my theory of morality, intentions matter just as much as the action in it of itself. "The LORD does not see as man does. For man sees the outward appearance, but the LORD sees the heart." 1 Samuel 16:7.

I could go on and on, but this idea is relatively intuitive and while many don't consciously think this way it isn't far from our minds.

54.12 *Morality and Religion

54.13 *The Ethics of Knowledge

54.14 Recursive Connection of Civilization & Virtue

Our capacity towards civilization and virtue are strictly linked in a depth rarely acknowledged.

Let's begin with the start, for civilization to happen, we must be able to do it well. This is the heart of virtue, while virtue can be connected with any doing any good thing well, the ability to be civilized and do civilization "well" is the very heart of it.

Now, how do we do civilization well? To make sure people do not become distrustful, honesty should be brought about. To bring about any positive outcome, discipline becomes a need.

To expect pro-social activities, empathy is not only an ideal but a requirement. While I won't explore every angle, it is clear where this leads. Our modern ideas of culture and morality is primary based upon how to build a civilization, a society.

Though, how to develop virtue on a mass scale? Society. Through community virtue is developed, the most obvious being how we enforce guilt and pride into those that exhibit virtues and vices. To teach them young, how ethical theory works.

A bit deeper, through habits. Virtues are all about who we are, our attributes, our character, presence, intellect, and much more. We become who we are through our habits; habits of action, ideas, thoughts, and much more. Society creates these habits, even when we don't realize this. To explore this idea, I will present a case study: a commonly criticized culture rule is to not curse. It may seem illogical, they are words, sometimes not even directed at the person offended. Now, before I begin on this analysis, I will say that this particular analysis will focus on social rules as habit forming implies virtue forming. It will ignore the dozens of other reasons and arguments.

From this idea of it being illogical, you can clearly see the reason by what the person is actually doing. They are *tempering* their own *impulses* for the sake of *social harmony* and *empathy* towards others. Having *interpersonal tact* and learning from *moral authority and teachers* without rebelling because you didn't see the big picture. These habits are both skill and virtue building so we may be civilized adults. This is also why it is so extremely emphasized during childhood.

Now, while these parts are extremely important, they aren't simply circular self-defining concepts. Virtues exist beyond civilization and their primary justification being to mimic Christ and fulfill our duty to steward his creation.

54.14.1 *Social Rules and Morality**54.14.2 *Culture as Moral Education****54.14.3 Intensity of Duty***

While on a strictly moral view, ethics and their virtues are extremely important. I have gone over again and again, how these ideas should be pursued fanatically, but individually. While they should be taught to others, expecting purely moral actions from other is not something to be expected from them.

Virtues of civilization are another story. Some lenience is required, but too much leads to a denegation, a slippery slope. Things like duty, respect, loyalty,

54.15 Morality in the End of the World

Chapter 55

Personal Philosophy

"Only through the radical examination of one's own life and the reality surrounding may one gain authority over it. Through such authority their own happiness becomes clear. To have the freedom to be responsible for themselves and the integrity of their own mind. This is the principle and highest form of human action, to divine their own morality and values then shape themselves and reality into the image of their highest values." — AJ Cason

55.1 *God as Foundation

55.2 Will as Identity

All throughout history, the question of identify has been constant. Who we are: are we are personality, outward actions, inward thoughts, natural impulses, cognitive style?

I propose a different method, we are our will. That is our method of identity. This proposition has many consequences.

Before I get to that, I must define your will. Your will is your

highest self within you, that which pushes you to your highest ideals. The image that burns within you of what you could become, that whispers how you may achieve it. That is your will, your will to your own power over yourself.

First, it allows acting through purpose to be acting authentically. Many people find paradox between our need to live 'authentically'(a moronic ideal) and fulfilling their purpose and duty to others and themselves. Though, from my view these are the same. To act and bend your personality to your personal will, to fulfill your duty, isn't unauthentic, but rather the purest form of authenticity.

To 'act' and author yourself is your responsibility as a human, a creation of God the most high that has been made in his image as a being with a conscious mind. Don't live passively letting your whims and childhood define you, but rather your will is your definition.

Second, a devotion to your will is your responsibility. To 'falter' is to kill yourself and let a husk(personality, impulses, etc take the place of the human mind that God has put within you. To let the flesh control you is dishonorable, evil, disgusting, and never to be considered. It is not understandable, it is not acceptable, never surrender, never compromise against anything beyond your will.

This can be observed through giving up, compromising, complaining, etc. These are evils. People seem to accept complaining as a natural reaction to 'unfairness' and that you are nothing more than your "childhood traumas," don't become this subhuman husk. Become more, become a thinking human.

Four, to let others to your thinking for you is to outsource your soul. To degenerate God's creation. Think for yourself, think logically and analytically, derive your 'first principles' clearly.

55.3 *Fanatical Ethics

55.4 The Duty as an Actor

As mentioned in "Will as Identity" you must in a sense become an "actor" in your life. What I mean by this is that you must control your reactions (be civilized and dignified) to fulfill your duty.

Be cheerful within the risk of irksome tasks and weighty responsibilities, hold your head up high, never complain, don't profane what is sacred, and so much more.

One, through "acting" you will become that in which you pursue, virtuous. You will become happy, become moral, become dignified, become who your will decides you must be. It is through this "act" you will become that image of your highest self within you.

55.5 *The Aesthetic Pursuit of Truth

Chapter 56

Personal-Analysis

"The unexamined life is not worth living" — Socrates

56.1 Introduction

I will use this journaling of sorts to examine myself and my own personality.

56.2 Disgrace and Pride

(if anyone else reads this, don't pay too much mind. I use words differently than their actual meaning. I don't word things very well. I was simply trying to capture my own indescribable and esoteric and possibly failable thoughts in this moment on this topic, my actual feelings are very different than what can be interpreted through the word choice.)(Additional update: upon later analysis, I have found that these ideas of pride and disgrace are for the most part the idea of seeing my earlier mentioned "personal values" in a light very similar to traditional morality. This doesn't fully explain it and I will continue to

work on refining these ideas beyond the esoteric concentration currently presented.)

Why is it that you are so accustomed to the use of disgrace in your speech and why does it effect you in such a manner. The idea of 'disgracing' yourself is so vital to your worldview, it effects everything; morals, interactions with other, and so much more. It isn't even like you are all that effected by the thoughts of others. In fact the most abnormal and extreme feelings of self-disgust and disgrace are related to a refection of the thoughts of other. Why is it that I feel like to see other, value their opinions, and let them influence men to disgrace myself. It is an odd thing, yet it is so very natural. It influences everything, and it is so much more extreme when I am in isolation as I am now. What is it about the concept of being influenced by others to be so repulsive, so disgusting that I won't just let it influence my actions but I will try to persuade others my way is best with pride. With this eternal pride that locks me in, says that I am always all of the way in all things. There are so many things to think about in this discussion so I will try to take it piece by piece and hopefully add some things.(also, what it with the strange switching from dialogue with an external, internal, and this explanation style?)”

Well, disgust is a natural motivating force. It motivates far better than most, it is consuming. It involves fear, pride, righteous anger, annoyance, and so many more extreme and strong emotions. It motives far more than most, and it comes so easily, especially for someone like me. There is no greater fear than falling in ones own eyes. My disgust of others is just who I am, it is what makes me who I am. It disconnects me from the petty emotions around me. I mean, can people really say that those are better; to be insecure, jealous, vapid, empty, etc? Is it really better? Take insecurity, it is such a strong emotion that consumes people so very easily, making them bicker and fight, making them claw at each other with their broken paws

that do very little besides hurt themselves and hurt others by their own volition of nonsensical blight. My 'pride' can rise me above that. If I do not see them as anything, then for what reason do I have to be insecure about this vapid collection of dust. Now I am not truly some raging narcissist who sees no one beyond and object for my own use. I see people as they are, I care for others, value them. I just simply do not value them in the way others prescribe them. I don't see them as threats, competition, just broken monkeys in need of assistance (and that I am the same way and can occasionally use assistance myself, just less than most.) I still see them, I make friends just as easy, if not easier than most due to this.

This disgust is a strengthening endeavor that assistance me in many ways. My self-disgust pushes me further. It is what lead me to teach myself calculus in elementary, and has brought me here know with graduate level knowledge, research, and so much more. It has disconnected me from the vapid desires of others in their looks and whatever other nonsense. (though I will add, this disgust is not some obscene emotional self-distrain but rather a more abstract understanding that doesn't truly make me feel bad, just concous of 'wrong doing.' More so like a passion against, I feel no negative emotions about myself)

Though I must acknowledge that this does not make me better, it makes me better within my own eyes, my own values, not others. My values have no greater truth than that they are my own. So my disgust should always remain abstract, looking down at the ideas not the people. I have done great at this, never prescribing habits or concepts to people. Though as of late I have had trouble with this, seeing people beyond the moment. Being conscious of their deniers and thoughts, it can create some discontent. I should remain as I was, seeing people only in the instant. Nothing more. I am the only conscious being within the confines of my own mind, for it is better this way. You cannot accuse a rock of moral failure, only

a man. For which I am the only man that I see, the only man that I know, I am to constituent for the moral blame. The disgust should be within me and me alone seeing only me. For I am the only one held to my values and the only one who could suffer the breaking of them. For they are my values alone, that is the purpose of the disconnect. Why my values must be mine alone, for I am the only one to be connected to them, the only man in my eyes.

For this I also hold the burden of thought and reason. The thought behind my morals, my values, my beliefs, my religion, my reality. I see the world as my land for the conquest of my own knowledge to build up it all. That is the whole purpose of this book, to fully disconnect, to find myself fully and fully alone. Because, in the end that is all that matters in the abstract. For sure, I love my family, my friends, and my fellow man, but I love myself in the way that one can only love oneself, the expectation of my own measurement, the pride in my achievements, the disgust in my faults, the understanding of my beliefs. For one cannot love others without loving oneself, for love is derived from ones understanding of their own values, who they are in all of reality.

Now why are these feelings and thoughts all the more present in isolation. For when with others, those are my real feelings. In isolation I attempt to derive and find, deriving and finding the strange and unwieldy emotions of the mind does not come truly and with the same accuracy of that of physics. It comes in strange botches of thought that don't mean what they literally mean but can be described through thoughtful examination of the words, other words, and the actions of the person.

When did I become this, so thoughtful behind it all. Seeing the world beyond the material, seeing my thoughts beyond the exact. Maybe I really am chaining, in a way that must lead to the changing of even my most basic assumptions.(this last paragraph is really stupid)

56.3 On Ayn Rand

It would be obvious to admit that this section would be an analysis and critique of Ayn Rand's ideas, but like the more erudite among you will notice that this would go against the structure of this chapter. This is exactly correct, this will rather be an analysis about my propensity towards Ayn Rand's ideas.

While this may seem useless and without purpose, it isn't. It has been a strange psychological question about my enjoyment of Ayn Rand's novels even though I starkly disagree with actual philosophy and the fact that her books lack the many of the general characteristics that generally allow me to enjoy such novels. So what is it?

There are several reasons, though I will start with the most fundamental. Simply put, I see myself within the characters. Most particularly Howard Roark. Many of the descriptions of his won emotions and others description of me mirror is a strange way. While not absolutely similar, it builds off in a way more closely connected than any other fictional character. How he exists as an independent entity, not noticing others but living by their moral code not out of other means but as its own mean. Because of integrity above all. The way he is an act of moral striving rather than a disgusting abstraction for those even more disgusting to connect to. He lacks those pathetic neurotic tendencies that those around me let control and give authority over them.

Beyond that, he gives me ways to articulate my own personal feelings in a way that I have never seen before. Being "too proud to boast" by seeing both the criticism and complements of the world to be equally insignificant because it does not come from myself. To find pride, value, truth, within myself not within the pathetic world beneath me. Seeing another person see how pathetic the social validation games are, not is

disdain for others but as seeing it beneath myself.

The way happiness is the his natural order rather than some far flung ideal that is beyond, that negative emotions seem dulled by their pathetic attempt. That he is truly happy at all times, just as I am.

The way compromise, even in the slightest way seems to be evil. That this concept has been a driving force in my life thoroughly. Because it is a self-betrayal, a betrayal that can't even be thought about. An evil beyond belief and idea. Even a white lie, a broken ideal without real backing, a principle made as a child, and so much more seem evil and I can't figure out how others live with it. Though these compromises are never entertained long enough to feel anything beyond the knowledge of it.

The way others see him as cold and arrogant despite him clearly not being, due to there misunderstanding of him. Because he is independent, because he doesn't care about friends and what insignificant interactions they had, what their friend's did to annoy them. That they don't care about philosophy, physics, mathematics.

How he is both happy in isolation and with others. Equally, because the existence of others doesn't have that effect on him.

The way his creative and logical thoughts of Philosophy, physics, mathematics, and other intellectual topics are all that matter, well beyond the existence so often people confine themselves to.

Finally, he lacks all neurotic tendencies. No desire for complements, praise, people to soften their words, bend down. These neurotic tendencies are found everywhere, in everything. People, fictional characters, and others. Though I feel such thoughts so rarely. It is a impeded idea in almost all of fictional by its own virtue, but I never see it within my own mind. He like me is truly beyond these petty neuroticism, no capacity for

them at all. No vulnerabilities, no anxieties, no insecurities, no of it.

56.4 On the Pursuit of Thought

NFC, or Need for Cognition is a psychological concept seen clearly in this book. Though it may not be obvious the the extreme extent it is true.

Much of my ideas of philosophy, ethics, physics, and more may seem like a desire to know reality in its greatest extent (and this is certainly true), but in its most basic and primate way, it is my need to think.

I love thinking, it is my favorite thing. This is what drive me, my desire to satisfy that part of me. I think about anything complex enough; philosophy, ethics, physics, mathematic, coding, economics, political philosophy, literary analysis, world building, international relations theory, geopolitics, formal analysis, psychology, meta-cognition, and so much more. The more thinking required the better.

In fact, I love it the most when it is beyond me. When it takes me weeks to not even fully understand what questions to ask, when it feels beyond my comprehension, when I think for days and go no-where. I love it, the scavenger to knowledge, then to actually get it. For it to all fall into place just as it should.

Now very little does this, in fact most things just come. They are understood intuitively. Seem to basic to even consider. Even some of my other hobbies like psychology, politics, and such seem to basic, and most of the other things beyond my intellectual hobbies seem so basic as to not even give it time at all.

Another funny consequence, is despite my almost compulsion to efficiency, I still am drawn to complexity. Despite my

normally physicalistic and literal tendencies I am also drawn to the abstract. I have found this most easily seen in my obsession with using higher order math. I use tensor calculus, hamiltonian mechanics, geometric identities, when similar methods can do it. I obsess over these ideas when there are more practical matters.

56.5 Nietzsche's Sovereign Man and Morality

Upon further reflection, my chapter on disgrace and pride has much connection with Nietzsche's "sovereign man." Here I will first explain what that is, its connection, and divergence.

Before I begin, one thing with Nietzsche's writing is that it is up to controversial interpretation. Some say the sovereign man is sincere, other ironic, others meta-ironic, others see it simply as a literary device, and there are still other interpretations out there. Luckily though, for the purposes of this exercise it doesn't matter Nietzsche's intentions, only my intentions and ideas. For instance, Nietzsche's reasoning for presenting the sovereign individual differ from mine; which I will present later.

Now to actually begin with the analysis. Who is the sovereign individual? The sovereign individual is the archetype of Nietzsche's will to power(the *Übermensch* later takes its place as a further extreme but I care little for this idea.) It is an implied ideal in which a person creates mastery over their own life. To live their life in accordance with their values, that they themselves had created rather than inherited. They do this with their mastery over their own impulses; a mastery so extreme that they eventually shape their impulses into however they consciously wish. Further more, these values and 'morals' are created through aesthetic ideas, creativity, future-bound, and

affirmation to the power of life rather than traditional fear. Finally another key part in is surprisingly forgiveness, though not in the traditional sense. Rather than forgiveness caused by God or some other values it is a combination of the acknowledgment of the fact that most people are incapable of true moral thought, self-mastery, and to let go of their resentments and impulses; another key idea is simply the fact that not forgiving hurts the sovereign individual, Nietzsche's suggests that it is better to simply forget about the trespasses rather than hold on to resentment and call it responsibility to forgive when it is hidden resentment. Essentially the sovereign individual is the archetype of the promise made flesh(or word)

Now how does this relate? Well before I get to pride and disgrace, let's explore the philotimic virtue. The philotimic virtue highly mirrors the concept of creating your own values, setting your life and soul in pursuit of them, and finally holding yourself to them in a way that can be seen as fanatic to the outside. Also, creating these values individual of the existence of others(though the philotimic virtue includes the will and thought of God to yet be higher than the will to power of myself.) To live your life as you will it so. T

Now disgrace, the idea that my personal values that I have willed hold the same good and evil. That to betray them is evil. But both ideas have a similar theme, by not hiding from it but rather carrying the burden is guilt erased. By taking the conscious action of purifying oneself one becomes worthy of forgiveness(not God's forgiveness but my personal one) and because I have achieved this, while I carry momentary understandings of mistakes unlike the sovereign man, I hold no returning 'guilt' or resentment of any kind. No neurotic tendencies, no projections onto others, none.

Pride, this as mentioned earlier is the same as the earlier mentioned philotimic virtue. The idea of creation and happiness as a norm that has been willed by me as an action of my

pursuit of virtue and creation. This is where the disgrace truly comes in, it is reabsorbed, not as pain but as energy for the recreation of the self in the form of higher values.

Another key idea is that of forgiveness. Once again there is the connection between my idea and his. The fact that for the self; moral, ethical, value-driven, etc is not a same thing. It is a rapture of the very existence, but rather than waste energy on being sorry for oneself, one must use that energy on self-perfection. Though, forgiveness of others is another story. Other people are not capable of moral action, they are slaves to impulses, temperament, social constraints and much more. While people claim to be moral, much of there actions disgrace their ideals, their intentions align with physiological temperaments rather than values, and they abandon so much so easily. Their transgressions should be forgotten, or better yet not noticed at all.

Though, I am sure you are thinking about the differences(I mean I have been sprinkling them about these section) and while they are important to some extent. I don't want to go though a point by point refutation of Nietzsche ideas, especially when this analogy of the sovereign man is meant more an a psychological analogy to help explain intuitively my own personality rather than philosophical affirmation.(I disagree with Nietzsche on much)

One thing I will say though is the difference in end goal. For me, the sovereign man in a more refined sense(axiomatically driven, recursively made, God fearing, etc) is the final form. The *Uebermensch* is useless. The chaotic form of disparaging logic, reason, and God in forming morality disbanded the entire project and is just as disgusting as the slave morality of others.

56.6 Half-Growth and Half-Death

While many proclaim that my ideas of seeing people not as thinking being, not as moral agents is misanthropic and anti-social. That I should have more respect for others, that this lack of respect can lead to later personal problems with others. Though I disagree, first I will go over how I have seen others interact that makes me disagree, and my own personal evidence from my falterations with these values.(one thing I will add, is once again these are not true literal beliefs but analogies to explore esoteric identities.)

Think of how people treat children, then people they deem as equals, then themselves, and then those they deem as higher than them when they perceive a slight.

The child's slight is either ignored or the person takes conscious action to help them, not by expecting moral action as they would with others but by understanding the child's abilities and inabilities and working around them in a kind manner.(Bar extreme causes or grotesque individuals)

With others there exist a range of reactions to slights, but their is a clear differentiator. They expect others that are 'equals' to exist in a semi-moral manner(this is because most people hardly understand moral manner to begin with). They try to act in kindness and understand others faults and temperaments but when push comes to shove they resent others, they expect from others. This can come in a wide variety of ranges: resentment, anger, fear, annoyance, instability, and so much more. These reactions are almost always unproductive and hurt relationships, people and such. They also use the perceived moral responsibility to negate their own moral responsibility in many cases(under my analysis, this is where most pain received by the affliction of others comes from, either consciously or unconsciously)

Then from here it is easy to see that the extremes of these go up and up.

These are where a majority of the petty, vapid, and pathetic emotions I wrote about earlier come from.

Now what I suggest isn't as radical as it seems, it is just shifting the average person closer to where others see children, no one other than God as above me. That I intentionally analysis and 'handle' other people. I understand them both to better see them as pathetic not in a disgusting way but like a child. Also, to better handle them for interactions with them.

Though, one thing I didn't include in my earlier writing is that some people do come closer. Select people I know that I am close to enjoy(or don't depending on how you look at it) the responsibility of a human and conscious life within my own ideas.

Now, where does my growth and death come from? Well in recent time I have faltered, I have subconsciously found that I do expect things from others. These is so terrible in my eyes from several reasons. One, as I have mentioned earlier, this 'equality' is the source of much of our troubles in the modern age void of true horrors. Second, for my personality and moral theory this effects me in a much more extreme sense than it does other. From a shallow perspective I am an extreme puritanical person who is obviously very prone to disdain. Beyond that I hold some values in high regards that many don't hold at all. Though, more truly than that, my in my moral theory intentions mean everything and so very few people have truly pure intentions. Now I don't mean everyone is a selfish jerk(though many are) but beyond that many are husks of people that only follow moral theory because they don't comprehend any other way, others hedonists that find minor moral action as easier, others do it because of insecurities, and so on and so forth. The concept of doing things for the mere fact they are moral/-logical is lost on almost every person you will ever meet.

Here I will give an illustration. Last week(as of writing this section) I had an interaction. Someone had very condescendingly given me 'advice.' Telling me in a clearly condescending and antagonistic tone to remember to unroll my selves I had rolled up to wash my hands.(this is also after of many other similar interactions with this person) Now because I knew that to explain the ethical ideas of attempting to 'assert' fake authority on another in such efforts was immoral due to the fact that on a psychological basis that such things could very easily be comprehended as attacks on their own self-determination and competency. Also, that such actions could easily be interpreted(and likely truly) as things like need for superiority, low self-esteem, extreme lack of social awareness, control issues, projection of their own incompetence, desire for a reaction, or malignant/grandiose/competitive narcissism. Though I instead let it slide and said thank you, I didn't feel it would be worth it or would give any results on the matter. Now I didn't expect them to realize their fault or anything of that extreme. I expected them to say your welcome either as a empty social connection or an understanding of the fact that I understood their game and they would either give up or hopefully be filled with self-disgust over the conscious acknowledgment of why they did what they did. Instead they didn't make eye contact and instead "hhmf" at me. I still had the rational ability to understand that this person was either too stupid or immoral to understand whatever I wanted to say on the matter, so I decided not to. Though regardless I was filled with disdain. I was disgusted by such extreme evil. I know for many this doesn't seem like evil, but in the eyes of intentions it can only be logically assumed as. Those of lacking ability of intelligence to react in moral manners wouldn't make such an extreme mistake, those powered by resentment or anger would likely be filled with shame of such actions, and so on and so forth. Likely the only remaining interpretations would be that of malignant/competitive narcissism, desire for the reaction/-

pain of others, and other similar grotesque ideas. Now while I see the earlier stated ideas as immoral, these take special places, especially for someone like her who has the capacity of living in dignity.

Now this isn't some one shot, lately I have been more and more likely to feel disdain for others. I don't know if it is puberty, social connection, increases in closeness with others, or whatever. All I know is that while this is something that so many other have proclaimed as what would be the greatest thing to my self-perfection is rather the worst thing.

56.7 Lack of Aesthetic

When I am referring to aesthetics in this section, I do not mean philosophical aesthetics, as in valuing something over another; I mean traditional ideas of beauty in music, art, and natural affairs.

In this sense of the word, I lack almost all aesthetic values. I have no favorite color, no concept on beauty in most things, and no care for musics, visual art, or other 'artistic' concepts.

This doesn't mean I have none. I have some interest in the artistic understanding in complex and intriguing literature, I can also be temporary incapsed by complex art/music(though only as long as it takes for me to understand it, and if it is too abstract as to be useless I have little or no care for it); beyond that I can find beauty in mathematics, logic, efficiency, ideas, plans.

In relation with that I have no feelings of sentimentality, meek emotions, and other similar emotions.

Now, I have always naturally found all of these things as beneath me and childish(and still do to some extent) though I have been taking time to try to expand my horizons.

As I have been doing this, I have noticed some changes. I have thought in more lofty ways than before, been more open to some experiences, and even engage with emotions in a way previously unheard of. Now here I will leave with this, but later I will explore this in more detail. What is actually causing this change, Explore what the change actually is rigorously, and whether is it actually a good or bad thing?

56.8 Boredom

Just as NFC is a primary driver in my life, so is aversion to boredom.

The most obvious connection is simply for my NFC is grown and in a feedback loop with my aversion to boredom. I satisfy my boredom with challenging cognitive tasks. This is one of if not the largest driver in my self-education, research projects, writing this book, and so much more are driven by aversion to boredom.

A little beyond that is ambition. Just as I choice challenging cognitive tasks to satisfy my boredom, so do I choice other challenging tasks. Leading, planning, mentorship, responsibility, work, physical activity, and more. This is what pushes me in SVA, Slack, JROTC, Scouts, OA, and much more. These activities satisfy my boredom.

I will go even further and say this desire for challenge is primary from aversion to boredom, not fear, perfectionism, and other traditional psychological reasons for people pushing themselves further than most.

Now, my aversion to boredom isn't just in ambition and other productive means. The most obvious is in watching TV or non-educational videos. Though this is largely to simple to take any time analyzing.

Though there are other things to analysis, relations. First I will go over casual relations.

For some reason, as I will discuss in more detail later on, I switch between 'extroversion' and 'introversion' in a sense. When I am away from others I hold no desire to be with them, I satisfy my boredom through abstract thought, arguments with myself, physics, and other similar things. That this feels the most natural thing. Though when I am with others this switches. Getting lost in thought no longer feels natural but rather challenging and hard to focus in the same way. I naturally feel the way to satisfy my boredom is through conversation with others. It is an interesting development that I will likely look into further.

56.9 Why Physics

While I have many interest; math, coding, economics, psychology, philosophy, logic, and many more. Though, one interest stands above and beyond the others. Physics. Ever since I was a kid I have been obsessed with physics.

Now I am sure I can come up with some lofty reason on how physics is the nature of the universe in its purest form, or the most fundamental science of them all. While I am sure that it is part of the reason I like physics so much, it is not one of the primary.

The most easily observed is its balance of abstract thinking and practicality. It is one of the most abstract and purest forms of logic other than pure maths and some forms of philosophy, but unlike the others it has a more direct connection to real-world application. For instance, my work is working towards fusion reactors, something I think once working will be one of the most influential and greatest creations of the modern age. To combine both reduction to axioms and analytics and

contructionism of creating something practical.

Next is the challenge. Not only is physics itself extremely challenging, with some aspects of it taking months of studying to even comprehend it, it also combines many other hard disciplines; math, coding, engineering, and even metaphysics as times. This challenge is exhilarating and as mentioned elsewhere is one of the things that i most desire.

On combining other things, physics is a multi-disciplinary subject. Combining and using many of my other hobbies.

There are many more reasons but these are the main ones.

56.10 Lack of Resistance

As developed throughout this book, especially in "On the Pursuit of Thought" the idea of challenge as a goal.

The joy of an intellectual challenge specifically. To mull over a topic for hours. To have something beyond current comprehension. Something that doesn't make sense. Then all of a sudden explodes, not only to explain itself, but making connections all over. Like a flood.

This is my greatest joy, what I live for. Though, even as a teenager I am coming to limits. I will examine this in parts, first through my self-education and then through external world interactions.

What is going on personally is I fear that I won't feel this feeling. Many topics like economics, psychology, geopolitics, history, philosophy, literature, and more don't give this rush anymore.

As an example I will examine literature. In the past year I have found interest in literature, a topic I had previously missed. It was enticing, learning about the usages of symbolic characters(and figuring out who was and what was), setting as

a character, philosophy through stories, and so much more. I had that excitement, though lately I don't. While there is still more to learn, it isn't the same. Everything new, is obvious. No requirements of complex thought when the workings are already there, only new information to be feed into the pre-made algorithm.

Now, I still have some exiting things in the above fields and much, much more in abstract mathematics and physics, but I am only 16 at the moment. If I am already this far along now, what is to say in 10 years there will be much left, what about 40? I have already surpassed all traditional classes and all that is left is research articles and manuscripts, these will leave me for quite some time, but who knows how long that will be.

This also continues in my personal life, school has the intellectual engagement of watching paint dry, talking to people is a burden, leadership roles still have excitement but have less and less return,

56.11 Am I Misanthropic?*

My misanthropic like tendencies are clear throughout this book, but am I really misanthropic? No, well... maybe a little.

I do have many friends, friends that I enjoy. I like talking to people. I am truly an extrovert in nature.

Though, beneath that there is a disconnect. I don't truly enjoy talking but rather staving off boredom. While in isolation I generate distrust and disgust of others. Many people I dislike more than I dislike boredom and everyday the percentage of people within that camp grows.

56.12 Depth of Feeling

All of this discussion on emotions and psychology can leave a reader with the thoughts of a sensitive and emotional boy, but this is far from the case.

While I use extreme words to convey my ideas, the extremity is not there. All my life my feelings have been dulled. While others are overwhelmed by feelings, for me they are hard to observe. Like they don't fully touch me.

This isn't true for all emotions; my passion, excitement, fanaticism, and need for cognition are all real. This doesn't take away from my thesis.

Beyond these, I have always required conscious effort to hold on to emotions. While this may sound like lamentations, it is not. It is a great thing to forget anger and pain because of their insignificance. To never feel overwhelmed or anxious, to have control over my thoughts and actions, I am in control. I love that I am this way.

There was once a time that this made me feel inhuman. While others talked about their emotions, though that feeling like all has passed. Though, don't take this too far, I am by no means some robot devoid of all emotions, just simply that for any population with variance some people will be above or below average at things. For me I am simply far enough that observing others like me is a rare enough occurrence to think it truly is rare (but it really isn't.)

56.13 Language Usage*

Before I actually begin upon the main idea of this section I feel the best way to introduce this is to explain the inspiration of this section.

The inspiration is the oddest place possible; Ted Kaczynski, or the Unabomber.

I remember when I was a kid I always hated the idiom "Have your cake and eat it too." The reason for this is due to the fact that the have your cake is placed first, and this fit(in a way that doesn't fit with the idiom.), because you must have your cake to eat it. So, I started to switch it around to "Eat your cake and have it to" though never really liked that. Until I read that part of the reason the Unabomber was caught was due to his strange phrase "Eat your cake, and have it all the same." When I had read this my first thought was insult that I hadn't thought of that myself.

Once I thought of that I realized more connection. Some more 'normal.' Starting with a thoughtfully researched and observed effect being the fact that highly educated members of fields like physics, mathematics, logisticians, computer science, and philosophy use mathematical and logic terms in regular speech. This is because their thorough and clear definitions have clear applications. Axioms, manifolds, prior, bias(in a mathematical sense), fallacy, paradox, induction, inference, deduction abduction, ambiguity, equivocation, disjunction, mapping, inversion, duality, convergence, divergence, topology, entropy, singularity, resonance, phase transition, field, fractal, lattice, span, gradient, symmetry breaking, null, second order, and more. This isn't that odd considering that it is more wide spread, I mean a couple of those terms described above have even introduced themselves into colloquial speech(though used slightly differently.)

Another is using words that were created in academia, then introduced into regular speech that then changed it, in their original usage.

Though some are a bit stranger, mainly the fact that I simply create my own definitions for words or even create my own words.

So lets go through these one by one.

The first is very simple, these words are very useful in regular speech, convey abstract ideas easily, and can be applied easily. The mere fact that I understand them makes it only logical that I use it.

Next, the usage of academic definitions has a similar requirement. The fact that these academic definitions are generally more completely and logically defined than generally used word.

Finally, the changing of words and creation has a similar efficiency and logic.

Though, this brings new questions, why do I feel so comfortable with this, what criteria do I use for greater 'logic,' what is with this obsession with logic, why don't I just pick either change current words or create new ones instead of do a bit of both, and much more.

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56.14 Euphesus*

As I mentioned earlier, there has been change in me as of recent, and it has recently come to head with the advent of 'dating'(not actually dating the girl formally[will explain more later]Also not completely sure it is 'coming to head' just more extreme than before).

What I mean, is for the first time in a long time I feel conflicted. As strange as it may sound, I have been complete for a while now. I never feel internal conflict, neuroticism, second guessing myself, or anything of that such. I have for a very long time clearly defined my morals and values and thus never needed to second guess myself. While I am not perfect,

the few times I make mistakes, I clearly identify them after the fact and rectify them. There is no lingering guilt nor do I go back and forth between what I should have done or should do for the future.

Now, in the last year or so, I have had some rare conflicts; many of whom I have talked about in earlier sections. Though with this ambiguous relationship, it has become an almost daily thing.

I went back and forth between whether or not I should ask her out in the first place. When I did I had a lingering regret about doing it over text rather than in-person. Later when texting her after the fact, I felt the inexplicable desire to talk to her, without any real end-desire other than the conversation. First, I felt conflicted because I have never felt the such a desire, in fact I often was disgusted by others for this desire. It always seemed beneath me, while it wasn't against my values, for someone who has always been so 'complete' and whose entire personality and temperaments are recursively defined by myself the mere fact of something new is insane upon the face of it. It is like a self-betrayal which it took me a while to figure out what the betrayal was in the first place. Furthermore, this caused me to start questioning thing, why I didn't desire to talk to my friends about nothing in particular, should I start doing it, etc. It even lead me to try it. Then this started to make me think about how some random girl I am not even formally dating, has caused me to alter my own behavior. Just slightly, but nobody alters me even slightly.

Once I had finally gone through all of that and moved on, I decided I would talk to her over text. During that time I neurotically tried to when would be a good time and what to say. Once again, I don't do that. I don't care what other people think of me, I don't care if the message wasn't perfect, I don't feel neurotic over anything, certainly not a person. Yet I did, I thought about it. Finally, I texted her, it went nice.

I decided I should keep in touch, because I wouldn't see her in person for a month. So I decided to reduce ambiguity I would simply have a schedule of when to initiate a conversation. The number of days between initiations was meaningless and I knew I just needed some arbitrary number, yet I thought about what would be an embarrassingly long amount of time.

Later, came the day for my next initiated conversation. and once again I spent way to long figuring out what to say, but when I did, she hadn't responded. I didn't think much of it, due to the fact there were dozens of possible explanations for the lack of communication that were fine. That was until the next day, when she responded on a group text, but continued to ignore the individual one. Now even still, hours later I am still thinking of it. Still conflicted, wondering if asking her out was the right choice, or my number, or what I said. Wondering what my next steps should be.

While yes, I understand these emotions are normal for an adolescent boy of my age, but they aren't for me. Not only am I rarely conflicted about anything. I am much less about people. I am the kind of person who interacts with people for amusement or out of responsibility; deep down I don't care other people in the way required to feel these neurotic thoughts and emotions.

It is so strange, having the last couple years of my identity and foundation so clearly defined for everything. Only to bring out something new, something that doesn't fit. Now I don't know what to do; stay rigid or change (but change what)

Over the last month or so I have gotten over the visceral reaction. Now that I am back from my trip we can see each other in person, but haven't very much.

Chapter 57

Self-Authorship

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57.1 Scope and Understanding

It has always been a clear belief of mine that it is a persons duty and responsibility to define themselves though conscious thought and action. It has been something on the forefront of my mind. As mentioned in an earlier chapter on post-surgery and sicknesses, I haven't been the person I used to be.

Regardless it is a very important thing to be done. To clearly define the virtues that define how I should live my life and then make sure that I live in accordance to them.

I will begin defining who I want to be, much of it has already been defined in ethics, but personal ethics will be defined as well. Then, I will look into actions I can take.

57.2 Ideals of Becoming

57.2.1 Overview of Virtues

In becoming I define a set of categories of virtues that will be further defined.

Virtues Intellectual

Virtues Volitional

Virtues Executive

Virtues Stability

Virtues Adaptive

Virtues Social

While these are not equal categories nor exhaustive, they work as a good way to organize these essays for now.

57.2.1.0.1 Intellectual Virtues To be able to create oneself, one must be conscious of how to do so and who one wants to be. Some underlying things must be true.

I believe that someone(me) should have a world-view. One that is completely constaint, that brokers all areas, and is useful for further creation. A person should not simply listen vapidly and repeat the words of others without clear thought, but must have a formalized system to be able to not simply add information but be capable of sythesising it and using it in a broader context. To have a rigorous view on everything and logic behind it. This is the only true way to actually learn and function, without it you are simply repeating others with the variations of subconscious bias.

One must also have clear meta-cognitive skills, to question oneself if the only way to clearly understand something. While this can become natural through rigour, developing the natural tendency may speed of the process without formal rigour at every turn.

Knowledge and understanding is incredibly important. To use, develop oneself in the abstract sense, and to be able to abstract for future knowledge.

The ability to abstract things to develop clearer understanding of it or relating systems. Always look for the underlying mechanism beneath, not simply the beginning and top and leave.

Humility to understand that which is beyond.

I want to be someone in constant thought, of ethics, social responsibility, understanding the world around me. While maybe not the best decision, it is mine.

Also, have your ideas be your own, fit them within your formal world-view and build upon them.

Chapter 58

Philotimic

58.1 Intro

As I am moving on to college, one thing I note, beyond traditional internal self-authorship, there also exists the aspect of image. Image enforces behavior through wanting to be consistent, consistency builds virtue, image brings you together with like minded people, etc.

To continue to develop myself both internally and externally through consciously practiced drills and habits is my goal. The highest human good in my world view. This area is going to be used both to explore the virtues and habits I wish to do and my way of practicing them in order to work on it. Then, log my activity.

58.2 Overview

58.2.0.0.1 Self-Perfection In this process, I will make mistakes. The purpose is to grow. I tend to forget about mistakes as soon as resolved, thus, a plan would be to write down daily

errors, mistakes, way to improve, etc. I will likely put them in the drill log later in this chapter. Another option would be in the journal. I will figure out soon.

Some other drills would be to argue against my own position, Steelman other, and write about when my views change.

58.2.0.0.2 Outward image Before entering any professional room, take three seconds to check posture, breath, and warm neutral face.

Try to repeat people's names in order to help develop memory of them.

Whenever walking into a new room, come with a question or striking statement prepared.

I will also try to record 60 second introductions and work to improve them.

58.2.0.0.3 Speech, Vocabulary, Diction, and Rhetoric To improve my speech, I should work on trying to remove both filler words and vague quantifiers.

Three ways to practice this:

1: as I am developing my YouTube Videos, I will naturally develop these skills when working on the scripts reading them.

2: I can develop things that I wish to describe, explain, or define. Then, record myself speaking on them in order to see ways to naturally improve.

3: Lastly, take something that I said to another person and take some time each day to rewrite in a better form.

58.2.0.0.4 Academic Communication After stating a result, explain the scope and research.

when explaining or talking about a result. State the result,

evidence for said statement, and finally end with limitations of my claim.

Write down paragraph summaries of research papers.

58.2.0.0.5 Assumptions and Cognitive Habits Many people, including myself, rest ideas on assumptions we don't even know we have. I would like to work on this.

One way that I have already started is by writing my axiom book.

Another would be randomly taking random assumptions and either questioning them or proving them.

58.2.0.0.6 Bayes Reasoning Using Bayes reasoning to think more logically is Paramount. Getting to the habit is the problem.

One way would be to continuously be making predictions, guesses, conclusions, etc. Evaluate them using Bayes.

An example would be through every new room, evaluating people around me. Look at what they are doing, guess why, guess their job, age, height, and many more. When meeting people truly, try to come to a psychological profile of them of sorts.

58.2.0.0.7 Meta-analysis When evaluating my thinking, actions, plans, etc. Don't be vague, come up with an number.

58.2.0.0.8 Attention Evaluate wandering and conclude I am doing it. Don't feel shame and such, just fix and move on.

Try to be attentive even without deep work. Ponder, think on things, observe, to constantly be in a state of thought.

58.2.0.0.9 Observational awarness Many examples have already been said. Conclusions on people, observe things like exits, direction, fire extengushers.

58.2.0.0.10 Leadership The most obvious thing is to get into it and figure out ways to obtain leadership positions.

Some practice subjects could be mock meetings,

58.2.0.0.11 Dealines Every deadline or commitment, write it down to not forget. Get into the habit of using outlook and not relying on memory alone.

58.2.0.0.12 Body Language Straight posture, long gait, eyes straight, weight even, feet shoulder width apart, hands unclapsed and seen, use hands rationally, and no stress responses.

Videos of me explaining things can be helpful in this regard.

58.2.0.0.13 Memory Practice memory skills. For instance, now I want to work on LaTeX snippets and Physics library, so I will work on that.

58.2.0.0.14 Desicion making Evaluate desicions. The information used to make them, my confidence of the results, actual results, how good/bad of a disusing it was, etc.

58.2.0.0.15 Teaching Practice teaching as much as possible; YouTube, personal videos, or writing.

58.2.0.0.16 Schedule I have now written schedules, I want to follow them and by my nature develop them for the future.

Once a week I will work on refining them.

58.2.0.0.17 Humor Avoid too much self-deprecating humor as I have done in the past. Also, avoid making fun of others. Work on continuing my general ironic humor delivered in an ironical dry tone. I think that works well.

58.2.0.0.18 Writing Write more in this book. Even when I go to college, try to write something beyond lecture notes once a day.

I also plan to carry pocket note books with me.

58.2.0.0.19 Socratic Dialog In working on speaking, teaching, and debate. I would like to incorporate a kind of Socratic dialog with others. An obvious way to work on this is continuing to write essays. Another would be to actually practice it with others. How to slowly do that remains a challenge.

58.3 Drills Log

58.3.1 Short Explanation Ideas

Chapter 59

The Structure of Motion: A Philosophical Journey Through Hamiltonian Mechanics

“The Hamiltonian formalism of mechanics, especially in its canonical form, has been an inexhaustible source of inspiration for modern theoretical physics.” — Albert Einstein

59.1 Introduction

This chapter is planned to be an extreme examination of Hamiltonian mechanics, especially from a philosophical, logic, and abstract way. Eventually, it will go from the basic logic and axioms underlying it, to the metaphysics and epistemology that are derived, and finally to advanced and modern usages. The primary purpose of this exercise is for me to find the gaps in my knowledge and gain a more intuitive and deep understanding of these concepts. (It was also written in 10th grade before

the rest of the book so may not have the same standards.)

The reasoning for this is due to the extremely complex and abstract nature of such mechanics. To see the world through 'flows' instead of forces.

59.1.1 Physical Reality and Frameworks

Now, there are many theories, models, descriptions, framework, and so much more. Though, what really differentiates them? What makes one thing one and the other the other. While I won't truly go into extreme linguistic detail for every single one, I will go into moderate detail within the art to get the general idea down.

Let's focus on frameworks, because that is what a Hamiltonian is (others define it as schemes which is another great word and arguably more accurate, but I will refer to it as a framework because it has a more intuitive grasp). It is an entirely different framework than Newtonian. It comes from an entirely different perspective, looking into geometric identities rather than the causal relation of forces and such interactions. Also, other theories and identities are derived through it, independent of the actual reality of it. For instance, the flow of electromagnetic interactions, relativistic, or even gravitational are all derived from this concept of the flow of energies through sympathetic geometry.

Even beyond that, Hamiltonian mechanics can represent any function that changes; statistics, viruses, and many others. This is due, to that Hamiltonian's is a pure mathematic relational concept rather than based upon physical axioms. For instance, Newtonians is based upon the axiom of inertia. Though, I will add one interesting involvement is the idea of privileging Hamiltonian over Lagrangian and stating that Lagrangian is a subset of Hamiltonian rather than its own frame-

work.

Then this begins to ask our questions of scientific realism vs anti-realism(I found out the more neutral language is instrumentalism). Basically, what makes such theories more fundamentalist, predictive modeling or metaphysical truth. Further, this now engages the complexities of mathematical axiomatic reduction or physical.

Let's start with realism vs instrumentalism. I have already made my stance clear, having metaphysical truth in important but not in the way that non-metaphysical schemes should be thrown away. Though, this then brings about the question of what frameworks have greater metaphysical truth, but I will later look into this in later sections.

This now leads to the question of mathematical abstraction vs physical realism. Though, one thing I will add is that the physical axioms of Newton's laws are based upon falsified ideas like absolute space and time(but you can get around such semantics by employing Whewell's axioms of Mechanics[1st: "Every change is produced by a cause." 2nd says that: "Causes are measured by their effects." Finally, the third remain unchanged from Newton's formulation.] Now back to the main event, I feel that the structuralism of Hamiltonian is better due to the fact that when you break down physics to the quantum the classical forces, inertia, and such break down and you can see that the structure is what remains. That while forces still interact as abstract 'causes of change' through bosons, their classical and 'physical' realism fails to encapsulate their extreme complexity.

59.1.2 Philosophical Context*

The Kantian influence on Sir Hamilton is clear....

59.1.3 Notion of Time, States, and Trajectories

Let's start with time. I have already explained my thoughts on time, but here I will deepen my explanation. Time, is the factor on which states evolve. It is then evolution of entropy. This axiom is very simple and intuitive. While objective empirical proving the nature of time is extremely challenging due to its abstract nature. For a more in-depth analysis of the nature of time visit Chapter 1, section 3&7.

One thing I will discuss is time reversibility. Now, Hamiltonian mechanics works best for time reversal symmetries(it can still work in some use cases but rarely.). In fact, time reversal symmetry is a corner stone of much of modern physics, especially quantum mechanics, but this interferes with our intuitive grasp of reality, the second law of thermal dynamics, and even dark energy; in essence the emergent properties of reality seem to interfere with our 'fundamental mathematical' derivations. So I will examen these questions; both for their own state and because it continues our question of the fundamental nature of reality. I will also attempt to discuss imaginary time with regards to quantum mechanics.

Actually, upon further thought the nature of time will be further developed through after the derivation of analytical mechanics.

Next will the on trajectories; now like many other ideas presented within this book, that may seem simple but the derivation of the ideas will be thorough and show its true complexity. Within Hamiltonian mechanics, the trajectories hold special geometric identities due to the nature of motion, but what can we derive from this simple observations

59.1.4 Introduction to Phase Space

Another question that arises from a Hamiltonian view of meta-physics is Phase Space. The fact that, according to our theories, phase space almost seems real, but yet it is so very different than what we observe. Or, we think so.

Obviously, there is the idea of the conflict between the teleology that comes as a consequence and our view of causality. I will address this in a second.

A simpler idea is that, from a perspective view, change becomes fundamental and any identity becomes simply emergent. This isn't truly new, simply more emergent in phase space, so I will move on.

One new thing, is that Phase Space allows for entropy irreversibility. Traditional theories find time reversibility to be a corner stone, this going against it.

This is due to the fact that entropy increases

$$S = -k_B \sum_i p_i \log p_i$$

Where the volume of the phase space stays the same

$$Vol(\phi_t(\Omega_M)) = Vol(\Omega_M)$$

Lastly, the 6D phase space seems very different to ours. Having generalized position and momentum coordinates is odd. An instrumentalist view makes this simple, but a scientific realism makes this difficult. Sadly, I don't have the knowledge to come up with.

59.1.5 Teleology vs Causality

My privileging of causal relations has already become clear, but now I will justify my thoughts through a Hamiltonian view

point. While many claim that the nature of Hamiltonian mechanics necessitates a teleology, and my stated view on structuralism seems to contradict my views on causality, but I will address these claims categorically.

First, the nature of Hamiltonian mechanics necessitates a teleological view of the natural world. This statement will seemingly logical, lacks the tautological strength required of such an extreme statement. For one, Hamilton himself had a clear belief in causality(while this obviously isn't enough, I will explain why he and I see it that way.) First, the concepts of causal forces(forces are by definition causation principles) can be derived through the inverse gradient of potential energy. Second, the connection of the 'principle of least' action and teleology is flimsy at best. This is because their main argument is based upon intuition of mathematic alone not physical realism(I know this goes against stated structuralism but I will come back to that), the reason I state this against due to the fact that, mathematical intuition in the face of metaphysical truth. For instance, as stated earlier the teleological 'energy' is found as a derivation of causal force. Finally, returning to our arguments placed in the first chapter.

59.1.6 Role of Variational Principles in Mechanics

59.2 Axiomatic Mechanics

59.2.1 Axiomatic Derivation of ...

59.2.1.1 Introduction

For any mathematical physical axiomatic derivation, a mathematical derivation is first required. I won't go super in depth

but I will still go over it for rigors sake. I may go back and do a deeper dive but for now I will simply put these things.

59.2.1.2 Axiom of Real numbers

Why needed: Physical quantities like position, velocity, time, energy, and action are represented by real numbers. The calculus used in mechanics relies on the properties of real numbers.

Key axioms:

- Commutative, associative, and distributive properties for addition and multiplication.
- Existence of additive and multiplicative identities (0 and 1).
- Existence of additive and multiplicative inverses (for non-zero numbers).
- Completeness axiom: Every non-empty set of real numbers with an upper bound has a least upper bound (ensures continuity, critical for calculus).

Role: These axioms enable arithmetic operations, inequalities, and the construction of functions like the Lagrangian $L(q, \dot{q}, t)$.

59.2.1.3 Axiom of Euclidean Geometry

Why needed: If the system involves spatial coordinates (e.g., particles moving in 3D space), Euclidean geometry provides the framework for defining positions and distances.

Key axioms(informal):

- Points, lines, and planes exist.

- A straight line can be drawn between any two points.
- Distance between points is defined (e.g., via the Pythagorean theorem).

Role: Defines generalized coordinates q (e.g., Cartesian or polar coordinates) and kinetic energy terms like $\frac{1}{2}m\dot{x}^2$. Note: For abstract systems (e.g., in generalized coordinates), geometry may be less critical, but it's foundational for physical intuition.

59.2.1.4 Axiom of Set Theory (Basic)

Why needed: Sets are used to define the domain of variables (e.g., time $t \in \mathbb{R}$, coordinates $q \in \mathbb{R}^n$) and functions.

Key axioms(informal):

- Existence of sets: Sets can be formed to represent collections of objects (e.g., possible paths of a system).
- Union, intersection, and Cartesian product: Allow combining and manipulating sets.
- Axiom of choice (implicitly): Ensures a choice of path exists in variational problems.

Role: Provides the language for defining functions, spaces, and the configuration space of a system.

59.2.1.5 Axiom of Calculus (Differential and Integral Calculus)

Why needed: The principle of stationary action involves integrals (action $S = \int L dt$) and derivatives (in Euler-Lagrange equations).

Key concepts (built on axioms of real numbers):

- Limits: Define continuity and differentiability of functions like $L(q, \dot{q}, t)$.
- Derivatives: Partial derivatives (e.g., $\frac{\partial L}{\partial q}$) are used to describe rates of change.
- Integrals: The Riemann integral defines the action S .
- Fundamental theorem of calculus: Links derivatives and integrals, essential for variational calculus.

Role: Enables the formulation of the action and the variation $\delta S = 0$.

59.2.1.6 Axioms of Variational Calculus

Why needed: The principle of stationary action requires finding the path that makes the action stationary, which is a problem in variational calculus.

Key axioms(informal):

- Functional: The action S is a functional (a function of functions, e.g., paths $q(t)$).
- Variation: Small changes in the path $\delta q(t)$ are used to compute δS .
- Stationary condition: The path satisfies $\delta S = 0$, leading to the Euler-Lagrange equations.

Role: Provides the mathematical machinery to derive the equations of motion from the action.

59.2.1.7 Axioms of Linear Algebra (Basic)

Why needed: For systems with multiple coordinates or degrees of freedom, vectors and matrices describe generalized coordinates q_i and momenta p_i .

Key axioms:

- Vector space axioms: Addition and scalar multiplication of vectors (e.g., coordinates in $\mathbb{R}^n$).
- Linearity: Operations like partial derivatives in Hamilton's equations are linear.

Role: Supports the formulation of the Hamiltonian $H(q, p, t)$ and phase space (the space of (q, p)).

59.2.1.8 Axioms of Variational Calculus

Why needed: The principle of stationary action requires finding the path that makes the action stationary, which is a problem in variational calculus.

Key axioms(informal):

- Functional: The action S is a functional (a function of functions, e.g., paths $q(t)$).
- Variation: Small changes in the path $\delta q(t)$ are used to compute δS .
- Stationary condition: The path satisfies $\delta S = 0$, leading to the Euler-Lagrange equations.

Role: Provides the mathematical machinery to derive the equations of motion from the action.

59.2.1.9 Physical Assumptions (Not Strictly Mathematical Axioms)

While not mathematical axioms, certain physical assumptions are mathematically formalized:

- Time is continuous and one-dimensional ($t \in \mathbb{R}$). (Though Hamiltonians can work in other forms)
- Energy is well-defined: Kinetic energy T and potential energy V are functions of coordinates and velocities.
- Differentiability: The Lagrangian L is sufficiently smooth (at least twice differentiable) to allow partial derivatives
-

Role: These ensure the mathematical framework applies to physical systems.

59.2.2 Axiomatic Derivation of Hamiltonian Mechanics

Two hundred years ago, William Rowan Hamilton reformulated classical mechanics using the **principle of stationary action**, a single axiom that unifies the dynamics of physical systems. This principle states that the path taken by a system between two times makes the action S stationary.

The **action** is defined as:

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt,$$

where $L = T - V$ is the **Lagrangian**, with T as kinetic energy, V as potential energy, q as generalized coordinates, and $\dot{q}$ as their time derivatives.

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The system follows the path where $\delta S = 0$. Varying the action:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt = 0.$$

Integrating by parts on the second term:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \delta \dot{q} dt = \left. \frac{\partial L}{\partial \dot{q}} \delta q \right|_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q dt.$$

Since $\delta q = 0$ at t_1, t_2 , we get the **Euler-Lagrange equations**:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0.$$

Hamilton defined the **Hamiltonian** as:

$$H(q, p, t) = \sum_i \dot{q}_i p_i - L(q, \dot{q}, t),$$

where $p_i = \frac{\partial L}{\partial \dot{q}_i}$ are generalized momenta. Typically, $H = T + V$.

Using the Legendre transform, the dynamics are governed by **Hamilton's equations**:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

Example: For a particle in gravity, $L = \frac{1}{2}m\dot{h}^2 - mgh$. The momentum is $p = \frac{\partial L}{\partial \dot{h}} = m\dot{h}$. The Hamiltonian is:

$$H = \frac{p^2}{2m} + mgh.$$

Hamilton's equations yield:

$$\dot{h} = \frac{p}{m}, \quad \dot{p} = -mg,$$

reproducing the equation of motion $m\ddot{h} = -mg$.

The principle of stationary action simplifies dynamics, applies to diverse systems, and underpins modern physics, including quantum mechanics and relativity. Hamilton's work remains a cornerstone of theoretical physics.

59.2.3 *Hamiltonian Mechanics ad a Geometric Theory

59.2.4 *Poisson Brackets and the Algebra of Dynamics

59.2.5 *Canonical Transformations: Symmetry, Simplicity, an Structure

59.3 Philosophical Implications*

59.3.1 *Time in Essence

59.3.2 *Hamiltonian Constraints and the Problem with Time

59.3.3 Noether's Theorem

Noether's Theorem is a brilliant application of formal mathematics to show that conservation quantaties define conserved quantaties. Such as time symmetry defines energy conservation.

59.3.3.1 *Ontological Priority of Symmetry

As I keep coming back to, is symmetry more fundamental than more traditional physicallist things like energy or momentum.

59.3.3.2 Laws as Autonomous Structures

59.3.3.3 The Continual Definition of Energy

59.3.3.4 Nature of Time

59.4 Modern Hamiltonian

59.4.1 Modern Research

59.4.1.1 *Hamiltonian Formulation in Plasma Physics

59.4.1.2 *Hamiltonian and Metriplectic Mechanics

59.4.1.3 *Symplectic Integrators: Preserving the Structure of Nature

59.4.1.4 *HNNN(Hamiltonian Neural Networks)

59.4.2 Fun Mess Around

59.4.2.1 Non-Canonical Poisson Brackets

A Poisson bracket is a bilinear, antisymmetric operation $[F, G]$ that defines the time evolution of a functional F via $\dot{F} = [F, H]$, where H is the Hamiltonian. Non-canonical brackets arise when the phase space variables (e.g., density, velocity, magnetic field in MHD) do not follow the standard canonical structure $\{q_i, p_j\} = \delta_{ij}$.

Properties: The bracket must satisfy:

- Antisymmetry: $[F, G] = -[G, F]$
- Leibniz Rule: $[F, GH] = [F, G]H + G[F, H]$
- Jacobi Identity: $[F, [G, H]] + [G, [H, F]] + [H, [F, G]] = 0$

For this I will be defining my own Poisson bracket specifically for plasma thrusters.

For this I will obviously need to define the variables. Things like:

- Mass density: $\rho(\mathbf{r}, t)$
- Velocity field: $\mathbf{v}(\mathbf{r}, t)$
- Magnetic field: $\mathbf{B}(\mathbf{r}, t)$
- Entropy or internal energy: $s(\mathbf{r}, t)$ or $\epsilon(\rho, s)$

Now define the Hamiltonian, for this I will use the pre-established ideal-MHD, but in all reality for plasma thrusters I should use a more expanded model.

$$H[\rho, \mathbf{v}, \mathbf{B}, s] = \int \left(\frac{1}{2} \rho v^2 + \rho \epsilon(\rho, s) + \frac{B^2}{2\mu_0} \right) d^3x$$

Next, I must define the phase spaces and constraints. Incompressibility or magnetic field divergence.

Next I propose a bracket system, with many of them being

$$[F, G] = \int \sum_{i,j} \frac{\delta F}{\delta \xi_i} J_{ij} \frac{\delta G}{\delta \xi_j} d^3x$$

Such that Ideal MHD is:

$$[F, G] = - \int \left\{ \rho \left[\frac{\delta F}{\delta \rho}, \frac{\delta G}{\delta \mathbf{v}} \right] + \left[\frac{\delta F}{\delta \mathbf{v}}, \frac{\delta G}{\delta \mathbf{v}} \right] \cdot \left(\frac{\mathbf{B}}{\rho} \times \nabla \times \frac{\delta G}{\delta \mathbf{B}} \right) + \frac{1}{\rho} \left(\nabla \times \frac{\delta F}{\delta \mathbf{B}} \right) \cdot \left(\frac{\delta G}{\delta \mathbf{v}} \times \mathbf{B} \right) \right\}$$

Another approach is using lie-groups. The Lie-Poisson approach is a method to construct non-canonical Poisson brackets by reducing a canonical Hamiltonian system on a large phase space (e.g., particle coordinates) to a smaller phase space of collective variables (e.g., fluid fields in MHD). It is rooted in the symmetry properties of the system's configuration space, described by a Lie group.

...

The other way is through Casimir invariants. Casimir invariants are functionals C that commute with all functionals F under the Poisson bracket: $[C, F] = 0$. They are conserved quantities that arise from the degeneracy of the non-canonical bracket and provide constraints on the system's dynamics.

Degeneracy: Non-canonical Poisson brackets are degenerate, meaning their Poisson tensor J_{ij} has a non-trivial kernel. Casimirs live in this kernel, satisfying:

$$J_{ij} \frac{\delta C}{\delta \xi_j} = 0$$

Physical Role: Casimirs represent invariants tied to the system's topology or symmetries, such as helicity in MHD, which constrain the evolution of plasma in thrusters.

Identification: Casimirs are found by solving the functional equation $[C, F] = 0$ for all F , often using the Lie algebra's cohomology.

Chapter 60

Dialog

60.1 Introduction and Scope

60.2 Politics

60.2.0.0.1 AI is a Net Positive*

60.3 Philosophy

60.3.0.0.1 Who Should I Become(part 1, Broad) Q: Before we continue, let us narrow the question down to a pieces. Come up with various categories and sub categories.

R: ok

- Presence
 - Physical Presentation
 - Facial Expressions
 - Tone of Voice

- Body Language
 - Tone of “warmth”
 - Humor and Levity
 - Professionalism
- Speech
 - Vocabulary and Precision
 - Technical Explanation
 - Academic Writing Structure
 - Cross-audience communication
 - Socratic Dialogue
- Conversation and Social Interactions
 - Core skills
 - Networking
 - Situation scripts
- Mind and Cognition
 - Cognitive Habits
 - mathematical and physical thinking
 - Problem solving
 - Decision making
 - Sustained and Consistent focus
 - Observation
 - Bayes thinking
 - Emotional Regulation
- Character...

Q: Ok, that is enough. We don't need more than that. What would you say is the most fundamental on that list?

R: Character, our morals dictate our actions, how we interpret actions, and much more. It influences everything else while being independent of everything else.

Q: Good, we have already discussed that in detail in this book, what would you say the next most fundamental? For the purposes of this exercise we will use fundamental to mean building blocks towards our conversation, not any other meaning you could describe.

R: For that, I would have to say that presence. As your goal of presence effects sets your goals for speech and conversation. Mind is largely independent, but it is also independent of the others. More orthogonal than fundamental.

Q: I accept your argument. To further constrain this conversation, let us define what you mean by presence.

R: I mean who I wish to be processed as. In a general sense.

Q: Does that fit with your above categories

R: Sadly not, but it can be useful in other situations. I would go back and reorder our conversation, but let us move ahead.

Q: Wouldn't that challenge the exercise?

R: No, the point of the beginning was simply to define what we should talk about, now we know. Furthering it could be of use, but we can do it later without such hidden truths coming to trouble in this conversation.

Q: Ah, so, now define our conversation.

R: While I find the whole purpose is to change internally for the sake of overcoming myself. Something I have overlooked is others perception of me. Thus I should examine that. Here we will define broadly who I want to be perceived to be.

Q: Why?

R: Why I haven't thought of it before or why it matters?

Q: The latter.

R: Very simply, modern society is... well a society. To succeed and achieve my broader goals I rely on others to see me, to help, to teach, and to let me lead when the time comes. It is also as I have mentioned a human's duty to be pro-social within their society. Having a defining oneself so others can see allows other to allow them to do those things much more easily.

Q: I could certainly examine that short statement further, but let us move one. for an experiment, let us start with writing some general ideas who you want to be seen as.

R: Rigorous, consiouncious(in all of its definitions), conscious, questions assumptions, actively helpful, kind, helpful, friendly, obedient, reverent, independent but still a leader, curious, broad minded, disciplined, consistent, deep, "always", precise, high standards, active, confident, purposefull, makes others more capable, composed, long-term vision, and principled.

Q: Alright, we will reconvine in the morning.

R: Alright, I have returned.

Q: Looking at the above, in broad strokes what would you say the image you want to convey to be?

R: Competent, in control, and helpful

Q: Do any constraints arise?

R: Being seen as competent, in control, and all of the related properties tends to lean towards stoic and general neutral line. While being helpful and a friendly would obviously lean towards being warm and "friendly."

Q:Remember, whenever paradoxes arise, test you assumptions, as contradictions do not exist.

R: Yes, there is no true paradox, only the question of picking

a point on the spectrum and knowing when to move about that spectrum.

Q: Why is this?

60.3.0.0.2 My use of AI Q: to question the use of AI, let us first constrain ourselves to

60.4 Physics

60.4.0.0.1 The Laws of Physics are the same everywhere

Q: To examine the claim, we must first define out terms. So, if you would please explain what you mean by the Laws of Physics and "are the same."

R: Well, that is quite simple. The Laws of physics are the underlying mechanisms and devices that govern the Universe. Whether that be symmetries, quantum fields, structure, or something else has little barring on this topic. As for are the same, simply that there exists a spacial invariants across the universe.

Q: Why does the nature of the laws of physics have little baring on the conversation at hand?

R: All of the listed above have the same observable nature by definition. It is not like how the electron behaves differently than the atom, but rather more like the quantum field fluctuations and electron. If they had easily distinguishable differences that can be examined scientifically, such discussions would not be had.

Q: Ok, I accept your premises, now on to the claim itself. How exactly would one be able to verify that the laws of physics are spatially invariant if we use our current programme to continue research into the new?

R: While I consider that it is quite difficult, and while my axiom should still be held, a general skepticism is perfered. Now how to look out for it, modern advancements allow us to see areas separated by great distances. Inconsistencies across different regions of space would be quite easy to observe. Allowing one to question held assumptions.

Q: What about areas beyond our observation?

R: No objective way to prove it, but science is not about unprovable "what-ifs," it examines observables. As mentioned, it is taken as an axiom unless falsified.

Q: Yes, until falsified. You make it sound just like another theory. Yet, is it not common that another criteria for a theory would be to adhere to this axiom?

R: Yes, just as any theory must adhere to modern scientific consensus, until reason to doubt such underlying theories.

Q: I realize something as we talk. When referring to spacial changes in physics, you take a more conservative view than I. Focusing on the idea of variable pockets of continuum with slight changes. Alas, there could be massive changes. Changes in dimensions, symmetries, or even fundamental properties like momentum transfer.

R: As for fundamental properties, they are simply consequences of more fundamental properties like dimensions and symmetries. Possible changes have been thoroughly examined so one would be able to observe them and theoretically figure it out.

Q: What if the changes are to logic and mathematics? Rather than symmetry.

R: That is absurd.

Q: Why is that?

R: No possible mechanism could possibly do that, as logic cannot be effected by mechanisms.

Q: How could you know that? Do you have omniscient knowledge?

R: No, but deductive reasoning can show that math is divorced from physical reality and thus cannot be effected by physical mechanisms. The idea of logic being capable of changing by external forces is paradoxical. If logic changes, that would simply be a consequence of other logic, thus consistent with logic. Only new.

Q: So there could be new logic that we have no knowledge of?

R: Yes, but, such idyllic daydreaming is unrealistic.

Q: Why, what makes it any less unlikely than anything else?

R: There is no possible evidence of it.

Q: Would there be evidence of it?

R: Yes, changes in the laws of physics over time or space.

Q: You assume you understand this unknown, how could you? How could you know you could identify it with any for of accuracy? What makes you think the current mysteries aren't that our current understanding it reliant of completely false axioms?

R: Such avenues are being explored.

Q: In very limited forms. Not in the forms I have been suggesting.

R: Yes, but your forms are paradoxical.

Q: How so?

R: Your forms throw out induction. This is simply Hume's problem dressed up, but the solution stays the same. One cannot move forward without some basic axiom of the invariant nature of reality over some configuration space. Primary space and time.

Q: Why can't we move forward?

R: If one questions basic logic, how could one possibly come to any sort of understanding of observations.

Q: As you most certainly know, a lack of an idea is hardly evidence that such an idea is impossible.

R: Ok, ok... If one where to doubt logic, deduction goes out the window. If deduction goes out abduction goes out. If we have no abduction we cannot trust our senses nor our insurements. Thus no evidence, ways to evaluate evidence, nor any theories to understand any evidence. Science goes down with it.

Q: Such a fragile thing science is.

R: I know where you are going. This calls into question the validity of science, hanging on by an unprovable axiom. Yet it matters not. If science works, then it helps, if it doesn't, it has helped in the past and abandoning it would not do us any better.

Q: So pragmatism becomes the answer?

R: In the question of axioms, yes.

Q: Why not everything else?

R: Once we have the basic axioms, such hand waving is unnessecary as more rigorous derivations are possible.

Q: Why are these more rigerous methods preferable?

R: We have gone all of the way from spacial invariants to the foundations of epistemology, why don't we give it a rest.

Q: Isn't this what you wanted?

R: What are you going on about, you are me. This meta-humor makes no sense, especially as no one will ever read this.

Q: Do you want someone to read this and think you clever?

R: No, at least I had not thought of it until you(I) wrote that down... I still hold it that it be a personal choice rather than

the desire for external validation.

Q: Ok... I won't push it, back to where we where. Why are more rigerous methods perfered in epistemology. If you already concede that absolute truth is unattainable(I know you haven't already outwardly have, but you will get there as you are me and this is quite long[what in the world am I doing, writing to myself?Yes, that is the whole point((Golly I sound like I had back when I started Logos[[Is that a bad thing, isn't that what you are trying to do?Alright, that is enough]])))] Anyways, wouldn't a pragmatic view be more consistent and make more sense?

R: In many cases, a pragmatic world view can lead to idyllic views where idealism leads to pragmatic outcome. In science this is obviously true.

Q: That is hardly a view in the philosophical sense, but more in the aesthetic.

R: Then I take that on. Pragmatic in epistemology, instrumentalism in pragmatic thought, and scientific realist in ideals.

Q: That is quite complicated.

R: It may be, but so is the universe.

60.4.0.0.2 Computational Models and their Accuracy*

60.4.0.0.3 Further Development in Structural preserving methods

60.4.0.0.4 Explaining Gravitational Waves Q: Explain to me gravitational waves

R: Essentially, gravitational waves, also refereed to as gravitational radiation, are small perturbations in spacetime. Mathematically equivalent to waves.

Q: What causes them?

R: It is something called quadropole moments, where quadropole refers to motion where the distribution of the stress-energy density changes while the “center of mass” remains the same. Examples would be binary orbits or CCSN.

Q: Why quadropole?

R: Quadropole moment, refers to distribution, it is a higher dimension extension than monopoles and dipoles. Monopoles referring to changes in the cumulative, such as total mass. While dipoles would be change in center of mass. Obviously, mass is conserved so any change in it would be by its nature a higher dimensional pole. A dipole movement would be uniform and thus unable to preterb spacetime in wave form. While quadropoles would be non-uniform by definition.

Q: What about higher dimensional?

R: I am not sure what higher dimensional poles would be so I would have to look that up.

Q: Before we do that, let us try to examine it without that. Think about an abstraction of what we are looking at.

R: Well, we go from the integral, to the $1/n$ of the integral, to the function itself. The general idea is on the tip of my tounge but I keep think aout it going nowhere.

Q: Ok, look it up, but then tell me about what that un-earthed.

R: Multipoles generalize “shape descriptors.” Going from a scalar that tells you the total but nothing of movement, symmetry breaking, etc. Then adding movement, then flattening or elongating(quadropoles), then three-fold. They look at the asymmetry of the asymmetry of an earlier order. Taking that all quadpoles plus define the distribution changes while quadropoles alone look at compression and elongating the structure.

Q: Great, what does this say about gravitational waves?

R: Quadrupole plus tend to have the heaviest effect while later ones add correction terms.

Q: How do the above examples give these?

R: Take a binary orbit, the distribution of mass obviously changes, the matter appears in different areas. Taking the example of flattening or elongating, it effectively does that, imagine the curvature of spacetime rather than the point particles. This makes the earlier analogy of the sphere much more clear.

60.5 Mathematics

60.6 Research Culture

60.7 Broader Culture

60.8 General Conversing

Chapter 61

Clarification

61.1 Scope and Purpose

Essentially, to write down clarifications on things I had forgotten or realized I didn't understand like I thought I had.

61.2 Physics

61.3 Math

61.3.0.0.1 Multilinear A tensor is a multilinear map of vectors and covectors. What is a multilinear mapping?

It means it takes several variables(or vectors), takes the rest as constant and maps only one, and does that for all of them.

Chapter 62

Random Stuff

62.1 Euphesus, Continuation

62.1.1 Anaysis of the Person

First things first, I must define my scope and what I am going to do. Let's start with interests, values, motivations, writing and speaking style, her friends, personality patterns, context, how these influence her interactions with me, my interactions with her, and finally moving forward.

As for interest; politics, philosophy, theology, law(spesifically criminal and constiitutional), European history, art(mostly realism with charcoal), legos, cooking, and such. I have also learned recently she is suprisingly into Lego NinjaGo. Excited to ramble about character development, meta-analysis of the show, and plot points. She generally enjoys card games, conversations, simple and easy games like catch, and surreal or absurdist humor. The humor often time juxtaposes childish tone or manner of speaking with witty creation and topic.

Career, she has a bit of a somewhat range of ideas for career. Ranging from constitutional lawyer to homicide detective, with

constitutional lawyer being at the top. Beyond that I am not sure. A general enjoyment of law is very obvious but why is less so. There are some obvious ideas such as an interest in social order and moral responsibility.

For Values, she is a devote Christian with a quite conservative outlook on the world. Generally valuing the ten commandments, respect, protestant work ethic, and liberal arts. She is quite consistent, having a structured theory of value. Seeing sin in a conservative manner, everyone is doing it constantly and that even thoughts are a sin, but people are not slaves to such sin. Through action they can control even their thoughts, even if imperfectly. Also, when discussing morality she tends to lean towards the side of virtue ethics.

General personality patterns, she is generally introverted. As mentioned earlier has a surrealist sense of humor. Often times focused on other people despite this introvertedness. Slightly more on the side of spontaneity. Pretty shy and tend to downplay her own abilities and exaggerate others. Generally focuses on being needed to the point in there is little self-introspection in wants. Enjoys my questions on morality, philosophy, science, etc; saying that it makes her think in new ways about new things.

Speaking/Writing style: starting with writing, as in texting. Tends to use short grammatically correct sentences, quite often uses emojis to convey tone, generally sends multiple short messages rather than one longer one, uses a relatively surprising amount of em and en dashes, uses exclamation points, very direct and to the point, and writes quickly. As mentioned multiple times, generally uses surreal and witty humor during most of the conversation in rapid fire. On more in-depth conversations uses condensed pseudo-paragraphs.

Motivations, we haven't specifically spoken on motivations but we have on fears. She has specifically mentioned a fear of losing inner child/imagination/creativity and forgetting what

it is like to be a kid. Citing that she feels she is already doing this in her life.

Context, she grew up in a quite conservative home in Chicago, before moving to Tennessee. Her mother suffers from cancer that debilitates her in an extreme way. Much of what Keira does is to help her mother. Her mother is very sweet but with her disabilities is mostly focused on her daughter. I don't know much about her father, he is introverted, polite, and hard-working.

Part V

Journaling

Chapter 63

***Introduction**

Chapter 64

Plans

64.1 Now

64.1.1 Self-Education

elf-education has always been a great value of mine. As mentioned earlier, I became obsessed with learning Einstein's field equation in 3-4th grade. Beyond that I have spent much of my time reading books, watching YouTube videos, and more about topics that interest me. Though, late in 8th grade, this desire to teach myself grew a great deal. Before I can go about my future goals for teaching myself I must first go over what I have already down.(This list is not full, mainly lectures finished and books from lectures, this does not include books not finished, non-lecture based video education, research papers read, projects where I learned things for the project alone, and other similar programs)

Mathematics:

- AP Calculus AB and BC - Khan Academy
- Multivariable Calculus

- Khan Academy
- 18.03 Differential Equations - MIT OpenCourseWare
- Gilbert Strang on linear algebra - MIT OpenCourseWare
- Vector Calculus - Trevor Bazett
- Tensor Analysis - eigenchris
- Tensor Calculus - eigenchris
- Crash Course in Complex Analysis - Steve Burton
- Introduction to Applied Numerical Analysis - Richard W. Hamming
- Symplectic geometry & classical mechanics - Tobias Osborne
- Lie groups, algebras, brackets - Mathemaniac (more conceptual than mathematical)
- Differential geometry - Robert Davie
- Calculus of variations - Faculty of Khan
- Working on
 - Advanced Analytic Methods in Continuum Mathematics: Fundamentals for Science and Engineering - Hung Cheng
 - Discrete Differential Geometry - CMU
 - Differential Geometry for students of Numerical Electrodynamics - Alain Bossavit

Physics:

- 8.02 Physics II - MIT OpenCourseWare

- 8.03 Physics III - MIT OpenCourseWare
- 8.033 Relativity - MIT OpenCourseWare
- General Relativity - Stanford Online
- Introduction to plasma physics - USYD senior plasma physics lectures
- Introduction to electromagnetism - Griffiths
- Computational physics - Mark Newman
- 8.224 Exploring Black Holes: General Relativity and astrophysics - MIT OpenCourseWare
- Classical Dynamics of Particles and Systems - Thornton Marion
- Introduction to cosmology - Stanford Online
- Introduction to fusion energy and plasma physics course - PPPL
- Seminar: Fusion and plasma physics - MIT OCW
- Statistical Mechanics - Stanford Online
- Hamiltonian description for magnetic field lines in fusion plasmas: A tutorial - AIP
- Fusion economics: power density, materials and maintenance - PPPL Frontiers Colloquia
- Flash-X code tutorial, a users perspective - University of Chicago(skimmed)
- Flash-X user guide - Flash-x
- Flash4 User support(skimmed)

- Radiative Processes in Astrophysical Phenomena- George B. Rybicki, Alan P. Lightman
- Goldstien Classical Mechanics Lectures - Prof. Jacob Linder
- PiTP 2016 - Institute of Advanced study
- 2024 PPPL Graduate Summer School
- Hamiltonian description of the ideal Fluid - P.J.Morrison
- Quantum Mechanics with Applications - David B. Beard & George B. Beard
- "What IS Structure, How Do You Create or Recognize It, and How Can You Use It? or Metriplectic Dynamics: Using the 4-Bracket for Constructing Thermodynamically Consistent Models." - workshop Geometric Mechanics Formulations for Continuum Mechanics
- "On an Inclusive Curvature-Like Framework for Describing Dissipation: Metriplectic 4-Bracket Dynamics." - workshop Infinite Dimensional Geometry and Fluids
- Haven't finished
 - Theoretical Physics - Georg Joos
 - Advanced MHD with Applications to Laboratory and Astrophysical Plasmas - Cambridge
 - Classical Mechanics: A Modern Perspective - Sudarshan and N. Mukunda
 - The Interpretation of Structure From Motion - Ullman, S.
 - Radiative Processes in Astrophysics - George B. Rybicki & Alan P. Lightman

- Symplectic geometry & classical mechanics - Tobias Osborne

Coding:

- Computational physics - Mark Newman
- Python Numerical Methods - Berkeley
- Applied Numerical Methods - Crice Carnahan, H.A Luther, James O.Wilkes

Other:

- Management in engineering - MITOpenCourseware
- Principles of microeconomics - MITOpenCourseware
- Dynamic leadership: using improvisation in business - OpenCourseware
- Logic 1 - OpenCourseware
- Policy for science, technology, and innovation - MITX
- Reducing The Danger Of Nuclear Weapons And Proliferation- MITOpenCourseware

64.1.2 Research

- Egg drop simulation
- Numerical Methods for 3D compressed Plasma using Lattice Boltzmann(paper written)
- Simulated Annealing for plasma thrusters
- High Altitude EMP for primordial black holes(never finished)

- Flash-x simulations:
 - Basic Sedov
 - Running simple MHD simulation of my own design
- A Hamiltonian Framework on ICF Implosions Rocket Equation Based on Rayleigh-Taylor Instabilities
- Contributed to PlasmaPY
- Relativistic Rocket Equation(Newtonian)
- Relativistic rocket equation(Hamiltonian)
- Finite difference MHD Model in Fortran
- Computes velocity based upon Navier Stokes equation adapted to magnetic and electric fields
- Outputs graphical color field of velocity
- Asteroid game
 - Classical
 - Time
 - Survival
 - Newtonian(player and asteroids have gravity)
 - Dark Matter(Invisiable gravity fields)
 - Relativistic(time dilation, spacial contraction, black holes, etc)
 - Proximal Policy Optimization “AI” played game
- Hamiltonian based plasma thruster Fortran Numerical Code
- N body simulation

- Fluid based N-body simulation
- Death Star
 - Basic gravitational ‘planet’
 - MHD Plasma beam hitting
 - Basic evaporation and kinetic energy transfer
- OpenMHD usage
- Usage Gkyl
- Omega-X project
 - Inspired by FLASH-X, attempting to build a multi-physics plasma physics codebase using Metriplectic forumism rather than Lagrangian and Newtonian
 - I have successfully implemented the 1D thermofluid metriplectic discretization onto a discontinuous Galerkin subspace using an L^2 projection. Integrated using implicit midpoint rule.
 - Working on module registry & driver pattern

64.1.3 Other activities

I also had some other activities that I have done in high school, I won't go too much in them.

- SVA
- Boy Scouts
- Church activities
- JROTC
- FCA

- OA
- Etc

The OA, Order of the Arrow, I will continue into college.

64.2 Pre-University

64.2.1 Independent Learning

One thing I need to do is nail calculus completely. I am going to be taking the AP calc BC test without taking the course.

Beyond that, I am completing differential geometry. Once done I will finish Symplectic geometry. Then lie (brackets, algebra, group, and theory)

I also want to get into functional analysis.

I should also learn C++ beyond what I have done in the past.

64.3 Undergrad

64.3.1 Classes

64.3.1.1 Requirments

64.3.1.1.1 General

I have tested out of many of the general education requirements, so here is the remaining bit:

- Written communication(pick 1)
 - Honors ethical theory
 - Haslam knowledge

- Oral communication
 - A survey in Contemporary Physics
- Arts and humanities(pick 2)
 - Haslam Knowledge
 - honors intro to philosophy
 - Professional responsibility
- Civilization(pick 2)
 - Honors development of Western civilization
 - Honors history of world civilizations
 - University Honors
 - World Religions

64.3.1.1.2 Academic Physics Major

- Prerequisites
 - Intro Computer Science
 - Differential Equations 1
 - Matrix Algebra
- Core
 - Fundamentals of modern physics
 - Mechanics 1
 - Thermal Physics
 - Electronics Lab
 - Intro to quantum mechanics
 - Modern optics
 - Electricity and Magnetism

- Survey in modern physics
 - Modern Lab
- Academic
 - Mechanics 2
 - Intro to quantum 2
 - Electricity and magnetism 2
- Recommended
 - Mathematical Methods for scientist
 - Partial Differential Equations
 - Complex Variables
- Thesis in physics

64.3.1.1.3 Applied Maths Major

- Computer Science
- Core
 - Diff equations
 - Matrix Algebra
 - Intro to abstract mathematics
 - Math proficiency
- Breath
 - Honors abstract algebra 1
 - Honors Analysis 1
 - Numerical Analysis
 - Stochastic processes

- Applied breath(physics courses)
- Depth
 - Numerical Algebra

64.3.1.1.4 Computer Science Minor

- Intro comp sci
- data structures
- Logic design
- Numerical Algorithms
- Numerical Analysis
- Numerical Algebra
- Parallel Programming

64.3.1.1.5 Electives

- PHYS 441 - Introduction to Computational Physics
- PHYS 494 - Special Topics in Physics
- 2nd PHYS 493 - Research and Independent Study
- PHYS 531 - Mechanics(Maybe, graduate)
- PHYS 541 - Electromagnetic Theory(Maybe)
- PHYS 571 - Mathematical Methods in Physics I(maybe)
- PHYS 573 Numerical methods in physics(maybe)
- PHYS 615 Astrophysics(maybe)
- PHYS 643 Computational Physics(maybe)

- MATH 462 - Differential Geometry
- MATH 467 - Honors: Topology
- MATH 536 - Partial Differential Equations II
- MATH 567 - Riemannian Geometry I
- MATH 568 - Riemannian Geometry 2
- MATH 572 - Numerical Analysis II
- MATH 577 - Optimization
- MATH 579 - Seminar in Numerical Mathematics

64.3.1.1.6 Total

- Honors ethical theory(x)
- honors intro to philosophy(x)
- Professional responsibility(x)
- Honors development of Western civilization(x)
- Honors history of world civilizations(x)
- Intro Computer Science(x)
- Differential Equations 1(x)
- Matrix Algebra(x)
- Fundamentals of modern physics(x)
- Mechanics 1(x)
- Thermal Physics(x)
- Electronics Lab(x)

- Intro to quantum mechanics(x)
- Modern optics(x)
- Electricity and Magnetism(x)
- Survey in modern physics(x)
- Modern Lab(x)
- Mechanics 2(x)
- Intro to quantum 2 x
- Electricity and magnetism 2 x
- Mathematical Methods for scientist(x)
- Partial Differential Equations x
- Complex Variables x
- Thesis in physics (x)
- Intro to abstract mathematics x
- Honors abstract algebra 1(x)
- Honors Analysis 1 (x)
- Numerical Analysis x
- Stochastic processes (x)
- Numerical Algebra(x)
- data structures(x)
- Logic design(x)
- Parallel Computing (x)

- 35(-1, -4, maybe) main(13-15 electives)
- MATH 462 - Differential Geometry(1)(x)
- MATH 467 - Honors: Topology(2)(x)
- PHYS 441 - Introduction to Computational Physics(15)
- PHYS 493 - Research and Independent Study(3)(x)
- PHYS 531 - Classical Mechanics(4)(x)
- PHYS 541 - Electromagnetic Theory(5)(x)
- PHYS 571 - Mathematical Methods in Physics I(6)(x)
- PHYS 573 - Numerical Methods in Physics(7)(x)
- MATH 515 - Analytical Applied Mathematics I(8)(x)
- MATH 567 - Riemannian Geometry I(9) x
- MATH 568 - Riemannian Geometry II(10) x
- MATH 569 - Seminar in Topology and Geometry(11)
- MATH 574 - Finite Element Methods(12)
- MATH 578 - Introduction to Scientific Computing(13) (x)
- MATH 585 - Optimal Control Theory(14)
- PHYS 599 - Seminars(x)
- PHYS 593 - Independent Study(x)

64.3.1.2 Schedule

physics general math comp sci

64.3.1.3 First Year

1. Honors ethical theory
2. Honors intro to philosophy
3. Professional Responsibility
4. HR: Western Civilization
5. HR: World civilization
6. Intro computer science
7. Diff equations 1
8. Matrix Algebra
9. Fundamentals in modern physics
10. mechanics 1
11. Intro to abstract mathematics
12. PHYS 503 - Physics Colloquium

64.3.1.4 Second Year

1. Thermal Physics
2. Electronics Lab
3. Intro to quantum mechanics
4. Modern optics

5. Electricity and Magnetism
6. Survey in modern physics
7. Modern Lab
8. Mechanics 2
9. Intro to quantum 2
10. Electricity and magnetism 2
11. Partial Differential Equations
12. Numerical Analysis

64.3.1.5 Third year

1. Honors abstract algebra 1
2. Honors Analysis 1
3. Stochastic Processes
4. Numerical Algebra
5. Mathematical Methods in physics
6. Research & independent study
7. Differential Geometry
8. Honors: Topology
9. Classical Mechanics, grad

10. PHYS 541 - Electromagnetic Theory, grad
11. MATH 515 - Analytical Applied Mathematics I
12. Independent study and research

64.3.1.6 Fourth year

1. Thesis in physics
2. MATH 567 - Riemannian Geometry I
3. MATH 568 - Riemannian Geometry II
4. Numerical methods in physics
5. data structures
6. Logic design
7. Parallel Computing
8. PHYS 599 - Seminars
9. PHYS 513 - Problems in Theoretical Physics I
10. MATH 578 - Introduction to Scientific Computing
11. MATH 534 - Calculus of Variations
12. MATH 537 - Mathematical Principles of Continuum Mechanics I

64.3.1.7 Others

1. PHYS 514 - Problems in Theoretical Physics II
2. MATH 585 - Optimal Control Theory
3. MATH 574 - Finite Element Methods
4. PHYS 615 - Astrophysics and Cosmology(grad required)
5. PHYS 643 - Computational Physics(grad required)
6. MATH 515 - Analytical Applied Mathematics I
7. MATH 571 - Numerical Analysis I
8. MATH 572 - Numerical Analysis II

64.3.1.8 Possible summer

1. Graduate Reading in Mathematics
2. Research and independent study
3. graduate reading in physics
4. Special Problems

64.3.2 Goals for the first three weeks

1. Talk to Dr. Mezz
2. Create LaTeX libraries for class
3. Organize schedule template

as for organizations I am interested in

1. Society of Physics Students()
2. Book to build
3. Students for Philanthropy
4. Academic quizbowl
5. Alpha Omega Campus Ministry
6. American Nuclear Society
7. Canoe and Hiking club()
8. Christian Student Fellowship
9. Christian Volunteers()
10. Climbing club()
11. Cycling club
12. Data science
13. Forum on science, ethics, and policy
14. Hema at UT
15. Nuclear Managment
16. Martial Arts club
17. Model UN
18. Naturalist club
19. Out in STEM
20. Pickle ball
21. Astronomy
22. Caving

64.3.3 First Week

I want to walk to and time walk between and time from

- Physics
- Humanities
- Business
- Health and Recreation
- Recreational fitness
- Rocky top dining
- Stokely Hall
- Volenteer Hall

I would also like to explore the library, Wesleyan house,

I want to try biking around the area

I am required to go to Authoring your story, rocky top road map, and torch night

I would consider going to dining scavenger hunt and brunch

I will work on my schedule to come up with something more detailed to my current model.

64.4 *Grad

64.5 *Early career

64.6 *Late Career

Chapter 65

Schedule Templates

65.1 Preliminary Semester 1

Time	Monday	Tuesday	Wednesday	Thursday
6:00				
6:30				
7:00				
7:30				
8:00				
8:30				
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10:00				

Time	Friday	Saturday	Sunday
6:00			
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10:00			

Chapter 66

Journal

This section will hold daily journaling. Essentially it will be day-to-day notes and plans. These will then be synthesized into the previous chapters in this book.

66.1 2025-11-27

66.2 Emacs

This is my first journal write using EMACS, for that has been my project at the moment. Creating a doom Emacs with Org-mode enhancements, AUCTeX/pdf tools. Org-roam(graph+server), Org-bable for multiple languages(FORTRAN, python, Julia, Jupyter, and shell), LSP for coding, Magit(for git), projectile/treemacs(for using projects), a daily journal that auto inserts into my $\LaTeX$ book, and create Org nodes and graphs for chapters and journals.

For the most part I have this set up. I need to edit AUCTeX so that it uses latexmc rather than latex, so that it will run multiple times for the book-like structure. I think I have more

dependencies I need for the languages. I still have more to learn to get comfortable with this program. Also, I am not sure that the program auto inserts my journals.

```
print("2+2 =", str(2+2))
```

66.3 2025-11-29

66.4 Axiom

As is very obvious, I am making a transfer from now on. I will have this section for day-to-day journaling and notes, while the previous sections for more concise and developed thought. I won't change the already made stuff. This new This takes me to Emacs, I have figured out how to incorporate Org files into my $\text{\LaTeX}$ book.

I do need to work on attaching it to the table of context. I believe I should have this completed by the end of the day.

66.5 Emacs

I am learning much about how to use Emacs, as I have mentioned above I have been working to better understand and use Emacs. I have created two new function's. One to make IDs for Org files written outside of Emacs and also to turn Org files into $\text{\LaTeX}$ to use.

66.6 Media

66.6.1 Zettelkasten

I was originally having this section be a Zettelkasten-like notes taking. Though after more consideration, I have reconsidered. While a similar linking will be useful, I am unlikely going to have the same extreme modularity that is accustomed to the system. Thus, I am unlikely to use it fully, but some use is likely.

66.7 2025-12-03

66.8 Links to Book Chapters

- `coding` `Axiom` `roam:Omega-X` `roam:plans` `roam:personal_analysis`

66.9 Emacs

66.9.1 Emacs updates

For this journal, remember that the command is `SPC n r d t`. Beyond that I have fully completed the `Org` $\rightarrow$ `LaTeX` file. Though, as useful as these programs can be it does raise the question of if I should use it. I have begun to question its usefulness.

66.9.2 Emacs vs Vim

I have found that as I have created these things, I have realized I can substitute a lot of the programs by simple scripting and

vim. While Vim’s speed and ease of use has its use. I do think the scripting of Emacs it also quite fast, ease of moving between files, and even the spell check. Beyond that the integration of things coding languages is potentially very useful once I gain the hang of it.

66.9.3 **Org vs Tex**

One question I have been looking into is whether I should write my notes directly in $\LaTeX$ or continue to write in an Org file and export to $\LaTeX$ after the fact. **Org**

Pros	Cons
Org babel to run commands	Easy errors
Org-capture to quickly make notes	whitespace inconsistencies
Graph view	Hard to write math with
Easy connections	Difficulty with more complex $\LaTeX$
Easy to use syntax	Degregation of performance over time
Easy to use Hierarchy	

Tex

Pros	Cons
Absolute control	more verbose
none of the easy errors	Have to script new stuff
Can be scripted to simulate Org-capture-like	No babel
More freedom of use in $\LaTeX$ commands	

66.9.4 Future

Currently my plan is to have continue to use Org for journaling, though I may revisit this idea later on.

With this I will continue to learn how to use Org/L^AT_EX integration and use. Also some of syntax of Org. I also need to learn more about running code and using it in Emacs. I currently only use it for notes.

- note-book enviornment(like jupiter)
- build-up the graph view
- Agenda
- linking
- task automation

66.10 Omega-X

66.10.1 Arguments

Currently, I have the poor practice of setting variables in a module and ignoring using arguments or such things. I plan to rectify this. This is for several reasons; to protect from errors and to allow for MPI(my next step).

66.10.2 Updates

I need to continue on a couple of sets. Create a IO files, finish the evolution driver, and have ways to analyze the data.

66.11 Future Ideas

I have a couple ideas for future projects:

66.11.1 Categorical Metriplectic

Using category theory to set up a fully and complete formal theory of the Metriplectic system. This is highly ambitious considering my limited knowledge in category theory, but it is a field that is thoroughly explored and thus I believe it is within my reach to accomplish.

66.11.2 Cason Brackets Dynamics(name pending)

I believe I can integrate metriplectic systems to Nambu mechanics. As far as I know this is a completely novel goal and thus would be my own. Let's see with the names: Cason Formalism, Cason Mechanics, Cason dynamics, Cason framework, Cason Analysis, Casonian Dynamics, Casonian Mechanics, Cason's Method, Cason's Dynamics, Casonic Dynamics, The Cason Principle, Cason Bracket Dynamics, Cason system, Cason Framework for hybrid Dynamics. I know this seems overtly narcissistic, trying hard to come up with a name based upon my own; I don't really care.

66.11.3 Metriplectic PINN

I have read several papers using Metriplectic dynamics to create PINN(Physics-informed-neural-networks).

66.12 Media

66.12.1 Star Trek

I forgot how much I like Star Trek next Gen, especially Picard.

66.13 Self-analysis

66.13.1 Stress & TV

For some reason, watching TV where the show lets you in on how stupid an idea is and yet the character does it stresses me out. Especially with lying and illicit activity. When they lie and it spirals out, then they lie more. I don't like it. Messes with me.

66.14 2025-12-04

66.15 Links to books

- plans

66.16 Projects

I have recently come into more free time and would like to use it to my advantage to increase my abilities and capacity; though, I am having trouble considering what to do, in what order, and how to block my time.

66.16.1 Formal projects

The actual projects I want to work on are continuing:

- FORTRAN Omega-X project([Omega-X]finish it, develop parallel computing, develop TUI, continue documentation, profile it, add higher order methods, more time stepping methods, add modules for working with additional systems, data analysis, visualization, GPU acceleration, etc)
- Cason Bracket dynamics: synthesis Nambu and Metriplectic and incorporate into Omega-X
- Categorical Theory of Metriplectic Dynamics: Self explanatory
- Metriplectic PINN: PINN based upon PINNs
- Emacs Building: I want to build workflows and other things in Emacs to make stuff easier for me(coding workflow, see data, writing book stuff)

66.16.2 Skills

There are several skills I want to continue to develop:

- Julia coding: right now in the basics
- C++: similar to Julia
- Emacs: I want to get more comfortable with the key-bindings and creating commands to make things simple
- Parallel computing: most clearly can gain this through my Omega-X project
- Org/TeX: Learning how to use both org and tex so they work together.

66.16.3 Plan

I am thinking schedule it:

1. develop a plan for the day in Org, experiment with Org/ $\text{\LaTeX}$ integration to gain that experience in that regard.
 - Incorporate Julia whenever possible
2. Theoretic study - work on one a day
 - Categorical Theory of Metriplectic Dynamics
 - Cason Bracket Dynamics
3. Programming Project - work on one maybe two a day
 - Develop Omega-X project
 - finish it
 - develop parallel computing
 - develop TUI
 - continue documentation
 - profile it
 - add higher order methods
 - more time stepping methods
 - add modules for working with additional systems
 - data analysis, visualization
 - GPU acceleration
 - Develop Emacs tools
 - Create Julia REPL in Emacs and keybinding to make it simple
 - custom $\text{\LaTeX}$ keys
 - Set FORTRAN build and run
4. Skill Development - work on one or two a day

- Julia
 - symbolic Poisson/Nambu brackets solver
 - ODE solver using the SciML interface
 - GPU-accelerated metriplectic evolution
 - Category theory for dynamics
- C++
 - Simple library
 - SIMD-accelerated
 - learn expression templates
 - GPU version using CUDA or HIP
 - Plug-in architecture
 - simple MHD solver
- FORTRAN
 - Coarrays
 - Polymorphic OOP with type-bound procedures
 - Generics (Fortran 2023)
 - MPI
 - GPU Offloading
 - Memory alignment + SIMD-aware design

66.17 2025-12-05

66.18 Links to Book Chapters

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66.19 Plans for Today:

I have been currently working on my school work; getting caught up with that. I have also contacted NSCC about the dual en-

rollment program to make sure I am not missing anything.

For my next steps:

- Programming Project: Omega-X
 - First: I am going to build a type binding for Omega-X's initialize subdriver.
 - Second: formalize `declarevars` module to work for this new update
 - Third: Move C++ binding to a safe place(C++/FORTRAN?[either fits])
- Theoretic study: Catogoric Metriplectic:
 - Create $\text{\LaTeX}$ notes sections
 - Write list of possible papers to read
 - Read one and write notes
- Programming/skill: EMACS building/Julia
 - Develop Julia REPL in EMACS
 - Play around with it once in

66.19.1 Omega-X Notes

As I mentioned in the above section my first task is to develop a type system for `driverinit.F90`. To create the types:

```
module types_make
  !individual types
  type :: name_of_type_1
    real :: variable_in_type_1
  end type name_of_type_1
```

```
type :: name_of_type_2
  real :: variable_in_type_2
end type name_of_type_2

! higher level type
type :: higher_level_type
  type(name_of_type_1)
  type(name_of_type_2)
end type higher_level_type
end module types_make
```

The to use it:

```
program use_types
  use types_make

  type(higher_level_type) :: H_T_L

  ! initialize fields
  H_T_L_of_type_2%variable_in_type_2 = 1.0
```

I ended up doing less than I imagined. I created the type system and corrected the arguments for many things. My next steps should be to develop the

1. declare_{vars}
2. fix uses of modules
3. Change python code to work with new type arguments.

This is what I can do tommorow.

66.19.2 Emacs Build

While working on Omega-X I figured I should focus on FORTRAN integration before Julia, I also saw a cool calculator thing

I will set up and shouldn't take long.

Calculator:

- Get in M-x calc or C-x * or M-x full-calc
- It is in reverse polish notation: operators follow their operands
- t to get to trail
- i to enter
- u to undo
- DD to redo
- TAD to swithc
- DEL to delete one

There is also a ~~El~~TeX embeded system:

- C-x * e → Enter calc-embedded mode.
- C-x * r → Replace the marked expression with its result.
- C-x * q → Quick evaluation of inline expressions.

calc-embedded C-x * e Enter embedded mode on the current region (or insert delimiters if none). calc-embedded-word C-x * w Evaluate the word at point as a Calc expression. calc-embedded-activate C-x * a Activate embedded formulas in the current buffer. calc-embedded-new-formula C-x * n Insert a new embedded formula at point. calc-embedded-select C-x * s Select the next embedded formula in the buffer. calc-embedded-edit C-x * e (when inside formula) Edit the embedded formula in Calc. calc-embedded-update C-x * r Replace/update the formula with its evaluated result. calc-embedded-update-all C-x * u Update all embedded formulas in the buffer.

calc-embedded-clear C-x * c Clear (remove) the result part of an embedded formula. calc-embedded-clear-all C-x * C Clear results for all formulas in the buffer. calc-embedded-duplicate C-x * d Duplicate the current formula. calc-embedded-next C-x * n Jump to the next embedded formula. calc-embedded-previous C-x * p Jump to the previous embedded formula. calc-embedded-reload C-x * l Reload Calc settings for embedded mode.

Now I have created the FORTRAN stuff:

1. Fortran-lang LSP
2. Fortran-indent improvements and compile.el
3. custom build commands
4. Org babel

66.19.3 Categorical Metreplectic

66.20 2026-01-03

66.21 Links to Book Chapters

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66.22 Emacs

66.22.1 TABS

I have recently added the tab like system. Here are the commands: New tab tab-new C-x t 2 Close tab tab-close C-x t 0 Switch tab tab-next / tab-previous C-x t o / C-x t O Rename

tab tab-rename M-x tab-rename Duplicate tab tab-duplicate
M-x tab-duplicate

So that's that.

66.23 2026-02-05

66.24 Links to Book Chapters

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66.25 Plans

As I mention in “Self-development” at this time I am working to return to normal. For this I have to re-establish and create habits to be the person I want to be.

66.26 Schedule

66.26.1 9:00 - 11:00

- Wake-up
- Prepare for the day
- Hair-cut

66.26.2 11:00 - 1:00

- Exercise
- Eat Lunch

66.26.3 1:00 - 4:00

- Do Calculus review exam

66.26.4 4:00 - 5:00

- Read
 - The Karamazov Brothers
 - Numerical Hamiltonian

66.26.5 5:00 - 5:45

- Eat

66.26.6 5:45 - 6:00

- Math discussion post

66.26.7 6:00 - 7:00

- Project
 - Numerical Hamiltonian
 - Website
 - * One idea is to create a kinda to do website based upon my schedule of that day.
 - Writing
 - Other

66.26.8 7:30 - 8:30

- Blank period for extra time to push things back

66.26.9 8:30 - 10:00

- Watch TV

66.27 2026-02-16**66.28 Bible Reading****66.28.1 Genesis 41:17-42:17**

God protects Joseph through establishing gifts within him. He does the same for us.

Why does Joseph test his brothers?

66.28.2 Matthew 13:24-46

The parables of Jesus, establish stories and myth as the methodology of developing the abstract concepts of God's will.

It refers to the weeds as everything that causes sin and all who do evil, what exactly does this mean? Everyone does evil and everyone has within that which tempts them. Taking out the "weeds" prematurely will destroy the wheat.

The final stories is about how the kingdom of God gives everything for the grace of those who are lost.

66.28.3 Psalm 18:1-15

Love of God

66.28.4 Proverbs 4:1-6

Community, family, and God fosters wisdom. Wisdom that protects you.

66.29 Today

Today was a good day. While I wasn't feeling my absolute best, I was productive. Not as in I did much work, but I was able to actively engage in things I enjoy. I wrote a large section on the fundamentals of metaphysics, some other things, got back into reading the bible, and read from Numerical Hamiltonian Problems.

The main thing was the writing, it has been a while sense I have been able to have sustained effort on such a complex and in-depth subject. Much of my stuff is rederived, but that is kinda the point.

66.30 2026-02-17

66.31 Bible reading

66.31.1 Old testament

Why is Joseph testing with them even after hearing what they had to say while also giving them money

66.31.2 New Testament

The kingdom of God takes in all, but not all remain.

Interesting how such a radical take on ones word is used.

66.31.3 Psalm

God rescues you and protects you from the enemies that are greater than you

66.31.4 Proverbs

Pursue wisdom over all other virtues

66.32 Ideas for projects

While I am still not in the headspace for in-depth research like before. I can still do things.

66.33 Schedule**66.33.1 Coding**

- Simple Julia
- Play around with parallel computing
- Learn some optimization

66.33.2 Writing

- Continue Axiom Project
- More on the metaphysical implications of duelist frame works
- More physics writing outside of metaphysics

- Deriving basic equations in other formalism's(Nambu, metriplectic, etc)
 - Classifications of poisson manifolds
 - Understanding of Lie-Poisson reductions
 - Chaos theory and integrability
 - Numerical theory
 - Infinite dimensional Hamiltonian systems
 - Noether's theorem
 - * Understanding on metaphysical lens
 - * understanding mathematically
 - * General relativity breaking it
 - Constructing 4-bracket metriplectic
 - Metriplectic VS Generic
 - Contact Geometry
- Review papers

66.33.3 Math/Physics

- continue reading my books

66.34 April 14

Today has been somewhat successful. What I need to do is focus on simply things that are productive and enjoyable. I will break it into a couple of parts here.

Things I need to do:

- School work

- Fall Fellowship

Simple projects

- LaTeX sty file to make writing easier
- Abbreviation in emacs to make writing easier

Reading/videos

- Bible
- Relativistic Quantum Field Theory
- Numerical Hamiltonian Problems
- Handbook on the Philosophy of physics
- Advanced GR by Wandu